In the clinic

Hiatus hernia

At the level of the esophageal hiatus, the diaphragm may be lax, allowing the fundus of the stomach to herniate into the posterior mediastinum (Fig. 4.136). This typically causes symptoms of acid reflux. Ulceration may occur and may produce bleeding and anemia. The diagnosis is usually made by barium studies or endoscopy. Treatment in the first instance is by medical management, although surgery may be necessary.

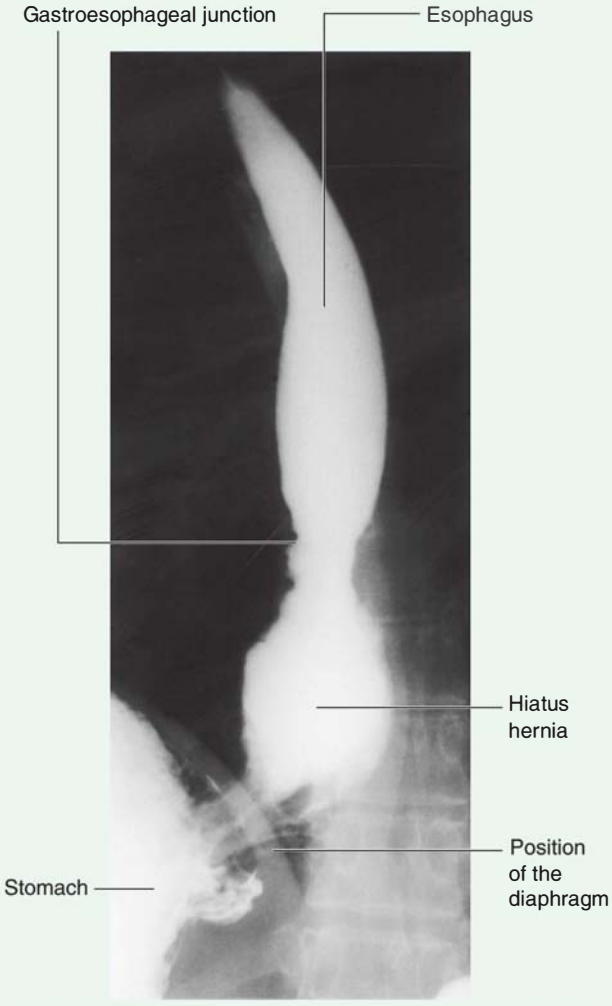


Fig. 4.136 Lower esophagus and upper stomach showing a hiatus hernia. Radiograph using barium.

Viscera

Kidneys

The bean-shaped kidneys are retroperitoneal in the posterior abdominal region (Fig. 4.137). They lie in the extraperitoneal connective tissue immediately lateral to the vertebral column. In the supine position, the kidneys extend from approximately vertebra TXII superiorly to vertebra LIII inferiorly, with the right kidney somewhat lower than the left because of its relationship with the liver. Although they are similar in size and shape, the left kidney is a longer and more slender organ than the right kidney, and nearer to the midline.

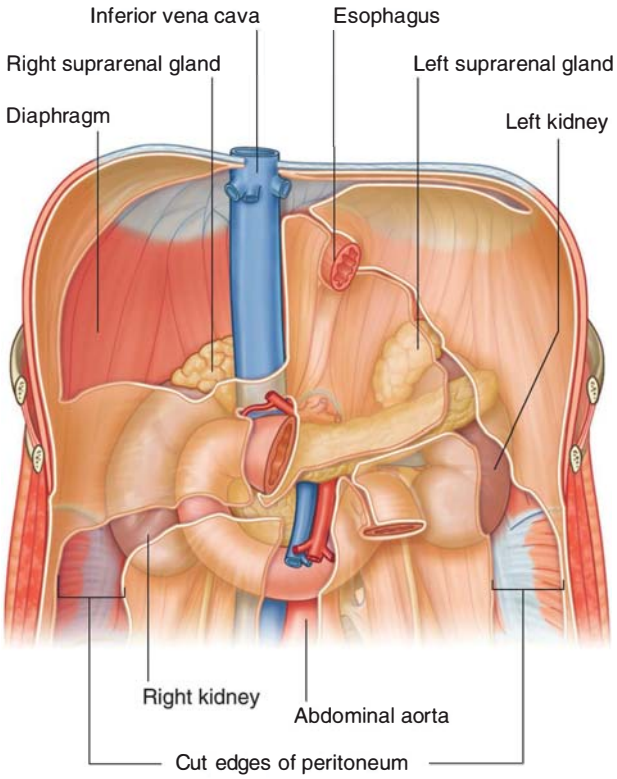


Fig. 4.137 Retroperitoneal position of the kidneys in the posterior abdominal region.

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Relationships to other structures

The anterior surface of the right kidney is related to numerous structures, some of which are separated from the kidney by a layer of peritoneum and some of which are directly against the kidney (Fig. 4.138):

- A small part of the superior pole is covered by the right suprarenal gland.
- Moving inferiorly, a large part of the rest of the upper part of the anterior surface is against the liver and is separated from it by a layer of peritoneum.
- Medially, the descending part of the duodenum is retroperitoneal and contacts the kidney.
- The inferior pole of the kidney, on its lateral side, is directly associated with the right colic flexure and, on its medial side, is covered by a segment of the intraperitoneal small intestine.

The anterior surface of the left kidney is also related to numerous structures, some with an intervening layer of peritoneum and some directly against the kidney (Fig. 4.138):

- A small part of the superior pole, on its medial side, is covered by the left suprarenal gland.
- The rest of the superior pole is covered by the intraperitoneal stomach and spleen.
- Moving inferiorly, the retroperitoneal pancreas covers the middle part of the kidney.
- On its lateral side, the lower half of the kidney is covered by the left colic flexure and the beginning of the

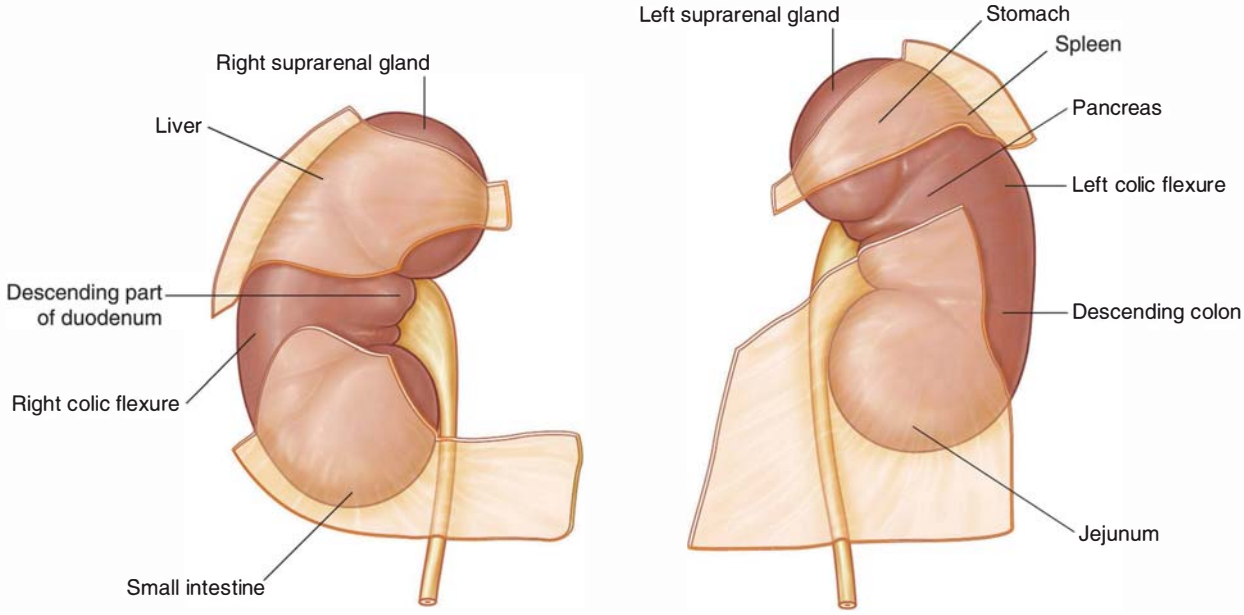


Fig. 4.138 Structures related to the anterior surface of each kidney.

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descending colon, and, on its medial side, by the parts of the intraperitoneal jejunum.

Posteriorly, the right and left kidneys are related to similar structures (Fig. 4.139). Superiorly is the diaphragm and inferior to this, moving in a medial to lateral direction, are the psoas major, quadratus lumborum, and transversus abdominis muscles.

The superior pole of the right kidney is anterior to rib XII, while the same region of the left kidney is anterior to ribs XI and XII. The pleural sacs and specifically, the costodiaphragmatic recesses therefore extend posterior to the kidneys.

Also passing posterior to the kidneys are the subcostal vessels and nerves and the iliohypogastric and ilio-inguinal nerves.

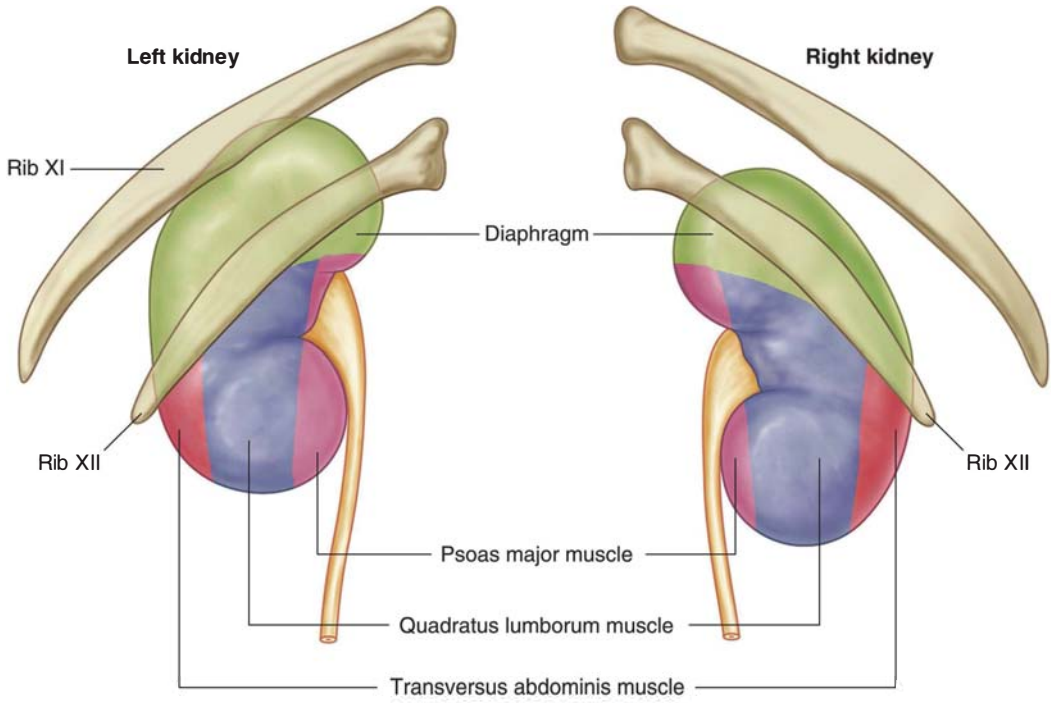


Fig. 4.139 Structures related to the posterior surface of each kidney.

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Renal fat and fascia

The kidneys are enclosed in and associated with a unique arrangement of fascia and fat. Immediately outside the renal capsule, there is an accumulation of extraperitoneal fat—the **perinephric fat** (**perirenal fat**), which completely surrounds the kidney (Fig. 4.140). Enclosing the perinephric fat is a membranous condensation of the extraperitoneal fascia (the **renal fascia**). The suprarenal glands are also enclosed in this fascial compartment, usually separated from the kidneys by a thin septum. The renal fascia must be incised in any surgical approach to this organ.

At the lateral margins of each kidney, the anterior and posterior layers of the renal fascia fuse (Fig. 4.140). This fused layer may connect with the transversalis fascia on the lateral abdominal wall.

Above each suprarenal gland, the anterior and posterior layers of the renal fascia fuse and blend with the fascia that covers the diaphragm.

Medially, the anterior layer of the renal fascia continues over the vessels in the hilum and fuses with the connective tissue associated with the abdominal aorta and the inferior vena cava (Fig. 4.140). In some cases, the anterior layer may cross the midline to the opposite side and blend with its companion layer.

The posterior layer of the renal fascia passes medially between the kidney and the fascia covering the quadratus lumborum muscle to fuse with the fascia covering the psoas major muscle.

Inferiorly, the anterior and posterior layers of the renal fascia enclose the ureters.

In addition to perinephric fat and the renal fascia, a final layer of **paranephric fat (pararenal fat)** completes the fat and fascias associated with the kidney (Fig. 4.140). This fat accumulates posterior and posterolateral to each kidney.

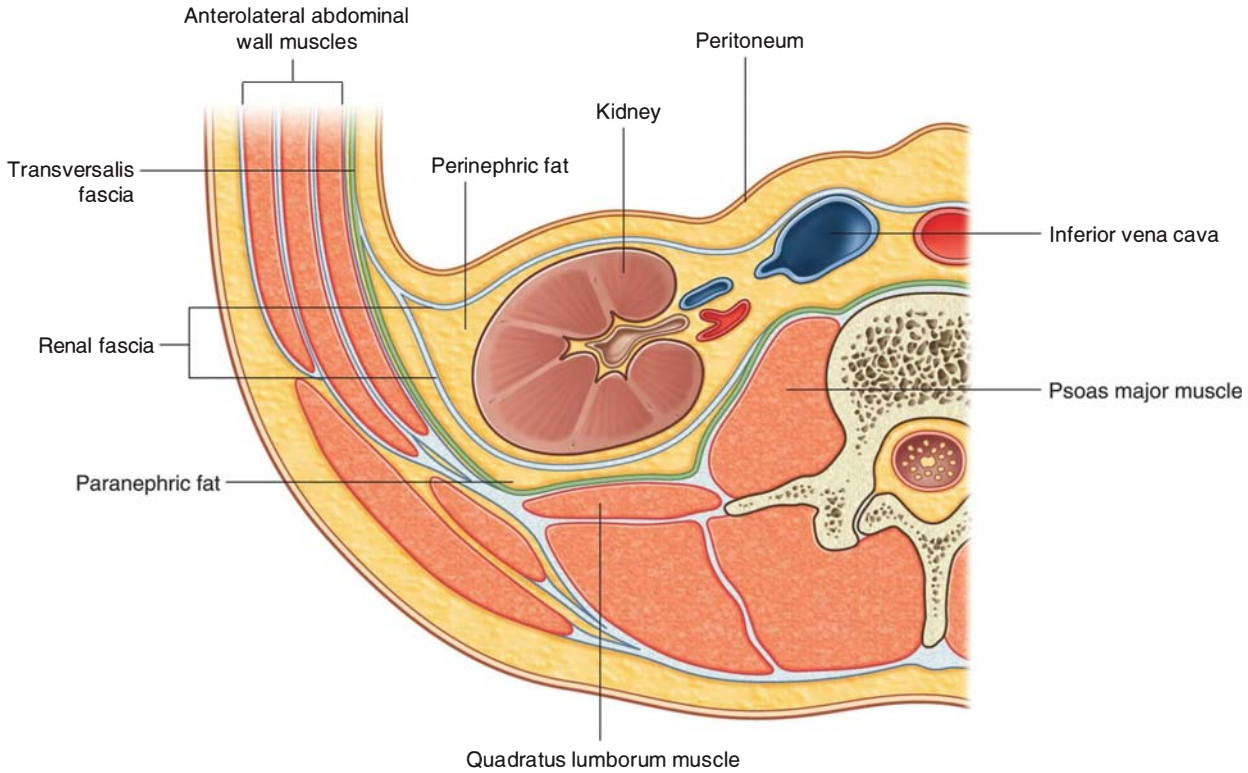


Fig. 4.140 Organization of fat and fascia surrounding the kidney.

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Kidney structure

Each kidney has a smooth anterior and posterior surface covered by a fibrous capsule, which is easily removable except during disease.

On the medial margin of each kidney is the **hilum of the kidney**, which is a deep vertical slit through which renal vessels, lymphatics, and nerves enter and leave the substance of the kidney (Fig. 4.141). Internally, the hilum is continuous with the renal sinus. Perinephric fat continues into the hilum and sinus and surrounds all structures.

Each kidney consists of an outer **renal cortex** and an inner renal medulla. The renal cortex is a continuous band of pale tissue that completely surrounds the renal medulla. Extensions of the renal cortex (the **renal columns**) project into the inner aspect of the kidney, dividing the renal medulla into discontinuous aggregations of triangular-shaped tissue (the **renal pyramids**).

The bases of the renal pyramids are directed outward, toward the renal cortex, while the apex of each renal pyramid projects inward, toward the **renal sinus**. The apical projection (**renal papilla**) is surrounded by a **minor calyx**.

The minor calices receive urine and represent the proximal parts of the tube that will eventually form the ureter (Fig. 4.141). In the renal sinus, several minor calices unite to form a **major calyx**, and two or three major calices unite to form the **renal pelvis**, which is the funnel-shaped superior end of the ureters.

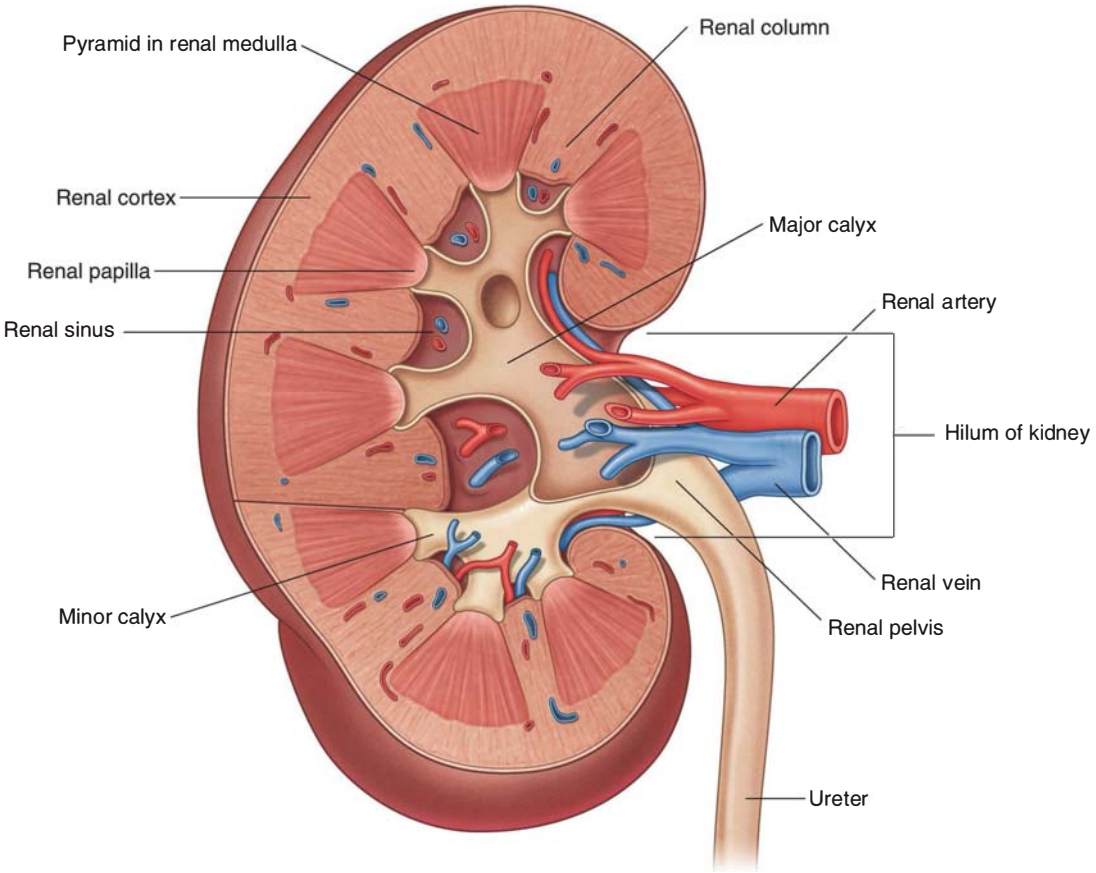


Fig. 4.141 Internal structure of the kidney.

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Renal vasculature and lymphatics

A single large **renal artery**, a lateral branch of the abdominal aorta, supplies each kidney. These vessels usually arise just inferior to the origin of the superior mesenteric artery between vertebrae LI and LII (Fig. 4.142). The **left renal artery** usually arises a little higher than the right, and the **right renal artery** is longer and passes posterior to the inferior vena cava.

As each renal artery approaches the renal hilum, it divides into anterior and posterior branches, which supply the renal parenchyma. Accessory renal arteries are common. They originate from the lateral aspect of the abdominal aorta, either above or below the primary renal arteries, enter the hilum with the primary arteries or pass directly into the kidney at some other level, and are commonly called **extrahilar arteries**.

Multiple renal veins contribute to the formation of the **left** and **right renal veins**, both of which are anterior to the renal arteries (Fig. 4.142). Importantly, the longer left renal vein crosses the midline anterior to the abdominal aorta and posterior to the superior mesenteric artery and can be compressed by an aneurysm in either of these two vessels.

The lymphatic drainage of each kidney is to the **lateral aortic (lumbar) nodes** around the origin of the renal artery.

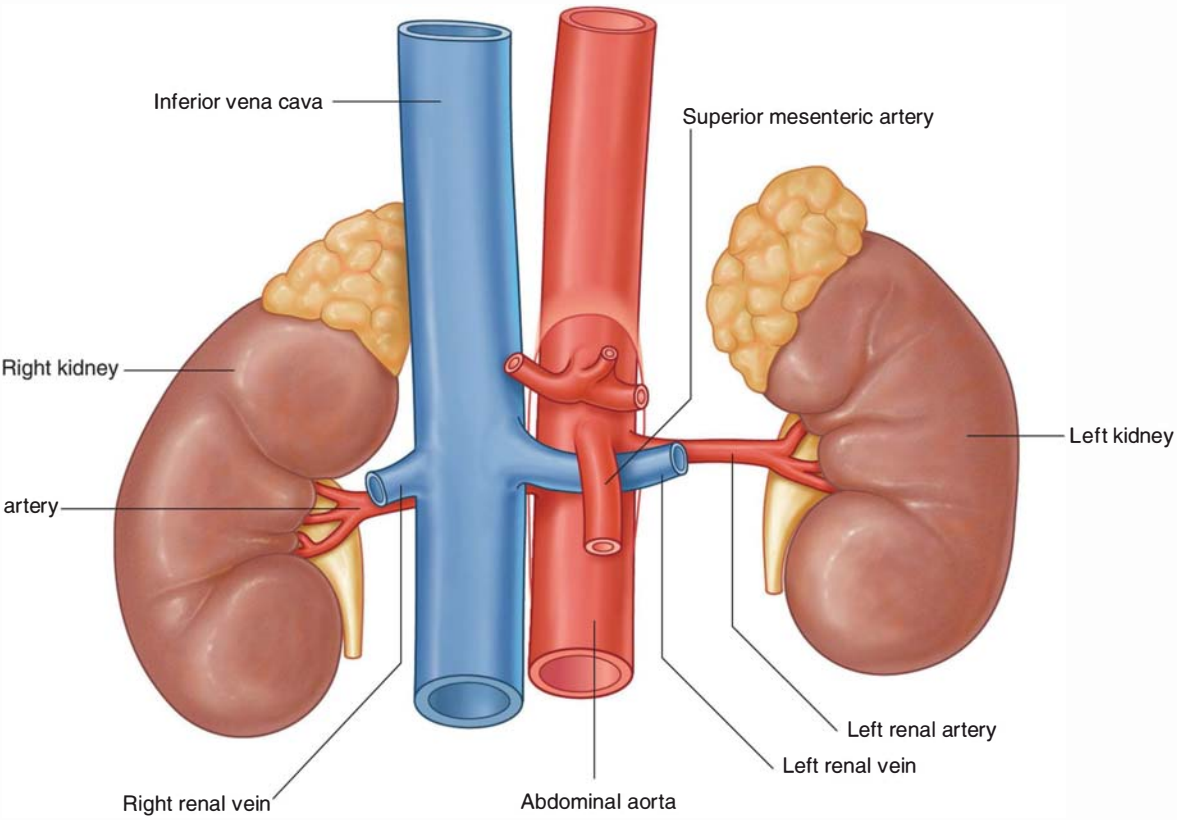


Fig. 4.142 Renal vasculature.

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Ureters

The ureters are muscular tubes that transport urine from the kidneys to the bladder. They are continuous superiorly with the renal pelvis, which is a funnel-shaped structure in the renal sinus. The renal pelvis is formed from a condensation of two or three major calices, which in turn are formed by the condensation of several minor calices (see Fig. 4.141). The minor calices surround a renal papilla.

The renal pelvis narrows as it passes inferiorly through the hilum of the kidney and becomes continuous with the ureter at the **ureteropelvic junction** (Fig. 4.143). Inferior to this junction, the ureters descend retroperitoneally on the medial aspect of the psoas major muscle. At the pelvic brim, the ureters cross either the end of the common iliac artery or the beginning of the external iliac artery, enter the pelvic cavity, and continue their journey to the bladder.

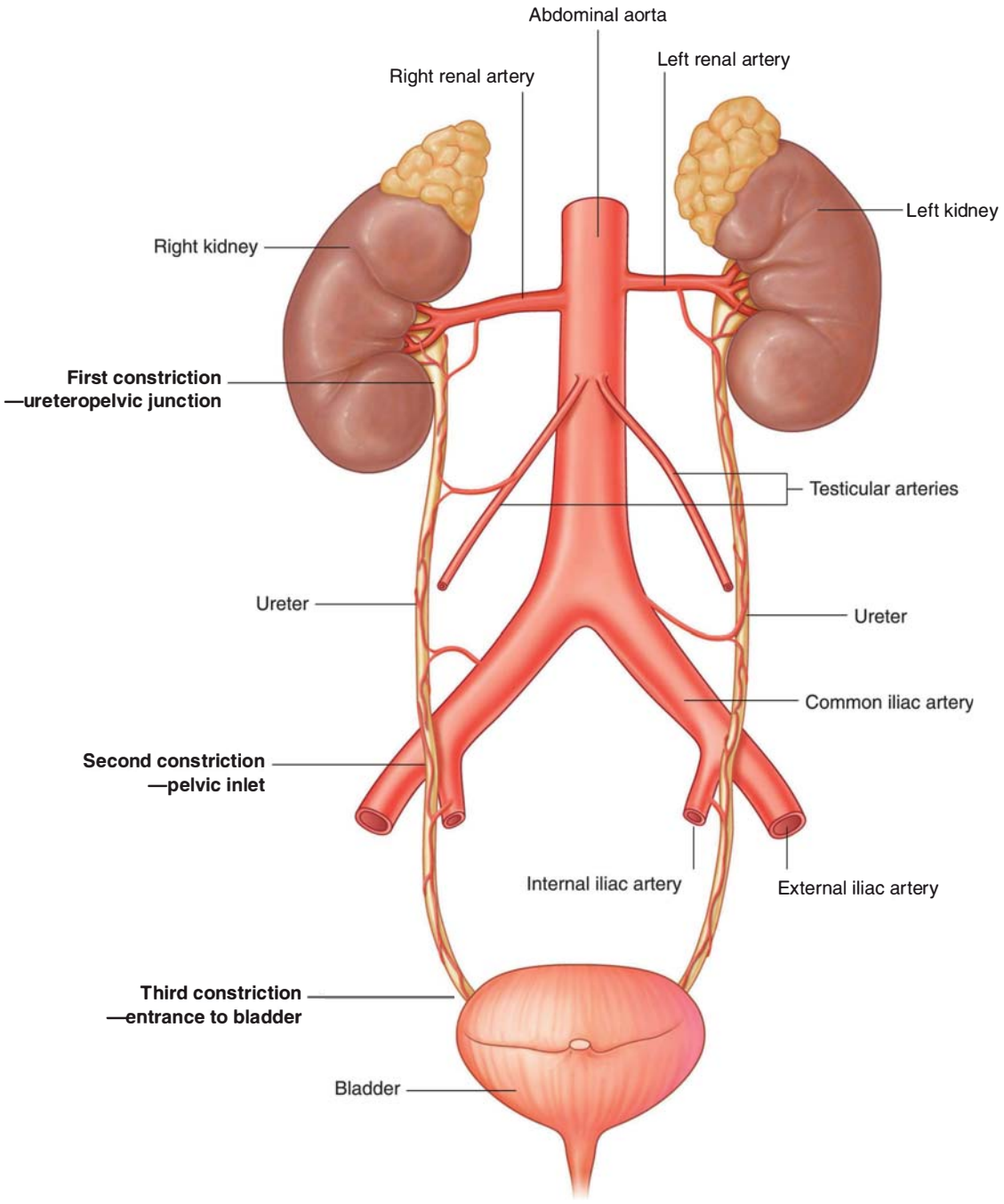


Fig. 4.143 Ureters.

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At three points along their course the ureters are constricted (Fig. 4.143):

- The first point is at the ureteropelvic junction.
- The second point is where the ureters cross the common iliac vessels at the pelvic brim.
- The third point is where the ureters enter the wall of the bladder.

Kidney stones can become lodged at these constrictions.

Ureteric vasculature and lymphatics

The ureters receive arterial branches from adjacent vessels as they pass toward the bladder (Fig. 4.143):

- The renal arteries supply the upper end.
- The middle part may receive branches from the abdominal aorta, the testicular or ovarian arteries, and the common iliac arteries.
- In the pelvic cavity, the ureters are supplied by one or more arteries from branches of the internal iliac arteries.

In all cases, arteries reaching the ureters divide into ascending and descending branches, which form longitudinal anastomoses.

Lymphatic drainage of the ureters follows a pattern similar to that of the arterial supply. Lymph from:

- the upper part of each ureter drains to the lateral aortic (lumbar) nodes,
- the middle part of each ureter drains to lymph nodes associated with the common iliac vessels, and
- the inferior part of each ureter drains to lymph nodes associated with the external and internal iliac vessels.

Ureteric innervation

Ureteric innervation is from the renal, aortic, superior hypogastric, and inferior hypogastric plexuses through nerves that follow the blood vessels.

Visceral efferent fibers come from both sympathetic and parasympathetic sources, whereas visceral afferent fibers return to T11 to L2 spinal cord levels. Ureteric pain, which is usually related to distention of the ureter, is therefore referred to cutaneous areas supplied by T11 to L2 spinal cord levels. These areas would most likely include the posterior and lateral abdominal wall below the ribs and above the iliac crest, the pubic region, the scrotum in males, the labia majora in females, and the proximal anterior aspect of the thigh.

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In the clinic

Urinary tract stones

Urinary tract stones (calculi) occur more frequently in men than in women, are most common in people aged between 20 and 60 years, and are usually associated with sedentary lifestyles. The stones are polycrystalline aggregates of calcium, phosphate, oxalate, urate, and other soluble salts within an organic matrix. The urine becomes saturated with these salts, and small variations in the pH cause the salts to precipitate.

Typically the patient has pain that radiates from the infrascapular region (loin) into the groin, and even into the scrotum or labia majora. Blood in the urine (**hematuria**) may also be noticed.

Infection must be excluded because certain species of bacteria are commonly associated with urinary tract stones.

The complications of urinary tract stones include infection, urinary obstruction, and renal failure. Stones may also develop within the bladder and produce marked irritation, causing pain and discomfort.

The diagnosis of urinary tract stones is based upon history and examination. Stones are often visible on abdominal radiographs. Special investigations include:

- ultrasound scanning, which may demonstrate the dilated renal pelvis and calices when the urinary system is obstructed; and
- an intravenous urogram, which will demonstrate the obstruction, pinpoint the exact level, and enable the surgeon to plan a procedure to remove the stone if necessary.

In the clinic

Urinary tract cancer

Most tumors that arise in the kidney are renal cell carcinomas. These tumors develop from the proximal tubular epithelium. Approximately 5% of tumors within the kidney are transitional cell tumors, which arise from the urothelium of the renal pelvis. Most patients typically have blood in the urine (hematuria), pain in the infrascapular region (loin), and a mass.

Renal cell tumors (Figs. 4.144 and 4.145) are unusual because not only do they grow outward from the kidney, invading the fat and fascia, but they also spread into the renal vein. This venous extension is rare for any other type of tumor, so, when seen, renal cell carcinoma should be suspected. In addition, the tumor may spread along the renal vein and into the inferior vena cava, and in rare cases can grow into the right atrium across the tricuspid valve and into the pulmonary artery.

Treatment for most renal cancers is surgical removal, even when metastatic spread is present, because some patients show regression of metastases.

Transitional cell carcinoma arises from the urothelium. The urothelium is present from the calices to the urethra

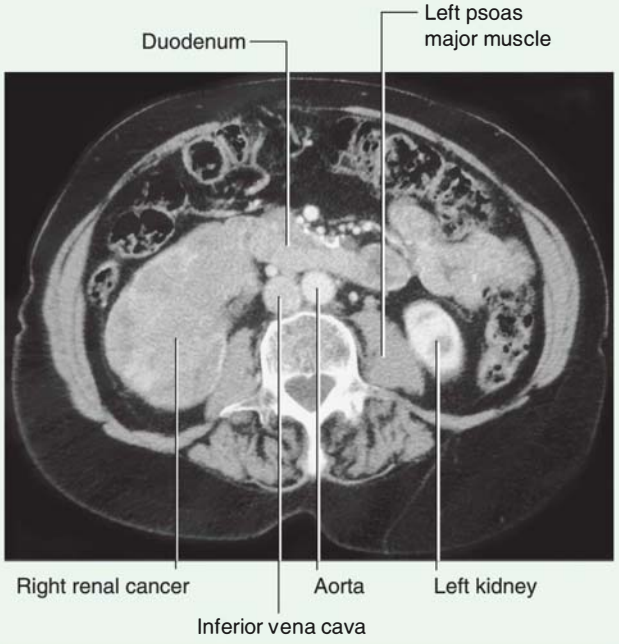


Fig. 4.144 Tumor in the right kidney growing toward, and possibly invading, the duodenum. Computed tomogram in the axial plane.

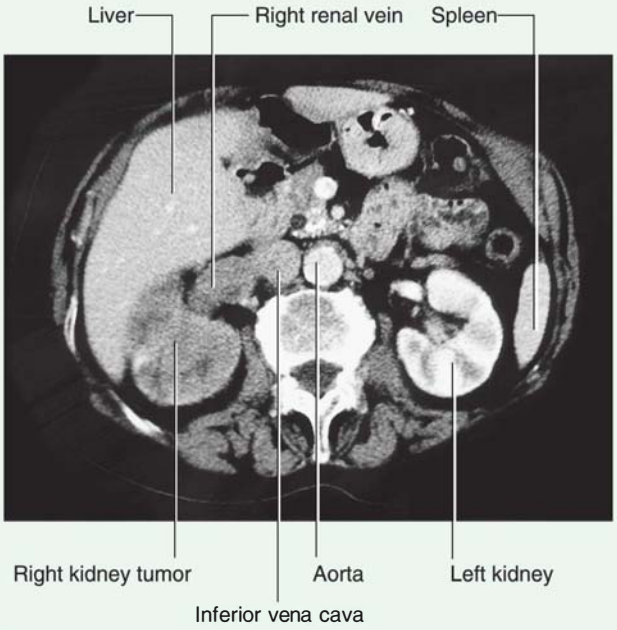


Fig. 4.145 Tumor in the right kidney spreading into the right renal vein. Computed tomogram in the axial plane.

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In the clinic—cont'd

and behaves as a "single unit." Therefore, when patients develop transitional carcinomas within the bladder, similar tumors may also be present within upper parts of the urinary tract. In patients with bladder cancer, the whole of the urinary tract must always be investigated to exclude the possibility of other tumors (Fig. 4.146).

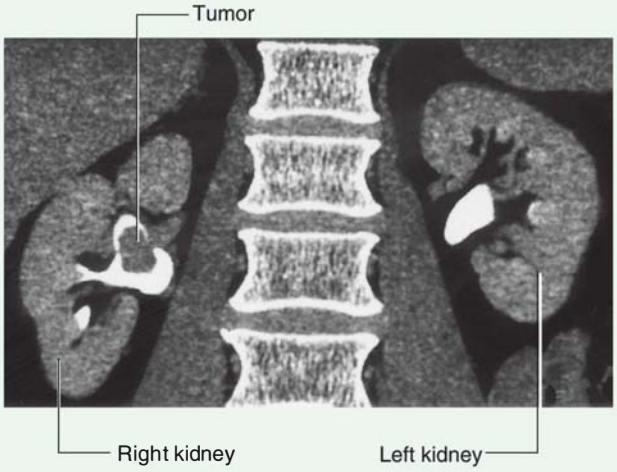


Fig. 4.146 Transitional cell carcinoma in the pelvis of the right kidney. Coronal computed tomogram reconstruction.

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In the clinic

Nephrostomy

A nephrostomy is a procedure where a tube is placed through the lateral or posterior abdominal wall into the renal cortex to lie within the renal pelvis. The function of this tube is to allow drainage of urine from the renal pelvis through the tube externally (Fig. 4.147).

The kidneys are situated on the posterior abdominal wall, and in thin healthy subjects may be only up to 2 to 3 cm from the skin. Access to the kidney is relatively straightforward, because the kidney can be easily visualized under ultrasound guidance. Using local anesthetic, a needle can be placed, under ultrasound direction, through the skin into the renal cortex and into the renal pelvis. A series of wires and tubes can be passed through the needle to position the drainage catheter.

The indications for such a procedure are many. In patients with distal ureteric obstruction the back pressure of urine within the ureters and the kidney significantly impairs the function of the kidney. This will produce renal failure and ultimately death. Furthermore, a dilated obstructed system is also susceptible to infection. In many cases, there is not only obstruction producing renal failure but also infected urine within the system.

In the clinic—cont'd



Fig. 4.147 This radiograph demonstrates a double-J stent (anteroposterior view). The superior aspect of the double-J stent is situated within the renal pelvis. The stent passes through the ureter, describing the path of the ureter, and the tip of the double-J stent is projected over the bladder, which appears as a slightly dense area on the radiograph.

In the clinic

Kidney transplant

Kidney transplantation began in the United States in the 1950s. Since the first transplant, the major problem for kidney transplantation has been tissue rejection. A number of years have passed since this initial procedure and there have been significant breakthroughs in transplant rejection medicine. Renal transplantation is now a common procedure undertaken in patients with end-stage renal failure.

Transplant kidneys are obtained from either living or deceased donors. The living donors are carefully assessed, because harvesting a kidney from a normal healthy individual, even with modern-day medicine, carries a small risk.

Deceased kidney donors are brain dead or have suffered cardiac death. The donor kidney is harvested with a small cuff of aortic and venous tissue. The ureter is also harvested.

(continues)

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In the clinic—cont'd

An ideal place to situate the transplant kidney is in the left or the right iliac fossa (Fig. 4.148). A curvilinear incision is made paralleling the iliac crest and pubic symphysis. The external oblique muscle, internal oblique muscle, transversus abdominis muscle, and transversalis fascia are divided. The surgeon identifies the parietal peritoneum but does not enter the peritoneal cavity. The parietal peritoneum is medially retracted to reveal the external iliac artery, external iliac vein, and bladder. In some instances the internal iliac artery of the recipient is mobilized and anastomosed directly as an end-to-end procedure onto the renal artery of the donor kidney. Similarly the internal iliac vein is anastomosed to the donor vein. In the presence of a small aortic cuff of tissue the donor artery is anastomosed to the recipient external iliac artery and similarly for the venous anastomosis. The ureter is easily tunneled obliquely through the bladder wall with a straightforward anastomosis.

The left and right iliac fossae are ideal locations for the transplant kidney because a new space can be created without compromise to other structures. The great advantage of this procedure is the proximity to the anterior abdominal wall, which permits easy ultrasound visualization of the kidney and Doppler vascular assessment. Furthermore, in this position biopsies are easily obtained. The extraperitoneal approach enables patients to make a swift recovery.

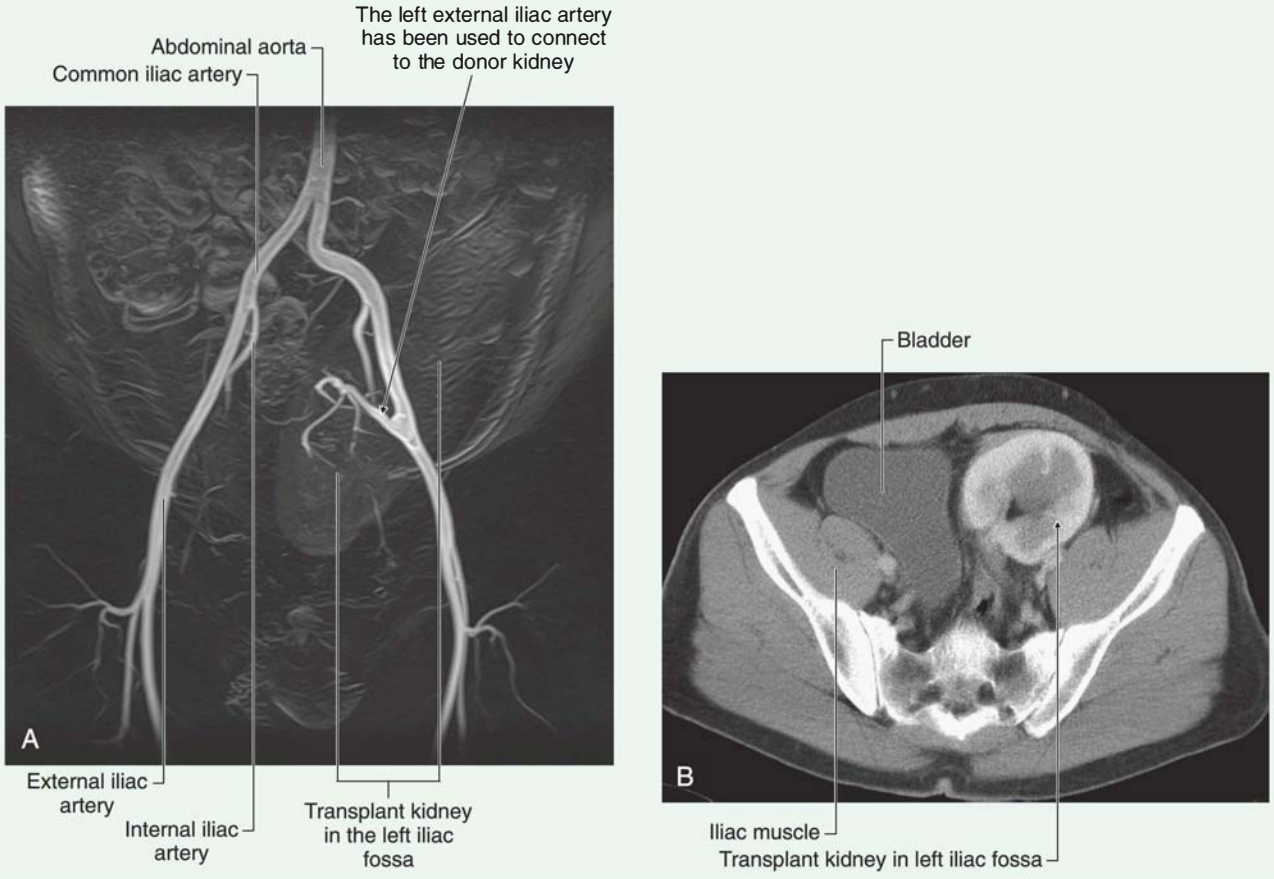


Fig. 4.148 Kidney transplant. **A.** This image demonstrates an MR angiogram of the bifurcation of the aorta. Attaching to the left external iliac artery is the donor artery for a kidney that has been transplanted into the left iliac fossa. **B.** Abdominal computed tomogram, in the axial plane, showing the transplanted kidney in the left iliac fossa.

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In the clinic

Investigation of the urinary tract

After an appropriate history and examination of the patient, including a digital rectal examination to assess the prostate in men, special investigations are required.

IVU (intravenous urogram)

An IVU is one of the most important and commonly carried out radiological investigations (Fig. 4.149). The patient is injected with iodinated contrast medium. Most contrast media contain three iodine atoms spaced around a benzene ring. The relatively high atomic number of iodine compared to the atomic number of carbon, hydrogen, and oxygen, attenuates the radiation beam. After intravenous injection, contrast media are excreted predominantly by glomerular filtration, although some are secreted by the renal tubules. This allows visualization of the collecting system as well as the ureters and bladder.

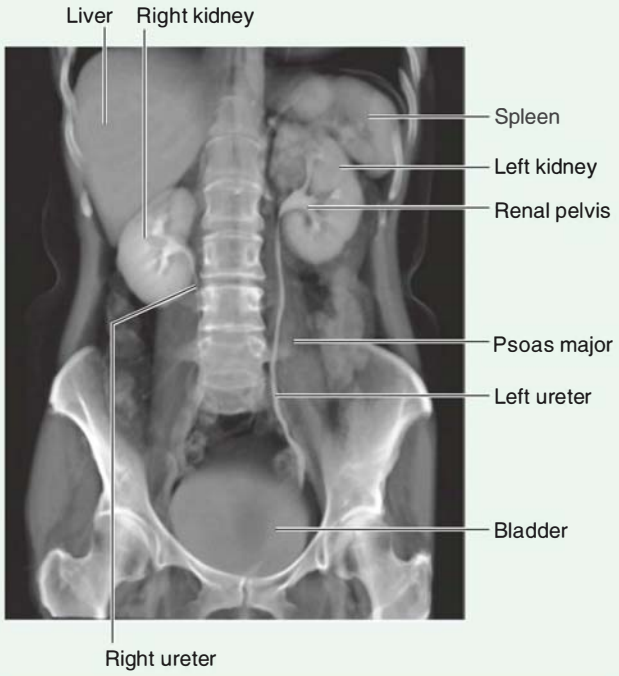


Fig. 4.149 Coronal view of 3-D urogram using multidetector computed tomography.

Ultrasound

Ultrasound can be used to assess kidney size and the size of the calices, which may be dilated when obstructed. Although the ureters are poorly visualized using ultrasound, the bladder can be easily seen when full. Ultrasound measurements of bladder volume can be obtained before and after micturition.

Computed tomography

Computed tomography can be used to assess the kidneys, ureters, bladder, and adjacent structures and is a powerful tool for staging primary urinary tract tumors.

Nuclear medicine

Nuclear medicine is an extremely useful tool for investigating the urinary tract because radioisotope compounds can be used to estimate renal cell mass and function and assess the parenchyma for renal scarring. These tests are often very useful in children when renal scarring and reflux disease is suspected.

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Suprarenal glands

The suprarenal glands are associated with the superior pole of each kidney (Fig. 4.150). They consist of an outer cortex and an inner medulla. The right gland is shaped like a pyramid, whereas the left gland is semilunar in shape and the larger of the two.

Anterior to the right suprarenal gland is part of the right lobe of the liver and the inferior vena cava, whereas anterior to the left suprarenal gland is part of the stomach, pancreas, and, on occasion, the spleen. Parts of the diaphragm are posterior to both glands.

The suprarenal glands are surrounded by the perinephric fat and enclosed in the renal fascia, though a thin septum separates each gland from its associated kidney.

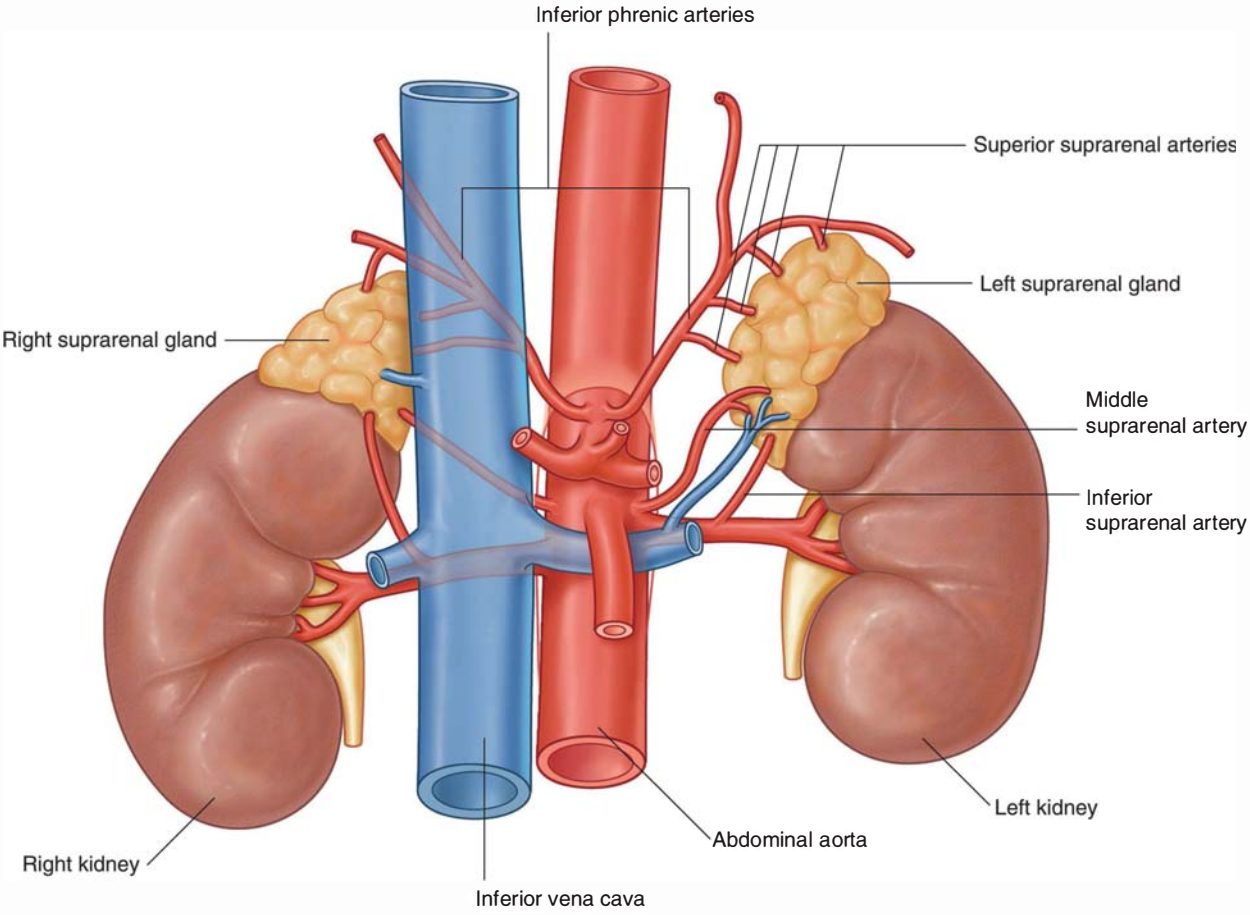


Fig. 4.150 Arterial supply to the suprarenal glands.

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Suprarenal vasculature

The arterial supply to the suprarenal glands is extensive and arises from three primary sources (Fig. 4.150):

- As the bilateral inferior phrenic arteries pass upward from the abdominal aorta to the diaphragm, they give off multiple branches (superior suprarenal arteries) to the suprarenal glands.
- A middle branch (middle suprarenal artery) to the suprarenal glands usually arises directly from the abdominal aorta.
- Inferior branches (inferior suprarenal arteries) from the renal arteries pass upward to the suprarenal glands.

In contrast to this multiple arterial supply is the venous drainage, which usually consists of a single vein leaving the hilum of each gland. On the right side, the **right suprarenal vein** is short and almost immediately enters the inferior vena cava, while on the left side, the **left suprarenal vein** passes inferiorly to enter the left renal vein.

Vasculature

Abdominal aorta

The abdominal aorta begins at the aortic hiatus of the diaphragm as a midline structure at approximately the lower level of vertebra TXII (Fig. 4.151). It passes downward on the anterior surface of the bodies of vertebrae LI to LIV, ending just to the left of midline at the lower level of vertebra LIV. At this point, it divides into the **right** and **left common iliac arteries**. This bifurcation can be visualized on the anterior abdominal wall as a point approximately 2.5 cm below the umbilicus or even with a line extending between the highest points of the iliac crest.

As the abdominal aorta passes through the posterior abdominal region, the prevertebral plexus of nerves and ganglia covers its anterior surface. It is also related to numerous other structures:

- Anterior to the abdominal aorta, as it descends, are the pancreas and splenic vein, the left renal vein, and the inferior part of the duodenum.

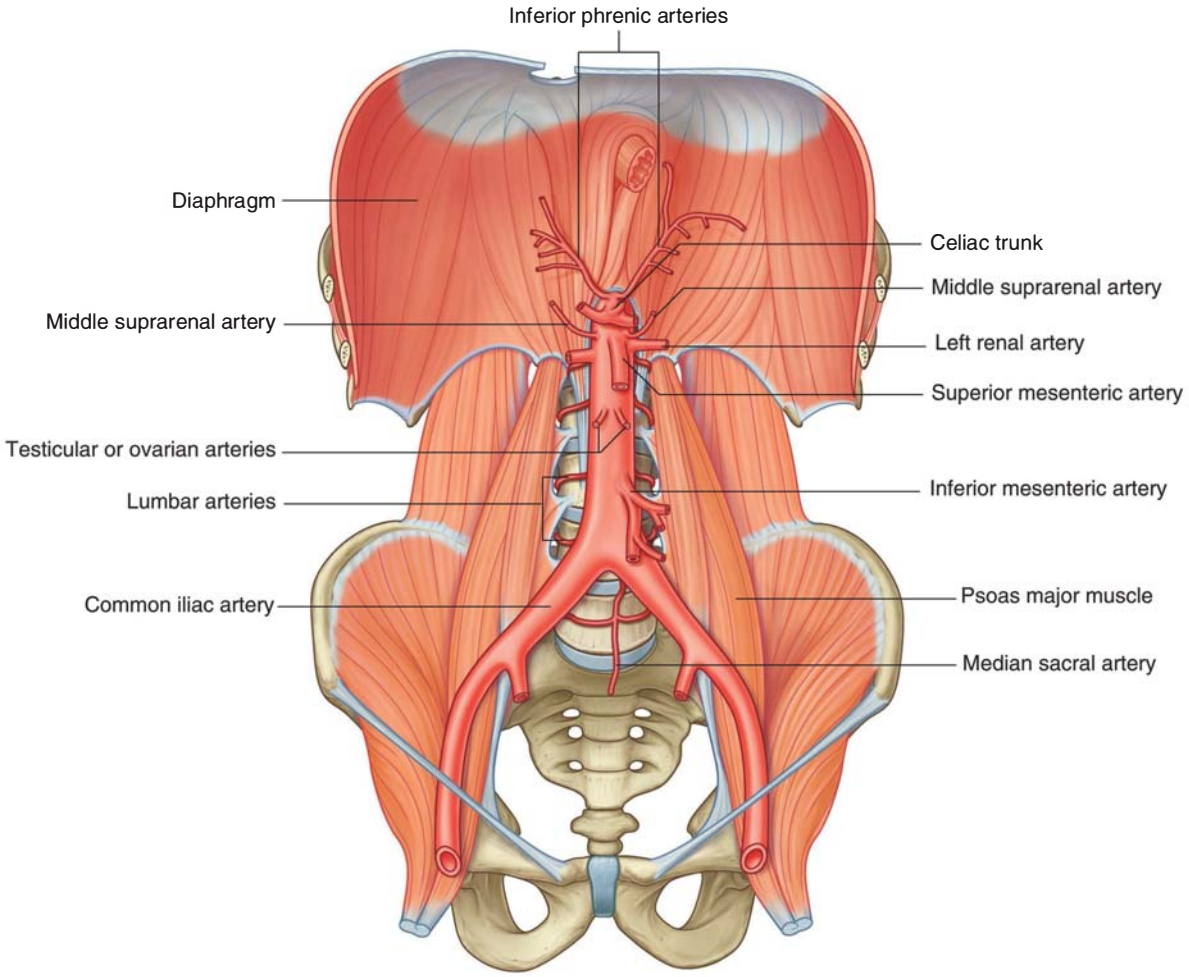


Fig. 4.151 Abdominal aorta.

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- Several lumbar veins cross it posteriorly as they pass to the inferior vena cava.
- On its right side are the cisterna chyli, thoracic duct, azygos vein, right crus of the diaphragm, and the inferior vena cava.
- On its left side is the left crus of the diaphragm.

Branches of the abdominal aorta (Table 4.3) can be classified as:

- visceral branches supplying organs,
- posterior branches supplying the diaphragm or body wall, or
- terminal branches.

Visceral branches

The visceral branches are either unpaired or paired vessels.

The three unpaired visceral branches that arise from the anterior surface of the abdominal aorta (Fig. 4.151) are:

- the celiac trunk, which supplies the abdominal foregut,
- the superior mesenteric artery, which supplies the abdominal midgut, and
- the inferior mesenteric artery, which supplies the abdominal hindgut.

The paired visceral branches of the abdominal aorta (Fig. 4.151) include:

- the middle suprarenal arteries—small, lateral branches of the abdominal aorta arising just above the renal arteries that are part of the multiple vascular supply to the suprarenal gland;
- the renal arteries—lateral branches of the abdominal aorta that arise just inferior to the origin of the superior mesenteric artery between vertebrae LI and LII, and supply the kidneys; and
- the testicular or ovarian arteries—anterior branches of the abdominal aorta that arise below the origin of the renal arteries, and pass downward and laterally on the anterior surface of the psoas major muscle.

Posterior branches

The posterior branches of the abdominal aorta are vessels supplying the diaphragm or body wall. They consist of the inferior phrenic arteries, the lumbar arteries, and the median sacral artery (Fig. 4.151).

Inferior phrenic arteries

The **inferior phrenic arteries** arise immediately inferior to the aortic hiatus of the diaphragm either directly from the abdominal aorta, as a common trunk from the abdominal aorta, or from the base of the celiac trunk (Fig. 4.151). Whatever their origin, they pass upward, provide some arterial supply to the suprarenal gland, and continue onto the inferior surface of the diaphragm.

Lumbar arteries

There are usually four pairs of **lumbar arteries** arising from the posterior surface of the abdominal aorta

Table 4.3 Branches of the abdominal aorta

Artery	Branch	Origin	Parts supplied
Celiac trunk	Anterior	Immediately inferior to the aortic hiatus of the diaphragm	Abdominal foregut
Superior mesenteric artery	Anterior	Immediately inferior to the celiac trunk	Abdominal midgut
Inferior mesenteric artery	Anterior	Inferior to the renal arteries	Abdominal hindgut
Middle suprarenal arteries	Lateral	Immediately superior to the renal arteries	Suprarenal glands
Renal arteries	Lateral	Immediately inferior to the superior mesenteric artery	Kidneys
Testicular or ovarian arteries	Paired anterior	Inferior to the renal arteries	Testes in male and ovaries in female
Inferior phrenic arteries	Lateral	Immediately inferior to the aortic hiatus	Diaphragm
Lumbar arteries	Posterior	Usually four pairs	Posterior abdominal wall and spinal cord
Median sacral artery	Posterior	Just superior to the aortic bifurcation, passes inferiorly across lumbar vertebrae, sacrum, and coccyx	
Common iliac arteries	Terminal	Bifurcation usually occurs at the level of LIV vertebra	

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(Fig. 4.151). They run laterally and posteriorly over the bodies of the lumbar vertebrae, continue laterally, passing posterior to the sympathetic trunks and between the transverse processes of adjacent lumbar vertebrae, and reach the abdominal wall. From this point onward, they demonstrate a branching pattern similar to a posterior intercostal artery, which includes providing segmental branches that supply the spinal cord.

Median sacral artery

The final posterior branch is the **median sacral artery** (Fig. 4.151). This vessel arises from the posterior surface of the abdominal aorta just superior to the bifurcation and passes in an inferior direction, first over the anterior surface of the lower lumbar vertebrae and then over the anterior surface of the sacrum and coccyx.

In the clinic

Abdominal aortic stent graft

An abdominal aortic aneurysm is a dilatation of the aorta and generally tends to occur in the infrarenal region (the region at or below the renal arteries). As the aorta expands, the risk of rupture increases, and it is now generally accepted that when an aneurysm reaches 5.5 cm or greater an operation will significantly benefit the patient.

With the aging population, the number of abdominal aortic aneurysms is increasing. Moreover, with the increasing use of imaging techniques a number of abdominal aortic aneurysms are identified in asymptomatic patients.

For many years the standard treatment for repair was an open operative technique, which involved a large incision from the xiphoid process of the sternum to the symphysis pubis and dissection of the aneurysm. The aneurysm was excised and a tubular woven graft was sewn into place. Recovery may take a number of days, even weeks, and most patients would be placed in the intensive care unit after the operation.

Further developments and techniques have led to a new type of procedure being performed to treat abdominal aortic aneurysms—the endovascular graft. The idea of placing the graft into the aortic aneurysm and lining the dilated vessel is not new and was first described over 10 years ago. Since the original description the devices have been modified on a number of occasions (Fig. 4.152).

The technique involves surgically dissecting the femoral artery below the inguinal ligament. A small incision is made in the femoral artery and the preloaded compressed graft with metal support struts is passed on a large catheter into the abdominal aorta through the femoral artery. Using X-ray for guidance the graft is opened, lining the inside of the aorta. Limb attachments are made to the graft that extend into the common iliac vessels. This bifurcated tube device effectively excludes the abdominal aortic aneurysm.

This type of device is not suitable for all patients. Patients who receive this device do not need to go to the intensive care unit. Many patients leave the hospital within 24 to 48 hours. Importantly, this device can be used for patients who were deemed unfit for open surgical repair.

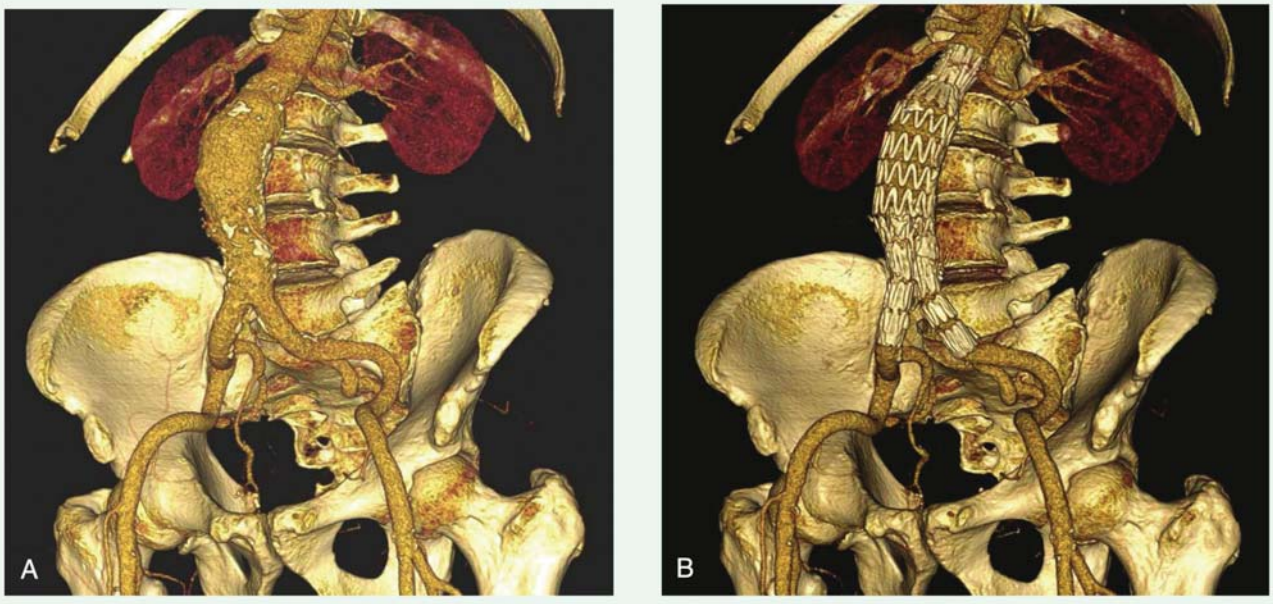


Fig. 4.152 Volume-rendered reconstruction using multidetector computed tomography of patient with an infrarenal abdominal aortic aneurysm before (**A**) and after (**B**) endovascular aneurysm repair. Note the image only demonstrates the intraluminal contrast and not the entire vessel. White patches in the aorta represent intramural calcium.

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Inferior vena cava

The inferior vena cava returns blood from all structures below the diaphragm to the right atrium of the heart. It is formed when the two common iliac veins come together at the level of vertebra LV, just to the right of midline. It ascends through the posterior abdominal region anterior to the vertebral column immediately to the right of the abdominal aorta (Fig. 4.153), continues in a superior direction, and leaves the abdomen by piercing the central tendon of the diaphragm at the level of vertebra TVIII.

During its course, the anterior surface of the inferior vena cava is crossed by the right common iliac artery, the root of the mesentery, the right testicular or ovarian artery, the inferior part of the duodenum, the head of the pancreas, the superior part of the duodenum, the bile duct, the portal vein, and the liver, which overlaps and on occasion completely surrounds the vena cava (Fig. 4.153).

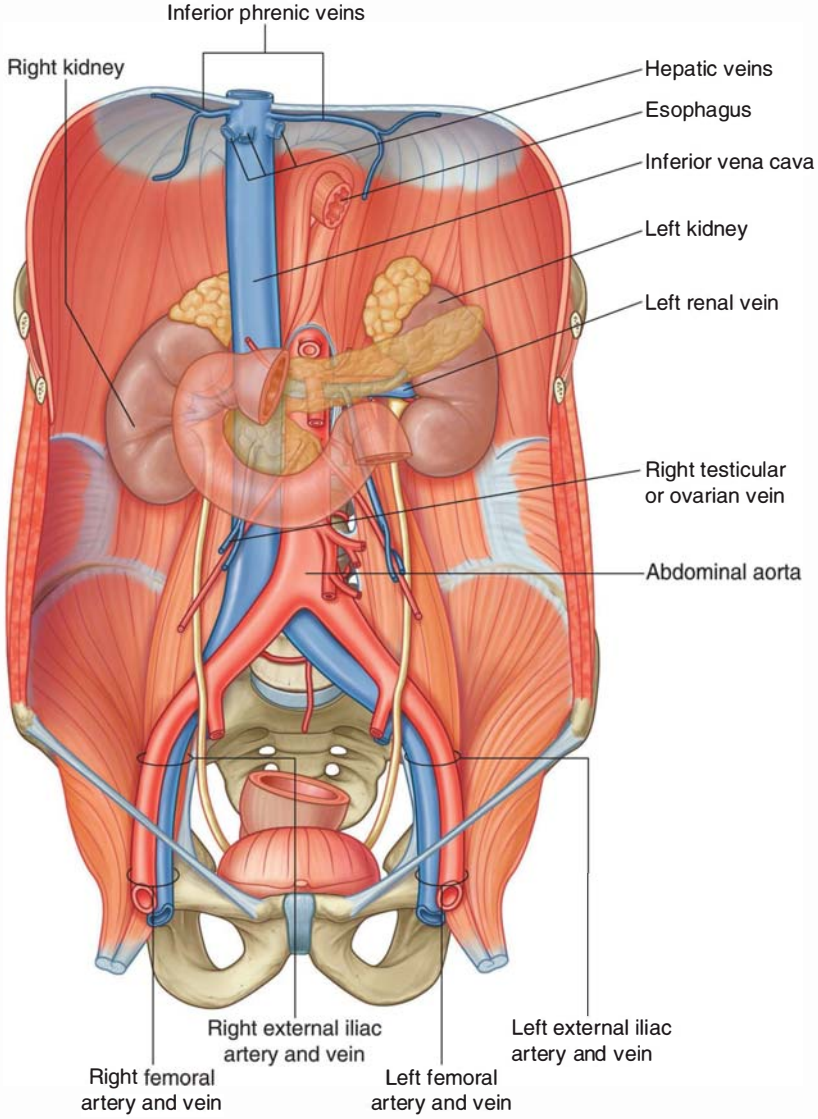


Fig. 4.153 Inferior vena cava.

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Tributaries to the inferior vena cava include the:

- common iliac veins,
- lumbar veins,
- right testicular or ovarian vein,
- renal veins,
- right suprarenal vein,
- inferior phrenic veins, and
- hepatic veins.

There are no tributaries from the abdominal part of the gastrointestinal tract, the spleen, the pancreas, or the gallbladder, because veins from these structures are components of the portal venous system, which first passes through the liver.

Of the venous tributaries mentioned above, the **lumbar veins** are unique in their connections and deserve special attention. Not all of the lumbar veins drain directly into the inferior vena cava (Fig. 4.154):

- The fifth lumbar vein generally drains into the iliolumbar vein, a tributary of the common iliac vein.
- The third and fourth lumbar veins usually drain into the inferior vena cava.

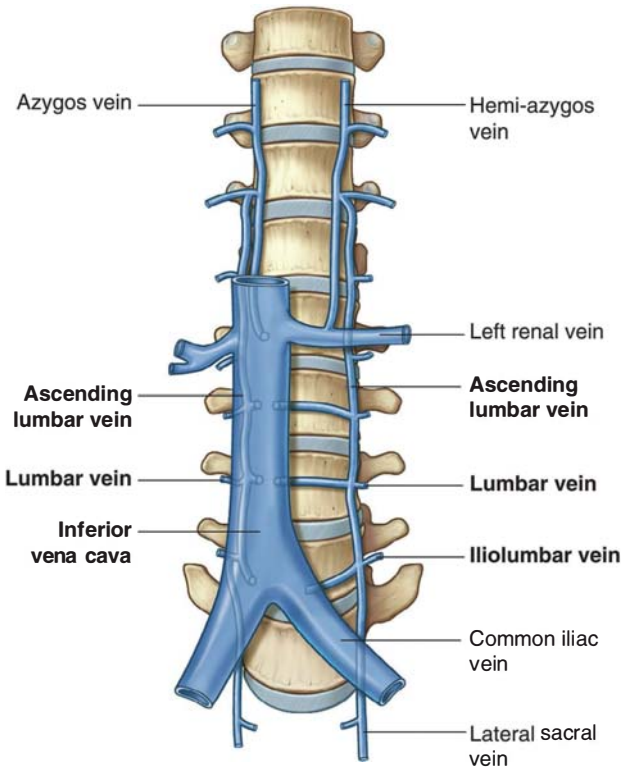


Fig. 4.154 Lumbar veins.

- The first and second lumbar veins may empty into the ascending lumbar veins.

The **ascending lumbar veins** are long, anastomosing venous channels that connect the common iliac, iliolumbar, and lumbar veins with the azygos and hemi-azygos veins of the thorax (Fig. 4.154).

If the inferior vena cava becomes blocked, the ascending lumbar veins become important collateral channels between the lower and upper parts of the body.

In the clinic

Inferior vena cava filter

Deep vein thrombosis is a potentially fatal condition where a clot (thrombus) is formed in the deep venous system of the legs and the veins of the pelvis. Virchow described the reasons for thrombus formation as decreased blood flow, abnormality of the constituents of blood, and abnormalities of the vessel wall. Common predisposing factors include hospitalization and surgery, the oral contraceptive pill, smoking, and air travel. Other factors include clotting abnormalities (e.g., protein S and protein C deficiency).

The diagnosis of deep vein thrombosis may be difficult to establish, with symptoms including leg swelling and pain and discomfort in the calf. It may also be an incidental finding.

In practice, patients with suspected deep vein thrombosis undergo a D-dimer blood test, which measures levels of a fibrin degradation product. If this is positive there is a high association with deep vein thrombosis.

The consequences of deep vein thrombosis are twofold. Occasionally the clot may dislodge and pass into the venous system through the right side of the heart and into the main pulmonary arteries. If the clots are of significant size they obstruct blood flow to the lung and may produce instantaneous death. Secondary complications include destruction of the normal valvular system in the legs, which may lead to venous incompetency and chronic leg swelling with ulceration.

The treatment for deep vein thrombosis is prevention. In order to prevent deep vein thrombosis, patients are optimized by removing all potential risk factors. Subcutaneous heparin may be injected and the patient wears compression stockings to prevent venous stasis while in the hospital.

In certain situations it is not possible to optimize the patient with prophylactic treatment, and it may be necessary to insert a filter into the inferior vena cava that traps any large clots. It may be removed after the risk period has ended.

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Lymphatic system

Lymphatic drainage from most deep structures and regions of the body below the diaphragm converges mainly on collections of lymph nodes and vessels associated with the major blood vessels of the posterior abdominal region (Fig. 4.155). The lymph then predominantly drains into the thoracic duct. Major lymphatic channels that drain different regions of the body as a whole are summarized in Table 4.4 (also see Chapter 1, pp. 29-30, for discussion of lymphatics in general).

Pre-aortic and lateral aortic or lumbar nodes (para-aortic nodes)

Approaching the aortic bifurcation, the collections of lymphatics associated with the two common iliac arteries and veins merge, and multiple groups of lymphatic vessels and nodes associated with the abdominal aorta and

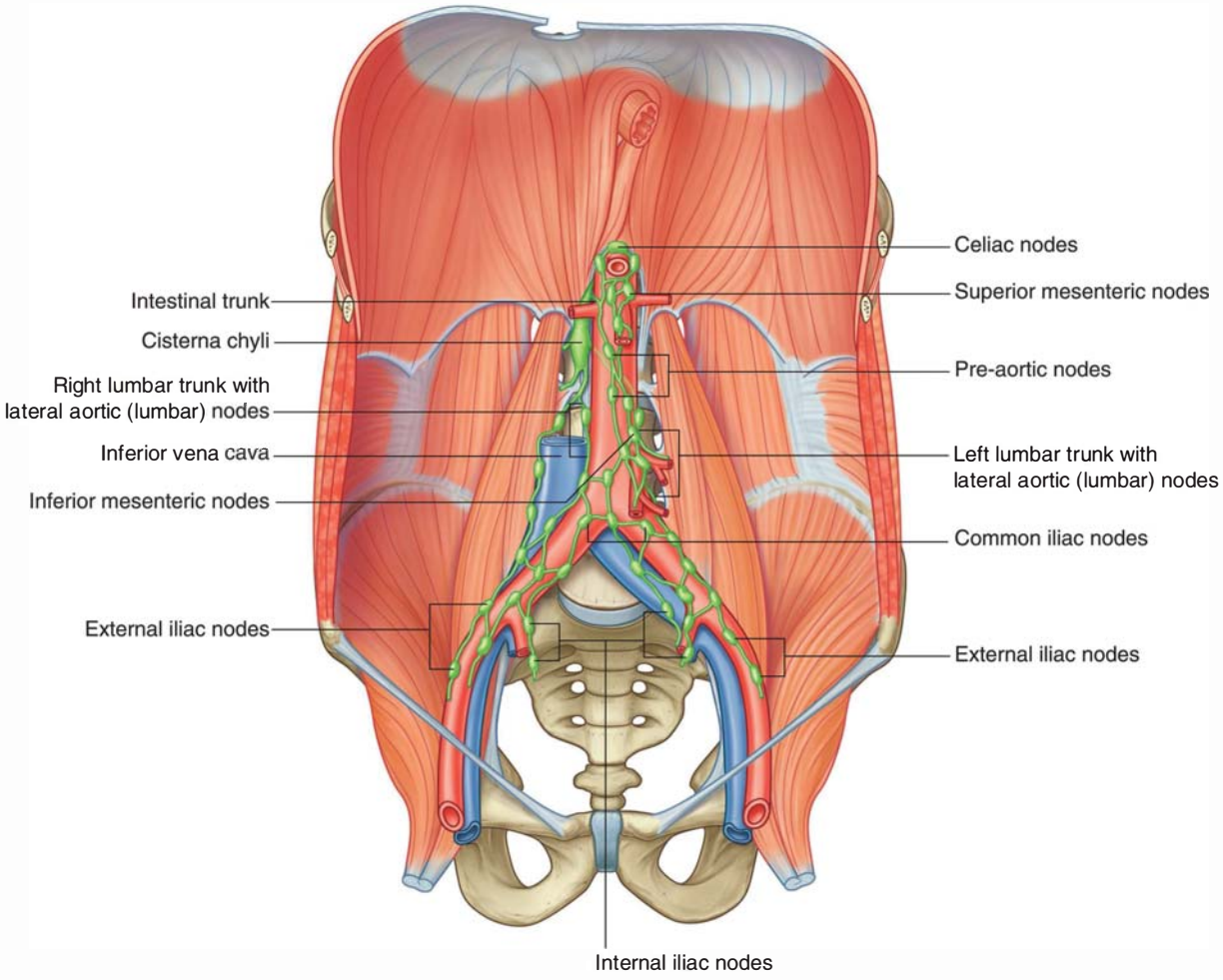


Fig. 4.155 Abdominal lymphatics.

Table 4.4 Lymphatic drainage

Lymphatic vessel	Area drained
Right jugular trunk	Right side of head and neck
Left jugular trunk	Left side of head and neck
Right subclavian trunk	Right upper limb, superficial regions of thoracic and upper abdominal walls
Left subclavian trunk	Left upper limb, superficial regions of thoracic and upper abdominal walls
Right bronchomediastinal trunk	Right lung and bronchi, mediastinal structures, thoracic wall
Left bronchomediastinal trunk	Left lung and bronchi, mediastinal structures, thoracic wall
Thoracic duct	Lower limbs, abdominal walls and viscera, pelvic walls and viscera, thoracic wall

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inferior vena cava pass superiorly. These collections may be subdivided into **pre-aortic nodes**, which are anterior to the abdominal aorta, and **right** and **left lateral aortic** or **lumbar nodes** (**para-aortic nodes**), which are positioned on either side of the abdominal aorta (Fig. 4.155).

As these collections of lymphatics pass through the posterior abdominal region, they continue to collect lymph from a variety of structures. The lateral aortic or lumbar lymph nodes (para-aortic nodes) receive lymphatics from the body wall, the kidneys, the suprarenal glands, and the testes or ovaries.

The pre-aortic nodes are organized around the three anterior branches of the abdominal aorta that supply the abdominal part of the gastrointestinal tract, as well as the spleen, pancreas, gallbladder, and liver. They are divided into celiac, superior mesenteric, and inferior mesenteric nodes, and receive lymph from the organs supplied by the similarly named arteries.

Finally, the lateral aortic or lumbar nodes form the right and left lumbar trunks, whereas the pre-aortic nodes form the intestinal trunk (Fig. 4.155). These trunks come together and form a confluence that, at times, appears as a saccular dilation (the cisterna chyli). This confluence of lymph trunks is posterior to the right side of the abdominal aorta and anterior to the bodies of vertebrae LI and LII. It marks the beginning of the thoracic duct.

In the clinic

Retroperitoneal lymph node surgery

From a clinical perspective, retroperitoneal lymph nodes are arranged in two groups. The pre-aortic lymph node group drains lymph from the embryological midline structures, such as the liver, bowel, and pancreas. The para-aortic lymph node group (the lateral aortic or lumbar nodes), on either side of the aorta, drain lymph from bilateral structures, such as the kidneys and adrenal glands. Organs embryologically derived from the posterior abdominal wall also drain lymph to these nodes. These organs include the ovaries and the testes (importantly, the testes do not drain lymph to the inguinal regions).

In general, lymphatic drainage follows standard predictable routes; however, in the presence of disease, alternate routes of lymphatic drainage will occur.

There are a number of causes for enlarged retroperitoneal lymph nodes. In the adult, massively enlarged lymph nodes are a feature of lymphoma, and smaller lymph node enlargement is observed in the presence of infection and metastatic malignant spread of disease (e.g., colon cancer).

The treatment for malignant lymph node disease is based upon a number of factors, including the site of the primary tumor (e.g., bowel) and its histological cell type. Normally, the primary tumor is surgically removed and the lymph node spread and metastatic organ spread (e.g., to the liver and the lungs) are often treated with chemotherapy and radiotherapy.

In certain instances it may be considered appropriate to resect the lymph nodes in the retroperitoneum (e.g., for testicular cancer).

The surgical approach to retroperitoneal lymph node resection involves a lateral paramedian incision in the midclavicular line. The three layers of the anterolateral abdominal wall (external oblique, internal oblique, and transversus abdominis) are opened and the transversalis fascia is divided. The next structure the surgeon sees is the parietal peritoneum. Instead of entering the parietal peritoneum, which is standard procedure for most intraabdominal operations, the surgeon gently pushes the parietal peritoneum toward the midline, which moves the intraabdominal contents and allows a clear view of the retroperitoneal structures. On the left, the para-aortic lymph node group is easily demonstrated, with a clear view of the abdominal aorta and kidney. On the right the inferior vena cava is demonstrated and has to be retracted to access the right para-aortic lymph node chain.

The procedure of retroperitoneal lymph node dissection is extremely well tolerated and lacks the problems of entering the peritoneal cavity (e.g., paralytic ileus). Unfortunately, a complication of a vertical incision in the midclavicular line is division of the segmental nerve supply to the rectus abdominis muscle. This produces muscle atrophy and asymmetrical proportions of the anterior abdominal wall.

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Nervous system in the posterior abdominal region

Several important components of the nervous system are in the posterior abdominal region. These include the sympathetic trunks and associated splanchnic nerves, the plexus of nerves and ganglia associated with the abdominal aorta, and the lumbar plexus of nerves.

Sympathetic trunks and splanchnic nerves

The sympathetic trunks pass through the posterior abdominal region anterolateral to the lumbar vertebral bodies, before continuing across the sacral promontory and into the pelvic cavity (Fig. 4.156). Along their course, small raised areas are visible. These represent collections of neuronal cell bodies—primarily postganglionic neuronal cell bodies—which are located outside the central nervous system. They are sympathetic paravertebral ganglia. There are usually four ganglia along the sympathetic trunks in the posterior abdominal region.

Also associated with the sympathetic trunks in the posterior abdominal region are the lumbar splanchnic nerves (Fig. 4.156). These components of the nervous system pass from the sympathetic trunks to the plexus of nerves and ganglia associated with the abdominal aorta. Usually two to four lumbar splanchnic nerves carry preganglionic sympathetic fibers and visceral afferent fibers.

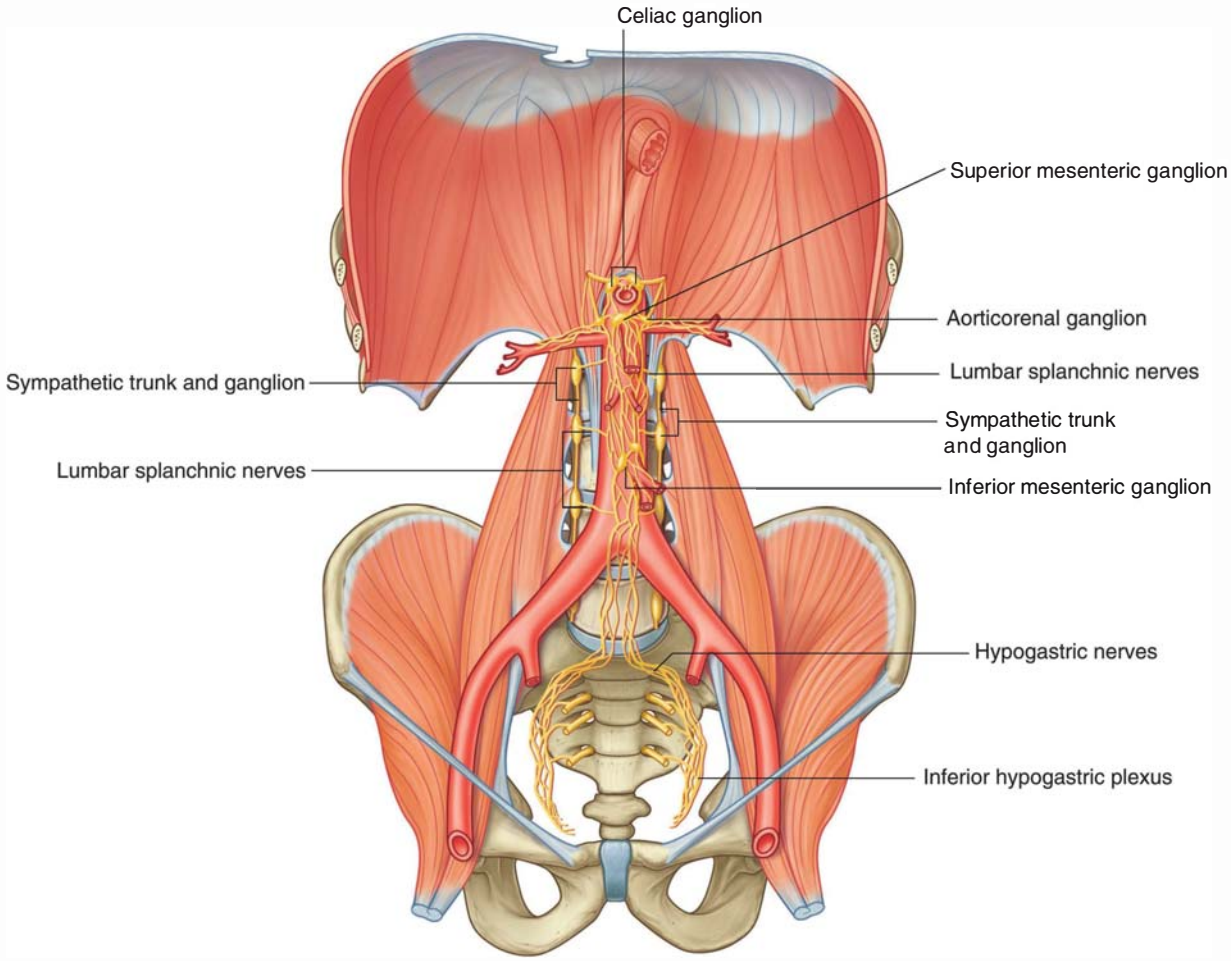


Fig. 4.156 Sympathetic trunks passing through the posterior abdominal region.

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Abdominal prevertebral plexus and ganglia

The abdominal prevertebral plexus is a network of nerve fibers surrounding the abdominal aorta. It extends from the aortic hiatus of the diaphragm to the bifurcation of the aorta into the right and left common iliac arteries. Along its route, it is subdivided into smaller, named plexuses (Fig. 4.157):

- Beginning at the diaphragm and moving inferiorly, the initial accumulation of nerve fibers is referred to as the celiac plexus—this subdivision includes nerve fibers associated with the roots of the celiac trunk and superior mesenteric artery.
- Continuing inferiorly, the plexus of nerve fibers extending from just below the superior mesenteric artery to the aortic bifurcation is the abdominal aortic plexus (Fig. 4.157).
- At the bifurcation of the abdominal aorta, the abdominal prevertebral plexus continues inferiorly as the superior hypogastric plexus.

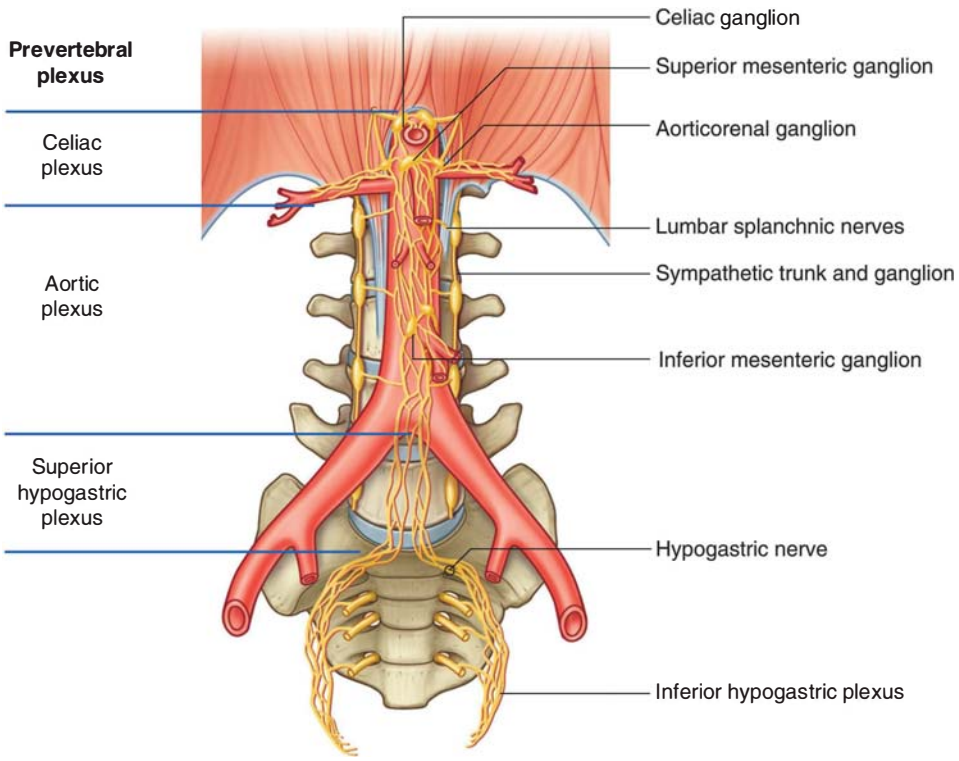


Fig. 4.157 Prevertebral plexus and ganglia in the posterior abdominal region.

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Throughout its length, the abdominal prevertebral plexus is a conduit for:

- preganglionic sympathetic and visceral afferent fibers from the thoracic and lumbar splanchnic nerves,
- preganglionic parasympathetic and visceral afferent fibers from the vagus nerves [X], and
- preganglionic parasympathetic fibers from the pelvic splanchnic nerves (Fig. 4.158).

Associated with the abdominal prevertebral plexus are clumps of nervous tissue (the **prevertebral ganglia**), which are collections of postganglionic sympathetic neuronal cell bodies in recognizable aggregations along the abdominal prevertebral plexus; they are usually named after the nearest branch of the abdominal aorta. They are therefore referred to as **celiac, superior mesenteric, aorticorenal,** and **inferior mesenteric ganglia** (Fig. 4.159). These structures, along with the abdominal prevertebral plexus, play a critical role in the innervation of the abdominal viscera.

Common sites for pain referred from the abdominal viscera and from the heart are given in Table 4.5.

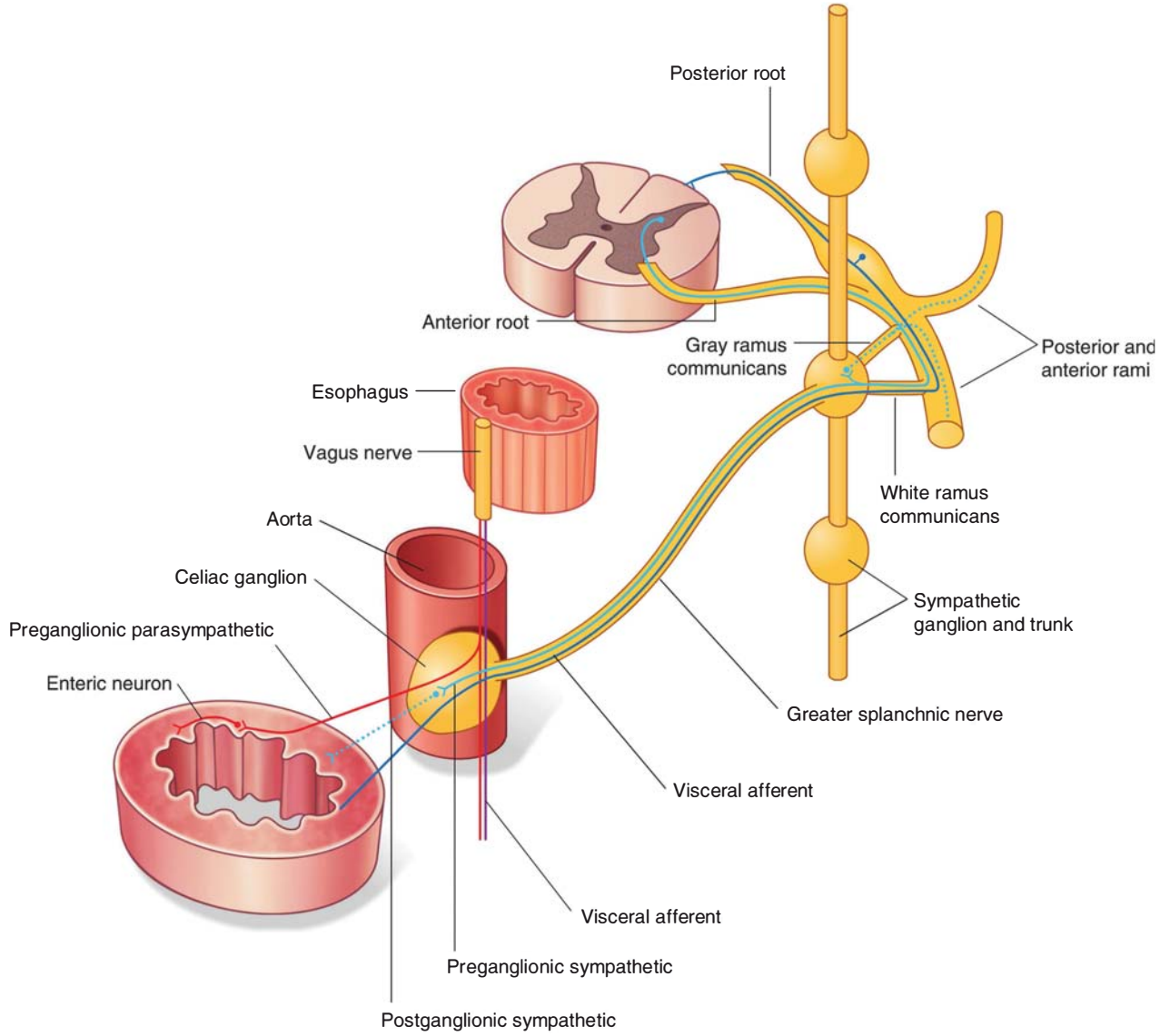


Fig. 4.158 Nerve fibers passing through the abdominal prevertebral plexus and ganglia.

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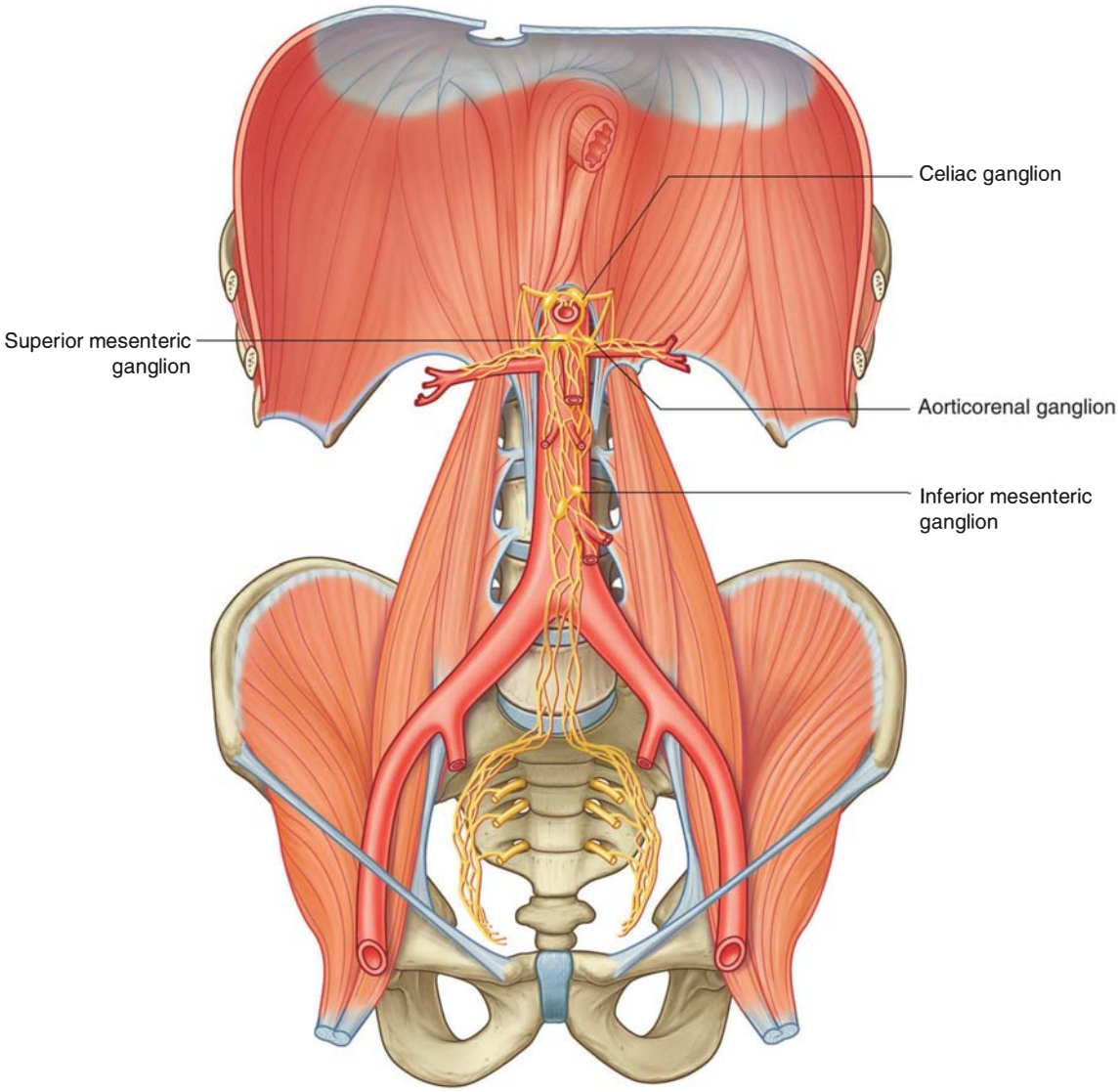


Fig. 4.159 Prevertebral ganglia associated with the prevertebral plexus.

Table 4.5 Referred pain pathways (visceral afferents)

Organ	Afferent pathway	Spinal cord level	Referral area
Heart	Thoracic splanchnic nerves	T1 to T4	Upper thorax and medial arm
Foregut (organs supplied by celiac trunk)	Greater splanchnic nerve	T5 to T9 (or T10)	Lower thorax and epigastric region
Midgut (organs supplied by superior mesenteric artery)	Lesser splanchnic nerve	T9, T10 (or T10, T11)	Umbilical region
Kidneys and upper ureter	Least splanchnic nerve	T12	Flanks (lateral regions)
Hindgut (organs supplied by inferior mesenteric artery) and lower ureter	Lumbar splanchnic nerves	L1, L2	Pubic region, lateral and anterior thighs, and groin

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Lumbar plexus

The lumbar plexus is formed by the anterior rami of nerves L1 to L3 and most of the anterior ramus of L4 (Fig. 4.160 and Table 4.6). It also receives a contribution from the T12 (subcostal) nerve.

Branches of the lumbar plexus include the iliohypogastric, ilio-inguinal and genitofemoral nerves lateral cutaneous nerve of the thigh (lateral femoral cutaneous) and femoral and obturator nerves. The lumbar plexus forms in the substance of the psoas major muscle anterior to its attachment to the transverse processes of the lumbar vertebrae (Fig. 4.161). Therefore, relative to the psoas major muscle, the various branches emerge either:

- anterior—genitofemoral nerve,
- medial—obturator nerve, or
- lateral—iliohypogastric, ilio-inguinal, and femoral nerves and the lateral cutaneous nerve of the thigh.

Iliohypogastric and ilio-inguinal nerves (L1)

The iliohypogastric and ilio-inguinal nerves arise as a single trunk from the anterior ramus of nerve L1 (Fig. 4.160). Either before or soon after emerging from the lateral border of the psoas major muscle, this single trunk divides into the iliohypogastric and the ilio-inguinal nerves.

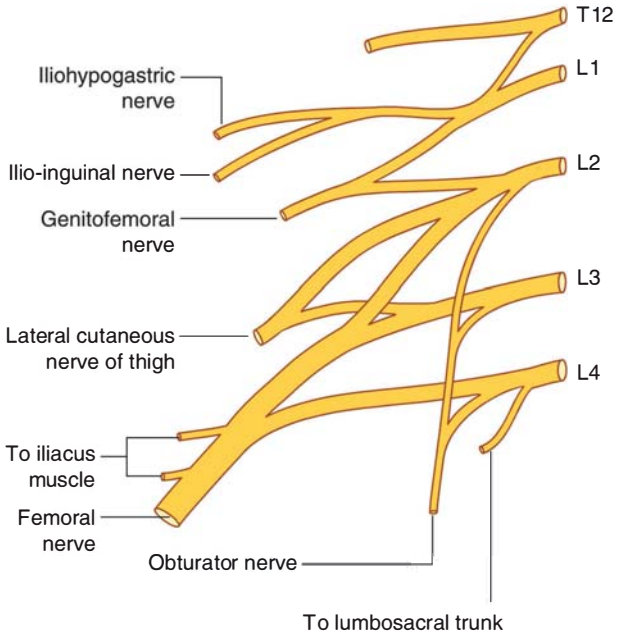


Fig. 4.160 Lumbar plexus.

Table 4.6 Branches of the lumbar plexus

Branch	Origin	Spinal segments	Function: motor	Function: sensory
Iliohypogastric	Anterior ramus L1	L1	Internal oblique and transversus abdominis	Posterolateral gluteal skin and skin in pubic region
Ilio-inguinal	Anterior ramus L1	L1	Internal oblique and transversus abdominis	Skin in the upper medial thigh, and either the skin over the root of the penis and anterior scrotum or the mons pubis and labium majus
Genitofemoral	Anterior rami L1 and L2	L1, L2	Genital branch—male cremasteric muscle	Genital branch—skin of anterior scrotum or skin of mons pubis and labium majus; femoral branch—skin of upper anterior thigh
Lateral cutaneous nerve of thigh	Anterior rami L2 and L3	L2, L3		Skin on anterior and lateral thigh to the knee
Obturator	Anterior rami L2 to L4	L2 to L4	Obturator externus, pectineus, and muscles in medial compartment of thigh	Skin on medial aspect of the thigh
Femoral	Anterior rami L2 to L4	L2 to L4	Iliacus, pectineus, and muscles in anterior compartment of thigh	Skin on anterior thigh and medial surface of leg

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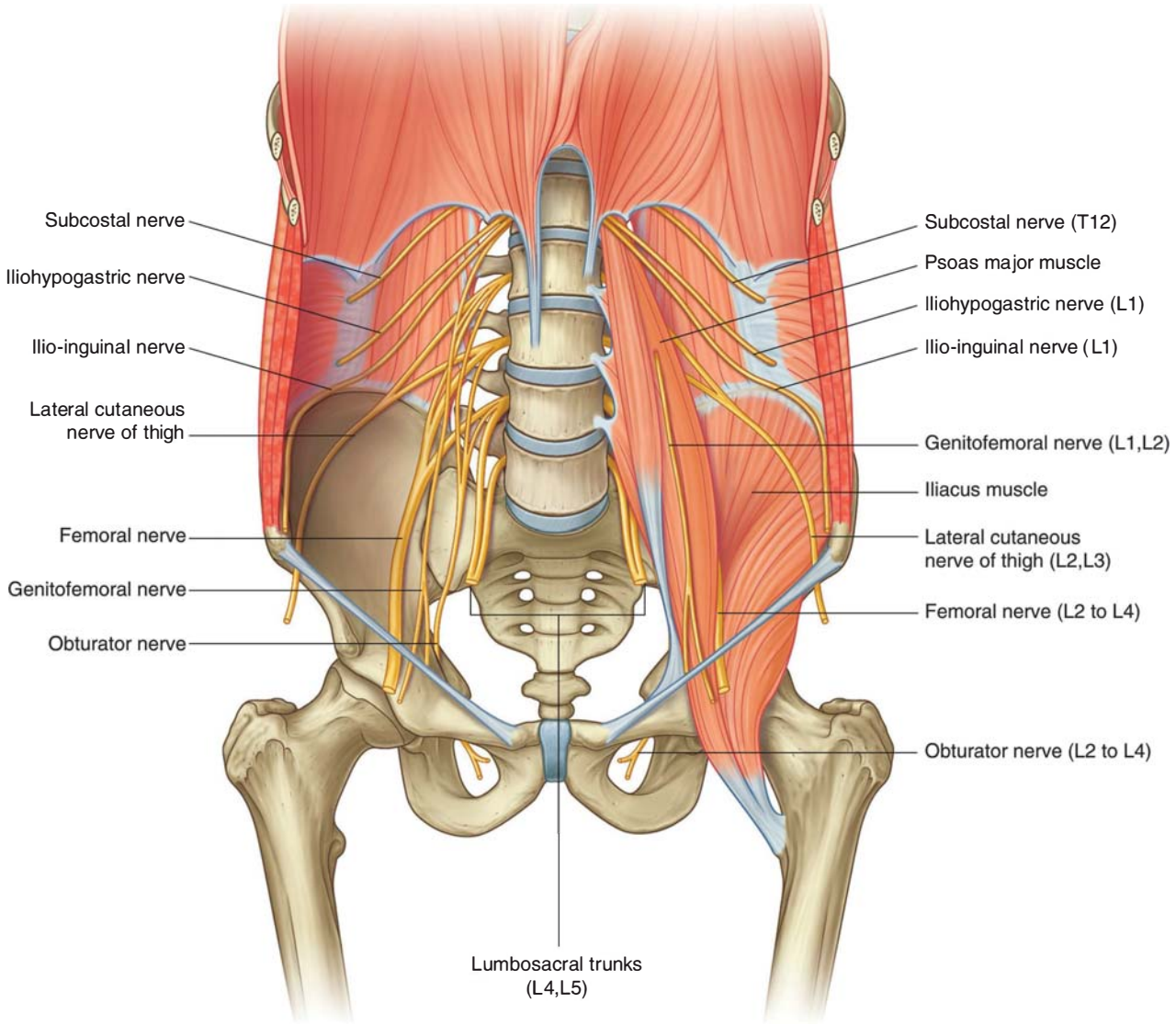


Fig. 4.161 Lumbar plexus in the posterior abdominal region.

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Iliohypogastric nerve

The **iliohypogastric nerve** passes across the anterior surface of the quadratus lumborum muscle, posterior to the kidney. It pierces the transversus abdominis muscle and continues anteriorly around the body between the transversus abdominis and internal oblique muscles. Above the iliac crest, a **lateral cutaneous branch** pierces the internal and external oblique muscles to supply the posterolateral gluteal skin (Fig. 4.162).

The remaining part of the iliohypogastric nerve (the **anterior cutaneous branch**) continues in an anterior direction, piercing the internal oblique just medial to the anterior superior iliac spine as it continues in an obliquely downward and medial direction. Becoming cutaneous, just above the superficial inguinal ring, after piercing the aponeurosis of the external oblique, it distributes to the skin in the pubic region (Fig. 4.162). Throughout its course, it also supplies branches to the abdominal musculature.

Ilio-inguinal nerve

The ilio-inguinal nerve is smaller than, and inferior to, the iliohypogastric nerve as it crosses the quadratus lumborum

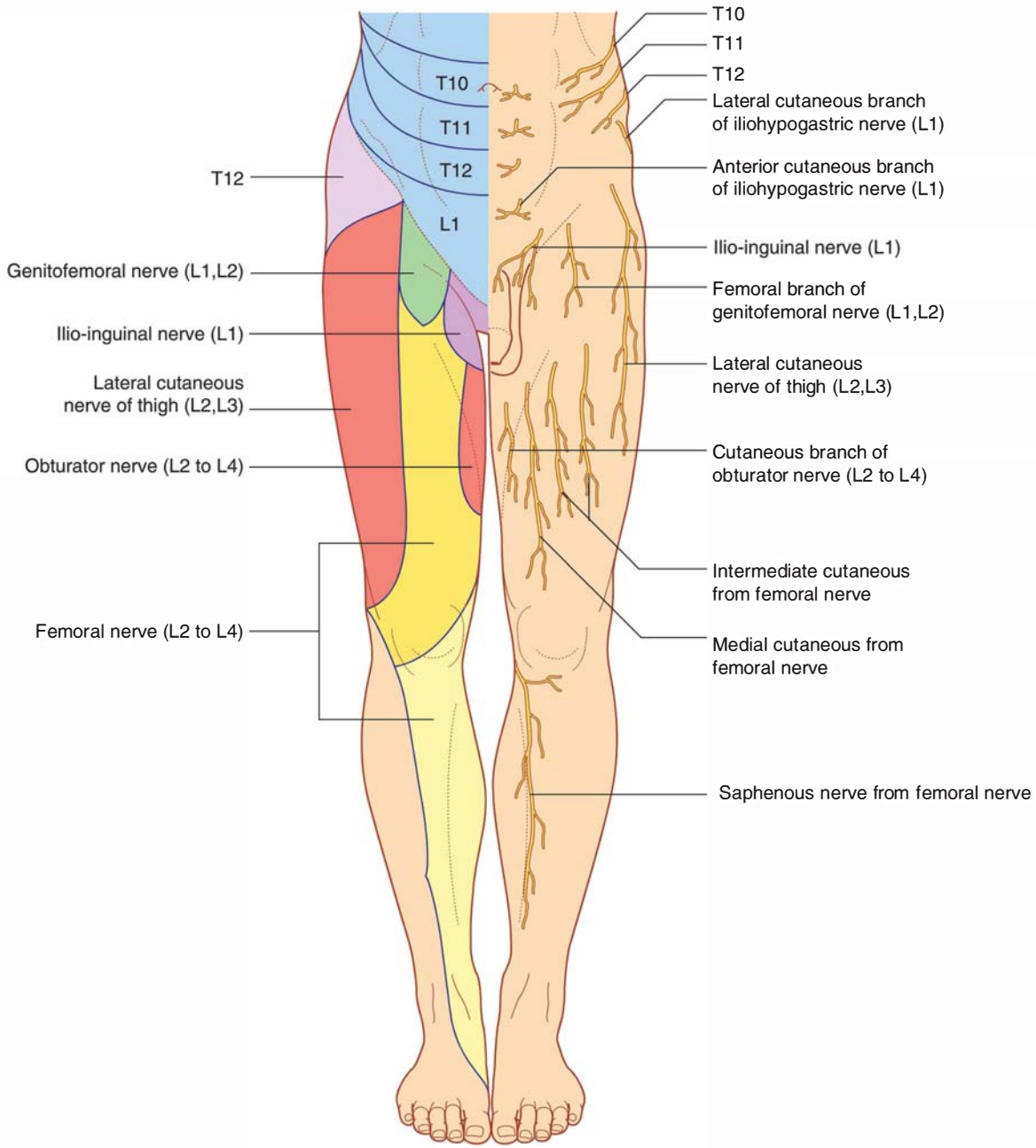


Fig. 4.162 Cutaneous distribution of the nerves from the lumbar plexus.

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muscle. Its course is more oblique than that of the iliohypogastric nerve, and it usually crosses part of the iliacus muscle on its way to the iliac crest. Near the anterior end of the iliac crest, it pierces the transversus abdominis muscle, and then pierces the internal oblique muscle and enters the inguinal canal.

The ilio-inguinal nerve emerges through the superficial inguinal ring, along with the spermatic cord, and provides cutaneous innervation to the upper medial thigh, the root of the penis, and the anterior surface of the scrotum in men, or the mons pubis and labium majus in women (Fig. 4.162). Throughout its course, it also supplies branches to the abdominal musculature.

Genitofemoral nerve (L1 and L2)

The genitofemoral nerve arises from the anterior rami of nerves L1 and L2 (Fig. 4.160). It passes downward in the substance of the psoas major muscle until it emerges on the anterior surface of the psoas major. It then descends on the surface of the muscle, in a retroperitoneal position, passing posterior to the ureter. It eventually divides into genital and femoral branches.

The **genital branch** continues downward and enters the inguinal canal through the deep inguinal ring. It continues through the canal and:

- in men, innervates the cremasteric muscle and terminates on the skin in the upper anterior part of the scrotum, and
- in women, accompanies the round ligament of the uterus and terminates on the skin of the mons pubis and labium majus.

The **femoral branch** descends on the lateral side of the external iliac artery and passes posterior to the inguinal ligament, entering the femoral sheath lateral to the femoral artery. It pierces the anterior layer of the femoral sheath and the fascia lata to supply the skin of the upper anterior thigh (Fig. 4.162).

Lateral cutaneous nerve of thigh (L2 and L3)

The lateral cutaneous nerve of the thigh arises from the anterior rami of nerves L2 and L3 (Fig. 4.160). It emerges from the lateral border of the psoas major muscle, passing obliquely downward across the iliacus muscle toward the anterior superior iliac spine (Fig. 4.162). It passes posterior to the inguinal ligament and enters the thigh.

The lateral cutaneous nerve of the thigh supplies the skin on the anterior and lateral thigh to the level of the knee (Fig. 4.162).

Obturator nerve (L2 to L4)

The obturator nerve arises from the anterior rami of nerves L2 to L4 (Fig. 4.160). It descends in the psoas major muscle, emerging from its medial side near the pelvic brim (Fig. 4.161).

The obturator nerve continues posterior to the common iliac vessels, passes across the lateral wall of the pelvic cavity, and enters the obturator canal, through which the obturator nerve gains access to the medial compartment of the thigh.

In the area of the obturator canal, the obturator nerve divides into **anterior** and **posterior branches**. On entering the medial compartment of the thigh, the two branches are separated by the obturator externus and adductor brevis muscles. Throughout their course through the medial compartment, these two branches supply:

- articular branches to the hip joint,
- muscular branches to the obturator externus, pectineus, adductor longus, gracilis, adductor brevis, and adductor magnus muscles,
- cutaneous branches to the medial aspect of the thigh, and
- in association with the saphenous nerve, cutaneous branches to the medial aspect of the upper part of the leg and articular branches to the knee joint (Fig. 4.162).

Femoral nerve (L2 to L4)

The femoral nerve arises from the anterior rami of nerves L2 to L4 (Fig. 4.160). It descends through the substance of the psoas major muscle, emerging from the lower lateral border of the psoas major (Fig. 4.161). Continuing its descent, the femoral nerve lies between the lateral border of the psoas major and the anterior surface of the iliacus muscle. It is deep to the iliacus fascia and lateral to the femoral artery as it passes posterior to the inguinal ligament and enters the anterior compartment of the thigh. Upon entering the thigh, it immediately divides into multiple branches.

Cutaneous branches of the femoral nerve include:

- medial and intermediate cutaneous nerves supplying the skin on the anterior surface of the thigh, and
- the saphenous nerve supplying the skin on the medial surface of the leg (Fig. 4.162).

Muscular branches innervate the iliacus, pectineus, sartorius, rectus femoris, vastus medialis, vastus intermedius, and vastus lateralis muscles. Articular branches supply the hip and knee joints.

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Surface anatomy

Abdomen surface anatomy

Visualization of the position of abdominal viscera is fundamental to a physical examination. Some of these viscera or their parts can be felt by palpating through the abdominal wall. Surface features can be used to establish the positions of deep structures.

Defining the surface projection of the abdomen

Palpable landmarks can be used to delineate the extent of the abdomen on the surface of the body. These landmarks are:

- the costal margin above and
- the pubic tubercle, anterior superior iliac spine, and iliac crest below (Fig. 4.163).

The costal margin is readily palpable and separates the abdominal wall from the thoracic wall.

A line between the anterior superior iliac spine and the pubic tubercle marks the position of the inguinal ligament, which separates the anterior abdominal wall above from the thigh of the lower limb below.

The iliac crest separates the posterolateral abdominal wall from the gluteal region of the lower limb.

The upper part of the abdominal cavity projects above the costal margin to the diaphragm, and therefore abdominal viscera in this region of the abdomen are protected by the thoracic wall.

The level of the diaphragm varies during the breathing cycle. The dome of the diaphragm on the right can reach as high as the fourth costal cartilage during forced expiration.

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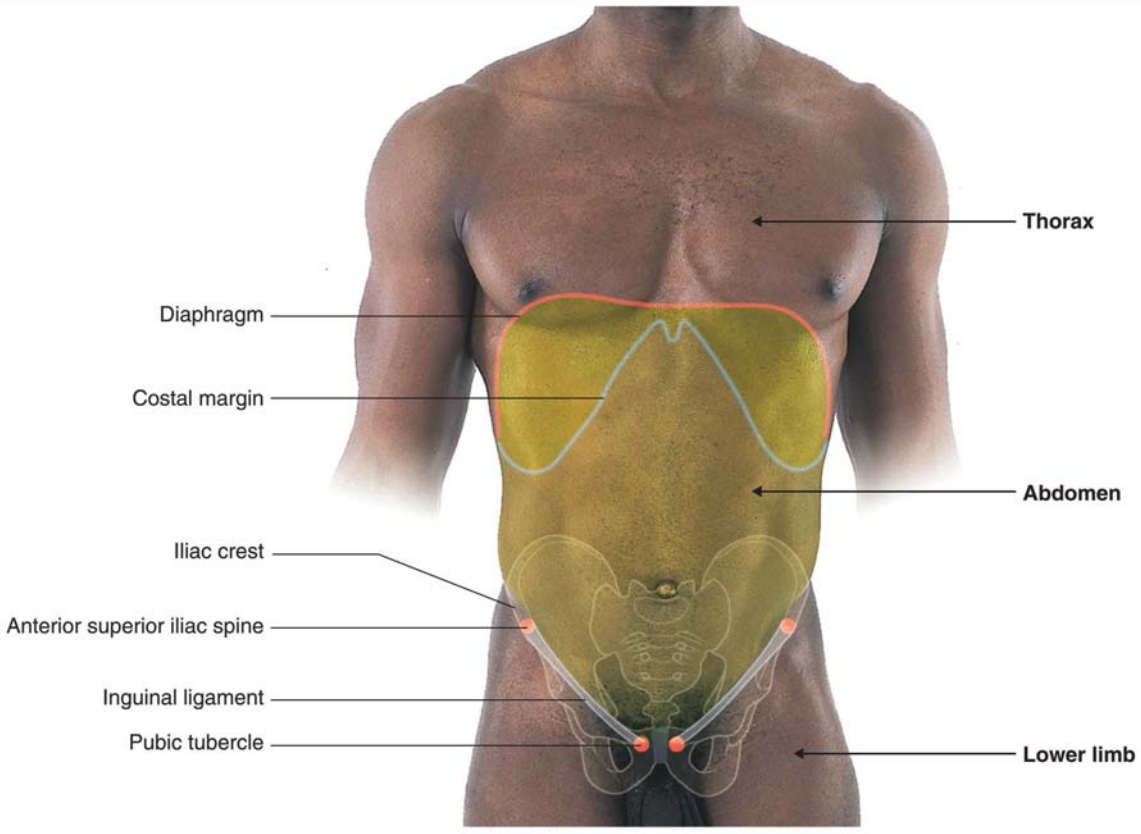


Fig. 4.163 Interior view of the abdominal region of a man. Palpable bony landmarks, the inguinal ligament, and the position of the diaphragm are indicated.

How to find the superficial inguinal ring

The superficial inguinal ring is an elongate triangular defect in the aponeurosis of the external oblique (Fig. 4.164). It lies in the lower medial aspect of the anterior abdominal wall and is the external opening of the inguinal canal. The inguinal canal and superficial ring are larger in men than in women:

- In men, structures that pass between the abdomen and the testis pass through the inguinal canal and superficial inguinal ring.

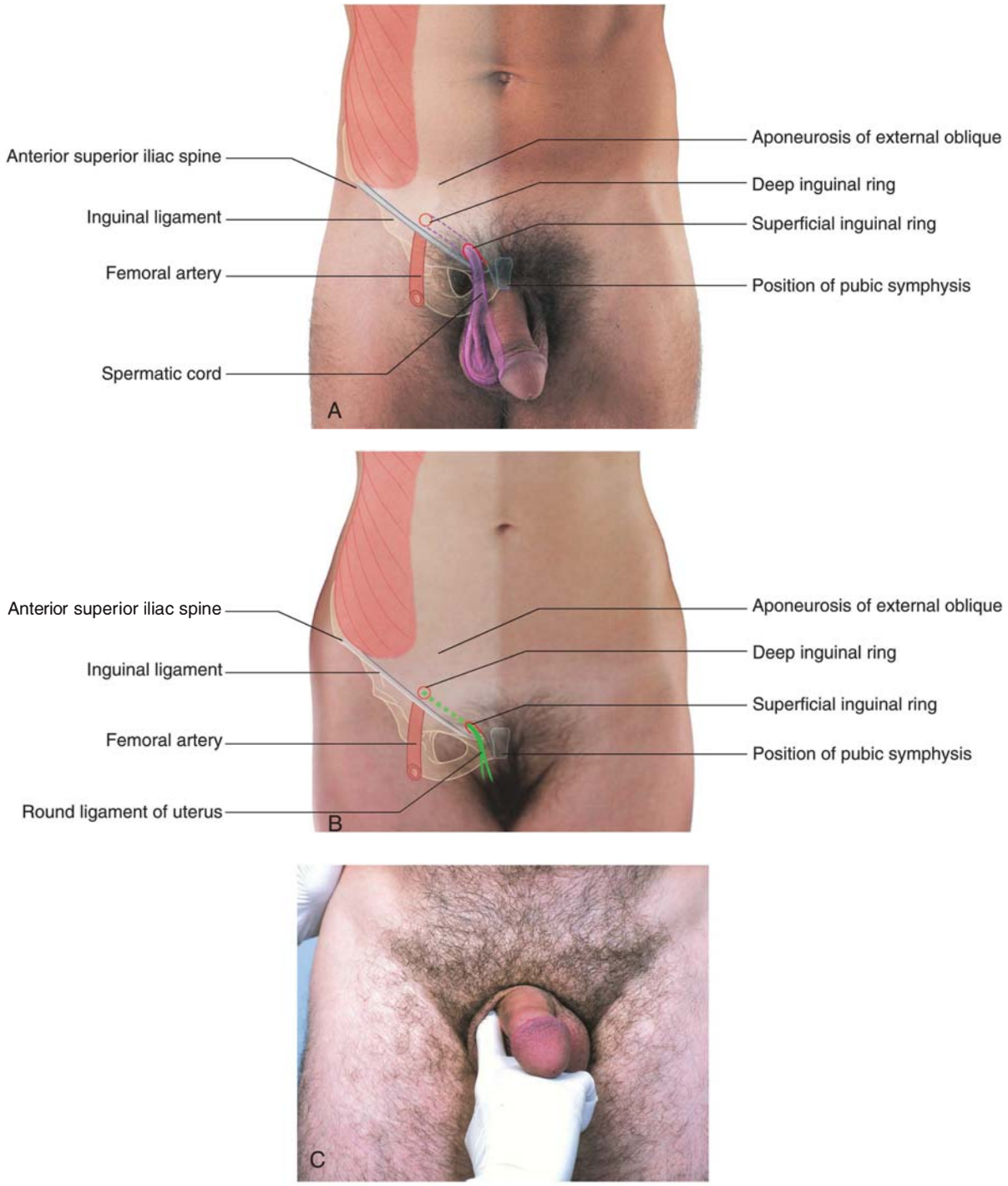


Fig. 4.164 Groin. **A.** In a man. **B.** In a woman. **C.** Examination of the superficial inguinal ring and related regions of the inguinal canal in a man.

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- In women, the round ligament of the uterus passes through the inguinal canal and superficial inguinal ring to merge with connective tissue of the labium majus.

The superficial inguinal ring is superior to the pubic crest and tubercle and to the medial end of the inguinal ligament:

- In men, the superficial inguinal ring can be easily located by following the spermatic cord superiorly to the lower abdominal wall—the external spermatic fascia of the spermatic cord is continuous with the margins of the superficial inguinal ring.
- In women, the pubic tubercle can be palpated and the ring is superior and lateral to it.

The deep inguinal ring, which is the internal opening to the inguinal canal, lies superior to the inguinal ligament, midway between the anterior superior iliac spine and pubic symphysis. The pulse of the femoral artery can be felt in the same position but below the inguinal ligament.

Because the superficial inguinal ring is the site where inguinal hernias appear, particularly in men, the ring and related parts of the inguinal canal are often evaluated during physical examination.

How to determine lumbar vertebral levels

Lumbar vertebral levels are useful for visualizing the positions of viscera and major blood vessels. The approximate positions of the lumbar vertebrae can be established using palpable or visible landmarks (Fig. 4.165):

- A horizontal plane passes through the medial ends of the ninth costal cartilages and the body of the LI vertebra—this transpyloric plane cuts through the body midway between the suprasternal (jugular) notch and the pubic symphysis.

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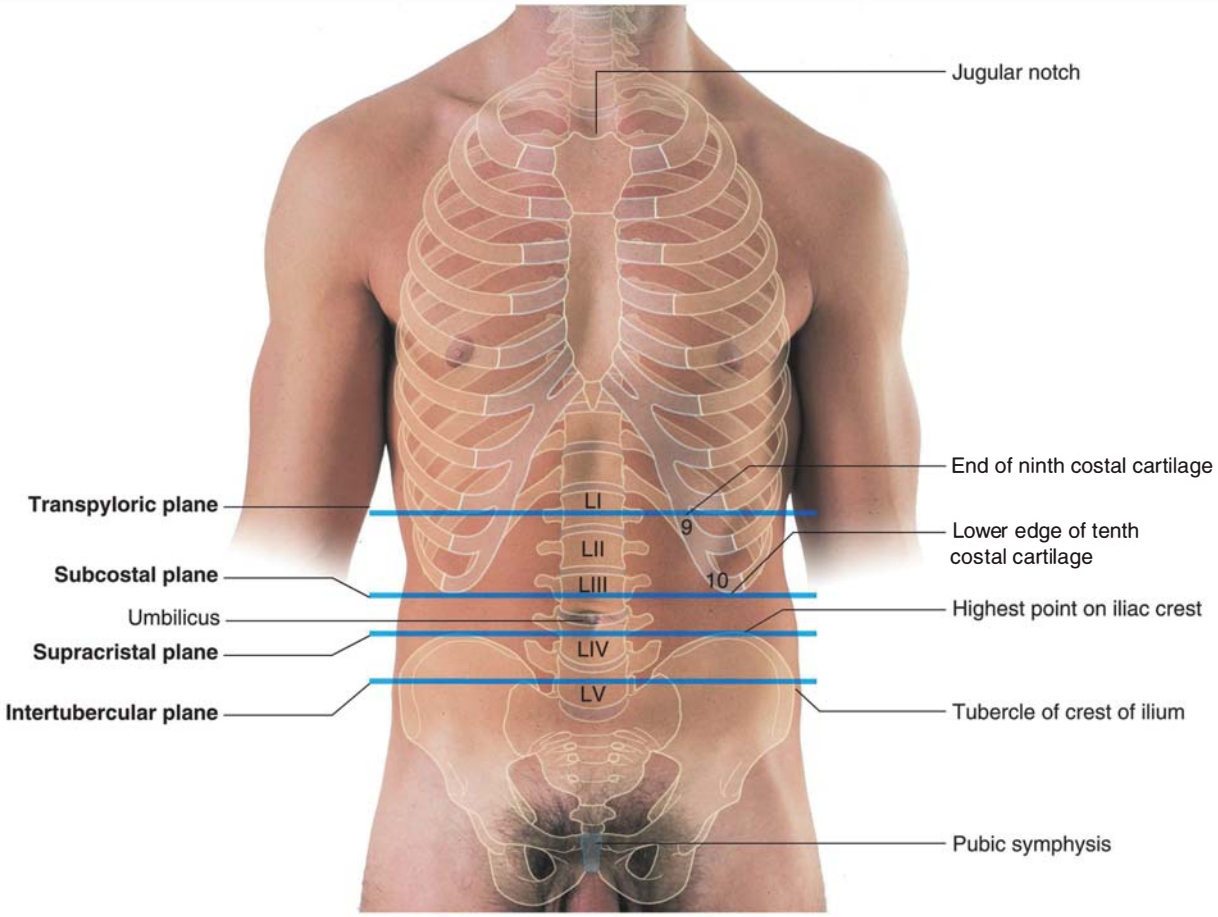


Fig. 4.165 Landmarks used for establishing the positions of lumbar vertebrae are indicated. Anterior view of the abdominal region of a man.

- A horizontal plane passes through the lower edge of the costal margin (tenth costal cartilage) and the body of the LIII vertebra—the umbilicus is normally on a horizontal plane that passes through the disc between the LIII and LIV vertebrae.
- A horizontal plane (supracristal plane) through the highest point on the iliac crest passes through the spine and body of the LIV vertebra;
- A plane through the tubercles of the crest of the ilium passes through the body of the LV vertebra.

Visualizing structures at the LI vertebral level

The LI vertebral level is marked by the transpyloric plane, which cuts transversely through the body midway between the jugular notch and pubic symphysis, and through the ends of the ninth costal cartilages (Fig. 4.166). At this level are:

- the beginning and upper limit of the end of the duodenum,
- the hila of the kidneys,
- the neck of the pancreas, and
- the origin of the superior mesenteric artery from the aorta.

The left and right colic flexures also are close to this level.

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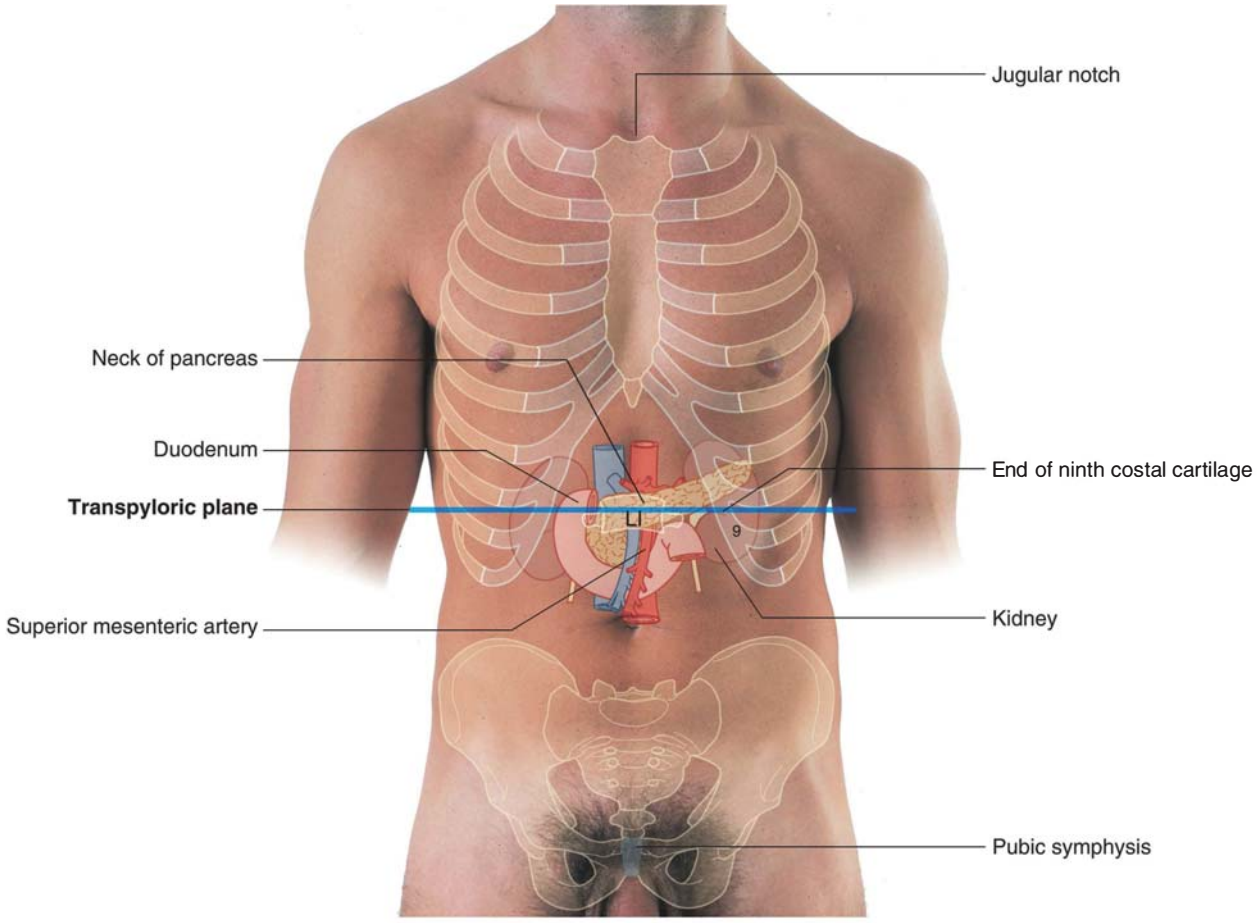


Fig. 4.166 LI vertebral level and the important viscera associated with this level. Anterior view of the abdominal region of a man.

Visualizing the position of major blood vessels

Each of the vertebral levels in the abdomen is related to the origin of major blood vessels (Fig. 4.167):

- The celiac trunk originates from the aorta at the upper border of the LI vertebra.
- The superior mesenteric artery originates at the lower border of the LI vertebra.
- The renal arteries originate at approximately the LII vertebra.
- The inferior mesenteric artery originates at the LIII vertebra.
- The aorta bifurcates into the right and left common iliac arteries at the level of the LIV vertebra.
- The left and right common iliac veins join to form the inferior vena cava at the LV vertebral level.

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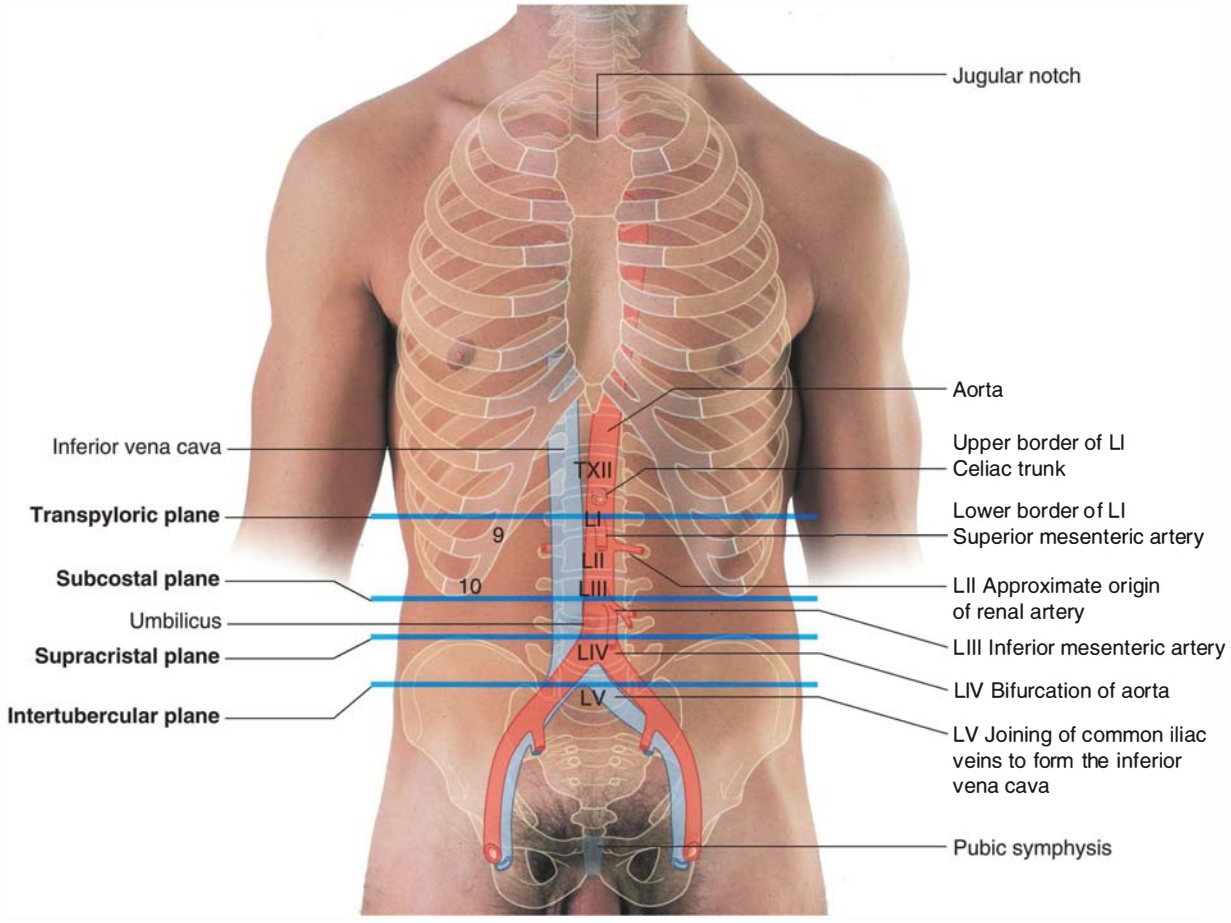


Fig. 4.167 Major vessels projected onto the body's surface. Anterior view of the abdominal region of a man.

Using abdominal quadrants to locate major viscera

The abdomen can be divided into quadrants by a vertical median plane and a horizontal transumbilical plane, which passes through the umbilicus (Fig. 4.168):

- The liver and gallbladder are in the right upper quadrant.
- The stomach and spleen are in the left upper quadrant.
- The cecum and appendix are in the right lower quadrant.
- The end of the descending colon and sigmoid colon are in the left lower quadrant.

Most of the liver is under the right dome of the diaphragm and is deep to the lower thoracic wall. The inferior margin of the liver can be palpated descending below the right costal margin when a patient is asked to inhale deeply. On deep inspiration, the edge of the liver can be felt "slipping" under the palpating fingers placed under the costal margin.

A common surface projection of the appendix is McBurney's point, which is one-third of the way up along a line from the right anterior superior iliac spine to the umbilicus.

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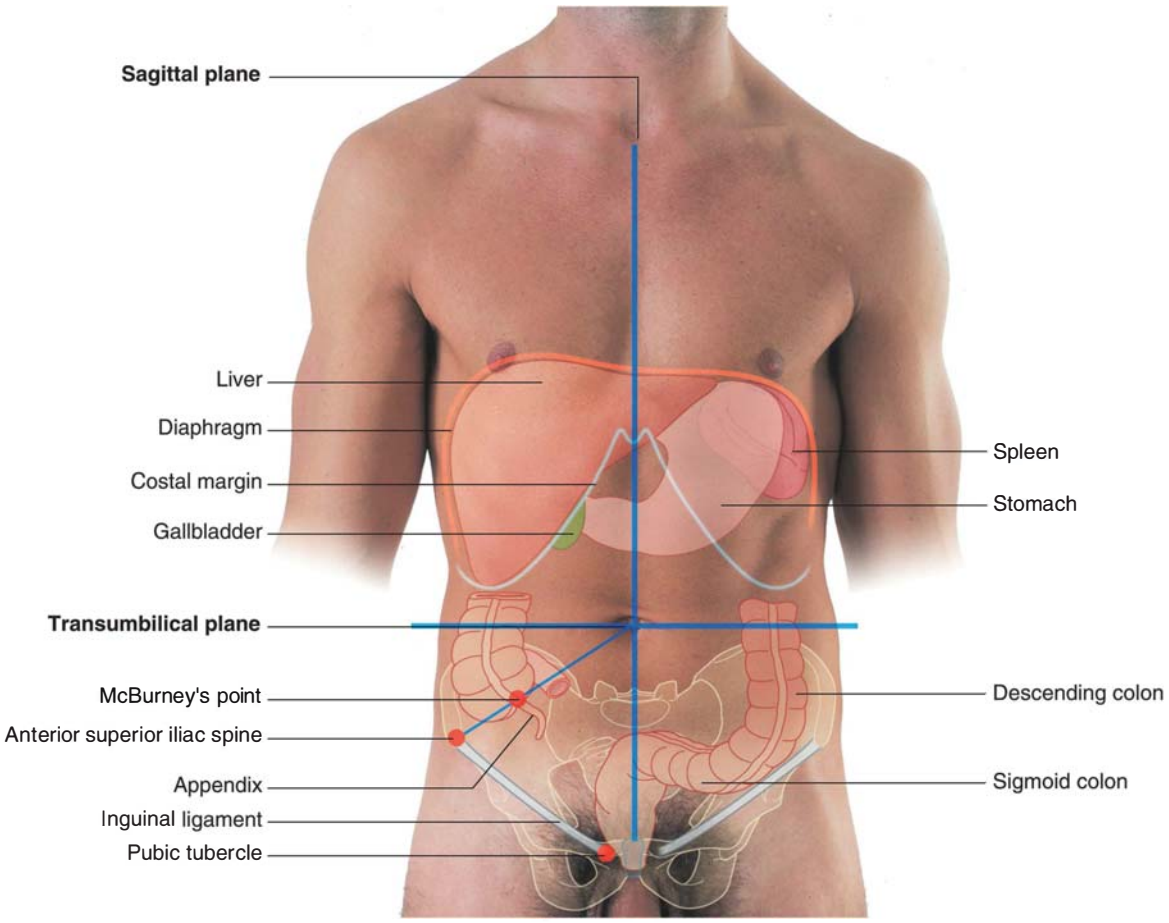


Fig. 4.168 Abdominal quadrants and the positions of major viscera. Anterior view of a man.

Defining surface regions to which pain from the gut is referred

The abdomen can be divided into nine regions by a midclavicular sagittal plane on each side and by the subcostal and intertubercular planes, which pass through the body transversely (Fig. 4.169). These planes separate the abdomen into:

- three central regions (epigastric, umbilical, pubic), and
- three regions on each side (hypochondrium, flank, groin).

Pain from the abdominal part of the foregut is referred to the epigastric region, pain from the midgut is referred to the umbilical region, and pain from the hindgut is referred to the pubic region.

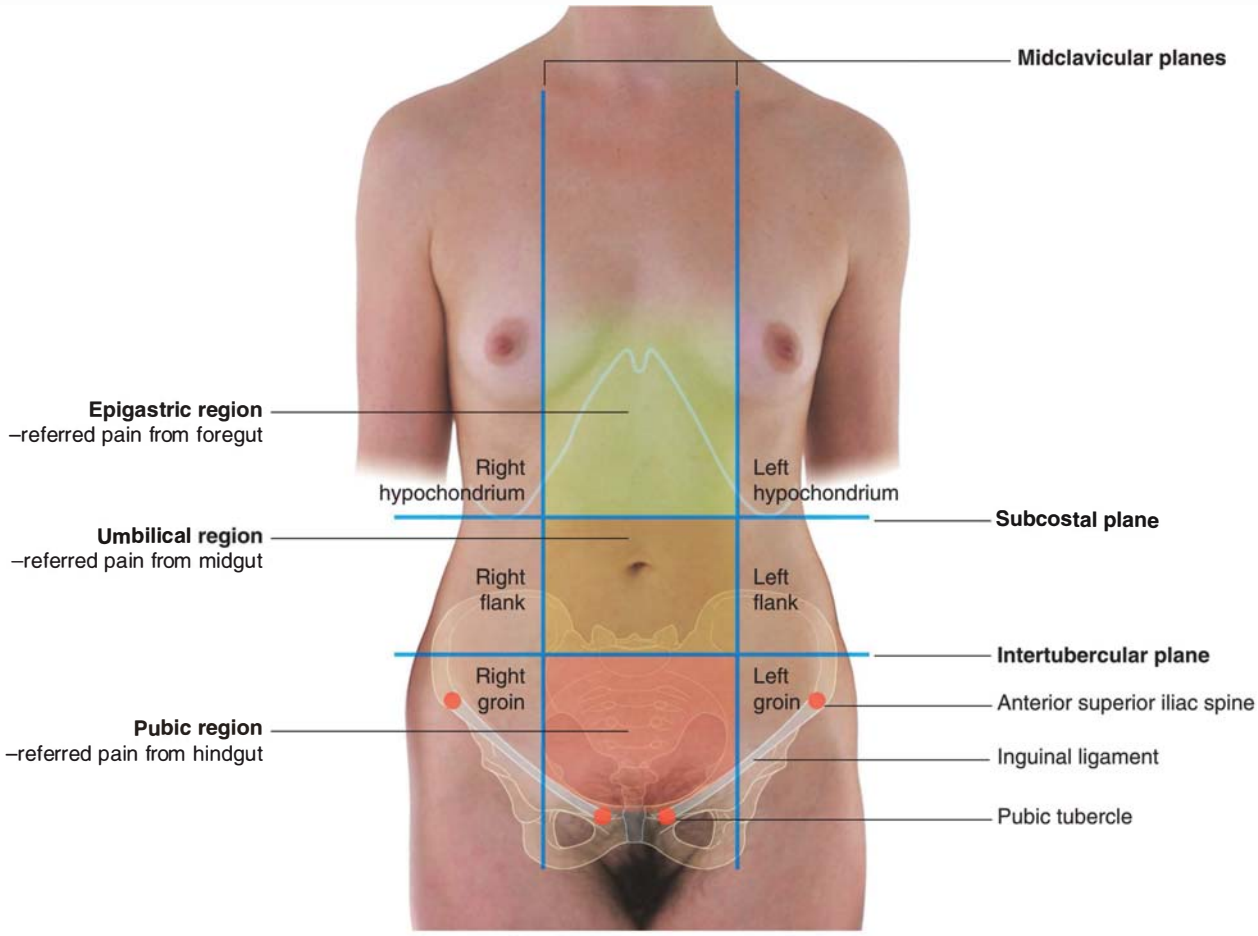


Fig. 4.169 The nine regions of the abdomen. Anterior view of a woman.

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Where to find the kidneys

The kidneys project onto the back on either side of the midline and are related to the lower ribs (Fig. 4.170):

- The left kidney is a little higher than the right and reaches as high as rib XI.
- The superior pole of the right kidney reaches only as high as rib XII.

The lower poles of the kidneys occur around the level of the disc between the LIII and LIV vertebrae. The hila of the kidneys and the beginnings of the ureters are at approximately the LI vertebra.

The ureters descend vertically anterior to the tips of the transverse processes of the lower lumbar vertebrae and enter the pelvis.

Where to find the spleen

The spleen projects onto the left side and back in the area of ribs IX to XI (Fig. 4.171). The spleen follows the contour of rib X and extends from the superior pole of the left kidney to just posterior to the midaxillary line.

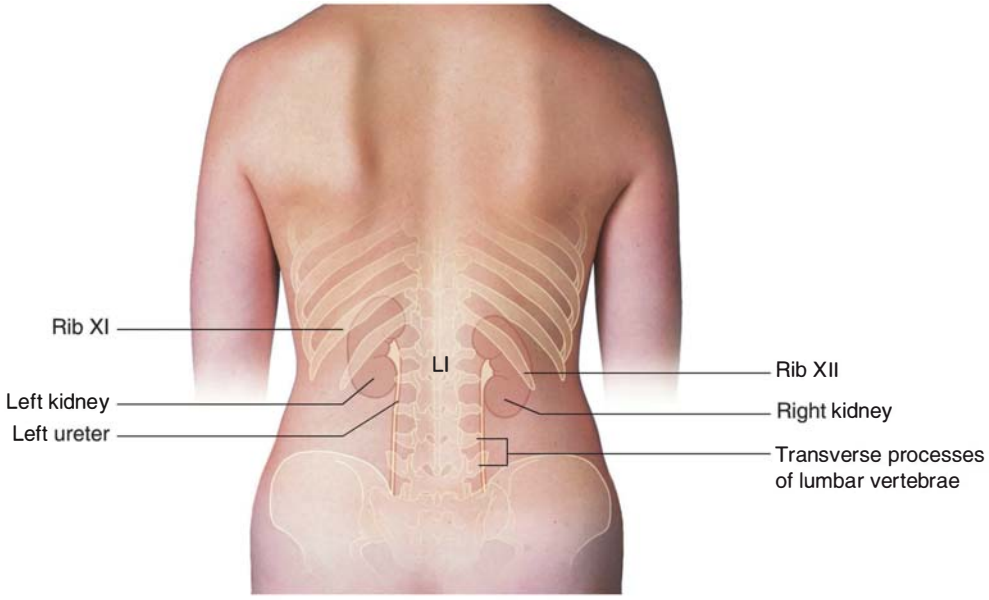


Fig. 4.170 Surface projection of the kidneys and ureters. Posterior view of the abdominal region of a woman.

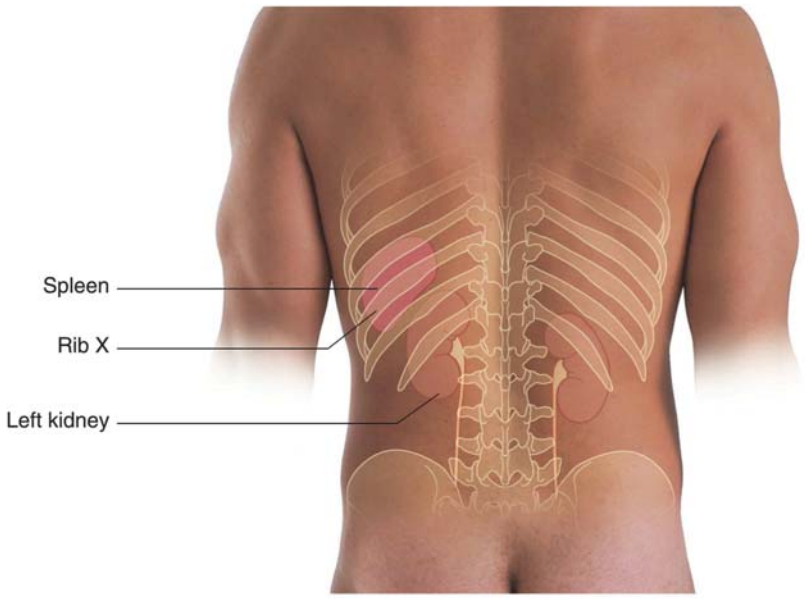


Fig. 4.171 Surface projection of the spleen. Posterior view of a man.

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Clinical cases

Case 1

TRAUMATIC RUPTURE OF THE DIAPHRAGM

A 45-year-old man had mild epigastric pain, and a diagnosis of esophageal reflux was made. He was given appropriate medication, which worked well. However, at the time of the initial consultation, the family practitioner requested a chest radiograph, which demonstrated a prominent hump on the left side of the diaphragm and old rib fractures.

The patient was recalled for further questioning.

He was extremely pleased with the treatment he had been given for his gastroesophageal reflux, but was concerned about being recalled for further history and examination. During the interview, he revealed that he had previously been involved in a motorcycle accident and had undergone a laparotomy for a "rupture." The patient did not recall what operation was performed, but was assured at the time that the operation was a great success.

The patient is likely to have undergone a splenectomy.

In any patient who has had severe blunt abdominal trauma (such as that caused by a motorcycle accident), lower left-sided rib fractures are an extremely important sign of appreciable trauma.

A review of the patient's old notes revealed that at the time of the injury the spleen was removed surgically, but it was not appreciated that there was a small rupture of the dome of the left hemidiaphragm. The patient gradually developed a hernia through which bowel could enter, producing the "hump" on the diaphragm seen on the chest radiograph.

Because this injury occurred many years ago and the patient has been asymptomatic, it is unlikely that the patient will come to any harm and was discharged.

Case 2

CHRONIC THROMBOSIS OF THE INFERIOR VENA CAVA

A medical student was asked to inspect the abdomen of two patients. On the first patient he noted irregular veins radiating from the umbilicus. On the second patient he noted irregular veins, coursing in a caudal to cranial direction, over the anterior abdominal wall from the groin to the chest. He was asked to explain his findings and determine the significance of these features.

In the first patient the veins were draining radially away from the periumbilical region. In normal individuals, enlarged veins do not radiate from the umbilicus. In patients with portal hypertension the portal venous pressure is increased as a result of hepatic disease. Small collateral veins develop at and around the obliterated umbilical vein. These veins pass through the umbilicus and drain onto the anterior abdominal wall, forming a portosystemic anastomosis. The eventual diagnosis for this patient was cirrhosis of the liver.

The finding of veins draining in a caudocranial direction on the anterior abdominal wall in the second patient is not typical for veins on the anterior abdominal wall. When veins are so prominent, it usually implies that there is an obstruction to the normal route of venous drainage and an alternative route has been taken. Typically, blood from the lower limbs and the retroperitoneal organs drains into the inferior vena cava and from here to the right atrium of the heart. This patient had a chronic thrombosis of the inferior vena cava, preventing blood from returning to the heart by the "usual" route.

Blood from the lower limbs and the pelvis may drain via a series of collateral vessels, some of which include the superficial inferior epigastric veins, which run in the superficial fascia. These anastomose with the superior, superficial, and deep epigastric venous systems to drain into the internal thoracic veins, which in turn drain into the brachiocephalic veins and the superior vena cava.

After the initial inferior vena cava thrombosis, the veins of the anterior abdominal wall and other collateral pathways hypertrophy to accommodate the increase in blood flow.

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Case 3

LIVER BIOPSY IN PATIENTS WITH SUSPECTED LIVER CIRRHOSIS

A 55-year-old man developed severe jaundice and a massively distended abdomen. A diagnosis of cirrhosis of the liver was made, and further confirmatory tests demonstrated that the patient had significant ascites (free fluid within the peritoneal cavity). A liver biopsy was necessary to confirm the cirrhosis, but there was some debate about how this biopsy should be obtained (Fig. 4.172).

In patients with cirrhosis it is important to determine the extent of the cirrhosis and the etiology.

History, examination, and blood tests are useful and are supported by complex radiological investigations. To begin treatment and determine the prognosis, a sample of liver tissue must be obtained. However, there are important issues to consider when taking a liver biopsy from a patient with suspected cirrhosis.

One issue is liver function.

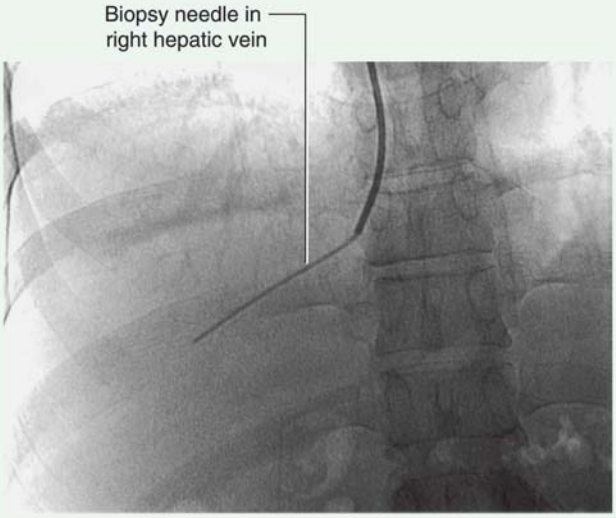


Fig. 4.172 Transjugular liver biopsy needle in the right hepatic vein. Radiograph.

The liver function of patients with suspected liver disease is poor, as demonstrated by the patient's jaundice—an inability to conjugate bilirubin. Importantly, because some liver products are blood-clotting factors involved in the clotting cascade, the blood-clotting ability of patients with severe liver disease is significantly impaired. These patients therefore have a high risk of bleeding.

Another issue is the presence of ascites.

Normally the liver rests against the lateral and anterior abdominal walls. This direct contact can be useful for care after a liver biopsy has been obtained. After the procedure, the patient lies over the region where the biopsy has been obtained and the weight of the liver stems any localized bleeding. When patients have significant ascites, the liver cannot be compressed against the walls of the abdomen and blood may pour freely into the ascitic fluid.

The patient has ascites, so another approach for a liver biopsy must be considered.

The patient was referred to the radiology department for a transjugular liver biopsy.

The skin around the jugular vein in the neck was anesthetized. Access was obtained through insertion of a needle and a guidewire. The guidewire was advanced through the right internal jugular vein and into the right brachiocephalic vein. It entered the superior vena cava, was passed along the posterior wall of the atrium, and entered the superior aspect of the inferior vena cava. A catheter was inserted over the wire and directed into the right hepatic vein. Using a series of dilators, the hole was enlarged and a biopsy needle was placed over the wire and into the right hepatic vein. The liver was biopsied through the right hepatic vein and the biopsy sample was removed. A simple suture was used to close the internal jugular vein in the neck, and minor compression stemmed any blood flow.

Assuming that the biopsy needle does not penetrate the liver capsule, it is not important how much the patient bleeds from the liver, because this bleeding will enter the hepatic vein and is immediately returned to the circulation.

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Case 4

HODGKIN'S LYMPHOMA

A 30-year-old man had a diffuse and poorly defined epigastric mass. Further examination revealed asymmetrical scrotal enlargement.

As part of her differential diagnosis, the resident considered the possibility that the man had testicular cancer with regional abdominal para-aortic nodal involvement (the lateral aortic, or lumbar, nodes).

A primary testicular neoplasm is the most common tumor in men between the ages of 25 and 34 and accounts for between 1% and 2% of all malignancies in men. A family history of testicular cancer and maldescent of the testis are strong predisposing factors.

Spread of the tumor is typically to the lymph node chains that drain the testes.

The testes develop from structures adjacent to the renal vessels in the upper abdomen, between the transversalis fascia and the peritoneum. They normally migrate through the inguinal canals into the scrotum just before birth. The testes take with them their arterial supply, their venous drainage, their nerve supply, and their lymphatics.

A computed tomography scan revealed a para-aortic lymph node mass in the upper abdomen and enlarged lymph nodes throughout the internal and common iliac lymph node chains.

Assuming the scrotal mass was a carcinoma of the testes, which would normally drain into the lateral aortic (lumbar) nodes in the upper abdomen, it would be very unusual for iliac lymphadenopathy to be present.

Further examination of the scrotal mass was required.

A transillumination test of the scrotum on the affected side was positive. An ultrasound scan revealed normal right and left testes and a large fluid collection around the right testis. A diagnosis of a right-sided hydrocele was made.

Scrotal masses are common in young males, and determining the exact anatomical site of the scrotal mass is of utmost clinical importance. Any mass that arises from the testis should be investigated to exclude testicular cancer. Masses that arise from the epididymis and scrotal lesions, such as fluid (hydrocele) or hernias, are also clinically important but are not malignant.

The ultrasound scan revealed fluid surrounding the testis, which is diagnostic of a hydrocele. Simple cysts arising from and around the epididymis (epididymal cysts) can be easily defined.

A diagnosis of lymphoma was suspected.

Lymphoma is a malignant disease of lymph nodes. Most lymphomas are divided into two specific types, namely Hodgkin's lymphoma and non-Hodgkin's lymphoma. If caught early the prognosis following radical chemotherapy is excellent.

The patient underwent a biopsy, which was performed from the posterior approach. He was placed in the prone position in the computed tomography (CT) scanner. A fine needle with a special cutting device was used to obtain a lymph node sample. A left-sided approach was used because the inferior vena cava is on the right side and the nodes were in the para-aortic regions (i.e., the biopsy needle would have to pass between the inferior vena cava and the aorta from a posterior approach, which is difficult). The skin was anesthetized using local anesthetic at the lateral border of the quadratus lumborum muscle. The needle was angled at approximately 45° within the quadratus lumborum muscle and entered the retroperitoneum to lie beside the left-sided para-aortic lymph nodes. Because this procedure is performed using CT guidance, the operator can advance the needle slowly, taking care not to "hit" other retroperitoneal structures.

A good biopsy was obtained and the diagnosis was Hodgkin's lymphoma. The patient underwent chemotherapy and 2 years later is in full remission and leads an active life.

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Case 5

INGUINAL HERNIA

A 35-year-old man had a soft mass approximately 3 cm in diameter in the right scrotum. The diagnosis was a right indirect inguinal hernia.

What were the examination findings?

The mass was not tender and the physician was not able to "get above it." The testes were felt separate from the mass, and a transillumination test (in which a bright light is placed behind the scrotum and the scrotal sac is viewed from the front) was negative. (A positive test occurs when the light penetrates through the scrotum.)

When the patient stood up, a positive cough "impulse" was felt within the mass.

After careful and delicate maneuvering, the mass could be massaged into the inguinal canal, so emptying from the scrotum. When the massaging hand was removed, the mass recurred in the scrotum.

An indirect inguinal hernia enters the inguinal canal through the deep inguinal ring. It passes through the inguinal canal to exit through the superficial inguinal ring in the aponeurosis of the external oblique muscle. The hernia sac lies superior and medial to the pubic tubercle and enters into the scrotum within the spermatic cord.

A direct inguinal hernia passes directly through the posterior wall of the inguinal canal. It does not pass down the inguinal canal. If large enough, it may pass through the superficial inguinal ring and into the scrotum.

Case 6

URETERIC STONE

A 25-year-old man developed severe pain in the left lower quadrant of his abdomen. The pain was diffuse and relatively constant but did ease for short periods of time. On direct questioning the patient indicated that the pain was in the inguinal region and radiated posteriorly into his left infrascapular region (loin). A urine dipstick was positive for blood (hematuria).

A diagnosis of a ureteric stone (calculus) was made.

The patient's initial infrascapular pain, which later radiated to the left groin, relates to passage of the ureteric stone along the ureter.

The origin of the pain relates to ureteral distention.

A series of peristaltic waves along the ureter transport urine along the length of the ureter from the kidney to the bladder. As the ureteric stone obstructs the kidney, the ureter becomes distended, resulting in an exacerbation of the pain. The peristaltic waves are superimposed upon the distention, resulting in periods of exacerbation and periods of relief.

The pain is referred.

The visceral afferent (sensory) nerve fibers from the ureter pass into the spinal cord, entering the first and second lumbar segments of the spinal cord. Pain is thus referred to cutaneous regions innervated by somatic sensory nerves from the same spinal cord levels.

The patient was investigated by a CT scan.

Traditionally patients are investigated with a plain radiograph to look for the radiopaque stone (90% of renal stones are radiopaque).

An ultrasound scan may be useful to assess for pelvicaliceal dilatation and may reveal stones at the pelviureteral junction or the vesicoureteric junction. Ultrasound is also valuable for assessing other causes of obstruction (e.g., tumors at and around the ureteric orifices in the bladder).

Usually an intravenous urogram would be carried out to enable assessment of the upper urinary tracts and precise location of the stone.

Not infrequently, CT scans of the abdomen are also obtained. These scans not only give information about the kidneys, ureters, and bladder but also show the position of the stone and other associated pathology.

If this patient's infrascapular pain was on the right and predominantly within the right lower abdomen, appendicitis would also have to be excluded. A CT scan would enable differentiation of appendicitis and urinary colic.

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Case 7

INTRAABDOMINAL ABSCESS

A 27-year-old woman was admitted to the surgical ward with appendicitis. She underwent an appendectomy. It was noted at operation that the appendix had perforated and there was pus within the abdominal cavity. The appendix was removed and the stump tied. The abdomen was washed out with warm saline solution. The patient initially made an uneventful recovery, but by day 7 she had become unwell, with pain over her right shoulder and spiking temperatures.

This patient had developed an intraabdominal abscess.

Any operation on the bowel may involve peritoneal contamination with fecal contents and fecal flora. This may not be appreciated at the time of the operation.

Over the postoperative period an inflammatory reaction ensued and an abscess cavity developed, filling with pus. Typically, the observation chart revealed a "swinging" pyrexia (fever).

The most common sites for abscess to develop are the pelvis and the hepatorenal recess.

When a patient is in the supine position, the lowest points in the abdominal and pelvic cavities are the posterior superior aspect of the peritoneal cavity (the hepatorenal recess) and, in women, the recto-uterine pouch (pouch of Douglas).

The shoulder pain suggested that the abscess was in the hepatorenal recess and that the pain was referred from the diaphragm.

The motor and sensory innervation of the diaphragm is from nerves C3 to C5. The somatic pain sensation from the parietal peritoneum covering the undersurface of the diaphragm is carried into the spinal cord by the phrenic nerve (C3 to C5) and is interpreted by the brain as coming from skin over the shoulder—a region supplied by other somatic sensory nerves entering the same levels of the spinal cord as those from the diaphragm.

A chest radiograph demonstrated elevation of the right hemidiaphragm.

This elevation of the right hemidiaphragm was due to the pus tracking from the hepatorenal space around the lateral and anterior aspect of the liver to sit on top of the liver in a subphrenic position. An ultrasound scan demonstrated this collection of fluid. The abscess cavity could be clearly seen by placing the ultrasound probe between ribs XI and XII. The inferior border of the right lower lobe lies at rib X in the midaxillary line. When the probe is placed between ribs XI and XII the ultrasound waves pass between the intercostal muscles and the parietal pleura laterally on the chest wall, and continue through the parietal pleura overlying the diaphragm into the cavity of the abscess, which lies below the diaphragm.

Drainage was not by an intercostal route. Instead, using CT guidance and local anesthesia, a subcostal drain was established and 1 liter of pus was removed (Fig. 4.173). It is important to bear in mind that placing a drain through the pleural cavity into the abdominal cavity effectively allows intraabdominal pus to pass into the thoracic cavity, and that this may produce an empyema (pus in the pleural space).

The patient made a slow and uneventful recovery.

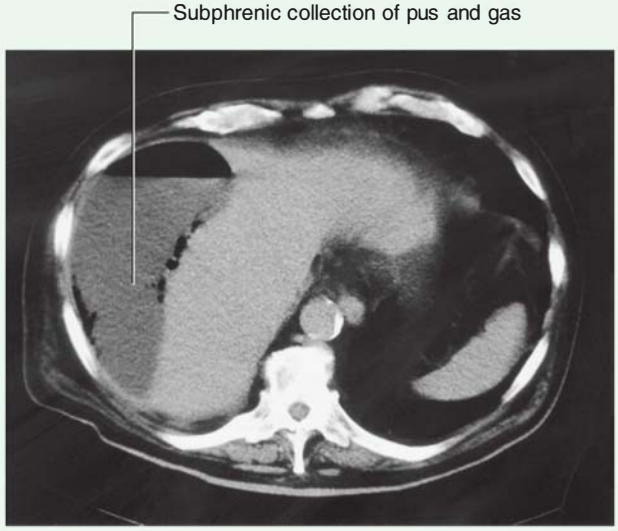


Fig. 4.173 Subphrenic collection of pus and gas. Computed tomogram in the axial plane.

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Case 8

COMPLICATIONS OF AN ABDOMINOPERINEAL RESECTION

A 45-year-old man developed a low-grade rectal carcinoma just above the anorectal margin. He underwent an abdominoperineal resection of the tumor and was left with a left lower abdominal colostomy (see below). Unfortunately, the man's wife left him for a number of reasons, including lack of sexual desire. He "turned to drink" and over the ensuing years developed cirrhosis. He was brought into the emergency room with severe bleeding from enlarged veins around his colostomy. An emergency transjugular intrahepatic portosystemic shunt was created, which stopped all bleeding (Figs 4.174 and 4.175). He is now doing well in a rehabilitation program.

A colostomy was necessary because of the low site of the tumor.

Carcinoma of the colon and rectum usually develops in older patients, but some people do get tumors early in life. Most tumors develop from benign polyps, which undergo malignant change. As the malignancy develops it invades through the wall of the bowel and then metastasizes to local lymphatics. The tumor extends within the wall for a few centimeters above and below its origin. Lymphatic spread is to local and regional lymph nodes and then to the pre-aortic lymph node chain. These drain eventually into the thoracic duct.

When this man was assessed for surgery, the tumor was so close to the anal margin that resection of the sphincters was necessary to be certain that the tumor margins were clear. The bowel cannot be joined to the anus without sphincters because the patient would be fecally incontinent. At surgery the tumor was excised, including the locoregional lymph node chains and the peritumoral fat around the rectum.

The free end of the sigmoid colon was brought through a hole in the anterior abdominal wall. The bowel was then carefully sutured to the anterior abdominal wall to allow placement of a bag to collect the feces. This is a colostomy.

Contrary to their usual immediate negative reaction to having a bag on the anterior abdominal wall, most patients cope extremely well, especially if they have been cured of cancer.

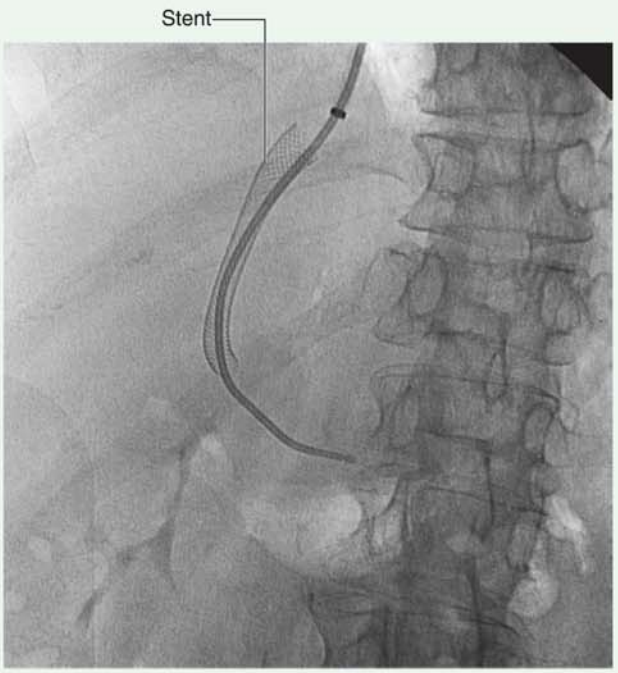


Fig. 4.174 Position of a transjugular intrahepatic portosystemic shunt stent. Radiograph.



Fig. 4.175 Functioning transjugular intrahepatic portosystemic shunt. Venogram.

(continues)

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Case 8 *(continued)*

This patient's pelvic nerves were damaged. The radical pelvic surgical dissection damaged the pelvic parasympathetic nerve supply necessary for erection of the penis. Unfortunately, this was not well explained to the patient, which in some part led to the failure of his relationship. With any radical surgery in the pelvis, the nerves that supply the penis or clitoris may be damaged, so interfering with sexual function.

This patient was bleeding from stomal varices.

As he developed a serious drinking problem, his liver became cirrhotic and this damaged the normal liver architecture. This in turn increased the blood pressure in the portal vein (**portal hypertension**).

In patients with portal hypertension small anastomoses develop between the veins of the portal system and the veins of the systemic circulation. These portosystemic anastomoses are usually of little consequence; however, at the gastroesophageal junction, they lie in a submucosal and mucosal position and are subject to trauma. Torrential hemorrhage may occur from even minor trauma, and death may ensue following blood loss. These varices require urgent treatment, which includes injecting sclerosant substances, banding, and even surgical ligation.

Fortunately, most of the other portosystemic anastomoses are of relatively little consequence. In patients with colostomies, small veins may develop between the veins of the large bowel (portal system drainage) and cutaneous veins on the anterior abdominal wall (systemic veins). If these veins become enlarged because of portal hypertension, they are subject to trauma as feces are passed through the colostomy. Torrential hemorrhage may ensue if they are damaged.

A procedure was carried out to lower the portal pressure.

To reduce the pressure in the portal vein in this patient, several surgical procedures were considered. These included sewing the side of the portal vein onto the inferior vena cava (**portacaval shunt**) and sewing the splenic vein onto the renal vein (a **splenorenal shunt**). These procedures, however, require a large abdominal incision and are extremely complex. As an alternative, it was decided to create a transjugular intrahepatic portosystemic shunt.

Creating a **transjugular intrahepatic portosystemic shunt** is a relatively new technique that may be carried out under local anesthesia. Using a right internal jugular approach, a long needle is placed through the internal jugular vein, the superior vena cava, and the right atrium, into the inferior vena cava. The right hepatic vein is cannulated and, with special steering wires, a needle is passed through the hepatic substance directly into the right branch of the portal vein. A small balloon is passed over the wire and through the hepatic substance and is inflated. After the balloon has been removed, a metallic stent (a flexible wire tube) is placed across this tract in the liver to keep it open. Blood now freely flows from the portal vein into the right hepatic vein, creating a portosystemic shunt.

As a result of this procedure, the pressure in this patient's portal system is lower and similar to that of the systemic venous system, so reducing the potential for bleeding at the portosystemic anastomoses (i.e., the colostomy).

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Case 9

CARCINOMA OF THE HEAD OF THE PANCREAS

A 52-year-old woman visited her family physician with complaints of increasing lethargy and vomiting. The physician examined her and noted that compared to previous visits she had lost significant weight. She was also jaundiced, and on examination of the abdomen a well-defined 10-cm rounded mass was palpable below the liver edge in the right upper quadrant (Fig. 4.176).

The clinical diagnosis was carcinoma of the head of the pancreas.

It is difficult to appreciate how such a precise diagnosis can be made clinically when only three clinical signs have been described.

The patient's obstruction was in the distal bile duct.



Fig. 4.176 Tumor in the head of the pancreas. Computed tomogram in the axial plane.

When a patient has jaundice, the causes are excessive breakdown of red blood cells (prehepatic), hepatic failure (hepatic jaundice), and posthepatic causes, which include obstruction along the length of the biliary tree.

The patient had a mass in her right upper quadrant that was palpable below the liver; this was the gallbladder.

In healthy individuals, the gallbladder is not palpable. An expanded gallbladder indicates obstruction either within the cystic duct or below the level of the cystic duct insertion (i.e., the bile duct).

The patient's vomiting was related to the position of the tumor.

It is not uncommon for vomiting and weight loss (cachexia) to occur in patients with a malignant disease. The head of the pancreas lies within the curve of the duodenum, primarily adjacent to the descending part of the duodenum. Any tumor mass in the region of the head of the pancreas is likely to expand and may encase and invade the duodenum. Unfortunately, in this patient's case, this happened, producing almost complete obstruction. Further discussion with the patient revealed that she was vomiting relatively undigested food soon after each meal.

A CT scan demonstrated further complications.

In the region of the head and neck of the pancreas are complex anatomical structures, which may be involved with a malignant process. The CT scan confirmed a mass in the region of the head of the pancreas, which invaded the descending part of the duodenum. The mass extended into the neck of the pancreas and had blocked the distal part of the bile duct and the pancreatic duct. Posteriorly the mass had directly invaded the portal venous confluence of the splenic and superior mesenteric veins, producing a series of gastric, splenic, and small bowel varices.

This patient underwent palliative chemotherapy, but died 7 months later.

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Case 10

CAVAL OBSTRUCTION

A 62-year-old man came to the emergency department with swelling of both legs and a large left varicocele (enlarged and engorged varicose veins around the left testis and within the left pampiniform plexus of veins).

The patient was known to have a left renal cell carcinoma and was due to have this operated on the following week.

Anatomically it is possible to link all of the findings with the renal cell carcinoma by knowing the biology of the tumor.

Renal cell carcinoma tends to grow steadily and predictably. Typically, when the tumor is less than 3 to 4 cm, it remains confined to the kidney. Large tumors have the propensity to grow into the renal vein, the inferior vena cava and the right atrium and through the heart into the pulmonary artery.

The tumor grew into the renal vein.

As the tumor grew into the renal vein it blocked off all tributaries draining into the vein, the largest of which is the left testicular vein. This blockage of the left testicular vein caused a dilation of the veins around the left testis (a varicocele occurred).

The swollen legs were accounted for by caval obstruction.

The tumor grew along the renal vein and into the inferior vena cava toward the heart. Renal tumors can grow rapidly; in this case the tumor grew rapidly into the inferior vena cava, occluding it. This increased the pressure in the leg veins, resulting in swelling and pitting edema of the ankles.

The patient unfortunately died on the operating table.

In this patient's case, a "tongue" of tumor grew into the inferior vena cava. At the time of surgery, the initial dissection mobilized the kidney on its vascular pedicle; however, a large portion of tumor became detached in the inferior vena cava. The tumor embolus passed through the right atrium and right ventricle and occluded the pulmonary artery. This could not be cleared at the time of surgery, and the patient succumbed.

Case 11

DIVERTICULAR DISEASE

A 65-year-old businessman came to the emergency department with severe lower abdominal pain that was predominantly central and left sided. He had pain radiating into the left loin, and he also noticed he was passing gas and fecal debris as he urinated.

A CT scan of his abdomen and pelvis was performed (Fig. 4.177).

The CT scan demonstrated a collection of fluid (likely a pelvic abscess) in the left iliac fossa. Associated with this collection of fluid was significant bowel wall thickening of the sigmoid colon and multiple small diverticula arising throughout the sigmoid colon. Gas was present in the bladder. An obstruction was noted in the left ureter and the left pelvicalyceal system.

The patient underwent an urgent operation.

As the surgeons entered into the abdominal cavity through a midline incision, the tissues in the left iliac

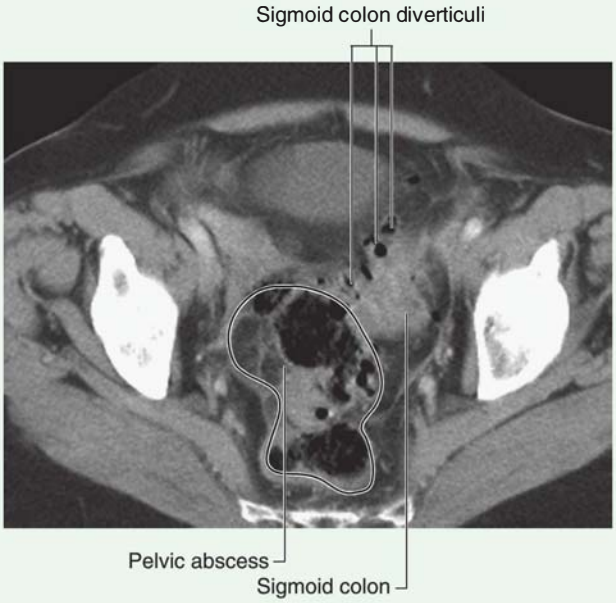


Fig. 4.177 A computed tomogram, in the axial plane, of the pelvis demonstrates a loop of sigmoid colon with numerous diverticula and a large abscess in the pelvic cavity.

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Case 11 *(continued)*

fossa were significantly inflamed. The surgeon used his hand to mobilize the sigmoid colon and entered a cavity from which there was a "whoosh" of pus as indicated on the CT scan. The pus was washed out and drained. The sigmoid colon was remarkably thickened and inflamed and stuck to the dome of the bladder. Careful finger dissection revealed a small perforation in the dome of the bladder, allowing the passage of fecal material and gas into the bladder and producing the patient's symptoms of pneumaturia and fecaluria. The sigmoid colon was resected. The rectal stump was oversewn and the descending colon was passed through the anterior abdominal wall to form a colostomy. The bladder was catheterized and the small hole in the dome of the bladder was oversewn.

The patient had a difficult postoperative period in the intensive care unit where he remained pyrexial and septic. The colostomy began to function well.

An ultrasound was performed and demonstrated the continued dilatation in the left kidney, and the patient underwent a nephrostomy.

Under ultrasound guidance a drainage catheter was placed into the renal pelvis through the renal cortex on the left. A significant amount of pus was drained from the renal tract initially; however, after 24 hours urine passed freely.

The likely cause for the obstruction was the inflammation around the distal ureter on the left. It is also possible that a small ureteric perforation occurred, allowing bacteria to enter the urinary tract.

The patient made a further uneventful recovery with resumption of normal renal function and left the hospital.

On return to the surgeon in the outpatient clinic some weeks later, the patient did not wish to continue with his colostomy and bag. Further to discussion, surgery was planned to "rejoin" the patient.

At operation the colostomy was "taken down" and the rectal stump was identified. There was, however, a significant gap between the bowel ends. To enable the bowel to be sutured, the descending colon was mobilized from the posterior abdominal wall. An anastomosis was performed and the patient left the hospital 1 week later and currently remains well.

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Case 12

ENDOLEAK AFTER ENDOVASCULAR REPAIR OF ABDOMINAL AORTIC ANEURYSM

A 72-year-old man was brought to the emergency department with an abdominal aortic aneurysm (an expansion of the infrarenal abdominal aorta). The aneurysm measured 10 cm, and after discussion with the patient it was scheduled for repair.

The surgical and endovascular treatment options were explained to the patient.

Treatment of abdominal aortic aneurysms has been, for many years, an operative procedure where the dilatation (ballooning) of the aorta is resected and a graft is sewn into position. A modern option is to place a graft to line the aneurysm from within the artery (endovascular aneurysm repair). In this technique the surgeon dissects the femoral artery and makes a small hole in it. The graft is compressed within a catheter and the catheter is passed through the femoral artery and the iliac arterial system into the distal abdominal aorta. The graft can then be released inside the aorta, effectively relining it to prevent further expansion of the aneurysm.

Occasionally the relined aneurysm may continue to enlarge after the endovascular graft has been placed and a cause needs to be identified.

A Doppler ultrasound investigation of the abdomen and a CT scan revealed there was flow between the endovascular lining and the wall of the aneurysm.

The likely sources for this bleeding were assessed.

The graft usually begins below the level of the renal arteries and divides into two limbs that end in the common iliac arteries. The aneurysm may continue to be fed from any vessels between the graft and the aneurysm wall. These vessels can include the lumbar arteries and the inferior mesenteric artery. Interestingly, blood usually flows from the abdominal aorta into the inferior mesenteric artery and the lumbar arteries; however, with the changes in flow dynamics with the graft in place,

(continues)

Case 12 *(continued)*

blood may flow in the opposite direction through these branches, thereby leading to enlargement of the aneurysm.

Blood flow was from the superior mesenteric artery into the aneurysm sac.

Above the level of the graft the superior mesenteric artery arises normally. From the right colic and middle colic branches a marginal branch around the colon anastomoses, in the region of the splenic flexure, with marginal branches from the inferior mesenteric artery (this can become a hypertrophied vessel known as the marginal artery of Drummond). In this situation, blood passed retrogradely into the inferior mesenteric artery, filling the aneurysm and allowing it to remain pressurized and expand.

The inferior mesenteric artery was ligated **laparoscopically** and the aneurysm failed to expand further. Over the ensuing 6 months the aneurysm contracted. The patient remains fit and healthy, with two small scars in the groin.

Case 13

METASTATIC LESIONS IN THE LIVER

A 44-year-old woman had been recently diagnosed with melanoma on the toe and underwent a series of investigations.

Melanoma (properly called malignant melanoma) can be an aggressive form of skin cancer that spreads to lymph nodes and multiple other organs throughout the body. The malignant potential is dependent upon its cellular configuration and also the depth of its penetration through the skin.

The patient developed malignant melanoma in the foot, which spread to the lymph nodes of the groin. The inguinal lymph nodes were resected; however, it was noted on follow-up imaging that the patient had developed two metastatic lesions within the right lobe of the liver.

Surgeons and physicians considered the possibility of removing these lesions.

A CT scan was performed that demonstrated the lesions within segments V and VI of the liver (Fig. 4.178).

The segmental anatomy of the liver is important because it enables the surgical planning for resection.

The surgery was undertaken and involved identifying the portal vein and the confluence of the right and left hepatic ducts. The liver was divided in the imaginary principal plane of the middle hepatic vein. The main hepatic duct and biliary radicals were ligated and the right liver was successfully resected.

The segments remaining included the left lobe of the liver.

The patient underwent a surgical resection of segments V, VI, VII, and VIII. The remaining segments included IVa, IVb, I, II, and III. It is important to remember that the lobes of the liver do not correlate with the hepatic volume. The left lobe of the liver contains only segments II and III. The right lobe of the liver contains segments IV, V, VI, VII, and VIII. Hence, cross-sectional imaging is important when planning surgical segmental resection.

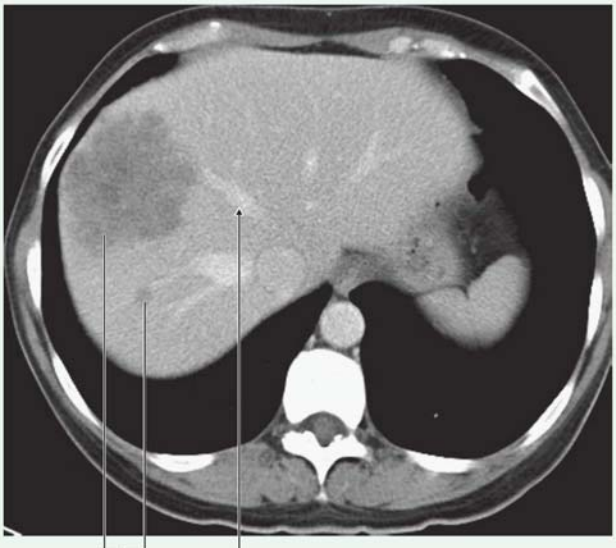


Fig. 4.178 This postcontrast computed tomogram, in the axial plane, demonstrates two metastases situated within the right lobe of the liver. The left lobe of the liver is clear. The larger of the two metastases is situated to the right of the middle hepatic vein, which lies in the principal plane of the liver dividing the left and right sides of the liver.

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5

Pelvis and Perineum

ADDITIONAL LEARNING RESOURCES for Chapter 5, Pelvis and Perineum, on STUDENT CONSULT (www.studentconsult.com):

- Image Library—illustrations of pelvic and perineal anatomy, Chapter 5
- Self-Assessment—National Board style multiple-choice questions, Chapter 5
- Short Questions—these are questions requiring short responses, Chapter 5
- Interactive Surface Anatomy—interactive surface animations, Chapter 5
- Medical Clinical Case Studies, Chapter 5
  - Pelvic kidney
  - Varicocele
- Clinical Cases, Chapter 5
  - Varicocele
  - Sciatic nerve compression
  - Pelvic kidney
  - Left common iliac artery obstruction
  - Latrogenic ureteric injury
  - Ectopic pregnancy
  - Uterine tumor

Free Online Self-Study Course: Anatomy and Embryology

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- Embryology modules 68 through 70

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Conceptual overview

GENERAL DESCRIPTION

The pelvis and perineum are interrelated regions associated with the pelvic bones and terminal parts of the vertebral column. The pelvis is divided into two regions:

- The superior region related to upper parts of the pelvic bones and lower lumbar vertebrae is the **false pelvis** (**greater pelvis**) and is generally considered part of the abdomen (Fig. 5.1).
- The **true pelvis** (**lesser pelvis**) is related to the inferior parts of the pelvic bones, sacrum, and coccyx, and has an inlet and an outlet.

The bowl-shaped **pelvic cavity** enclosed by the true pelvis consists of the pelvic inlet, walls, and floor. This cavity is continuous superiorly with the abdominal cavity and contains elements of the urinary, gastrointestinal, and reproductive systems.

The perineum (Fig. 5.1) is inferior to the floor of the pelvic cavity; its boundaries form the **pelvic outlet**. The perineum contains the external genitalia and external openings of the genitourinary and gastrointestinal systems.

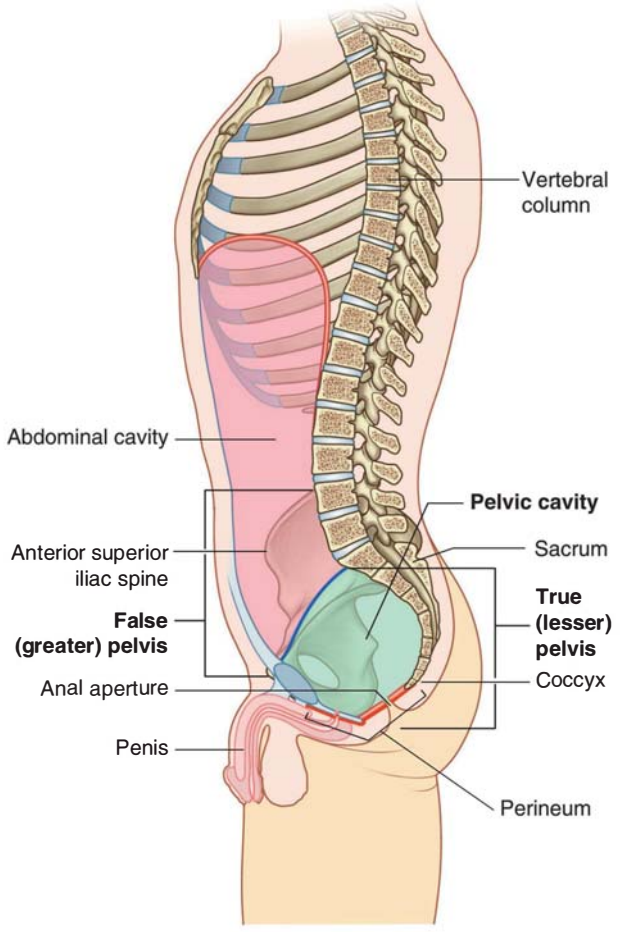


Fig. 5.1 Pelvis and perineum.

FUNCTIONS

Contains and supports the bladder, rectum, anal canal, and reproductive tracts

Within the pelvic cavity, the bladder is positioned anteriorly and the rectum posteriorly in the midline.

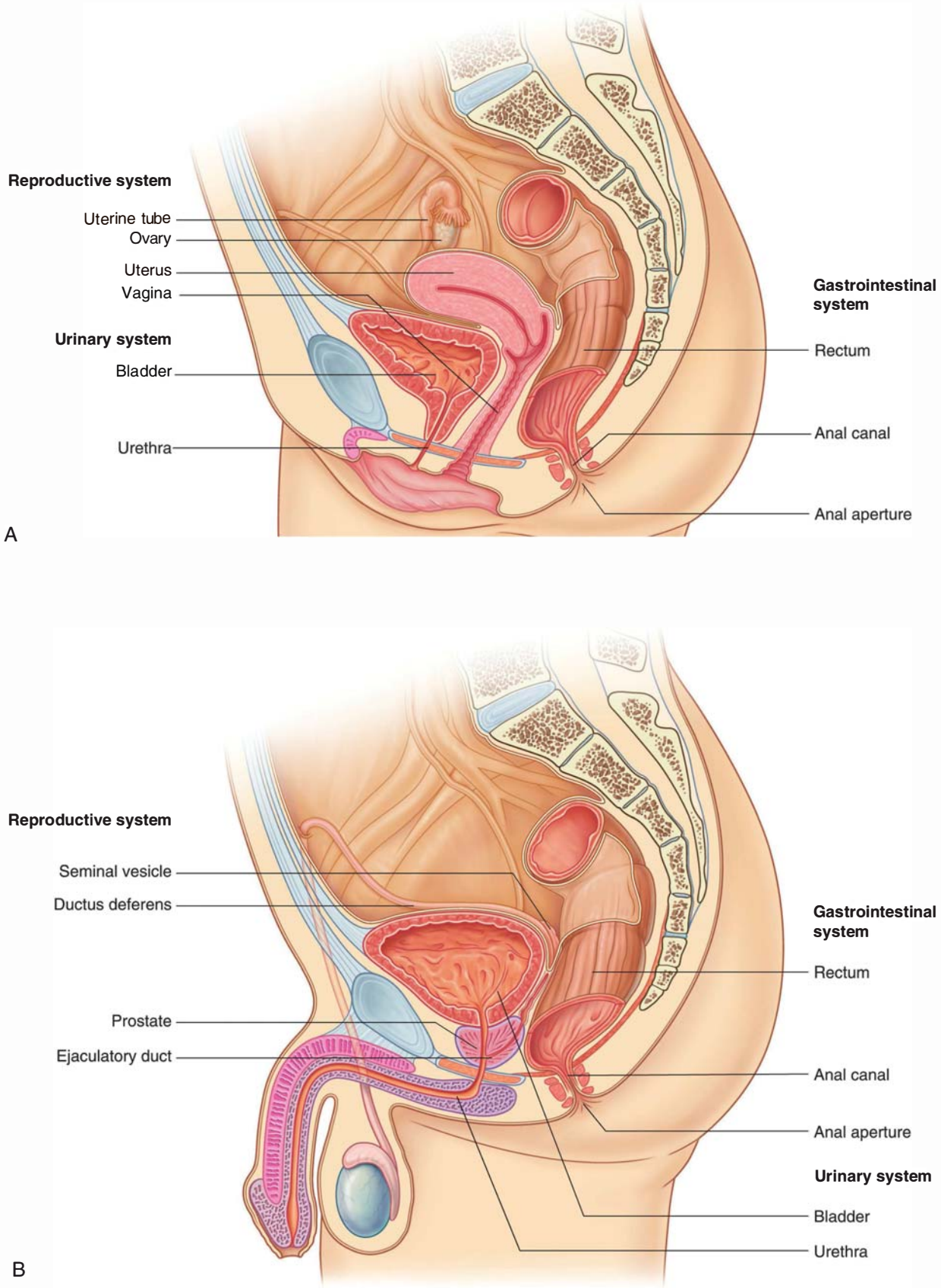
As it fills, the bladder expands superiorly into the abdomen. It is supported by adjacent elements of the pelvic bone and by the pelvic floor. The urethra passes through the pelvic floor to the perineum, where, in women, it opens externally (Fig. 5.2A) and in men it enters the base of the penis (Fig. 5.2B).

Continuous with the sigmoid colon at the level of vertebra SIII, the rectum terminates at the anal canal, which penetrates the pelvic floor to open into the perineum. The anal canal is angled posteriorly on the rectum. This flexure is maintained by muscles of the pelvic floor and is relaxed during defecation. A skeletal muscle sphincter is associated with the anal canal and the urethra as each passes through the pelvic floor.

The pelvic cavity contains most of the reproductive tract in women and part of the reproductive tract in men.

- In women, the vagina penetrates the pelvic floor and connects with the uterus in the pelvic cavity. The uterus is positioned between the rectum and the bladder. A uterine (fallopian) tube extends laterally on each side toward the pelvic wall to open near the ovary.
- In men, the pelvic cavity contains the site of connection between the urinary and reproductive tracts. It also contains major glands associated with the reproductive system—the prostate and two seminal vesicles.

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Fig. 5.2 The pelvis and perineum contain and support terminal parts of the gastrointestinal, urinary, and reproductive systems. **A.** In women. **B.** In men.

Anchors the roots of the external genitalia

In both genders, the roots of the external genitalia, the clitoris and the penis, are firmly anchored to:

- the bony margin of the anterior half of the pelvic outlet, and
- a thick, fibrous, perineal membrane, which fills the area (Fig. 5.3).

The roots of the external genitalia consist of erectile (vascular) tissues and associated skeletal muscles.

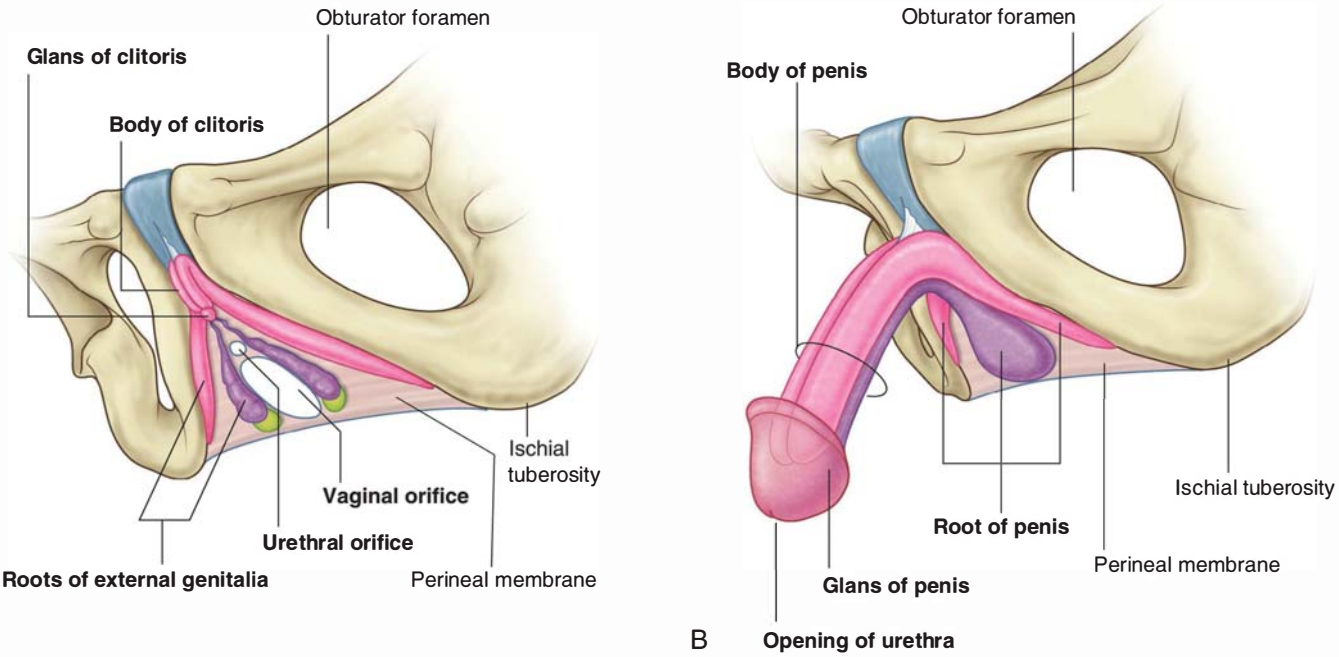


Fig. 5.3 The perineum contains and anchors the roots of the external genitalia. **A.** In women. **B.** In men.

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COMPONENT PARTS

Pelvic inlet

The pelvic inlet is somewhat heart shaped and completely ringed by bone (Fig. 5.4). Posteriorly, the inlet is bordered by the body of vertebra SI, which projects into the inlet as the sacral **promontory**. On each side of this vertebra, wing-like transverse processes called the **alae** (**wings**) contribute to the margin of the pelvic inlet. Laterally, a prominent rim on the pelvic bone continues the boundary of the inlet forward to the pubic symphysis, where the two pelvic bones are joined in the midline.

Structures pass between the pelvic cavity and the abdomen through the pelvic inlet.

During childbirth, the fetus passes through the pelvic inlet from the abdomen, into which the uterus has expanded during pregnancy, and then passes through the pelvic outlet.

Pelvic walls

The walls of the true pelvis consist predominantly of bone, muscle, and ligaments, with the sacrum, coccyx, and inferior half of the pelvic bones forming much of them.

Two ligaments—the **sacrospinous** and the **sacrotuberous ligaments**—are important architectural elements of the walls because they link each pelvic bone to the sacrum and coccyx (Fig. 5.5A). These ligaments also convert two notches on the pelvic bones—the **greater** and **lesser sciatic notches**—into foramina on the lateral pelvic walls.

Completing the walls are the **obturator internus** and **piriformis** muscles (Fig. 5.5B), which arise in the pelvis and exit through the sciatic foramina to act on the hip joint.

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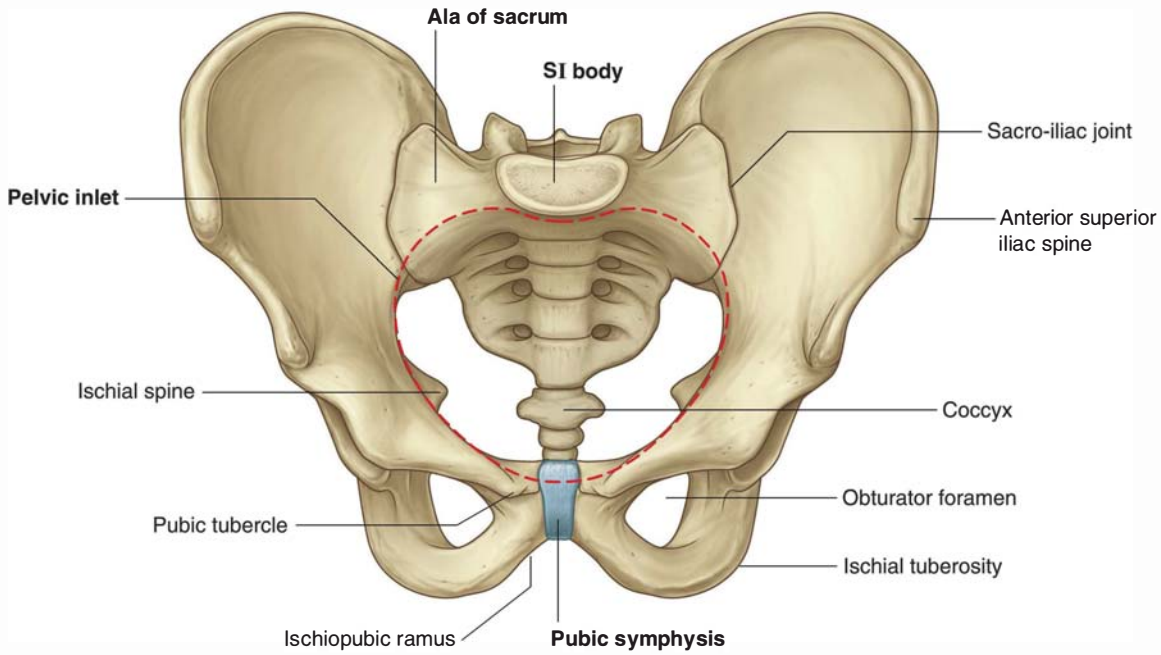


Fig. 5.4 Pelvic inlet.

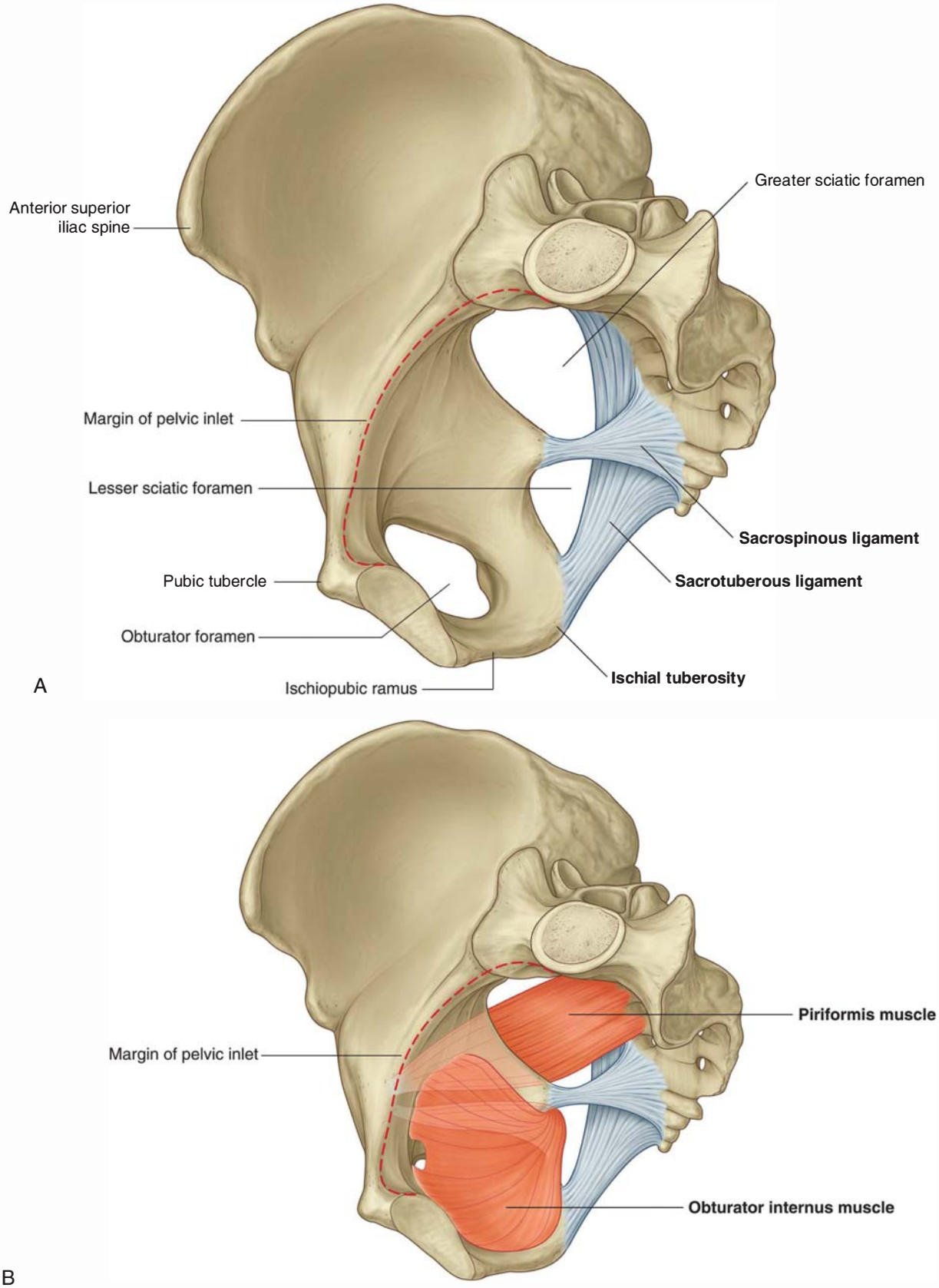


Fig. 5.5 Pelvic walls. **A.** Bones and ligaments of the pelvic walls. **B.** Muscles of the pelvic walls.

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Pelvic outlet

The diamond-shaped pelvic outlet is formed by both bone and ligaments (Fig. 5.6). It is limited anteriorly in the midline by the pubic symphysis.

On each side, the inferior margin of the pelvic bone projects posteriorly and laterally from the pubic symphysis to end in a prominent tuberosity, the **ischial tuberosity**. Together, these elements construct the pubic arch, which forms the margin of the anterior half of the pelvic outlet. The sacrotuberous ligament continues this margin posteriorly from the ischial tuberosity to the coccyx and sacrum. The pubic symphysis, ischial tuberosities, and coccyx can all be palpated.

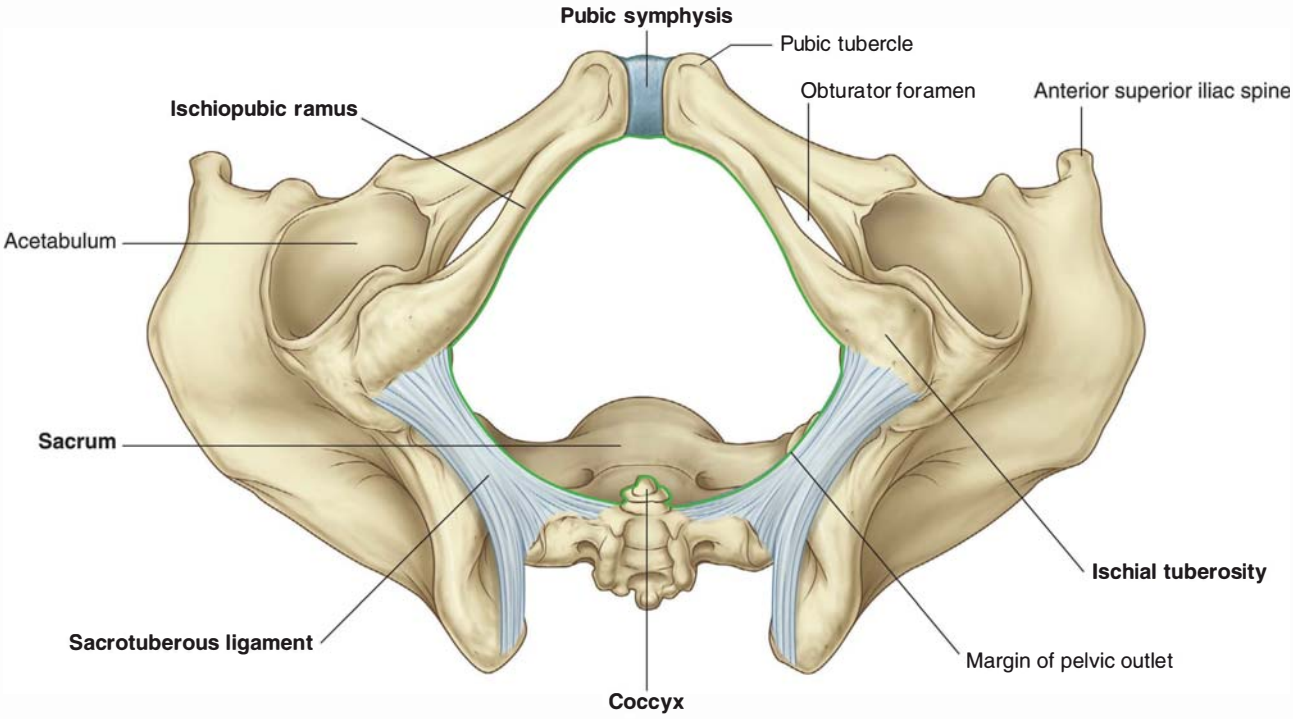


Fig. 5.6 Pelvic outlet.

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Pelvic floor

The pelvic floor, which separates the pelvic cavity from the perineum, is formed by muscles and fascia (Fig. 5.7).

Two **levator ani** muscles attach peripherally to the pelvic walls and join each other at the midline by a connective tissue raphe. Together they are the largest components of the bowl- or funnel-shaped structure known as the **pelvic diaphragm**, which is completed posteriorly by the **coccygeus muscles**. These latter muscles overlie the sacrospinous ligaments and pass between the margins of the sacrum and the coccyx and a prominent spine on the pelvic bone, the **ischial spine**.

The pelvic diaphragm forms most of the pelvic floor and in its anterior regions contains a U-shaped defect, which is associated with elements of the urogenital system.

The anal canal passes from the pelvis to the perineum through a posterior circular orifice in the pelvic diaphragm.

The pelvic floor is supported anteriorly by:

- the perineal membrane, and
- muscles in the **deep perineal pouch**.

The **perineal membrane** is a thick, triangular fascial sheet that fills the space between the arms of the pubic arch, and has a free posterior border (Fig. 5.7). The deep perineal pouch is a narrow region superior to the perineal membrane.

The margins of the U-shaped defect in the pelvic diaphragm merge into the walls of the associated viscera and with muscles in the deep perineal pouch below.

The vagina and the urethra penetrate the pelvic floor to pass from the pelvic cavity to the perineum.

Pelvic cavity

The pelvic cavity is lined by peritoneum continuous with the peritoneum of the abdominal cavity that drapes over the superior aspects of the pelvic viscera, but in most regions, does not reach the pelvic floor (Fig. 5.8A).

The pelvic viscera are located in the midline of the pelvic cavity. The bladder is anterior and the rectum is posterior. In women, the uterus lies between the bladder and rectum (Fig. 5.8B). Other structures, such as vessels and nerves, lie deep to the peritoneum in association with the pelvic walls and on either side of the pelvic viscera.

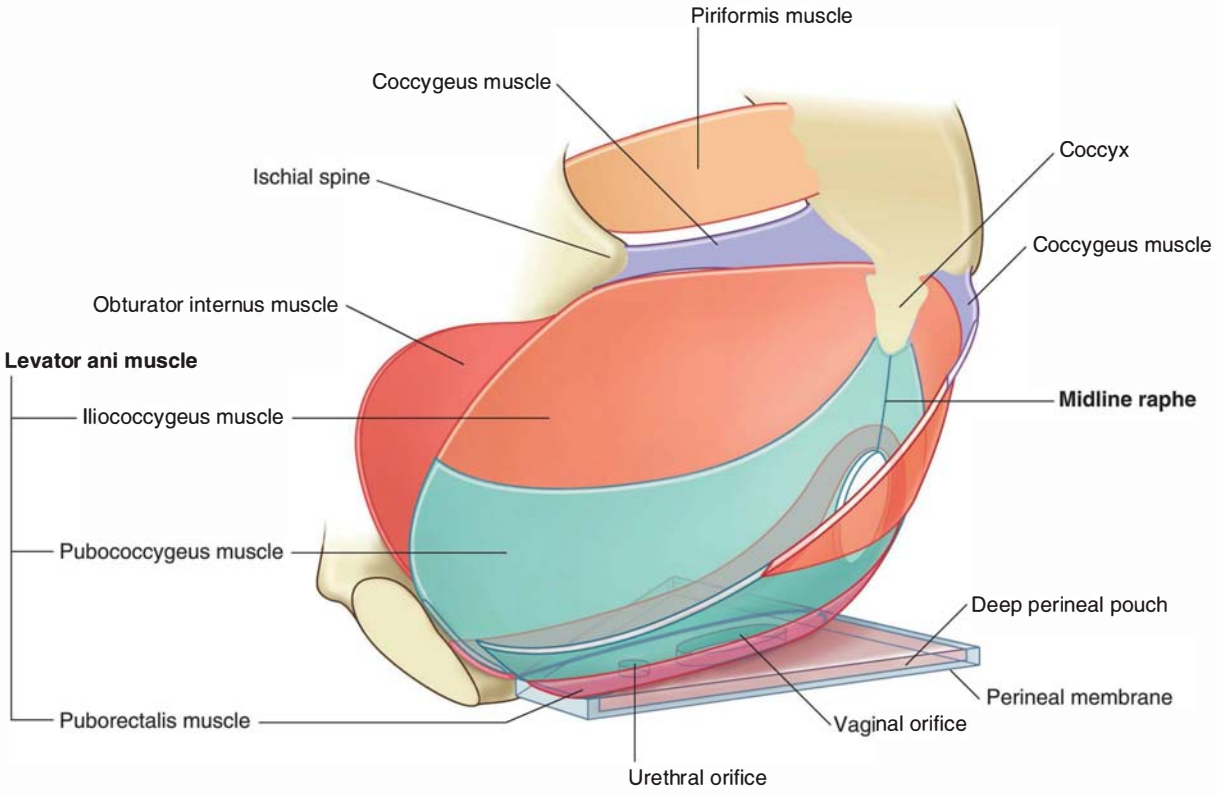


Fig. 5.7 Pelvic floor.

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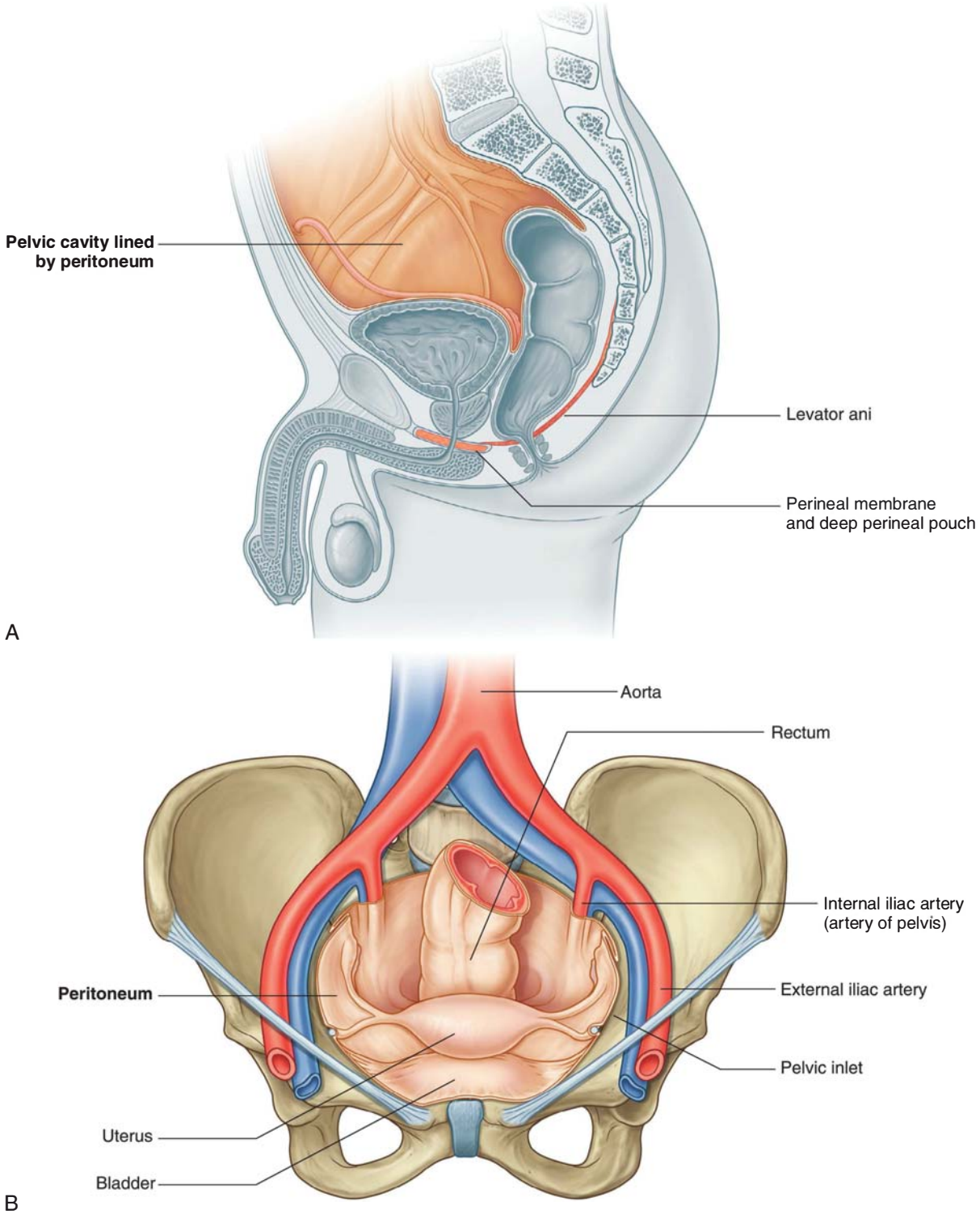


Fig. 5.8 Pelvic cavity and peritoneum. **A.** In men (sagittal section). **B.** In women (anterior view).

Perineum

The perineum lies inferior to the pelvic floor between the lower limbs (Fig. 5.9). Its margin is formed by the pelvic outlet. An imaginary line between the ischial tuberosities divides the perineum into two triangular regions.

- Anteriorly, the **urogenital triangle** contains the roots of the external genitalia and, in women, the openings of the urethra and the vagina (Fig. 5.9A). In men, the distal part of the urethra is enclosed by erectile tissues and opens at the end of the penis (Fig. 5.9B).
- Posteriorly, the **anal triangle** contains the anal aperture.

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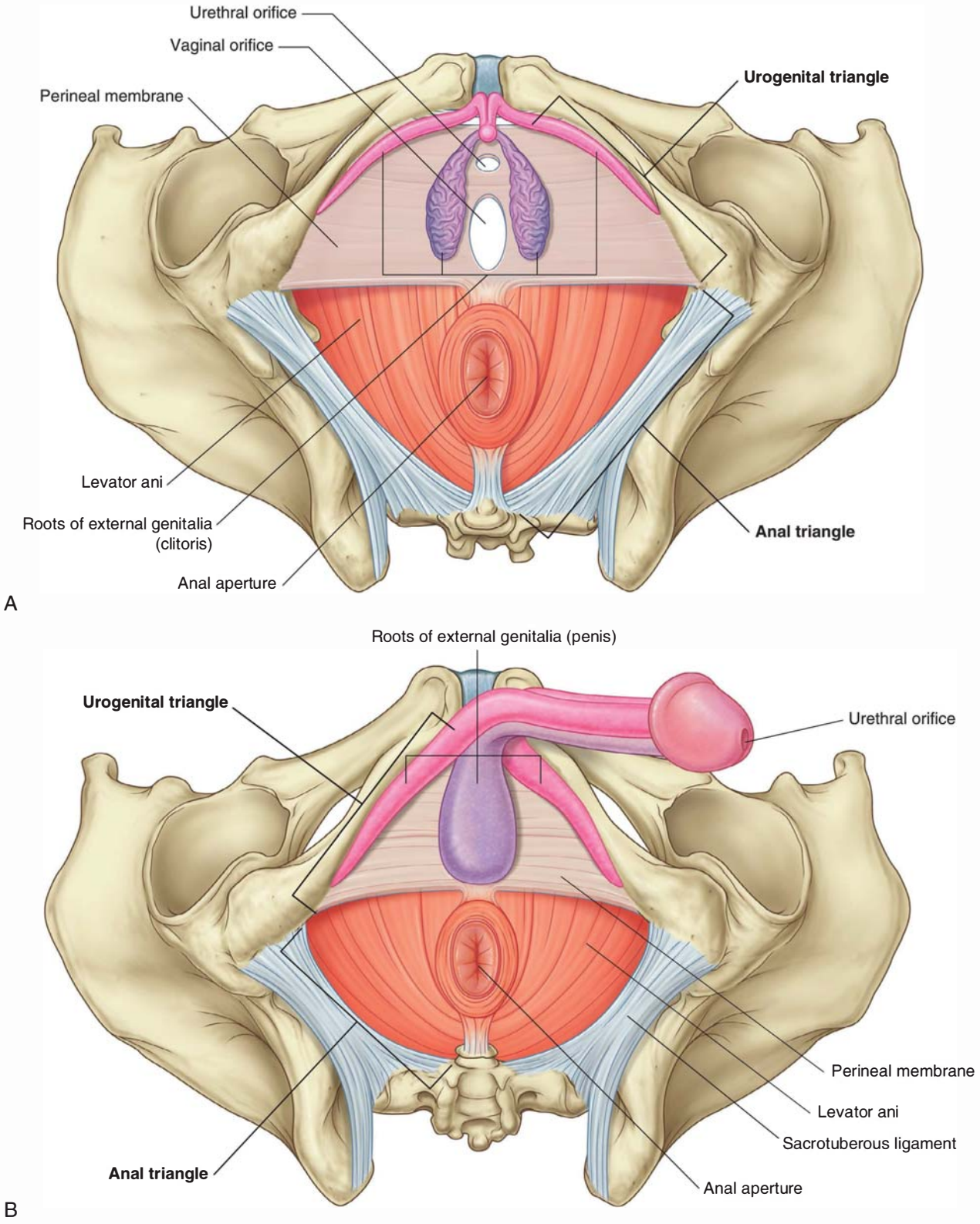


Fig. 5.9 Perineum. **A.** In women. **B.** In men.

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RELATIONSHIP TO OTHER REGIONS

Abdomen

The cavity of the true pelvis is continuous with the abdominal cavity at the pelvic inlet (Fig. 5.10A). All structures passing between the pelvic cavity and abdomen, including major vessels, nerves, and lymphatics, as well as the sigmoid colon and ureters, pass via the inlet. In men, the ductus deferens on each side passes through the anterior abdominal wall and over the inlet to enter the pelvic cavity. In women, ovarian vessels, nerves, and lymphatics pass through the inlet to reach the ovaries, which lie on each side just inferior to the pelvic inlet.

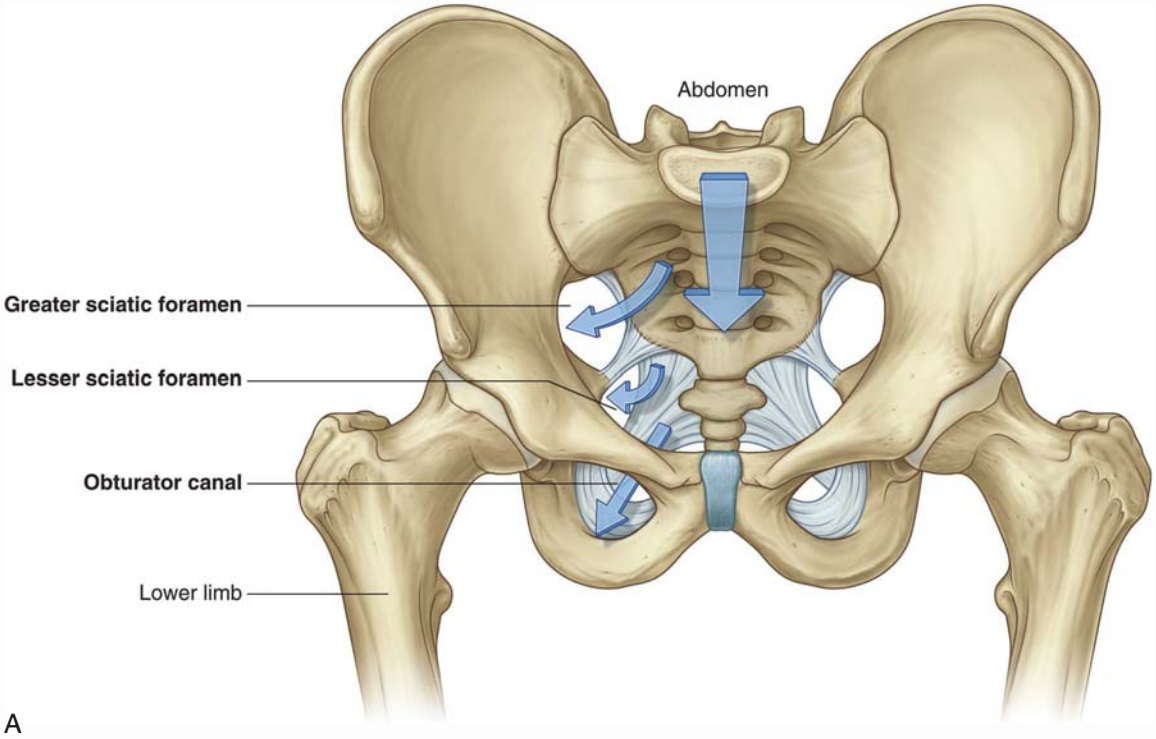


Fig. 5.10 Areas of communication between the true pelvis and other regions. **A.** Between the true pelvis, abdomen, and lower limb.

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Lower limb

Three apertures in the pelvic wall communicate with the lower limb (Fig. 5.10A):

- the obturator canal,
- the greater sciatic foramen, and
- the lesser sciatic foramen.

The obturator canal forms a passageway between the pelvic cavity and the adductor region of the thigh, and is formed in the superior aspect of the obturator foramen, between bone, a connective tissue membrane, and muscles that fill the foramen.

The lesser sciatic foramen, which lies inferior to the pelvic floor, provides communication between the gluteal region and the perineum (Fig. 5.10B).

The pelvic cavity also communicates directly with the perineum through a small gap between the pubic symphysis and the perineal membrane (Fig. 5.10B).

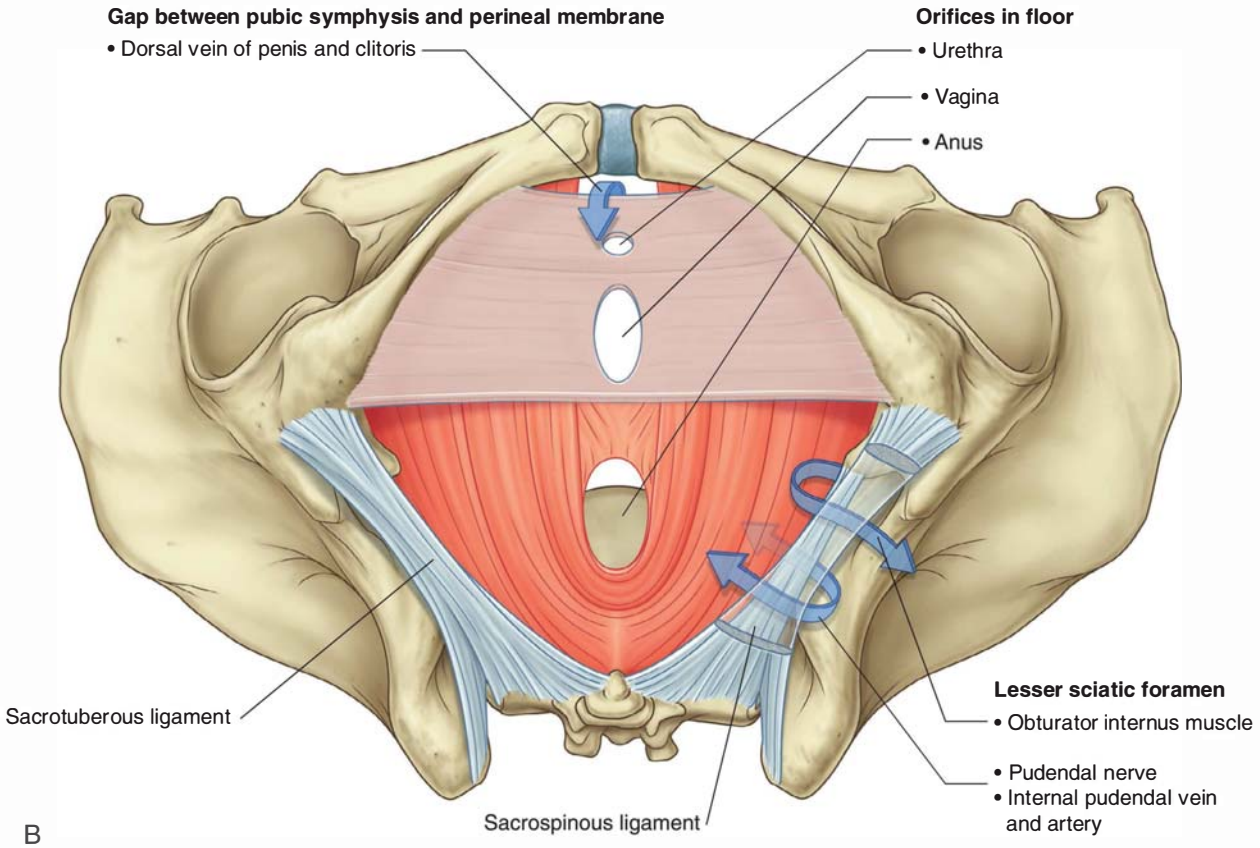


Fig. 5.10, cont'd B. Between the perineum and other regions.

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KEY FEATURES

The pelvic cavity projects posteriorly

In the anatomical position, the anterior superior iliac spines and the superior edge of the pubic symphysis lie in the same vertical plane (Fig. 5.11). Consequently, the pelvic inlet is angled 50°–60° forward relative to the horizontal plane, and the pelvic cavity projects posteriorly from the abdominal cavity.

Meanwhile, the urogenital part of the pelvic outlet (the pubic arch) is oriented in a nearly horizontal plane, whereas the posterior part of the outlet is positioned more vertically. The urogenital triangle of the perineum therefore faces inferiorly, while the anal triangle faces more posteriorly.

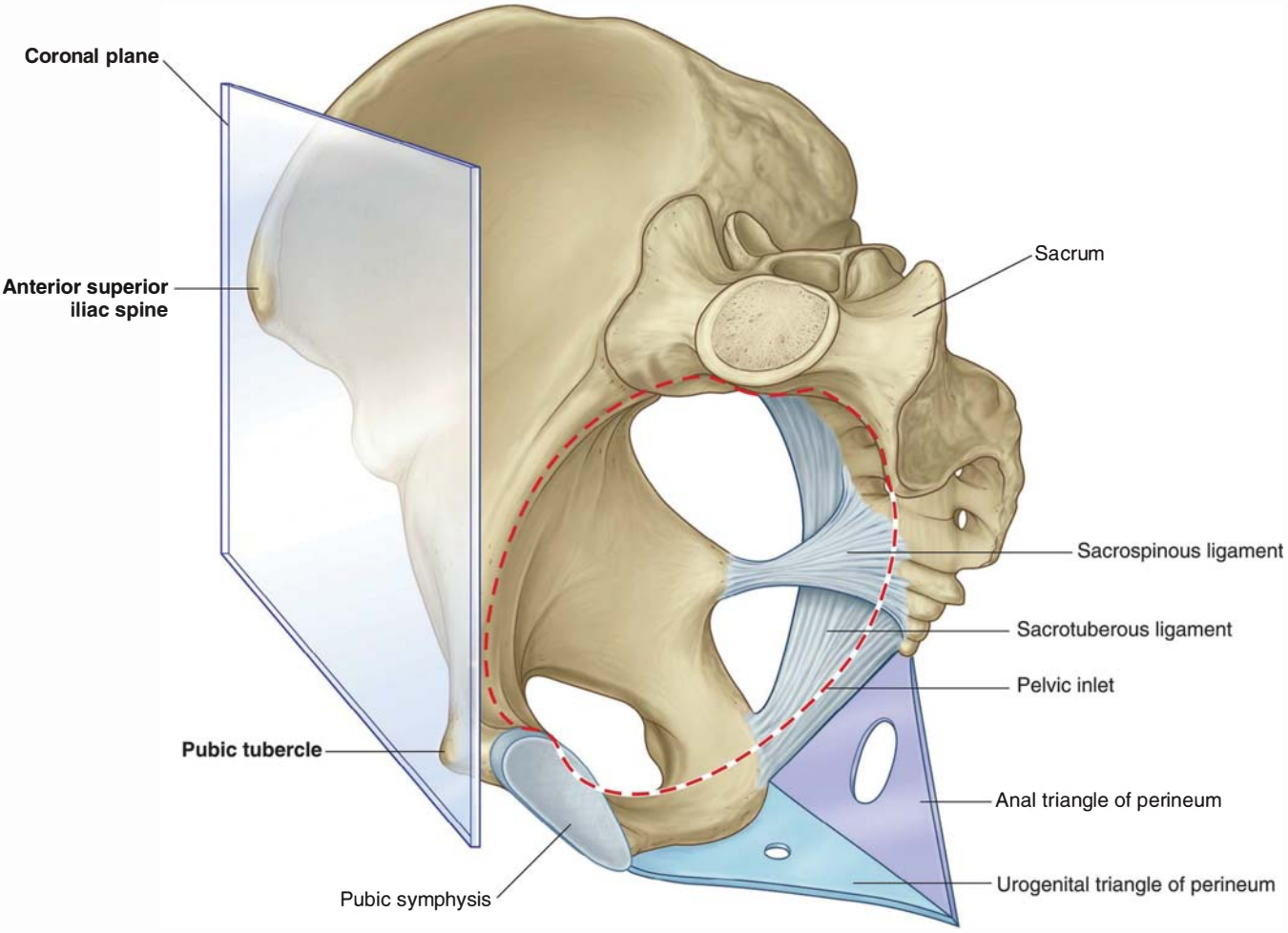


Fig. 5.11 Orientation of the pelvis and pelvic cavity in the anatomical position.

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Important structures cross the ureters in the pelvic cavity

The ureters drain the kidneys, course down the posterior abdominal wall, and cross the pelvic inlet to enter the pelvic cavity. They continue inferiorly along the lateral pelvic wall and ultimately connect with the base of the bladder.

An important structure crosses the ureters in the pelvic cavity in both men and women—in women, the uterine artery crosses the ureter lateral to the cervix of the uterus (Fig. 5.12A), and in men, the ductus deferens crosses over the ureter just posterior to the bladder (Fig. 5.12B).

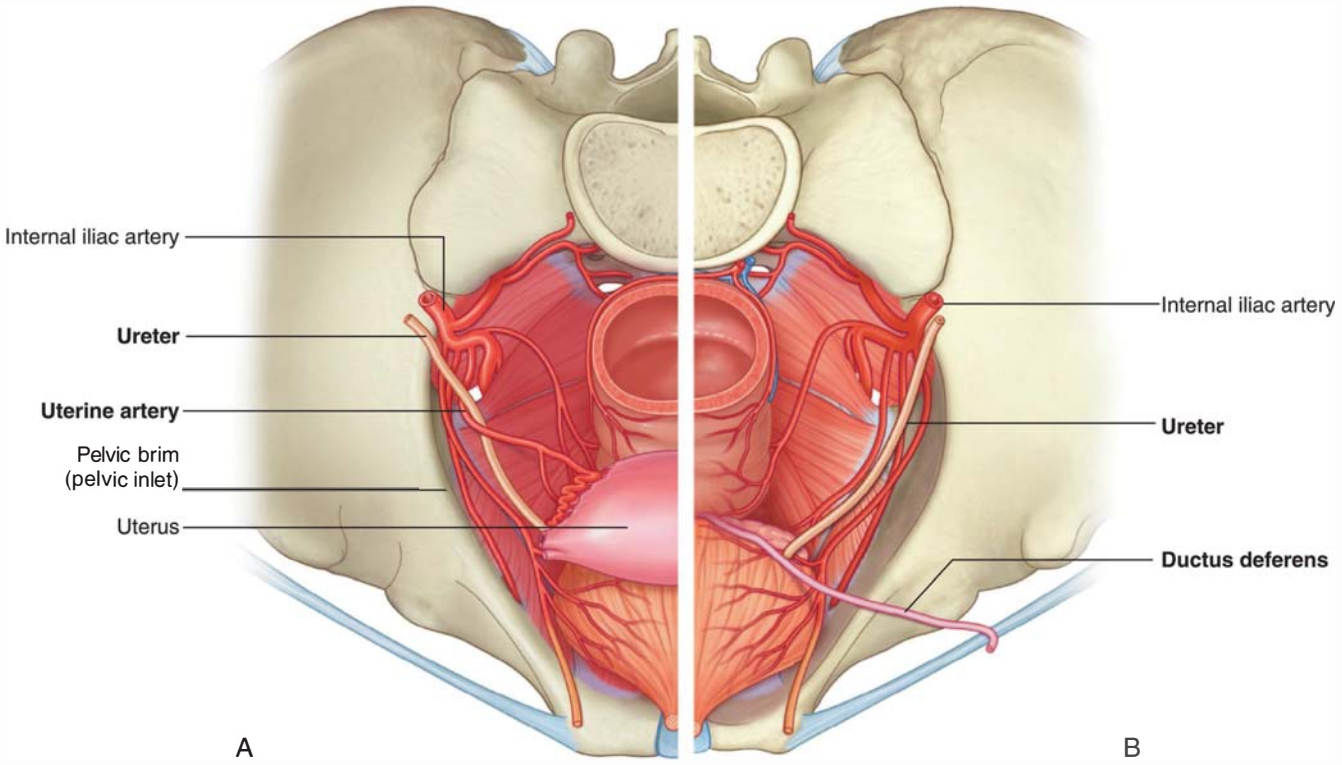


Fig. 5.12 Structures that cross the ureters in the pelvic cavity. **A.** In women. **B.** In men.

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The prostate in men and the uterus in women are anterior to the rectum

In men, the prostate gland is situated immediately anterior to the rectum, just above the pelvic floor (Fig. 5.13). It can be felt by digital palpation during a rectal examination.

In both sexes, the anal canal and the lower rectum also can be evaluated during a rectal examination by a clinician. In women, the cervix and lower part of the body of the uterus also are palpable. However, these structures can more easily be palpated with a bimanual examination where the index and middle fingers of a clinician's hand are placed in the vagina and the other hand is placed on the lower anterior abdominal wall. The organs are felt between the two hands. This bimanual technique can also be used to examine the ovaries and uterine tubes.

The perineum is innervated by sacral spinal cord segments

Dermatomes of the perineum in both men and women are from spinal cord levels S3 to S5, except for the anterior regions, which tend to be innervated by spinal cord level L1 by nerves associated with the abdominal wall (Fig. 5.14). Dermatomes of L2 to S2 are predominantly in the lower limb.

Most of the skeletal muscles contained in the perineum and the pelvic floor, including the external anal sphincter and external urethral sphincter, are innervated by spinal cord levels S2 to S4.

Much of the somatic motor and sensory innervation of the perineum is provided by the pudendal nerve from spinal cord levels S2 to S4.

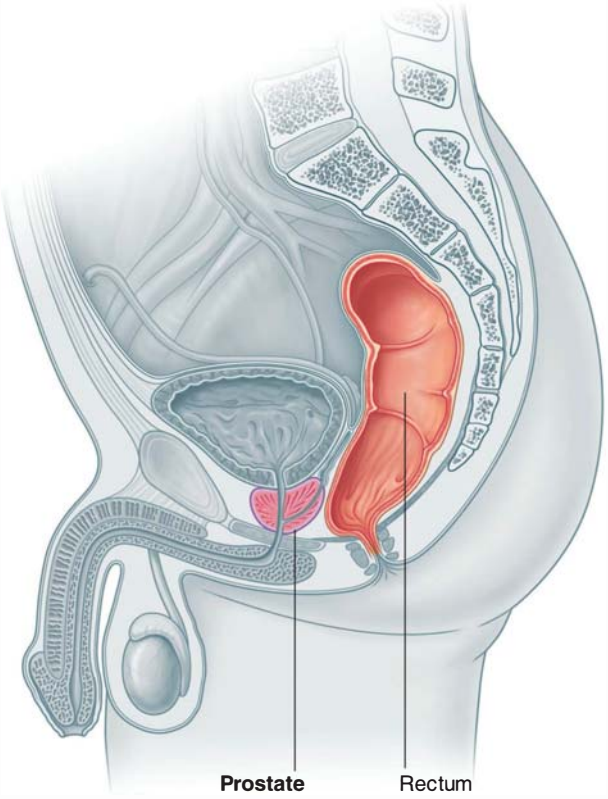


Fig. 5.13 Position of the prostate gland.

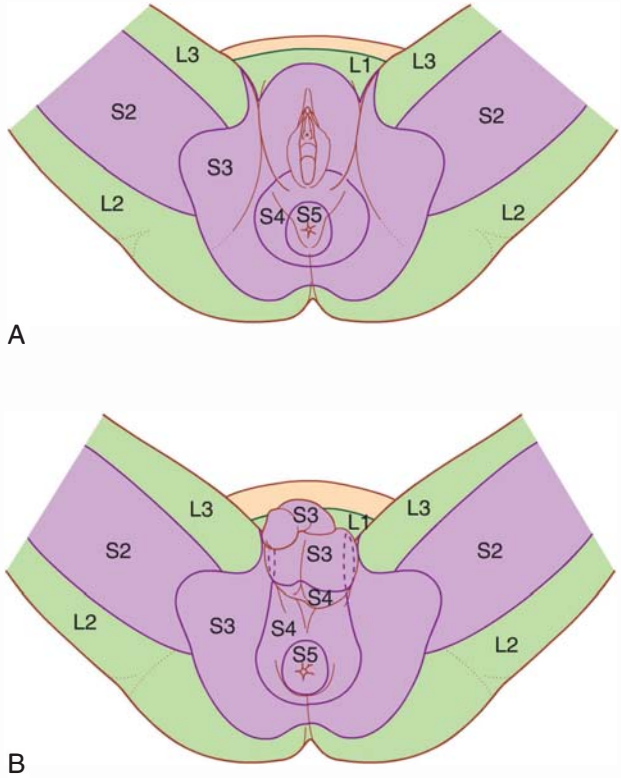


Fig. 5.14 Dermatomes of the perineum. **A.** In women. **B.** In men.

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Nerves are related to bone

The **pudendal nerve** is the major nerve of the perineum and is directly associated with the ischial spine of the pelvis (Fig. 5.15). On each side of the body, these spines and the attached sacrospinous ligaments separate the greater sciatic foramina from the lesser sciatic foramina on the lateral pelvic wall.

The pudendal nerve leaves the pelvic cavity through the greater sciatic foramen and then immediately enters the perineum inferiorly to the pelvic floor by passing around the ischial spine and through the lesser sciatic foramen (Fig. 5.15). The ischial spine can be palpated transvaginally in women and is the landmark for administering a pudendal nerve block.

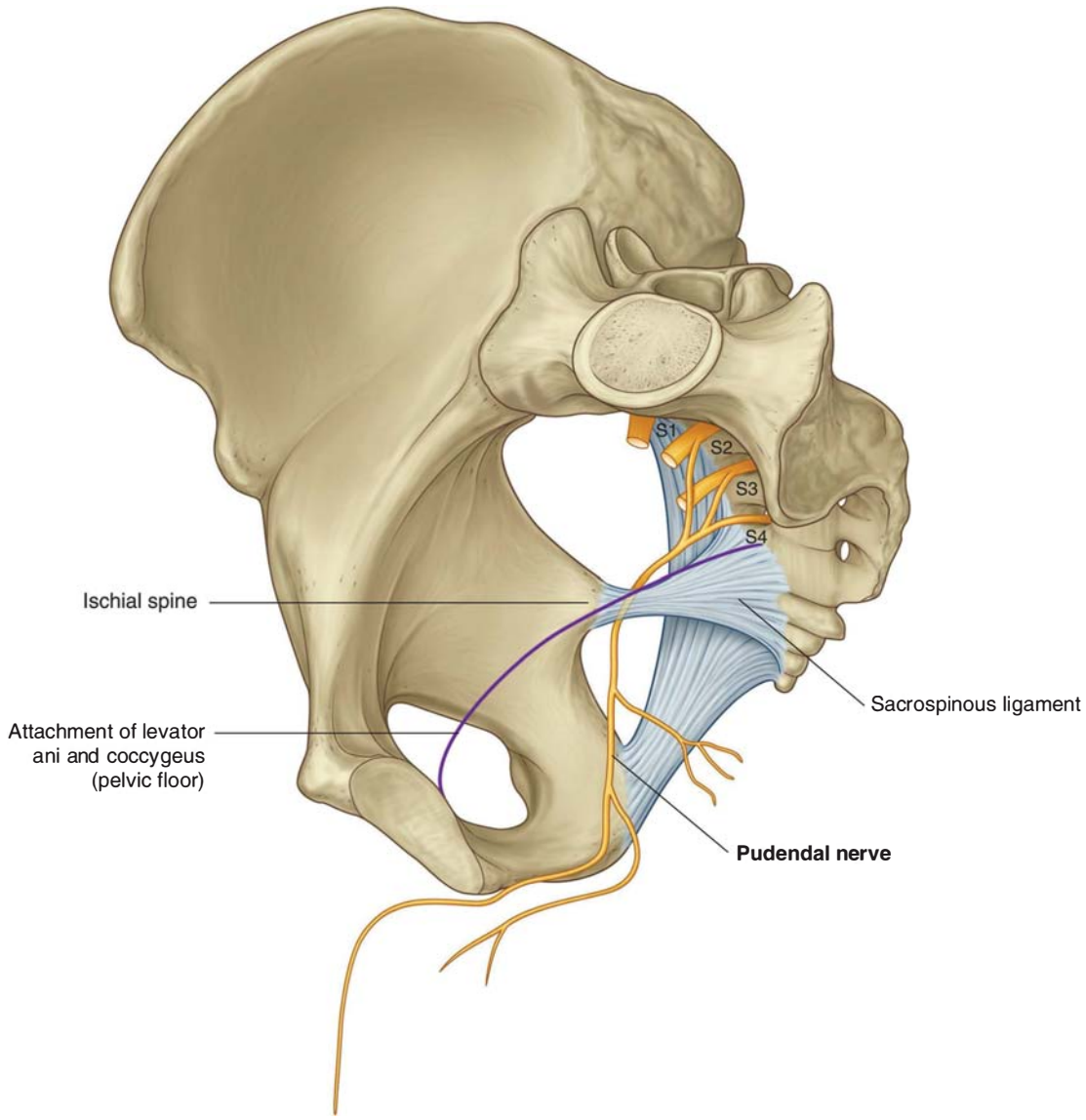


Fig. 5.15 Pudendal nerve.

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Parasympathetic innervation from spinal cord levels S2 to S4 controls erection

The parasympathetic innervation from spinal cord levels S2 to S4 controls genital erection in both women and men (Fig. 5.16). On each side, preganglionic parasympathetic nerves leave the anterior rami of the sacral spinal nerves and enter the **inferior hypogastric plexus** (pelvic plexus) on the lateral pelvic wall.

The two inferior hypogastric plexuses are inferior extensions of the abdominal prevertebral plexus that forms on the posterior abdominal wall in association with the abdominal aorta. Nerves derived from these plexuses penetrate the pelvic floor to innervate the erectile tissues of the clitoris in women and the penis in men.

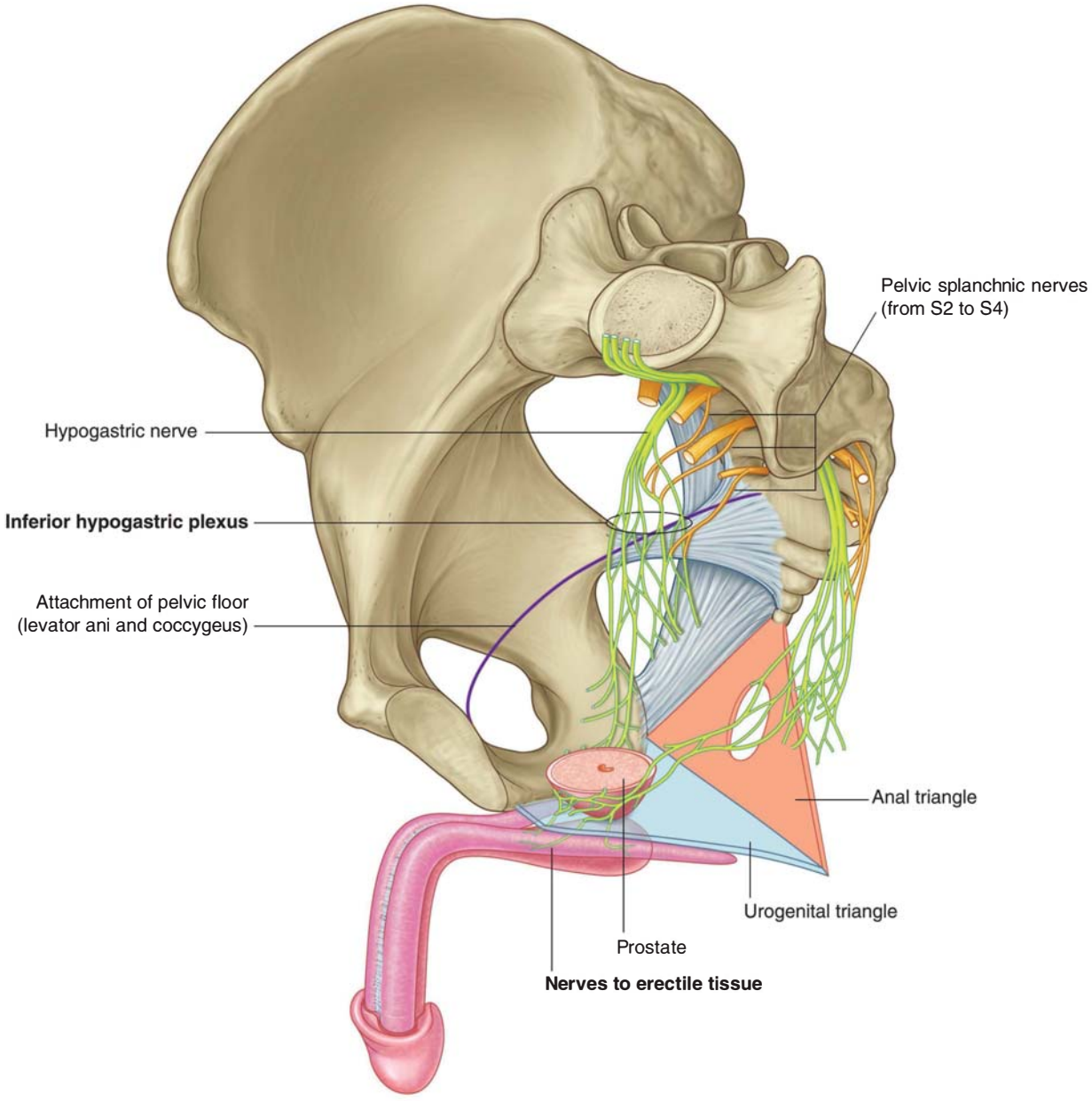


Fig. 5.16 Pelvic splanchnic nerves from spinal levels S2 to S4 control erection.

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Muscles and fascia of the pelvic floor and perineum intersect at the perineal body

Structures of the pelvic floor intersect with structures in the perineum at the **perineal body** (Fig. 5.17). This poorly defined fibromuscular node lies at the center of the perineum, approximately midway between the two ischial tuberosities. Converging at the perineal body are:

- the levator ani muscles of the pelvic diaphragm, and
- muscles in the urogenital and anal triangles of the perineum, including the skeletal muscle sphincters associated with the urethra, vagina, and anus.

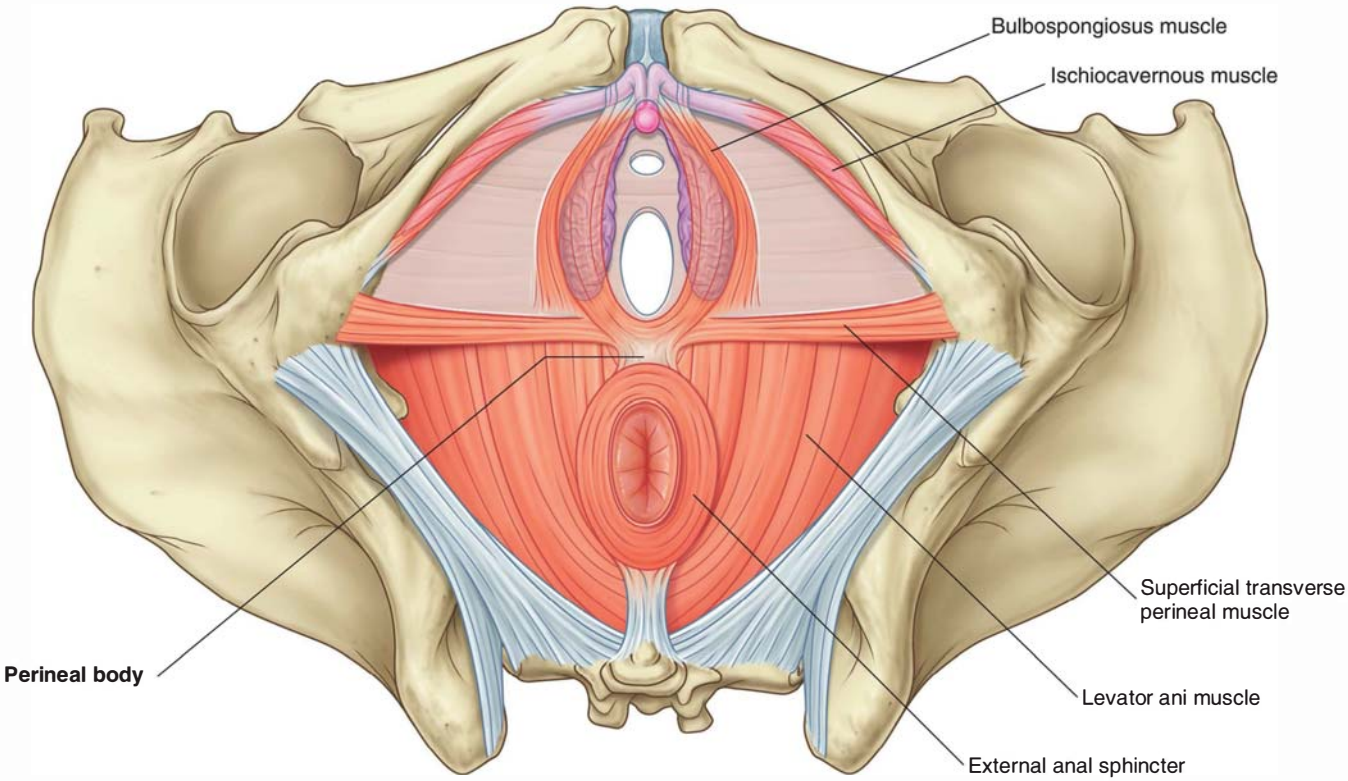


Fig. 5.17 Perineal body.

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The course of the urethra is different in men and women

In women, the urethra is short and passes inferiorly from the bladder through the pelvic floor and opens directly into the perineum (Fig. 5.18A).

In men the urethra passes through the prostate before coursing through the deep perineal pouch and perineal membrane and then becomes enclosed within the erectile tissues of the penis before opening at the end of the penis (Fig. 5.18B). The penile part of the male urethra has two angles:

- The more important of these is a fixed angle where the urethra bends anteriorly in the root of the penis after passing through the perineal membrane.
- Another angle occurs distally where the unattached part of the penis curves inferiorly—when the penis is erect, this second angle disappears.

It is important to consider the different courses of the urethra in men and women when catheterizing patients and when evaluating perineal injuries and pelvic pathology.

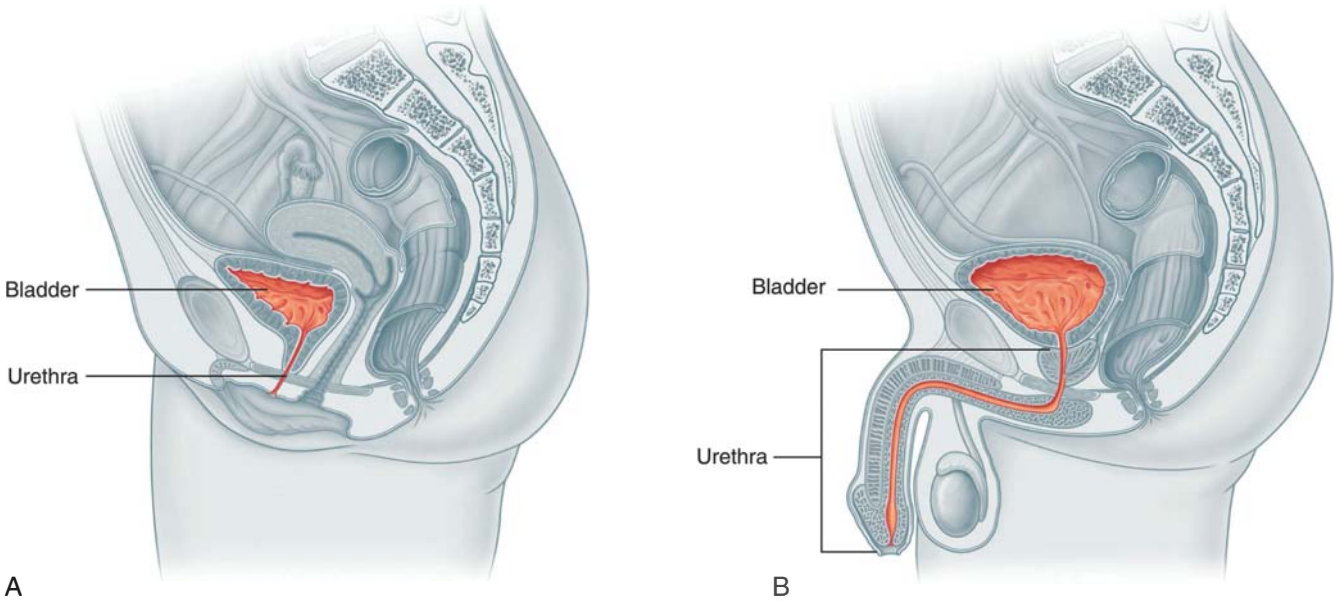


Fig. 5.18 Course of the urethra. **A.** In women. **B.** In men.

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Regional anatomy

The pelvis is the region of the body surrounded by the pelvic bones and the inferior elements of the vertebral column. It is divided into two major regions: the superior region is the false (greater) pelvis and is part of the abdominal cavity; the inferior region is the true (lesser) pelvis, which encloses the pelvic cavity.

The bowl-shaped pelvic cavity is continuous above with the abdominal cavity. The rim of the pelvic cavity (the pelvic inlet) is completely encircled by bone. The pelvic floor is a fibromuscular structure separating the pelvic cavity above from the perineum below.

The perineum is inferior to the pelvic floor and its margin is formed by the pelvic outlet. The perineum contains:

- the terminal openings of the gastrointestinal and urinary systems,
- the external opening of the reproductive tract, and
- the roots of the external genitalia.

PELVIS

Bones

The bones of the pelvis consist of the right and left pelvic (hip) bones, the sacrum, and the coccyx. The sacrum articulates superiorly with vertebra LV at the lumbosacral joint. The pelvic bones articulate posteriorly with the sacrum at the sacro-iliac joints and with each other anteriorly at the pubic symphysis.

Pelvic bone

The pelvic bone is irregular in shape and has two major parts separated by an oblique line on the medial surface of the bone (Fig. 5.19A):

- The pelvic bone above this line represents the lateral wall of the false pelvis, which is part of the abdominal cavity.
- The pelvic bone below this line represents the lateral wall of the true pelvis, which contains the pelvic cavity.

The linea terminalis is the lower two-thirds of this line and contributes to the margin of the pelvic inlet.

The lateral surface of the pelvic bone has a large articular socket, the **acetabulum**, which, together with the head of the femur, forms the hip joint (Fig. 5.19B).

Inferior to the acetabulum is the large **obturator foramen**, most of which is closed by a flat connective tissue membrane, the **obturator membrane**. A small obturator canal remains open superiorly between the membrane and adjacent bone, providing a route of communication between the lower limb and the pelvic cavity.

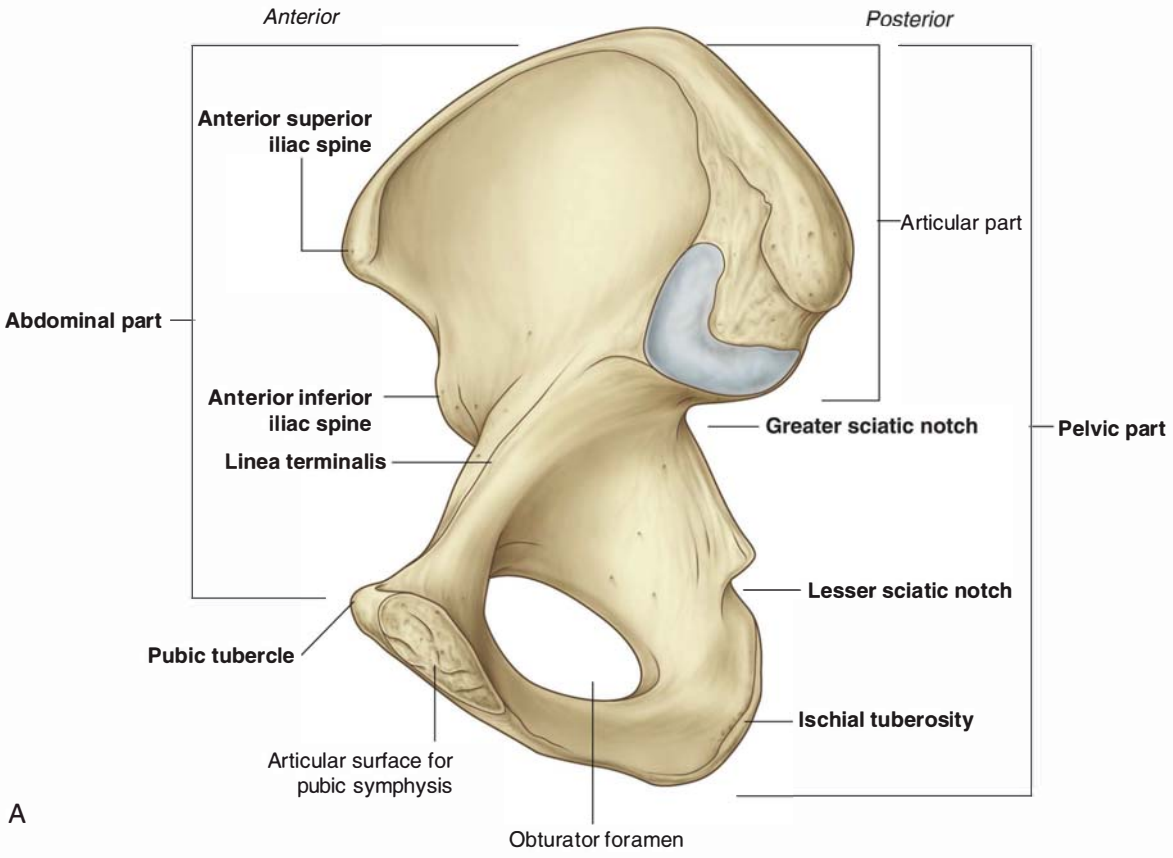
The posterior margin of the bone is marked by two notches separated by the **ischial spine**:

- the **greater sciatic notch**, and
- the **lesser sciatic notch**.

The posterior margin terminates inferiorly as the large **ischial tuberosity**.

The irregular anterior margin of the pelvic bone is marked by the **anterior superior iliac spine**, the **anterior inferior iliac spine**, and the **pubic tubercle**.

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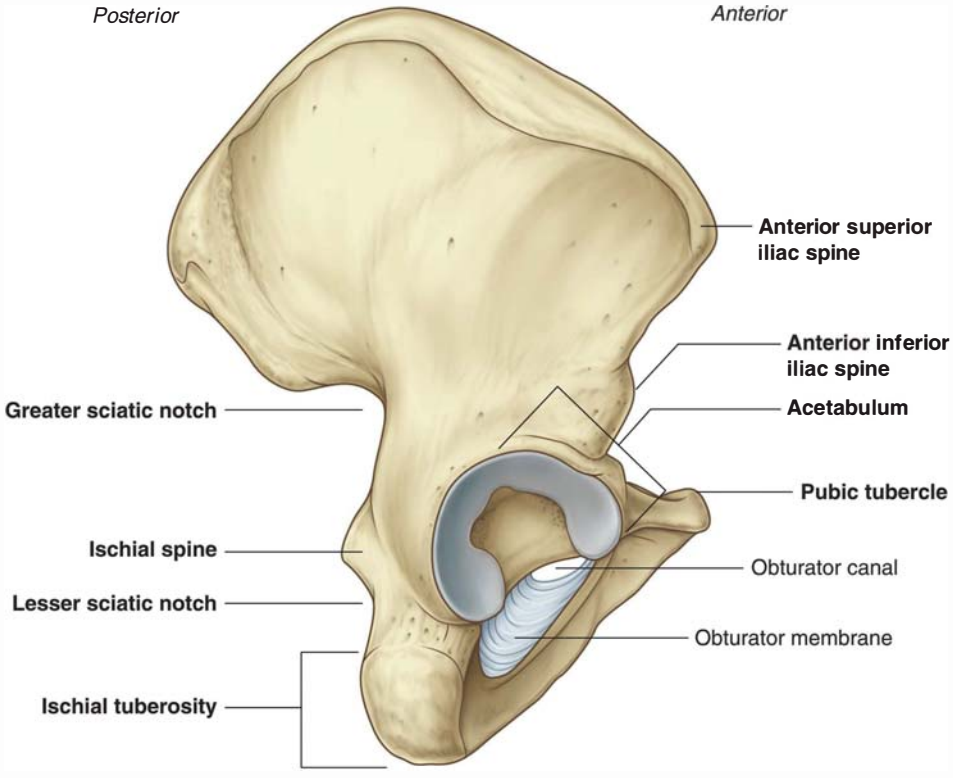


Fig. 5.19 Right pelvic bone. **A.** Medial view. **B.** Lateral view.

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Components of the pelvic bone

Each pelvic bone is formed by three elements: the ilium, pubis, and ischium. At birth, these bones are connected by cartilage in the area of the acetabulum; later, at between 16 and 18 years of age, they fuse into a single bone (Fig. 5.20).

Ilium

Of the three components of the pelvic bone, the **ilium** is the most superior in position.

The ilium is separated into upper and lower parts by a ridge on the medial surface (Fig. 5.21A).

- Posteriorly, the ridge is sharp and lies immediately superior to the surface of the bone that articulates with the sacrum. This sacral surface has a large L-shaped facet for articulating with the sacrum and an expanded, posterior roughened area for the attachment of the strong ligaments that support the sacro-iliac joint (Fig. 5.21).
- Anteriorly, the ridge separating the upper and lower parts of the ilium is rounded and termed the **arcuate line**.

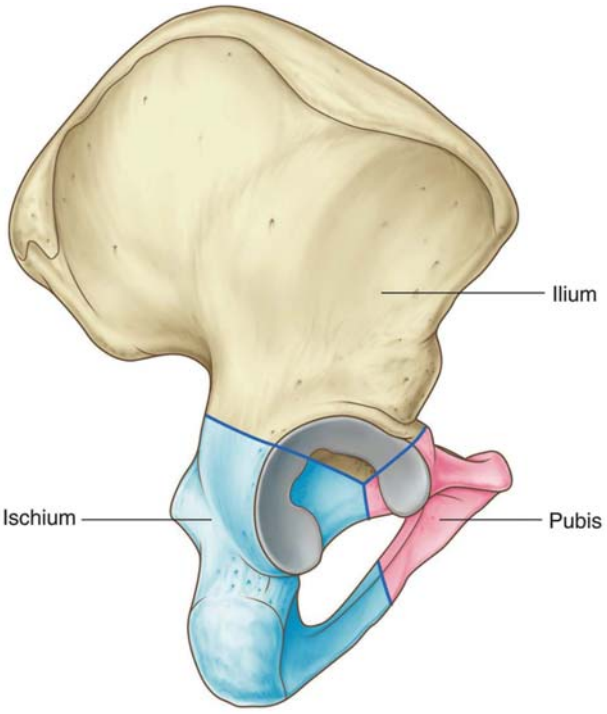


Fig. 5.20 Ilium, ischium, and pubis.

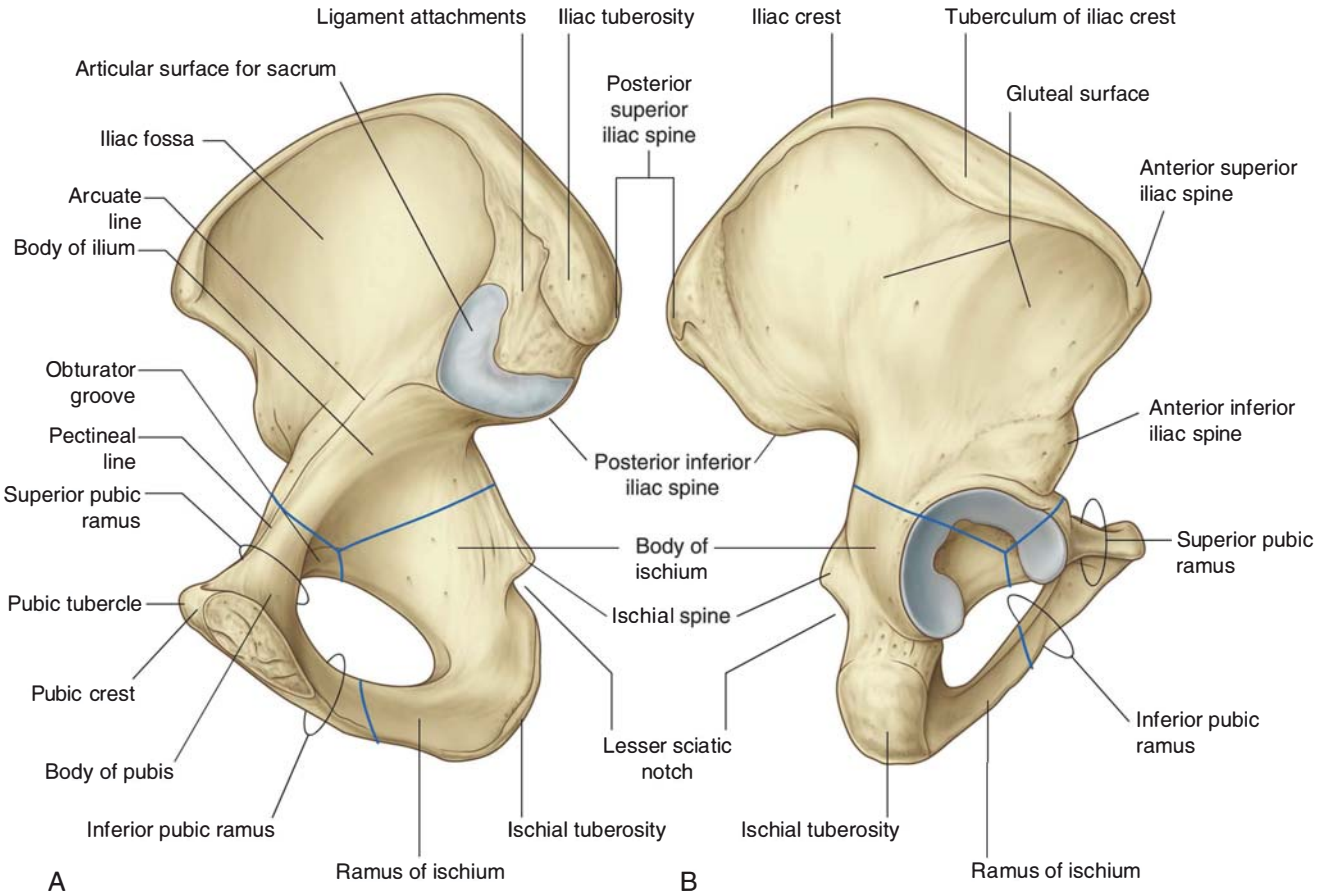


Fig. 5.21 Components of the pelvic bone. **A.** Medial surface. **B.** Lateral surface.

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The arcuate line forms part of the linea terminalis and the pelvic brim.

The portion of the ilium lying inferiorly to the arcuate line is the pelvic part of the ilium and contributes to the wall of the lesser or true pelvis.

The upper part of the ilium expands to form a flat, fan-shaped "wing," which provides bony support for the lower abdomen, or false pelvis. This part of the ilium provides attachment for muscles functionally associated with the lower limb. The anteromedial surface of the wing is concave and forms the **iliac fossa**. The external (gluteal) surface of the wing is marked by lines and roughenings and is related to the gluteal region of the lower limb (Fig. 5.21B).

The entire superior margin of the ilium is thickened to form a prominent crest (the **iliac crest**), which is the site of attachment for muscles and fascia of the abdomen, back, and lower limb and terminates anteriorly as the **anterior superior iliac spine** and posteriorly as the **posterior superior iliac spine**.

A prominent tubercle, the **tuberculum of the iliac crest**, projects laterally near the anterior end of the crest; the posterior end of the crest thickens to form the **iliac tuberosity**.

Inferior to the anterior superior iliac spine of the crest, on the anterior margin of the ilium, is a rounded protuberance called the **anterior inferior iliac spine**. This structure serves as the point of attachment for the rectus femoris muscle of the anterior compartment of the thigh and the iliofemoral ligament associated with the hip joint. A less prominent **posterior inferior iliac spine** occurs along the posterior border of the sacral surface of the ilium, where the bone angles forward to form the superior margin of the greater sciatic notch.

In the clinic

Bone marrow biopsy

In certain diseases (e.g., leukemia), a sample of bone marrow must be obtained to assess the stage and severity of the problem. The iliac crest is often used for such bone marrow biopsies. The iliac crest lies close to the surface and is easily palpated.

A bone marrow biopsy is performed by injecting anesthetic in the skin and passing a cutting needle through the cortical bone of the iliac crest. The bone marrow is aspirated and viewed under a microscope. Samples of cortical bone can also be obtained in this way to provide information about bone metabolism.

Pubis

The anterior and inferior part of the pelvic bone is the **pubis** (Fig. 5.21). It has a body and two arms (rami).

- The **body** is flattened dorsoventrally and articulates with the body of the pubic bone on the other side at the **pubic symphysis**. The body has a rounded pubic crest on its superior surface that ends laterally as the prominent **pubic tubercle**.
- The **superior pubic ramus** projects posterolaterally from the body and joins with the ilium and ischium at its base, which is positioned toward the acetabulum. The sharp superior margin of this triangular surface is termed the **pecten pubis** (**pectineal line**), which forms part of the linea terminalis of the pelvic bone and the pelvic inlet. Anteriorly, this line is continuous with the **pubic crest**, which also is part of the linea terminalis and pelvic inlet. The **superior pubic ramus** is marked on its inferior surface by the **obturator groove**, which forms the upper margin of the obturator canal.
- The inferior ramus projects laterally and inferiorly to join with the ramus of the ischium.

Ischium

The ischium is the posterior and inferior part of the pelvic bone (Fig. 5.21). It has:

- a large body that projects superiorly to join with the ilium and the superior ramus of the pubis, and
- a ramus that projects anteriorly to join with the inferior ramus of the pubis.

The posterior margin of the bone is marked by a prominent **ischial spine** that separates the lesser sciatic notch, below, from the greater sciatic notch, above.

The most prominent feature of the ischium is a large tuberosity (the **ischial tuberosity**) on the posteroinferior aspect of the bone. This tuberosity is an important site for the attachment of lower limb muscles and for supporting the body when sitting.

Sacrum

The sacrum, which has the appearance of an inverted triangle, is formed by the fusion of the five sacral vertebrae (Fig. 5.22). The base of the sacrum articulates with vertebra LV, and its apex articulates with the coccyx. Each of the lateral surfaces of the bone bears a large L-shaped facet for articulation with the ilium of the pelvic bone. Posterior to the facet is a large roughened area for the attachment of ligaments that support the sacro-iliac joint. The superior surface of the sacrum is characterized by the superior

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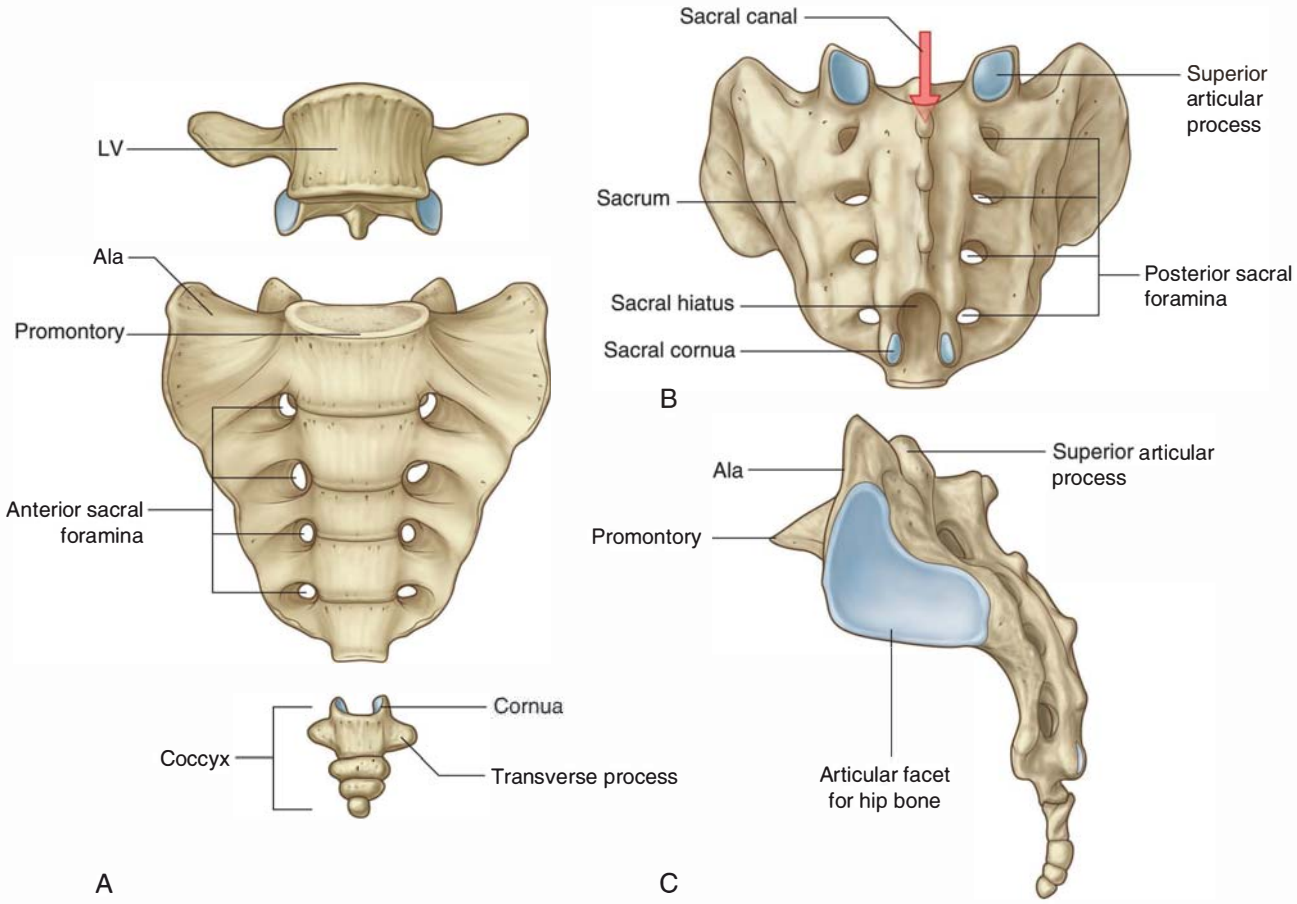


Fig. 5.22 Sacrum and coccyx. **A.** Anterior view. **B.** Posterior view. **C.** Lateral view.

aspect of the body of vertebra SI and is flanked on each side by an expanded wing-like transverse process termed the **ala**. The anterior edge of the vertebral body projects forward as the **promontory**. The anterior surface of the sacrum is concave; the posterior surface is convex. Because the transverse processes of adjacent sacral vertebrae fuse lateral to the position of the intervertebral foramina and lateral to the bifurcation of spinal nerves into posterior and anterior rami, the posterior and anterior rami of spinal nerves S1 to S4 emerge from the sacrum through separate foramina. There are four pairs of **anterior sacral foramina** on the anterior surface of the sacrum for anterior rami, and four pairs of **posterior sacral foramina** on the posterior surface for the posterior rami. The **sacral canal** is a continuation of the vertebral canal that terminates as the **sacral hiatus**.

Coccyx

The small terminal part of the vertebral column is the coccyx, which consists of four fused coccygeal vertebrae (Fig. 5.22) and, like the sacrum, has the shape of an inverted triangle. The base of the coccyx is directed superiorly. The superior surface bears a facet for articulation with the sacrum and two **horns**, or **cornua**, one on each side, that project upward to articulate or fuse with similar downward-projecting cornua from the sacrum. These processes are modified superior and inferior articular processes that are present on other vertebrae. Each lateral surface of the coccyx has a small rudimentary transverse process, extending from the first coccygeal vertebra. Vertebral arches are absent from coccygeal vertebrae; therefore no bony vertebral canal is present in the coccyx.

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In the clinic

Pelvic fracture

The pelvis can be viewed as a series of anatomical rings. There are three bony rings and four fibro-osseous rings. The major bony pelvic ring consists of parts of the sacrum, ilium, and pubis, which forms the pelvic inlet. Two smaller subsidiary rings are the obturator foramina. The greater and lesser sciatic foramina formed by the greater and lesser sciatic notches and the sacrospinous and sacrotuberous ligaments form the four fibro-osseous rings. The rings, which are predominantly bony (i.e., the pelvic inlet and the obturator foramina), are brittle rings. It is not possible to break one side of the ring without breaking the other side of the ring, which in clinical terms means that if a fracture is demonstrated on one side, a second fracture should always be suspected.

Fractures of the pelvis may occur in isolation; however, they usually occur in trauma patients and warrant special mention.

Owing to the large bony surfaces of the pelvis, a fracture produces an area of bone that can bleed significantly. A large hematoma may be produced, which can compress organs such as the bladder and the ureters. This blood loss may occur rapidly, reducing the circulating blood volume and, unless this is replaced, the patient will become hypovolemic and shock will develop.

Pelvic fractures may also disrupt the contents of the pelvis, leading to urethral disruption, potential bowel rupture, and nerve damage.

Joints

Lumbosacral joints

The sacrum articulates superiorly with the lumbar part of the vertebral column. The lumbosacral joints are formed between vertebra LV and the sacrum and consist of:

- the two **zygapophysial joints**, which occur between adjacent inferior and superior articular processes, and
- an intervertebral disc that joins the bodies of vertebrae LV and SI (Fig. 5.23A).

These joints are similar to those between other vertebrae, with the exception that the sacrum is angled posteriorly on vertebra LV. As a result, the anterior part of the intervertebral disc between the two bones is thicker than the posterior part.

The lumbosacral joints are reinforced by strong iliolumbar and lumbosacral ligaments that extend from the expanded transverse processes of vertebra LV to the ilium and the sacrum, respectively (Fig. 5.23B).

Sacro-iliac joints

The sacro-iliac joints transmit forces from the lower limbs to the vertebral column. They are synovial joints between the L-shaped articular facets on the lateral surfaces of the sacrum and similar facets on the iliac parts of the pelvic bones (Fig. 5.24A). The joint surfaces have an irregular contour and interlock to resist movement. The joints often

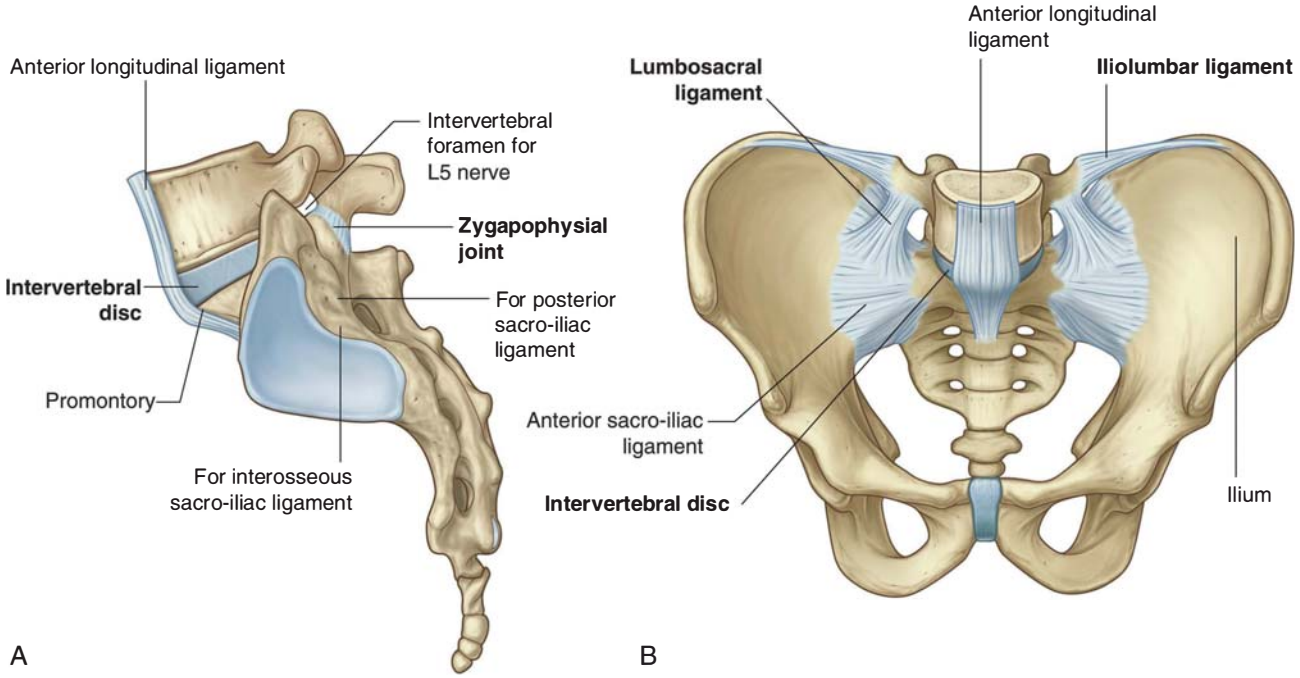


Fig. 5.23 Lumbosacral joints and associated ligaments. **A.** Lateral view. **B.** Anterior view.

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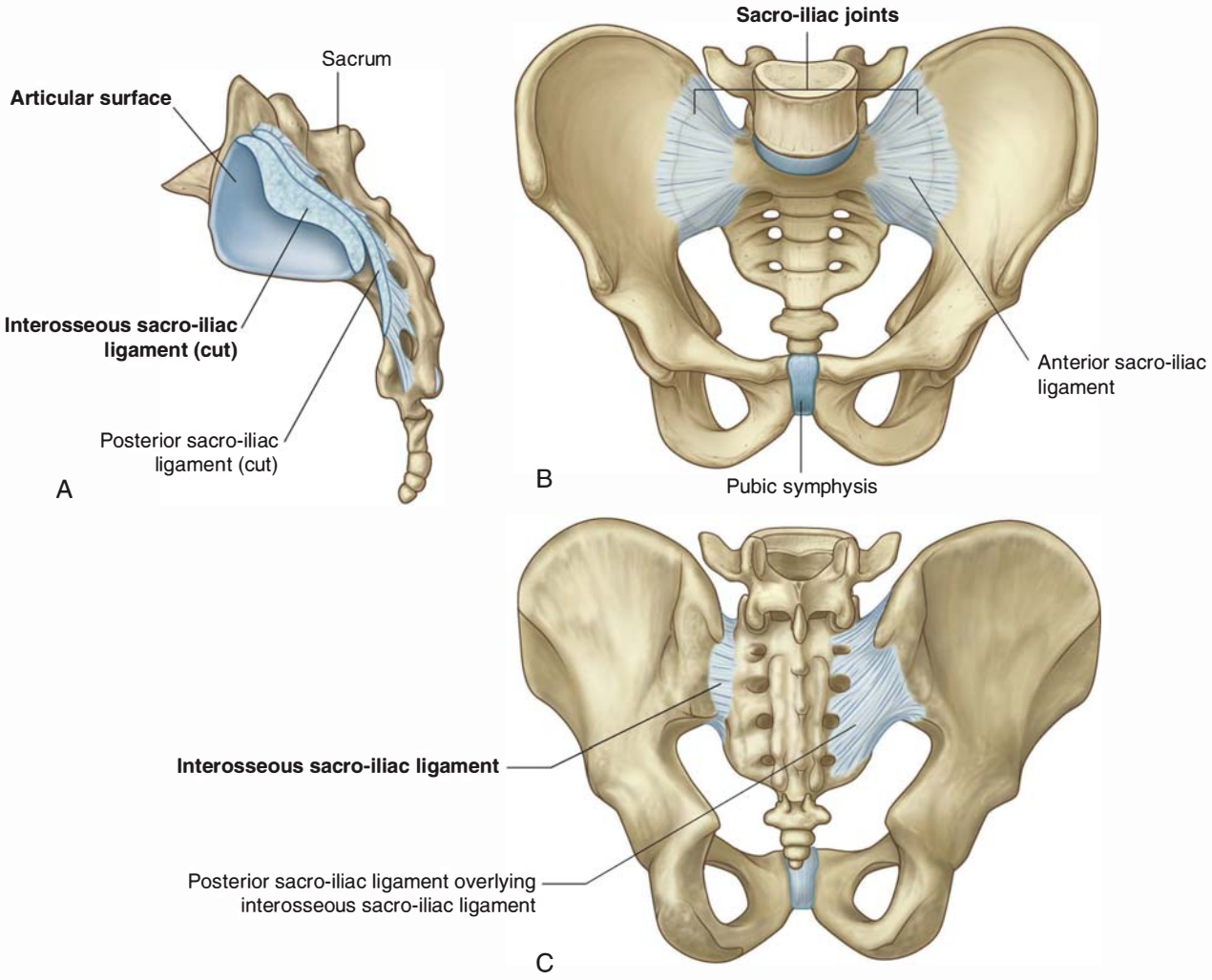


Fig. 5.24 Sacro-iliac joints and associated ligaments. **A.** Lateral view. **B.** Anterior view. **C.** Posterior view.

become fibrous with age and may become completely ossified.

Each sacro-iliac joint is stabilized by three ligaments:

- the **anterior sacro-iliac ligament**, which is a thickening of the fibrous membrane of the joint capsule and runs anteriorly and inferiorly to the joint (Fig. 5.24B);
- the **interosseous sacro-iliac ligament**, which is the largest, strongest ligament of the three, and is positioned immediately posterosuperior to the joint and attaches to adjacent expansive roughened areas on the ilium and sacrum, thereby filling the gap between the two bones (Fig. 5.24A,C); and
- the **posterior sacro-iliac ligament**, which covers the interosseous sacro-iliac ligament (Fig. 5.24C).

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Pubic symphysis joint

The pubic symphysis lies anteriorly between the adjacent surfaces of the pubic bones (Fig. 5.25). Each of the joint's surfaces is covered by hyaline cartilage and is linked across the midline to adjacent surfaces by fibrocartilage. The joint is surrounded by interwoven layers of collagen fibers and the two major ligaments associated with it are:

- the **superior pubic ligament**, located above the joint, and
- the **inferior pubic ligament**, located below it.

In the clinic

Common problems with the sacro-iliac joints

The sacro-iliac joints have both fibrous and synovial components, and as with many weight-bearing joints, degenerative changes may occur and cause pain and discomfort in the sacro-iliac region. In addition, disorders associated with the major histocompatibility complex antigen HLA-B27, such as rheumatoid arthritis, psoriasis, and inflammatory bowel disease, can produce specific inflammatory changes within these joints.

Orientation

In the anatomical position, the pelvis is oriented so that the front edge of the top of the pubic symphysis and the anterior superior iliac spines lie in the same vertical plane (Fig. 5.26). As a consequence, the pelvic inlet, which marks the entrance to the pelvic cavity, is tilted to face anteriorly, and the bodies of the pubic bones and the pubic arch are positioned in a nearly horizontal plane facing the ground.

Differences between men and women

The pelvises of women and men differ in a number of ways, many of which have to do with the passing of a baby through a woman's pelvic cavity during childbirth.

- The pelvic inlet in women is circular (Fig. 5.27A) compared with the heart-shaped pelvic inlet (Fig. 5.27B) in men. The more circular shape is partly caused by the less distinct promontory and broader alae in women.
- The angle formed by the two arms of the pubic arch is larger in women (80°–85°) than it is in men (50°–60°).
- The ischial spines generally do not project as far medially into the pelvic cavity in women as they do in men.

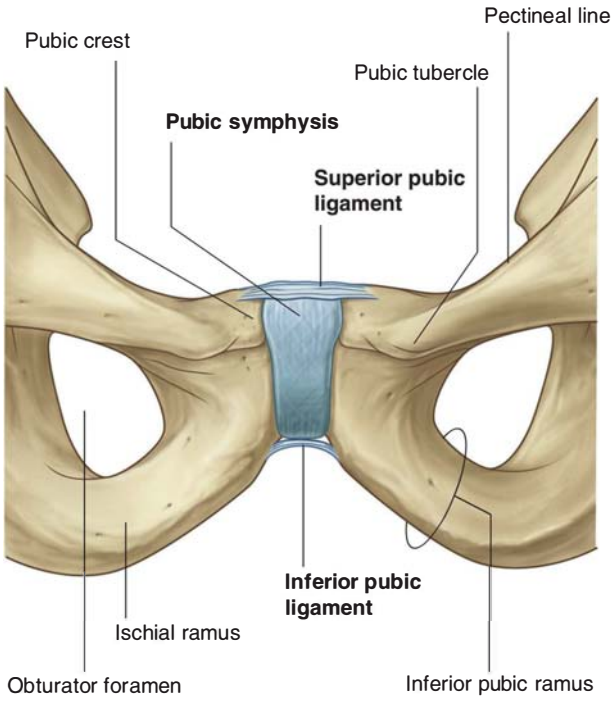


Fig. 5.25 Pubic symphysis and associated ligaments.

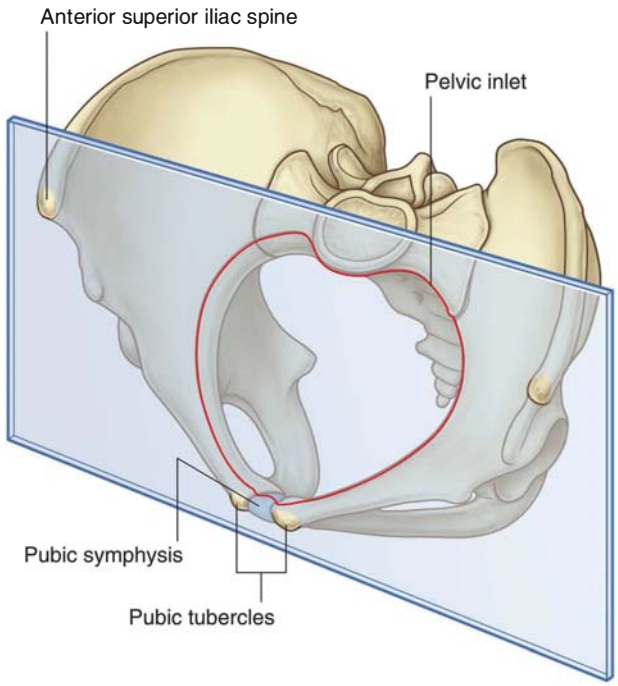


Fig. 5.26 Orientation of the pelvis (anatomical position).

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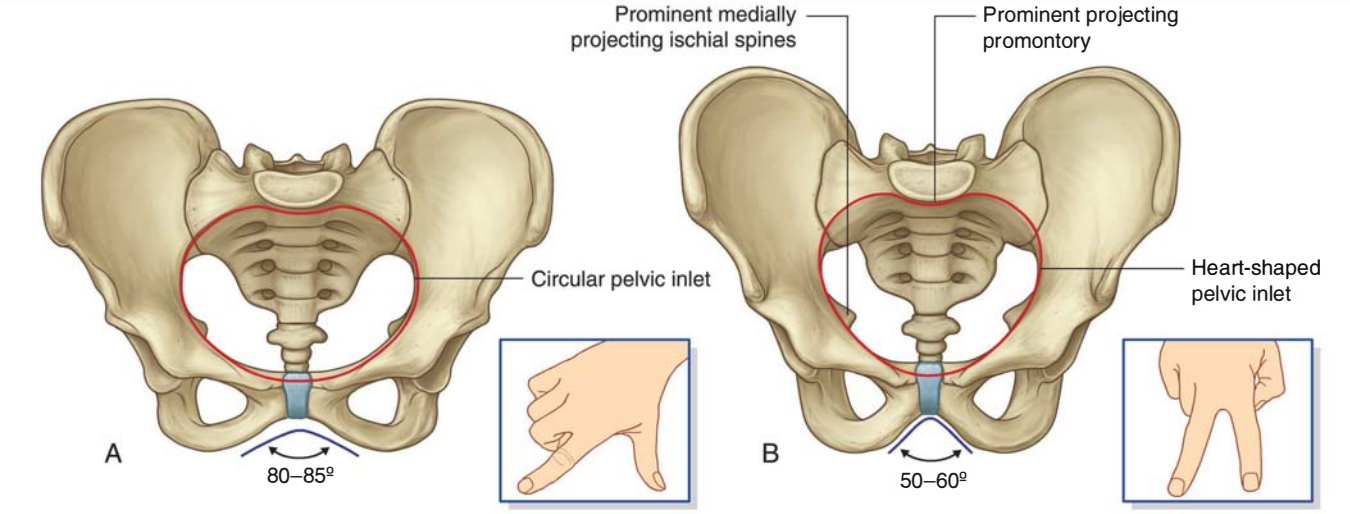


Fig. 5.27 Structure of the bony pelvis. **A.** In women. **B.** In men. The angle formed by the pubic arch can be approximated by the angle between the thumb and index finger for women and the angle between the index finger and middle finger for men as shown in the insets.

True pelvis

The true pelvis is cylindrical and has an inlet, a wall, and an outlet. The inlet is open, whereas the pelvic floor closes the outlet and separates the pelvic cavity, above, from the perineum, below.

Pelvic inlet

The pelvic inlet is the circular opening between the abdominal cavity and the pelvic cavity through which structures traverse between the abdomen and pelvic cavity. It is completely surrounded by bones and joints (Fig. 5.28). The promontory of the sacrum protrudes into the inlet, forming its posterior margin in the midline. On either side of the promontory, the margin is formed by the alae of the sacrum. The margin of the pelvic inlet then crosses the sacro-iliac joint and continues along the linea terminalis (i.e., the arcuate line, the pecten pubis or pectineal line, and the pubic crest) to the pubic symphysis.

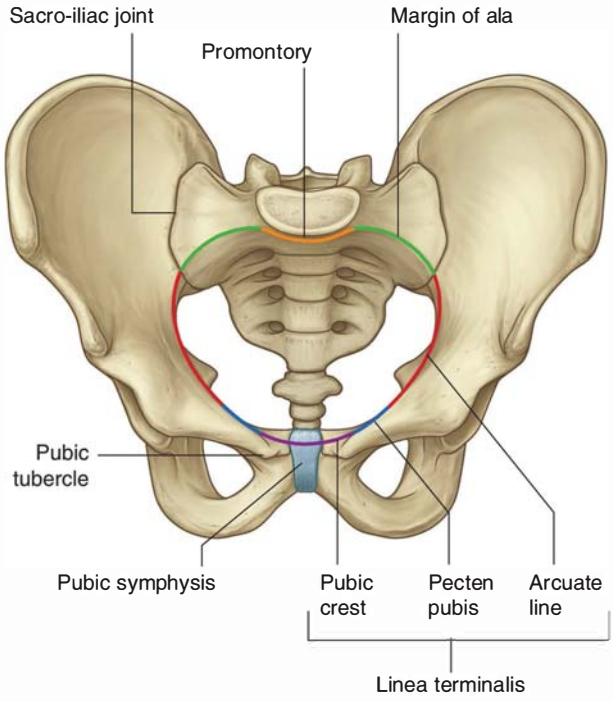


Fig. 5.28 Pelvic inlet.

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Pelvic wall

The walls of the pelvic cavity consist of the sacrum, the coccyx, the pelvic bones inferior to the linea terminalis, two ligaments, and two muscles.

Ligaments of the pelvic wall

The sacrospinous and sacrotuberous ligaments (Fig. 5.29A) are major components of the lateral pelvic walls that help define the apertures between the pelvic cavity and adjacent regions through which structures pass.

- The smaller of the two, the sacrospinous ligament, is triangular, with its apex attached to the ischial spine and its base attached to the related margins of the sacrum and the coccyx.
- The sacrotuberous ligament is also triangular and is superficial to the sacrospinous ligament. Its base has a broad attachment that extends from the posterior superior iliac spine of the pelvic bone, along the dorsal aspect and the lateral margin of the sacrum, and onto the dorsolateral surface of the coccyx. Laterally, the apex of the ligament is attached to the medial margin of the ischial tuberosity.

These ligaments stabilize the sacrum on the pelvic bones by resisting the upward tilting of the inferior aspect of the sacrum (Fig. 5.29B). They also convert the greater and lesser sciatic notches of the pelvic bone into foramina (Fig. 5.29A,B).

- The **greater sciatic foramen** lies superior to the sacrospinous ligament and the ischial spine.
- The **lesser sciatic foramen** lies inferior to the ischial spine and sacrospinous ligament between the sacrospinous and sacrotuberous ligaments.

Muscles of the pelvic wall

Two muscles, the obturator internus and the piriformis, contribute to the lateral walls of the pelvic cavity. These muscles originate in the pelvic cavity but attach peripherally to the femur.

Obturator internus

The obturator internus is a flat, fan-shaped muscle that originates from the deep surface of the obturator membrane and from associated regions of the pelvic bone that surround the obturator foramen (Fig. 5.30 and Table 5.1).

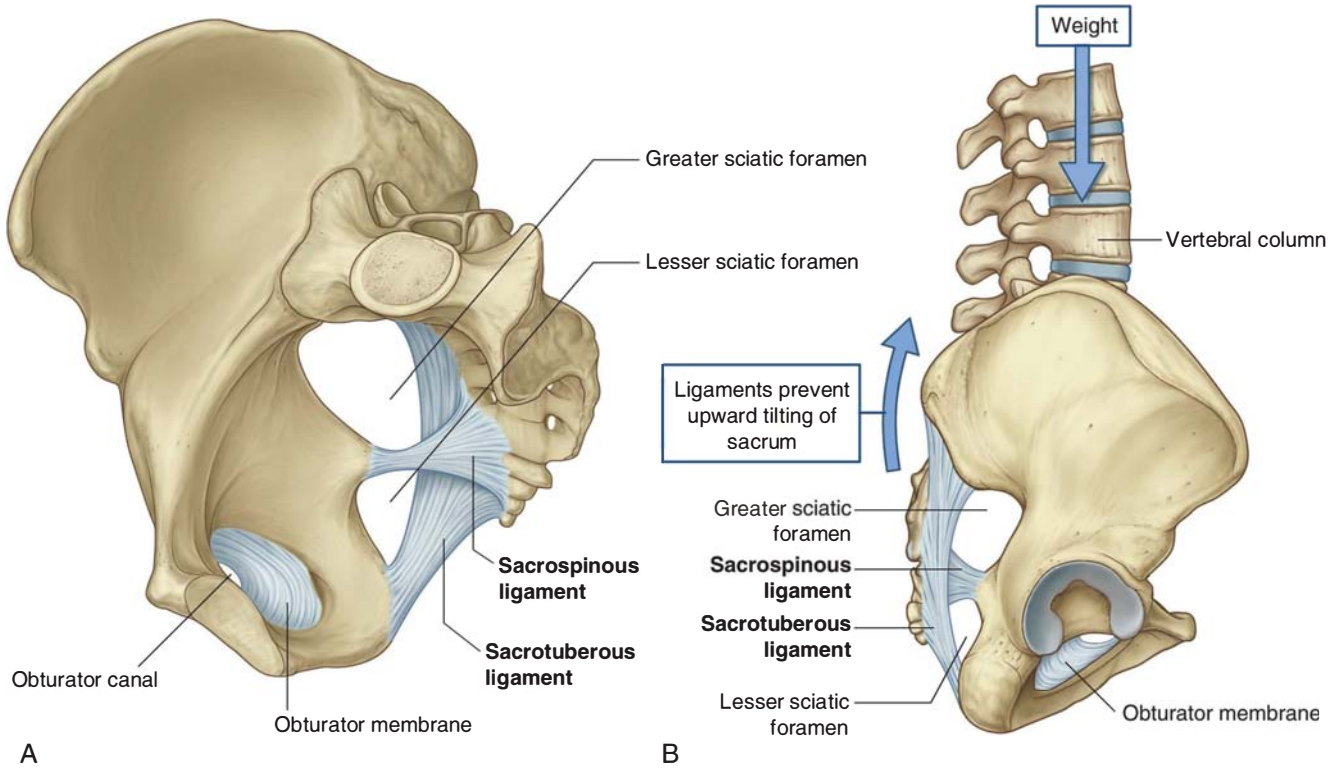


Fig. 5.29 Sacrospinous and sacrotuberous ligaments. **A.** Medial view of right side of pelvis. **B.** Function of the ligaments.

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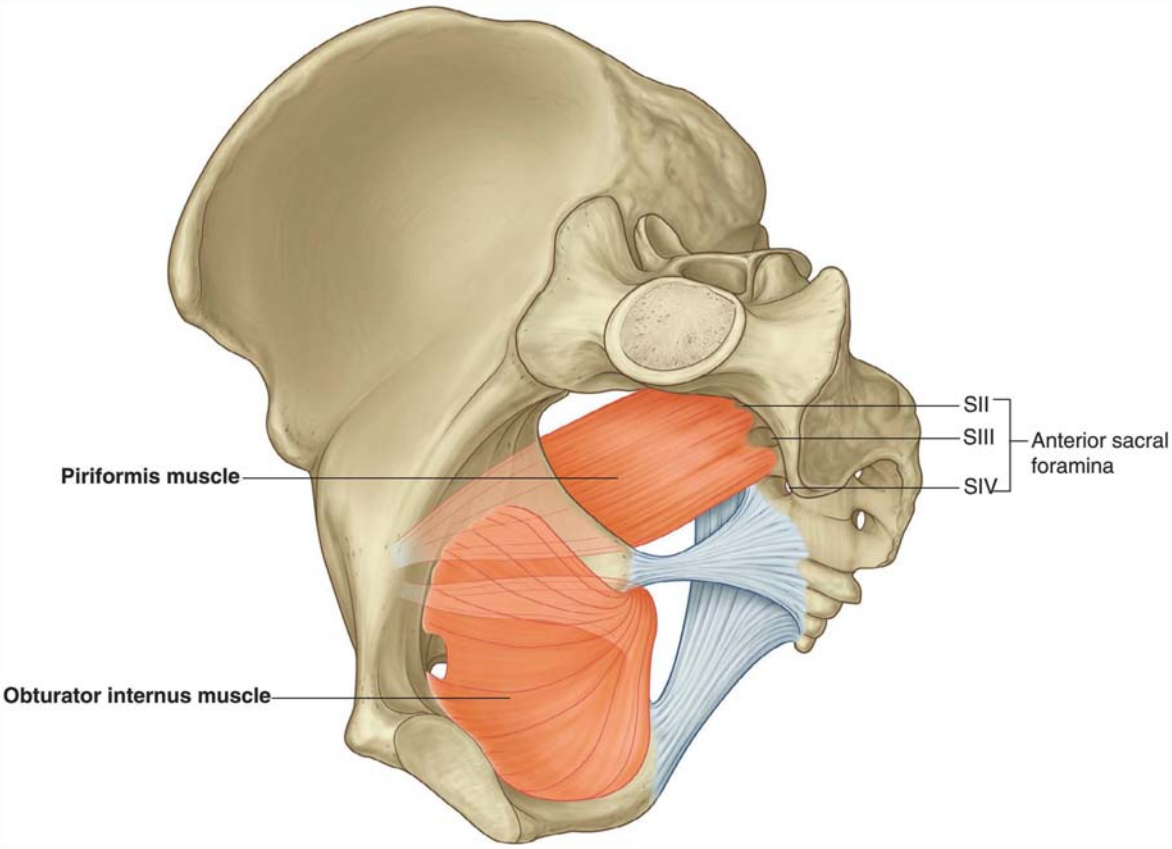


Fig. 5.30 Obturator internus and piriformis muscles (medial view of right side of pelvis).

Table 5.1 Muscles of the pelvic walls

Muscle	Origin	Insertion	Innervation	Function
Obturator internus	Anterolateral wall of true pelvis (deep surface of obturator membrane and surrounding bone)	Medial surface of greater trochanter of femur	Nerve to obturator internus **L5, SI**	Lateral rotation of the extended hip joint; abduction of flexed hip
Piriformis	Anterior surface of sacrum between anterior sacral foramina	Medial side of superior border of greater trochanter of femur	Branches from **SI**, and **S2**	Lateral rotation of the extended hip joint; abduction of flexed hip

The muscle fibers of the obturator internus converge to form a tendon that leaves the pelvic cavity through the lesser sciatic foramen, makes a 90° bend around the ischium between the ischial spine and ischial tuberosity, and then passes posterior to the hip joint to insert on the greater trochanter of the femur.

The obturator internus forms a large part of the anterolateral wall of the pelvic cavity.

Piriformis

The piriformis is triangular and originates in the bridges of bone between the four anterior sacral foramina. It passes laterally through the greater sciatic foramen, crosses the posterosuperior aspect of the hip joint, and inserts on the greater trochanter of the femur above the insertion of the obturator internus muscle (Fig. 5.30 and Table 5.1).

A large part of the posterolateral wall of the pelvic cavity is formed by the piriformis. In addition, this muscle separates the greater sciatic foramen into two regions, one above the muscle and one below. Vessels and nerves coursing between the pelvic cavity and the gluteal region pass through these two regions.

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Apertures in the pelvic wall

Each lateral pelvic wall has three major apertures through which structures pass between the pelvic cavity and other regions:

- the obturator canal,
- the greater sciatic foramen, and
- the lesser sciatic foramen.

Obturator canal

At the top of the obturator foramen is the obturator canal, which is bordered by the obturator membrane, the associated obturator muscles, and the superior pubic ramus (Fig. 5.31). The obturator nerve and vessels pass from the pelvic cavity to the thigh through this canal.

Greater sciatic foramen

The greater sciatic foramen is a major route of communication between the pelvic cavity and the lower limb (Fig. 5.31). It is formed by the greater sciatic notch in the pelvic bone, the sacrotuberous and the sacrospinous ligaments, and the spine of the ischium.

The piriformis muscle passes through the greater sciatic foramen, dividing it into two parts.

- The superior gluteal nerves and vessels pass through the foramen above the piriformis.
- Passing through the foramen below the piriformis are the inferior gluteal nerves and vessels, the sciatic nerve, the pudendal nerve, the internal pudendal vessels, the posterior femoral cutaneous nerves, and the nerves to the obturator internus and quadratus femoris muscles.

Lesser sciatic foramen

The lesser sciatic foramen is formed by the lesser sciatic notch of the pelvic bone, the ischial spine, the sacrospinous ligament, and the sacrotuberous ligament (Fig. 5.31). The tendon of the obturator internus muscle passes through this foramen to enter the gluteal region of the lower limb.

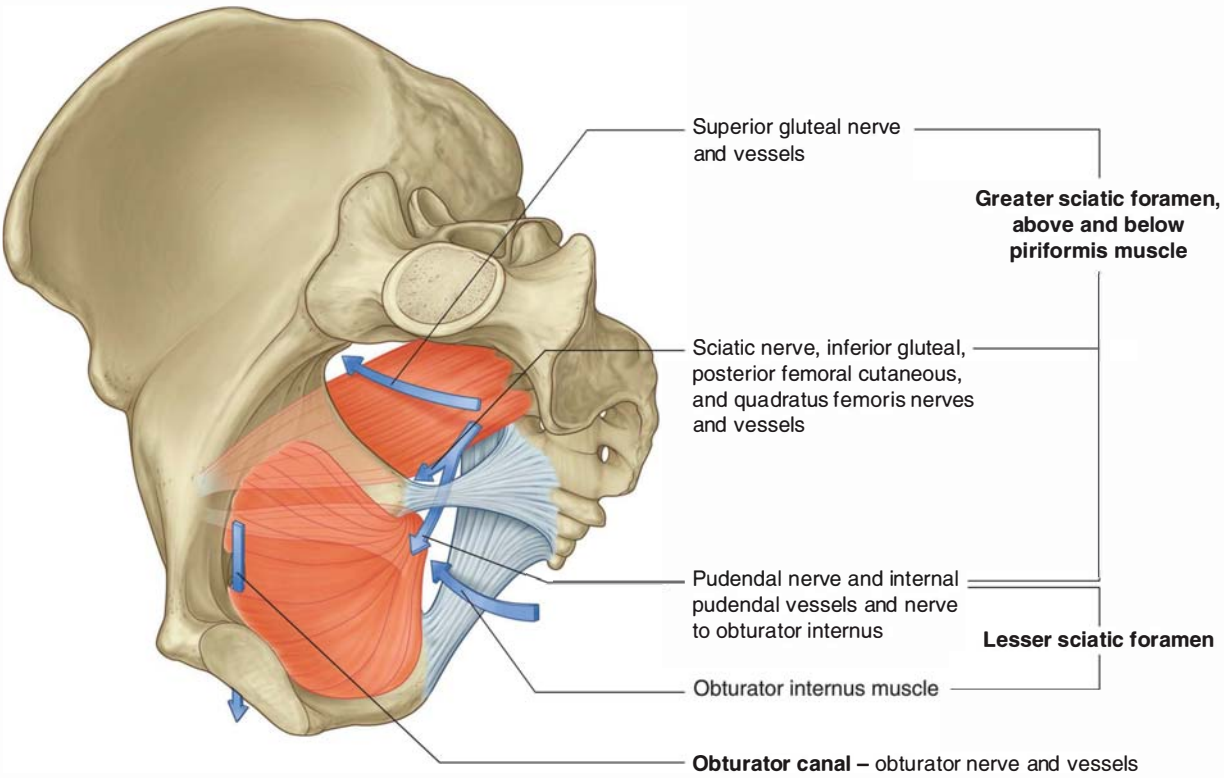


Fig. 5.31 Apertures in the pelvic wall.

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Because the lesser sciatic foramen is positioned below the attachment of the pelvic floor, it acts as a route of communication between the perineum and the gluteal region. The pudendal nerve and internal pudendal vessels pass between the pelvic cavity (above the pelvic floor) and the perineum (below the pelvic floor), by first passing out of the pelvic cavity through the greater sciatic foramen and then looping around the ischial spine and sacrospinous ligament to pass through the lesser sciatic foramen to enter the perineum.

Pelvic outlet

The pelvic outlet is diamond shaped, with the anterior part of the diamond defined predominantly by bone and the posterior part mainly by ligaments (Fig. 5.32). In the midline anteriorly, the boundary of the pelvic outlet is the pubic symphysis. Extending laterally and posteriorly, the boundary on each side is the inferior border of the body of the pubis, the inferior ramus of the pubis, the ramus of the ischium, and the ischial tuberosity. Together, the elements on both sides form the pubic arch.

From the ischial tuberosities, the boundaries continue posteriorly and medially along the sacrotuberous ligament on both sides to the coccyx.

Terminal parts of the urinary and gastrointestinal tracts and the vagina pass through the pelvic outlet.

The area enclosed by the boundaries of the pelvic outlet and below the pelvic floor is the **perineum**.

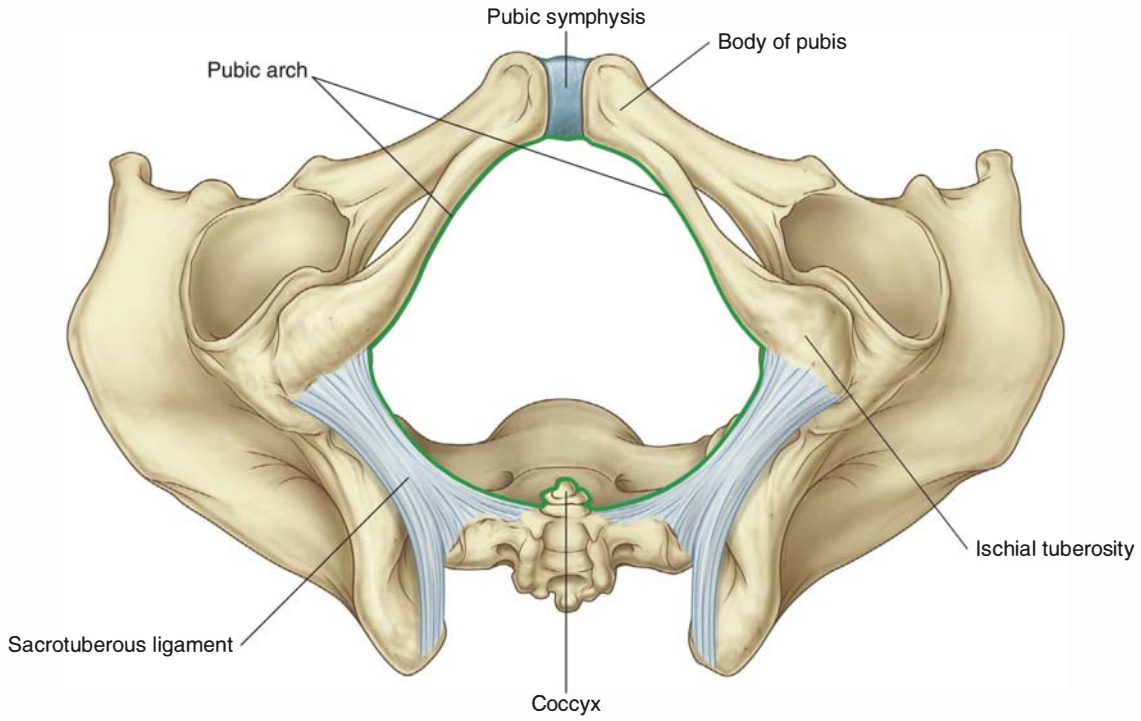


Fig. 5.32 Pelvic outlet.

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In the clinic

Pelvic measurements in obstetrics

Transverse and sagittal measurements of a woman's pelvic inlet and outlet can help in predicting the likelihood of a successful vaginal delivery. These measurements include:

- the sagittal inlet (between the promontory and the top of the pubic symphysis),
- the maximum transverse diameter of the inlet,
- the bispinous outlet (the distance between ischial spines), and
- the sagittal outlet (the distance between the tip of the coccyx and the inferior margin of the pubic symphysis).

These measurements can be obtained using magnetic resonance imaging, which carries no radiation risk for the fetus or mother (Fig. 5.33).

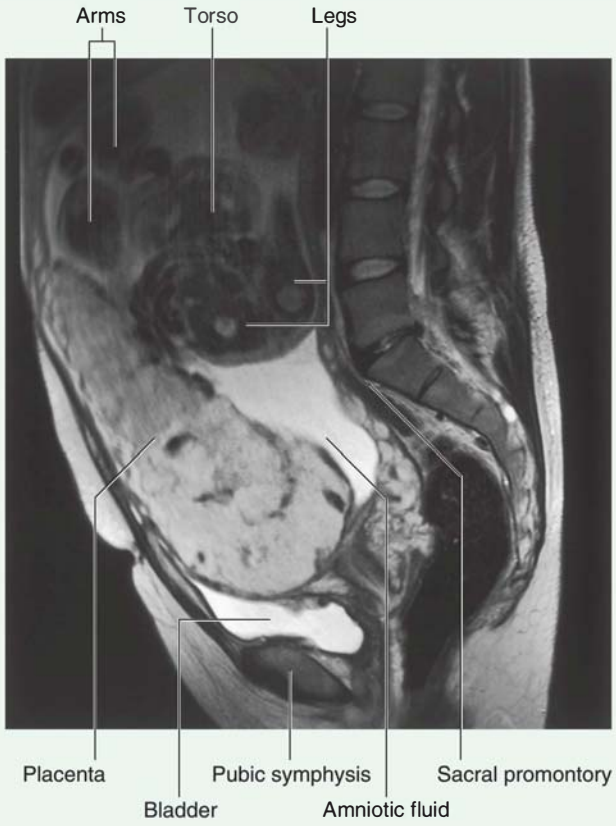


Fig. 5.33 Sagittal T2-weighted magnetic resonance image of the lower abdomen and pelvis of a pregnant woman.

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Pelvic floor

The pelvic floor is formed by the pelvic diaphragm and, in the anterior midline, the perineal membrane and the muscles in the deep perineal pouch. The pelvic diaphragm is formed by the levator ani and the coccygeus muscles from both sides. The pelvic floor separates the pelvic cavity, above, from the perineum, below.

The pelvic diaphragm

The pelvic diaphragm is the muscular part of the pelvic floor. Shaped like a bowl or funnel and attached superiorly to the pelvic walls, it consists of the levator ani and the coccygeus muscles (Fig. 5.34 and Table 5.2).

The pelvic diaphragm's circular line of attachment to the cylindrical pelvic wall passes, on each side, between the greater sciatic foramen and the lesser sciatic foramen. Thus:

- the greater sciatic foramen is situated above the level of the pelvic floor and is a route of communication between the pelvic cavity and the gluteal region of the lower limb; and
- the lesser sciatic foramen is situated below the pelvic floor, providing a route of communication between the gluteal region of the lower limb and the perineum.

Levator ani

The two levator ani muscles originate from each side of the pelvic wall, course medially and inferiorly, and join together in the midline. The attachment to the pelvic wall follows the circular contour of the wall and includes:

- the posterior aspect of the body of the pubic bone,
- a linear thickening called the **tendinous arch**, in the fascia covering the obturator internus muscle, and
- the spine of the ischium.

At the midline, the muscles blend together posterior to the vagina in women and around the anal aperture in both sexes. Posterior to the anal aperture, the muscles come

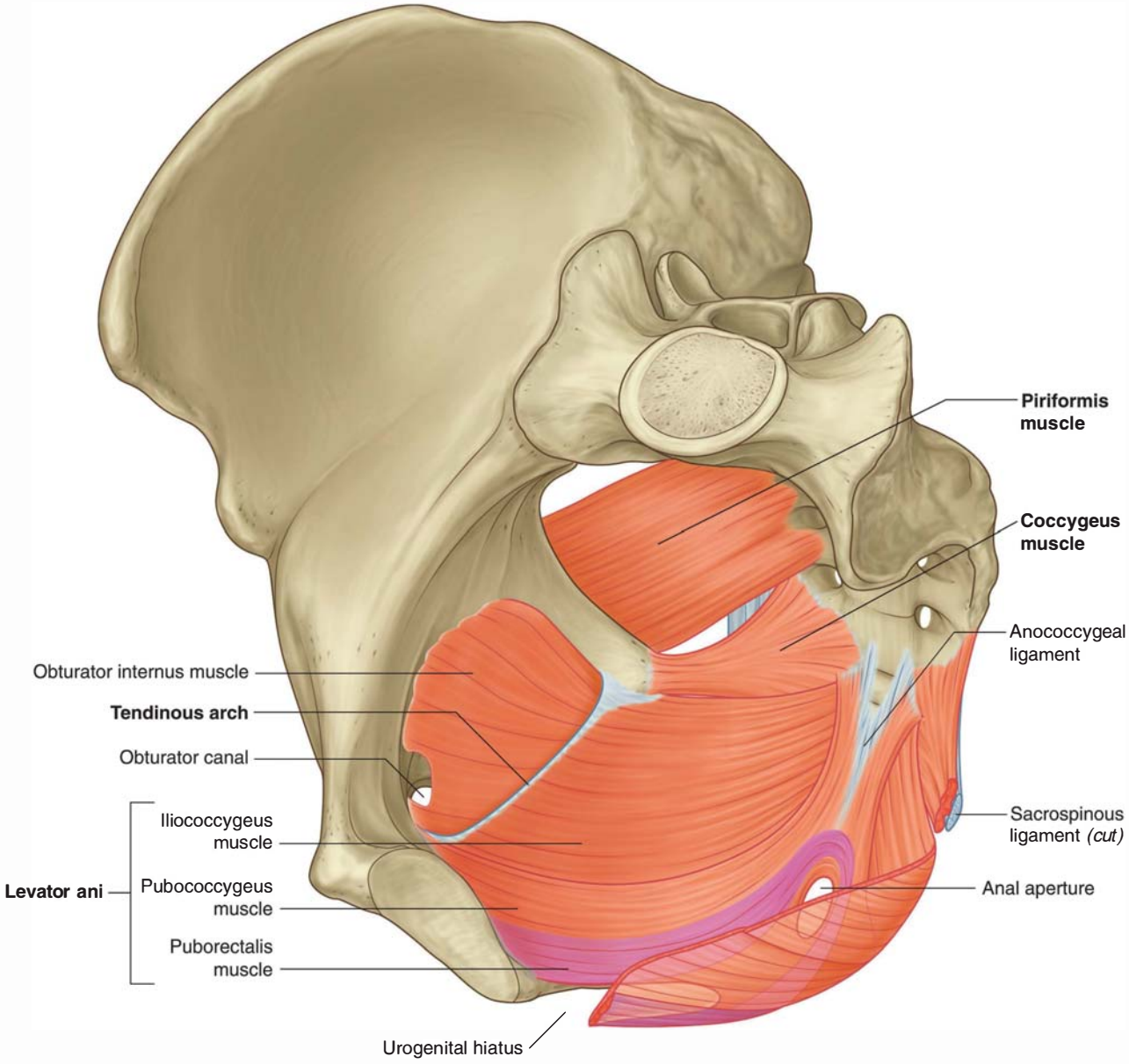


Fig. 5.34 Pelvic diaphragm.

Table 5.2 Muscles of the pelvic diaphragm

Muscle	Origin	Insertion	Innervation	Function
Levator ani	In a line around the pelvic wall beginning on the posterior aspect of the pubic bone and extending across the obturator internus muscle as a tendinous arch (thickening of the obturator internus fascia) to the ischial spine	The anterior part is attached to the superior surface of the perineal membrane; the posterior part meets its partner on the other side at the perineal body, around the anal canal, and along the anococcygeal ligament	Branches direct from the anterior ramus of S4, and by the inferior rectal branch of the pudendal nerve (S2 to S4)	Contributes to the formation of the pelvic floor, which supports the pelvic viscera; maintains an angle between the rectum and anal canal; reinforces the external anal sphincter and, in women, functions as a vaginal sphincter
Coccygeus	Ischial spine and pelvic surface of the sacrospinous ligament	Lateral margin of coccyx and related border of sacrum	Branches from the anterior rami of S3 and S4	Contributes to the formation of the pelvic floor, which suppo ts the pelvic viscera; pulls coccyx forward after defecation

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together as a ligament or raphe called the **anococcygeal ligament** (**anococcygeal body**) and attaches to the coccyx. Anteriorly, the muscles are separated by a U-shaped defect or gap termed the **urogenital hiatus**. The margins of this hiatus merge with the walls of the associated viscera and with muscles in the deep perineal pouch below. The hiatus allows the urethra (in both men and women), and the vagina (in women), to pass through the pelvic diaphragm (Fig. 5.34).

The levator ani muscles are divided into at least three collections of muscle fibers, based on site of origin and relationship to viscera in the midline: the pubococcygeus, the puborectalis, and the iliococcygeus muscles.

- The **pubococcygeus** originates from the body of the pubis and courses posteriorly to attach along the midline as far back as the coccyx. This part of the muscle is further subdivided on the basis of association with structures in the midline into the **puboprostaticus** (**levator prostatae**), the **pubovaginalis**, and the **puboanalis muscles**.
- A second major collection of muscle fibers, the **puborectalis** portion of the levator ani muscles, originates, in association with the pubococcygeus muscle, from the pubis and passes inferiorly on each side to form a sling around the terminal part of the gastrointestinal tract. This muscular sling maintains an angle or flexure, called the **perineal flexure**, at the anorectal junction. This angle functions as part of the mechanism that keeps the end of the gastrointestinal system closed.
- The final part of the levator ani muscle is the **iliococcygeus**. This part of the muscle originates from the fascia that covers the obturator internus muscle. It joins the same muscle on the other side in the midline to form a ligament or raphe that extends from the anal aperture to the coccyx.

The levator ani muscles help support the pelvic viscera and maintain closure of the rectum and vagina. They are innervated directly by branches from the anterior ramus of S4 and by branches of the pudendal nerve (S2 to S4).

Coccygeus

The two coccygeus muscles, one on each side, are triangular and overlie the sacrospinous ligaments; together they complete the posterior part of the pelvic diaphragm (Fig. 5.34 and Table 5.2). They are attached, by their apices, to the tips of the ischial spines and, by their bases, to the lateral margins of the coccyx and adjacent margins of the sacrum.

These coccygeus muscles are innervated by branches from the anterior rami of S3 and S4 and participate in supporting the posterior aspect of the pelvic floor.

In the clinic

Defecation

At the beginning of defecation, closure of the larynx stabilizes the diaphragm and intraabdominal pressure is increased by contraction of abdominal wall muscles. As defecation proceeds, the puborectalis muscle surrounding the anorectal junction relaxes, which straightens the anorectal angle. Both the internal and the external anal sphincters also relax to allow feces to move through the anal canal. Normally, the puborectal sling maintains an angle of about 90° between the rectum and the anal canal and acts as a "pinch valve" to prevent defecation. When the puborectalis muscle relaxes, the anorectal angle increases to about 130° to 140°.

The fatty tissue of the ischio-anal fossa allows for changes in the position and size of the anal canal and anus during defecation. During evacuation, the anorectal junction moves down and back and the pelvic floor usually descends slightly.

During defecation, the circular muscles of the rectal wall undergo a wave of contraction to push feces toward the anus. As feces emerge from the anus, the longitudinal muscles of the rectum and levator ani bring the anal canal back up, the feces are expelled, and the anus and rectum return to their normal positions.

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The perineal membrane and deep perineal pouch

The **perineal membrane** is a thick fascial, triangular structure attached to the bony framework of the pubic arch (Fig. 5.35A). It is oriented in the horizontal plane and has a free posterior margin. Anteriorly, there is a small gap (blue arrow in Fig. 5.35A) between the membrane and the **inferior pubic ligament** (a ligament associated with the pubic symphysis).

The perineal membrane is related above to a thin space called the **deep perineal pouch** (**deep perineal space**) (Fig. 5.35B), which contains a layer of skeletal muscle and various neurovascular elements.

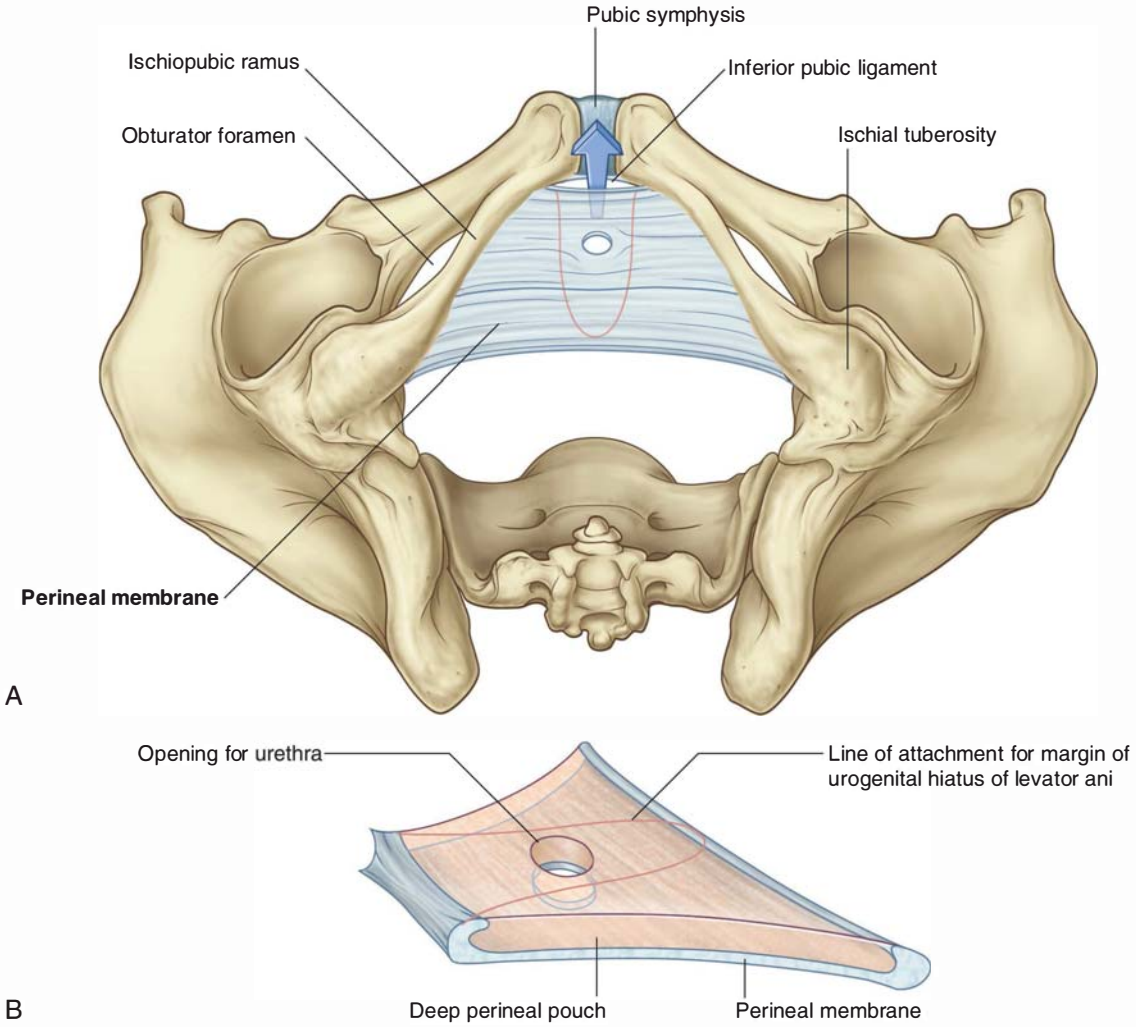


Fig. 5.35 Perineal membrane and deep perineal pouch. **A.** Inferior view. **B.** Superolateral view.

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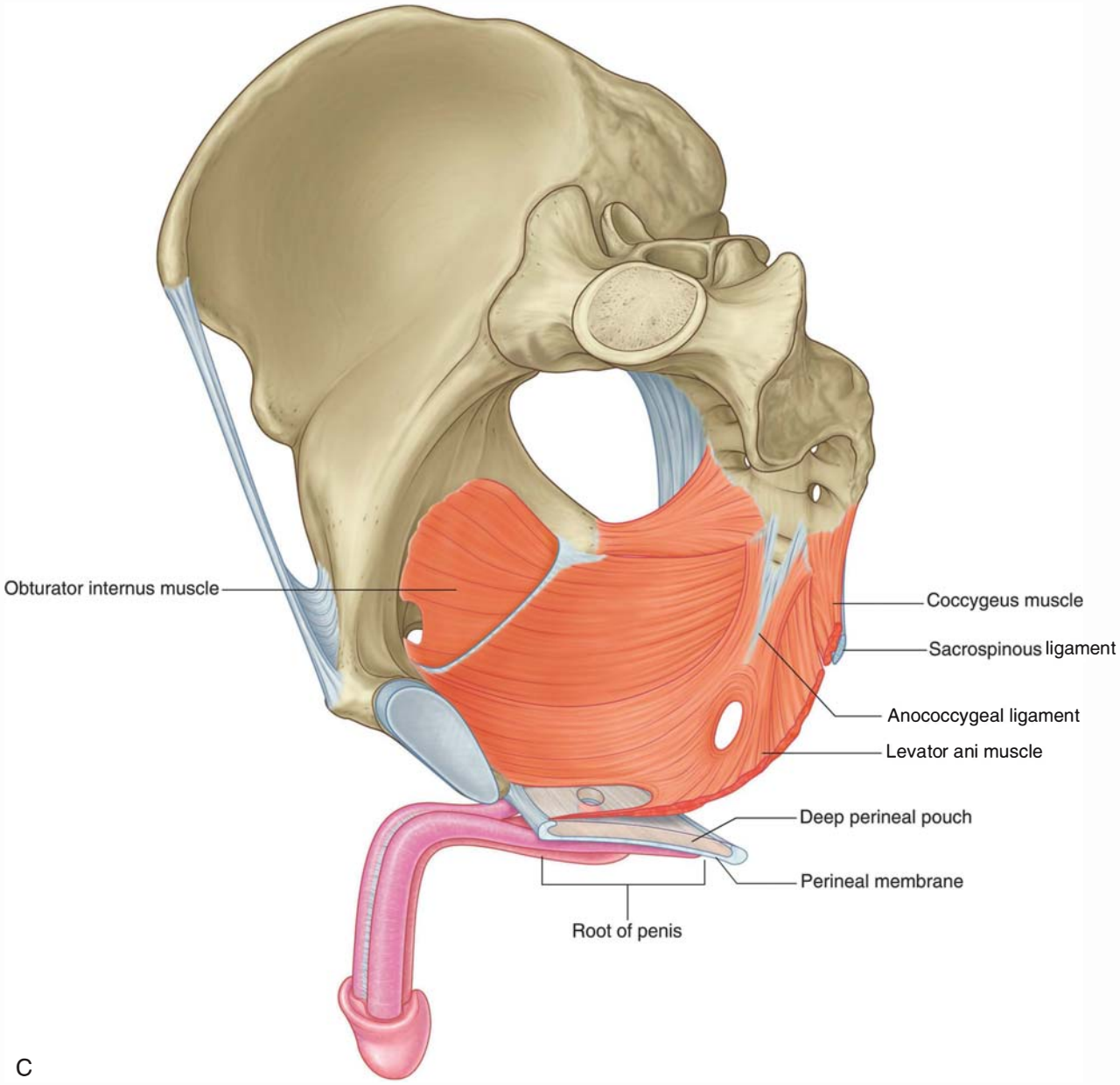


Fig. 5.35, cont'd Perineal membrane and deep perineal pouch. **C.** Medial view.

The deep perineal pouch is open above and is not separated from more superior structures by a distinct layer of fascia. The parts of the perineal membrane and structures in the deep perineal pouch, enclosed by the urogenital hiatus above, therefore contribute to the pelvic floor and support elements of the urogenital system in the pelvic cavity, even though the perineal membrane and deep perineal pouch are usually considered parts of the perineum.

The perineal membrane and adjacent pubic arch provide attachment for the roots of the external genitalia and the muscles associated with them (Fig. 5.35C).

The urethra penetrates vertically through a circular hiatus in the perineal membrane as it passes from the pelvic cavity, above, to the perineum, below. In women, the vagina also passes through a hiatus in the perineal membrane just posterior to the urethral hiatus.

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Within the deep perineal pouch, a sheet of skeletal muscle functions as a sphincter, mainly for the urethra, and as a stabilizer of the posterior edge of the perineal membrane (Fig. 5.36 and Table 5.3).

- Anteriorly, a group of muscle fibers surround the urethra and collectively form the **external urethral sphincter**.
- Two additional groups of muscle fibers are associated with the urethra and vagina in women. One group forms the **sphincter urethrovaginalis**, which surrounds the urethra and vagina as a unit. The second group forms the **compressor urethrae**, on each side, which originate from the ischiopubic rami and meet anterior to the urethra. Together with the external urethral sphincter, the sphincter urethrovaginalis and compressor urethrae facilitate closing of the urethra.
- In both men and women, a **deep transverse perineal muscle** on each side parallels the free margin of the perineal membrane and joins with its partner at the midline. These muscles are thought to stabilize the position of the perineal body, which is a midline structure along the posterior edge of the perineal membrane.

Perineal body

The perineal body is an ill-defined but important connective tissue structure into which muscles of the pelvic floor and

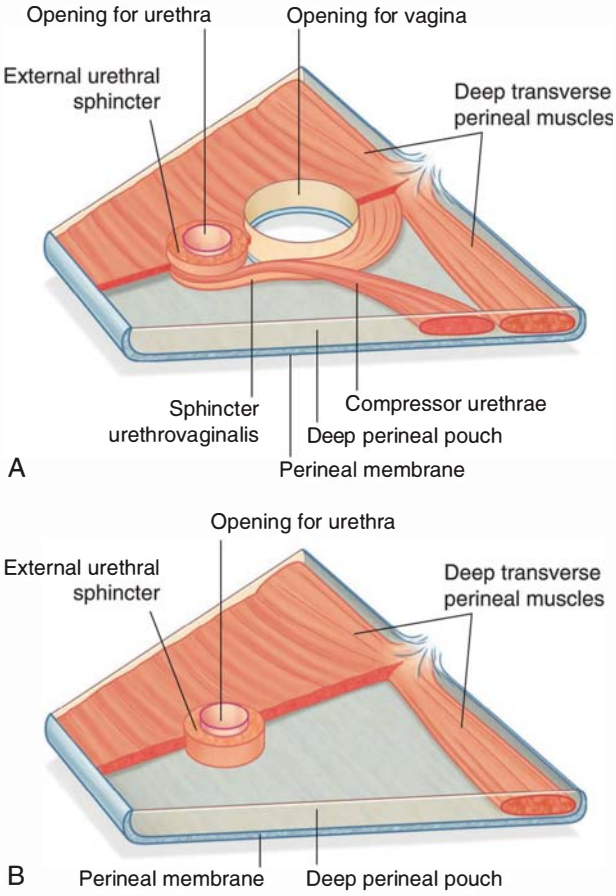


Fig. 5.36 Muscles in the deep perineal pouch. **A.** In women. **B.** In men.

Table 5.3 Muscles within the deep perineal pouch

Muscle	Origin	Insertion	Innervation	Function
External urethral sphincter	From the inferior ramus of the pubis on each side and adjacent walls of the deep perineal pouch	Surrounds membranous part of urethra	Perineal branches of the pudendal nerve (S2 to S4)	Compresses the membranous urethra; relaxes during micturition
Deep transverse perineal	Medial aspect of ischial ramus	Perineal body	Perineal branches of the pudendal nerve (S2 to S4)	Stabilizes the position of the perineal body
Compressor urethrae (in women only)	Ischiopubic ramus on each side	Blends with partner on other side anterior to the urethra	Perineal branches of the pudendal nerve (S2 to S4)	Functions as an accessory sphincter of the urethra
Sphincter urethrovaginalis (in women only)	Perineal body	Passes forward lateral to the vagina to blend with partner on other side anterior to the urethra	Perineal branches of the pudendal nerve (S2 to S4)	Functions as an accessory sphincter of the urethra (also may facilitate closing the vagina)

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the perineum attach (Fig. 5.37). It is positioned in the midline along the posterior border of the perineal membrane, to which it attaches. The posterior end of the urogenital hiatus in the levator ani muscles is also connected to it.

The deep transverse perineal muscles intersect at the perineal body; in women, the sphincter urethrovaginalis also attaches to the perineal body. Other muscles that connect to the perineal body include the external anal sphincter, the superficial transverse perineal muscles, and the bulbospongiosus muscles of the perineum.

In the clinic

Episiotomy

During childbirth the perineal body may be stretched and torn. Traditionally it was felt that if a perineal tear is likely, the obstetrician may proceed with an episiotomy. This is a procedure in which an incision is made in the perineal body to allow the head of the fetus to pass through the vagina. There are two types of episiotomies: a median episiotomy cuts through the perineal body, while a mediolateral episiotomy is an incision 45° from the midline. The maternal benefits of this procedure have been thought to be less traumatic to the perineum and to result in decreased pelvic floor dysfunction after childbirth. However, more recent evidence suggests that an episiotomy should not be performed routinely. Review of data has failed to show a decrease in pelvic floor damage with routine use of episiotomies.

Viscera

The pelvic viscera include parts of the gastrointestinal system, the urinary system, and the reproductive system.

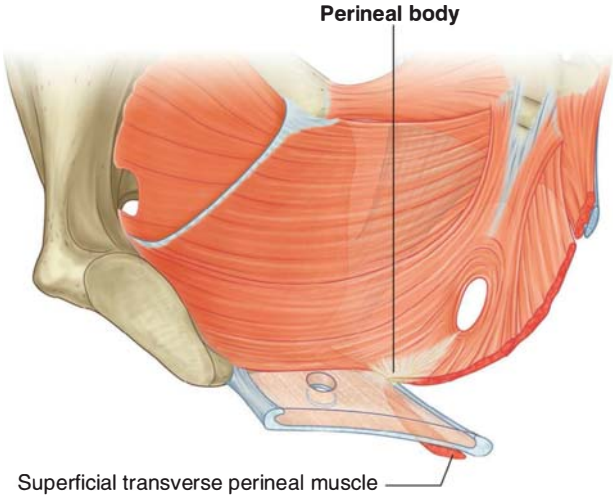


Fig. 5.37 Perineal body.

The viscera are arranged in the midline, from front to back; the neurovascular supply is through branches that pass medially from vessels and nerves associated with the pelvic walls.

Gastrointestinal system

Pelvic parts of the gastrointestinal system consist mainly of the rectum and the anal canal, although the terminal part of the sigmoid colon is also in the pelvic cavity (Fig. 5.38).

Rectum

The **rectum** is continuous:

- above, with the sigmoid colon at about the level of vertebra SIII, and
- below, with the anal canal as this structure penetrates the pelvic floor and passes through the perineum to end as the anus.

The rectum, the most posterior element of the pelvic viscera, is immediately anterior to, and follows the concave contour of the sacrum.

The anorectal junction is pulled forward (perineal flexure) by the action of the puborectalis part of the levator ani muscle, so the anal canal moves in a posterior direction as it passes inferiorly through the pelvic floor.

In addition to conforming to the general curvature of the sacrum in the anteroposterior plane, the rectum has three lateral curvatures; the upper and lower curvatures to the right and the middle curvature to the left. The lower part of the rectum is expanded to form the **rectal ampulla**. Finally, unlike the colon, the rectum lacks distinct taeniae coli muscles, omental appendices, and sacculations (haustra of the colon).

Anal canal

The **anal canal** begins at the terminal end of the rectal ampulla where it narrows at the pelvic floor. It terminates as the anus after passing through the perineum. As it passes through the pelvic floor, the anal canal is surrounded along its entire length by the internal and external anal sphincters, which normally keep it closed.

The lining of the anal canal bears a number of characteristic structural features that reflect the approximate position of the anococcygeal membrane in the fetus (which closes the terminal end of the developing gastrointestinal system in the fetus) and the transition from gastrointestinal mucosa to skin in the adult (Fig. 5.38B).

- The upper part of the anal canal is lined by mucosa similar to that lining the rectum and is distinguished by a number of longitudinally oriented folds known as

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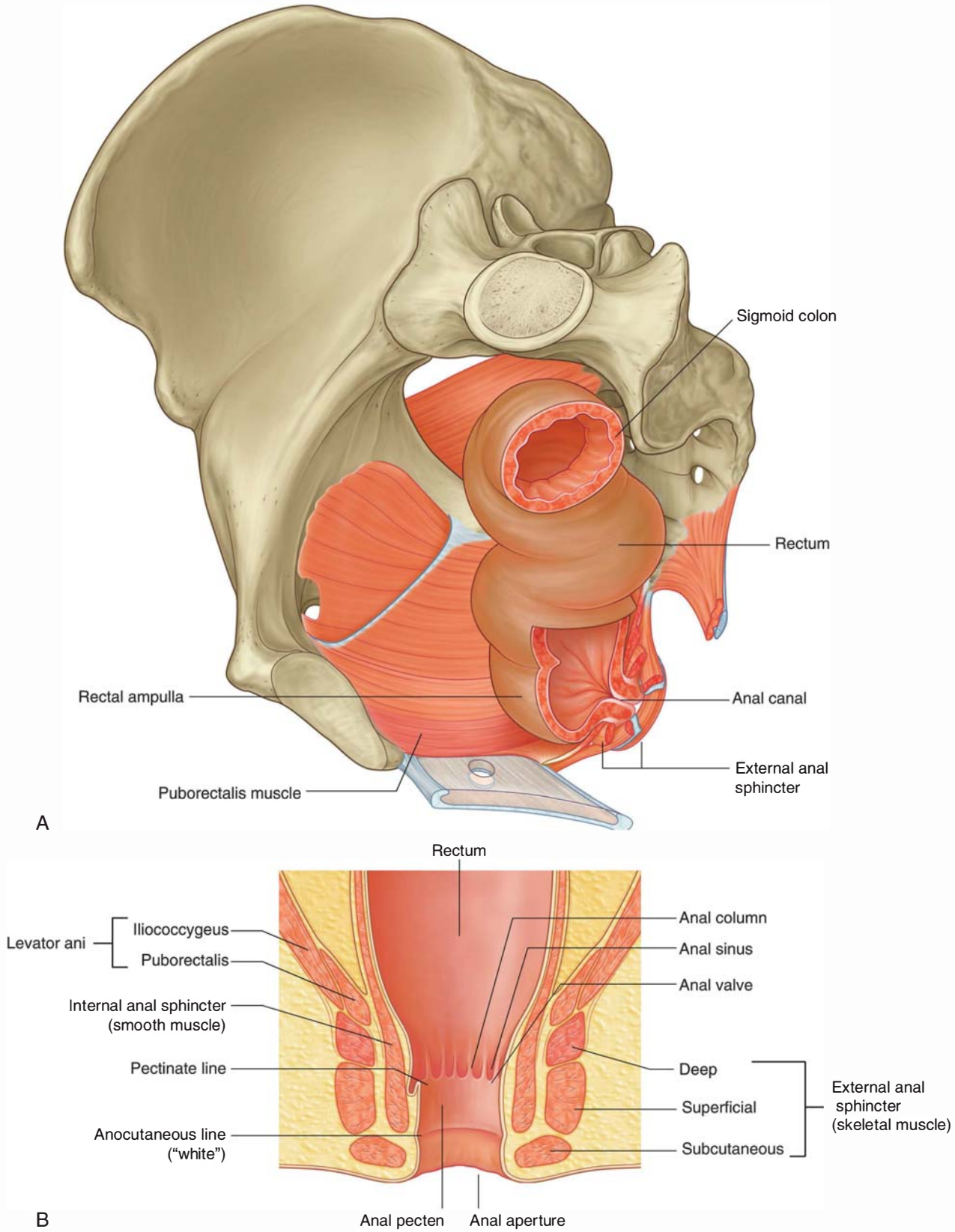


Fig. 5.38 Rectum and anal canal. **A.** Left pelvic bone removed. **B.** Longitudinal section.

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anal columns, which are united inferiorly by crescentic folds termed **anal valves**. Superior to each valve is a depression termed an **anal sinus**. The anal valves together form a circle around the anal canal at a location known as the **pectinate line**, which marks the approximate position of the anal membrane in the fetus.

- Inferior to the pectinate line is a transition zone known as the **anal pecten**, which is lined by nonkeratinized stratified squamous epithelium. The anal pecten ends inferiorly at the **anocutaneous line** ("white line"), or where the lining of the anal canal becomes true skin.

Given the position of the colon and rectum in the abdominopelvic cavity and its proximity to other organs, it is extremely important to accurately stage colorectal tumors: a tumor in the pelvis, for example, could invade the uterus or bladder. Assessing whether spread has occurred may involve ultrasound scanning, computed tomography, and magnetic resonance imaging.

In the clinic

Digital rectal examination

A digital rectal examination (DRE) is performed by placing the gloved and lubricated index finger into the rectum through the anus. The anal mucosa can be palpated for abnormal masses, and in women, the posterior wall of the vagina and the cervix can be palpated. In men, the prostate can be evaluated for any extraneous nodules or masses.

In many instances the digital rectal examination may be followed by proctoscopy or colonoscopy. An ultrasound probe may be placed into the rectum to assess the gynecological structures in females and the prostate in the male before performing a prostatic biopsy.

In the clinic

Carcinoma of the colon and rectum

Carcinoma of the colon and rectum (colorectum) is a common and often lethal disease. Recent advances in surgery, radiotherapy, and chemotherapy have only slightly improved 5-year survival rates.

The biological behavior of tumors of the colon and rectum is relatively predictable. Most of the tumors develop from benign polyps, some of which undergo malignant change. The overall prognosis is related to:

- the degree of tumor penetration through the bowel wall,
- the presence or absence of lymphatic dissemination, and
- the presence or absence of systemic metastases.

Urinary system

The pelvic parts of the urinary system consist of the terminal parts of the ureters, the bladder, and the proximal part of the urethra (Fig. 5.39).

Ureters

The ureters enter the pelvic cavity from the abdomen by passing through the pelvic inlet. On each side, the ureter crosses the pelvic inlet and enters the pelvic cavity in the area anterior to the bifurcation of the common iliac artery. From this point, it continues along the pelvic wall and floor to join the base of the bladder.

In the pelvis, the ureter is crossed by:

- the ductus deferens in men, and
- the uterine artery in women.

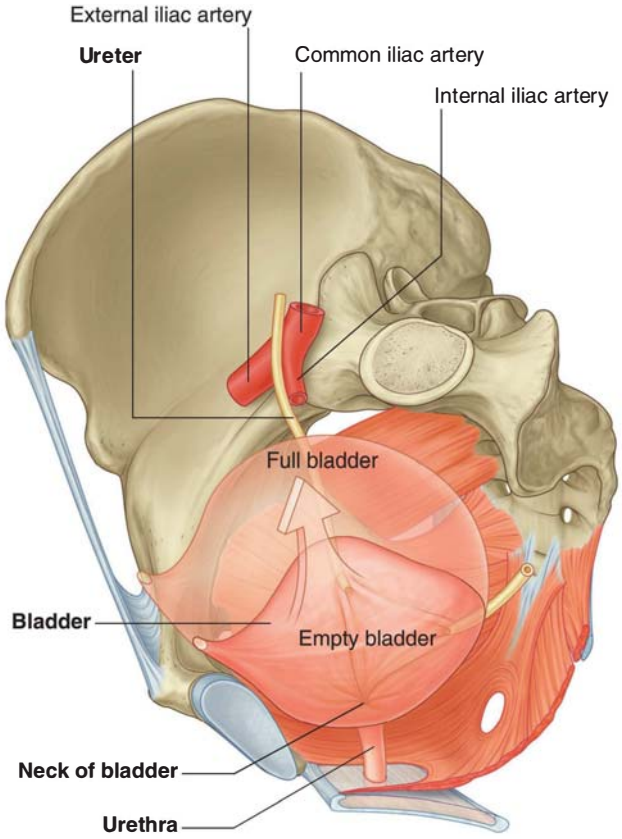


Fig. 5.39 Pelvic parts of the urinary system.

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Bladder

The bladder is the most anterior element of the pelvic viscera. Although it is entirely situated in the pelvic cavity when empty, it expands superiorly into the abdominal cavity when full (Fig. 5.39).

The empty bladder is shaped like a three-sided pyramid that has tipped over to lie on one of its margins (Fig. 5.40A). It has an apex, a base, a superior surface, and two inferolateral surfaces.

- The **apex** of the bladder is directed toward the top of the pubic symphysis; a structure known as the **median umbilical ligament** (a remnant of the embryological urachus that contributes to the formation of the bladder) continues from it superiorly up the anterior abdominal wall to the umbilicus.
- The **base** of the bladder is shaped like an inverted triangle and faces posteroinferiorly. The two ureters enter the bladder at each of the upper corners of the base, and the urethra drains inferiorly from the lower corner of the base. Inside, the mucosal lining on the base of the bladder is smooth and firmly attached to the underlying smooth muscle coat of the wall—unlike elsewhere in the bladder where the mucosa is folded and loosely attached to the wall. The smooth triangular area between the openings of the ureters and urethra on the inside of the bladder is known as the **trigone** (Fig. 5.40B).
- The **inferolateral surfaces** of the bladder are cradled between the levator ani muscles of the pelvic diaphragm and the adjacent obturator internus muscles above the attachment of the pelvic diaphragm. The superior surface is slightly domed when the bladder is empty; it balloons upward as the bladder fills.

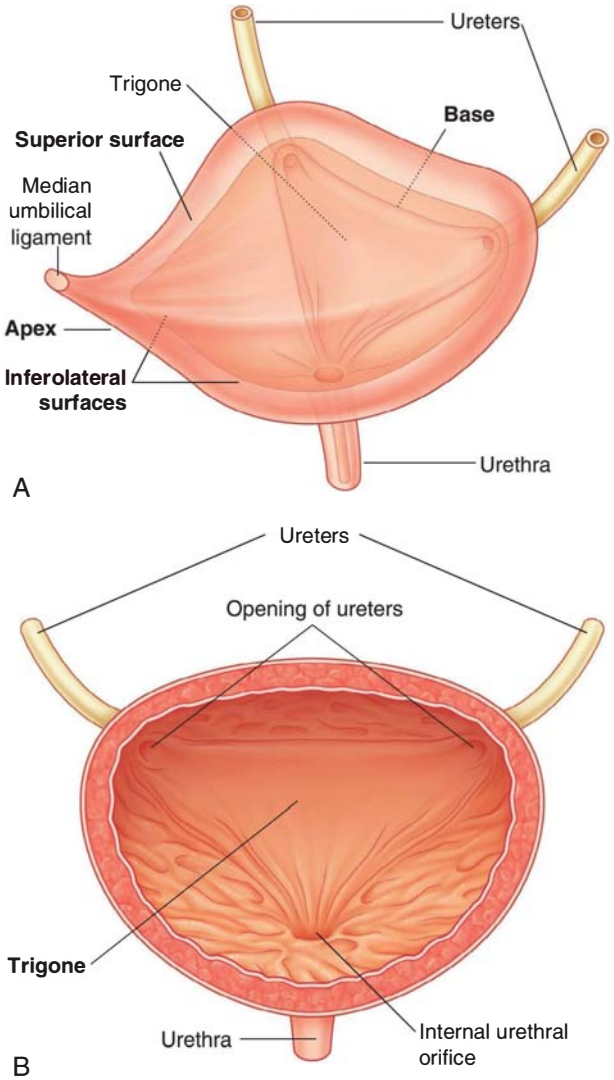


Fig. 5.40 Bladder. **A.** Superolateral view. **B.** The trigone. Anterior view with the anterior part of the bladder cut away.

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Neck of bladder

The neck of the bladder surrounds the origin of the urethra at the point where the two inferolateral surfaces and the base intersect.

The neck is the most inferior part of the bladder and also the most "fixed" part. It is anchored into position by a pair of tough fibromuscular bands, which connect the neck and pelvic part of the urethra to the posteroinferior aspect of each pubic bone.

- In women, these fibromuscular bands are termed **pubovesical ligaments** (Fig. 5.41A). Together with the perineal membrane and associated muscles, the levator ani muscles, and the pubic bones, these ligaments help support the bladder.
- In men, the paired fibromuscular bands are known as **puboprostatic ligaments** because they blend with the fibrous capsule of the prostate, which surrounds the neck of the bladder and adjacent part of the urethra (Fig. 5.41B).

Although the bladder is considered to be pelvic in the adult, it has a higher position in children. At birth, the bladder is almost entirely abdominal; the urethra begins approximately at the upper margin of the pubic symphysis. With age, the bladder descends until after puberty when it assumes the adult position.

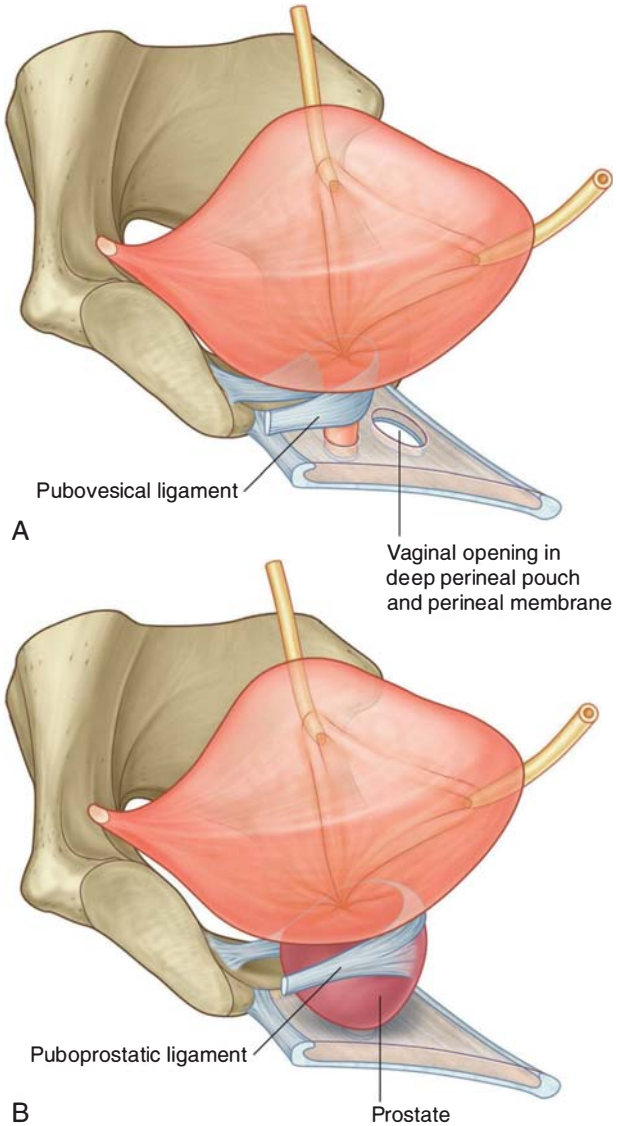


Fig. 5.41 Ligaments that anchor the neck of the bladder and pelvic part of the urethra to the pelvic bones. **A.** In women. **B.** In men.

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In the clinic

Bladder stones

In some patients, small calculi (stones) form in the kidneys. These may pass down the ureter, causing ureteric obstruction, and into the bladder (Fig. 5.42), where insoluble salts further precipitate on these small calculi to form larger calculi. Often, these patients develop (or may already have) problems with bladder emptying, which leaves residual urine in the bladder. This urine may become infected, which alters the pH of the urine, permitting further precipitation of insoluble salts.

If small enough, the stones may be removed via a transurethral route using specialized instruments. If the stones are too big, it may be necessary to make a suprapubic incision and enter the bladder retroperitoneally to remove them.

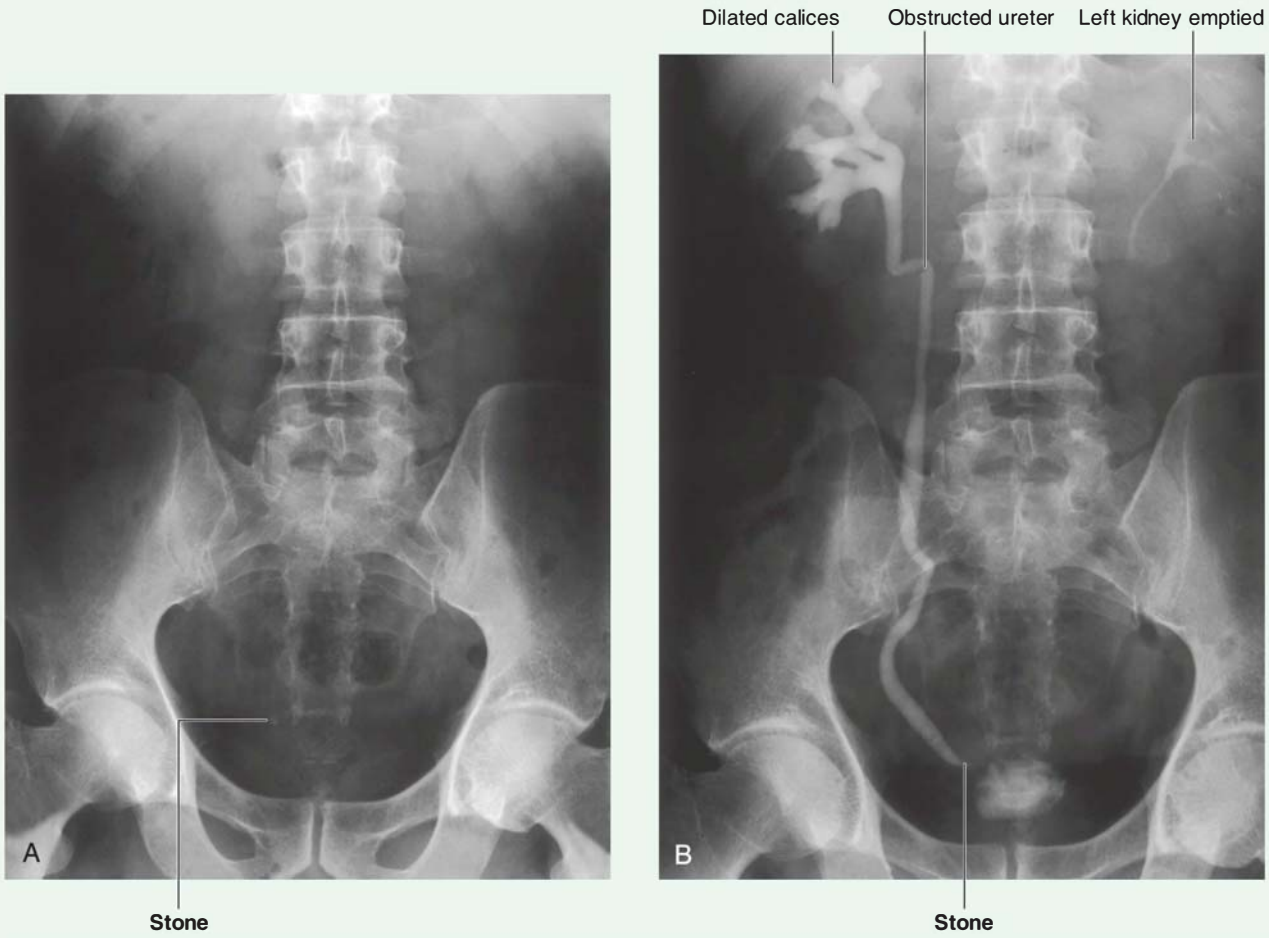


Fig. 5.42 Intravenous urogram demonstrating a stone in the lower portion of the ureter. **A.** Control radiograph. **B.** Intravenous urogram, postmicturition.

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In the clinic

Suprapubic catheterization

In certain instances it is necessary to catheterize the bladder through the anterior abdominal wall. For example, when the prostate is markedly enlarged and it is impossible to pass a urethral catheter, a suprapubic catheter may be placed.

The bladder is a retroperitoneal structure and when full lies adjacent to the anterior abdominal wall. Ultrasound visualization of the bladder may be useful in assessing the size of this structure and, importantly, differentiating this structure from other potential abdominal masses.

The procedure of suprapubic catheterization is straightforward and involves the passage of a small catheter on a needle in the midline approximately 2 cm above the pubic symphysis. The catheter passes easily into the bladder without compromise of other structures and permits free drainage.

In the clinic

Bladder cancer

Bladder cancer (Fig. 5.43) is the most common tumor of the urinary tract and is usually a disease of the sixth and seventh decades, although there is an increasing trend for younger patients to develop this disease.

Approximately one-third of bladder tumors are multifocal; fortunately, two-thirds are superficial tumors and amenable to local treatment.

Bladder tumors may spread through the bladder wall and invade local structures, including the rectum, uterus (in women), and lateral walls of the pelvic cavity. Prostatic involvement is not uncommon in male patients. The disease spreads via the internal iliac lymph nodes. Spread to distant metastatic sites rarely includes the lung.

Treatment for early-stage tumors includes local resection with preservation of the bladder. Diffuse tumors may be treated with local chemotherapy; more extensive tumors may require radical surgical removal of the bladder and, in men, the prostate.

Large bladder tumors may produce complications, including invasion and obstruction of the ureters. Ureteric obstruction can then obstruct the kidneys and induce kidney failure. Moreover, bladder tumors can invade other structures of the pelvic cavity.



Fig. 5.43 Intravenous urogram demonstrating a small tumor in the wall of the bladder.

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Urethra

The urethra begins at the base of the bladder and ends with an external opening in the perineum. The paths taken by the urethra differ significantly in women and men.

In women

In women, the urethra is short, being about 4 cm long. It travels a slightly curved course as it passes inferiorly through the pelvic floor into the perineum, where it passes through the deep perineal pouch and perineal membrane before opening in the vestibule that lies between the labia minora (Fig. 5.44A).

The urethral opening is anterior to the vaginal opening in the vestibule. The inferior aspect of the urethra is bound to the anterior surface of the vagina. Two small para-urethral mucous glands (**Skene's glands**) are associated with the lower end of the urethra. Each drains via a duct that opens onto the lateral margin of the external urethral orifice.

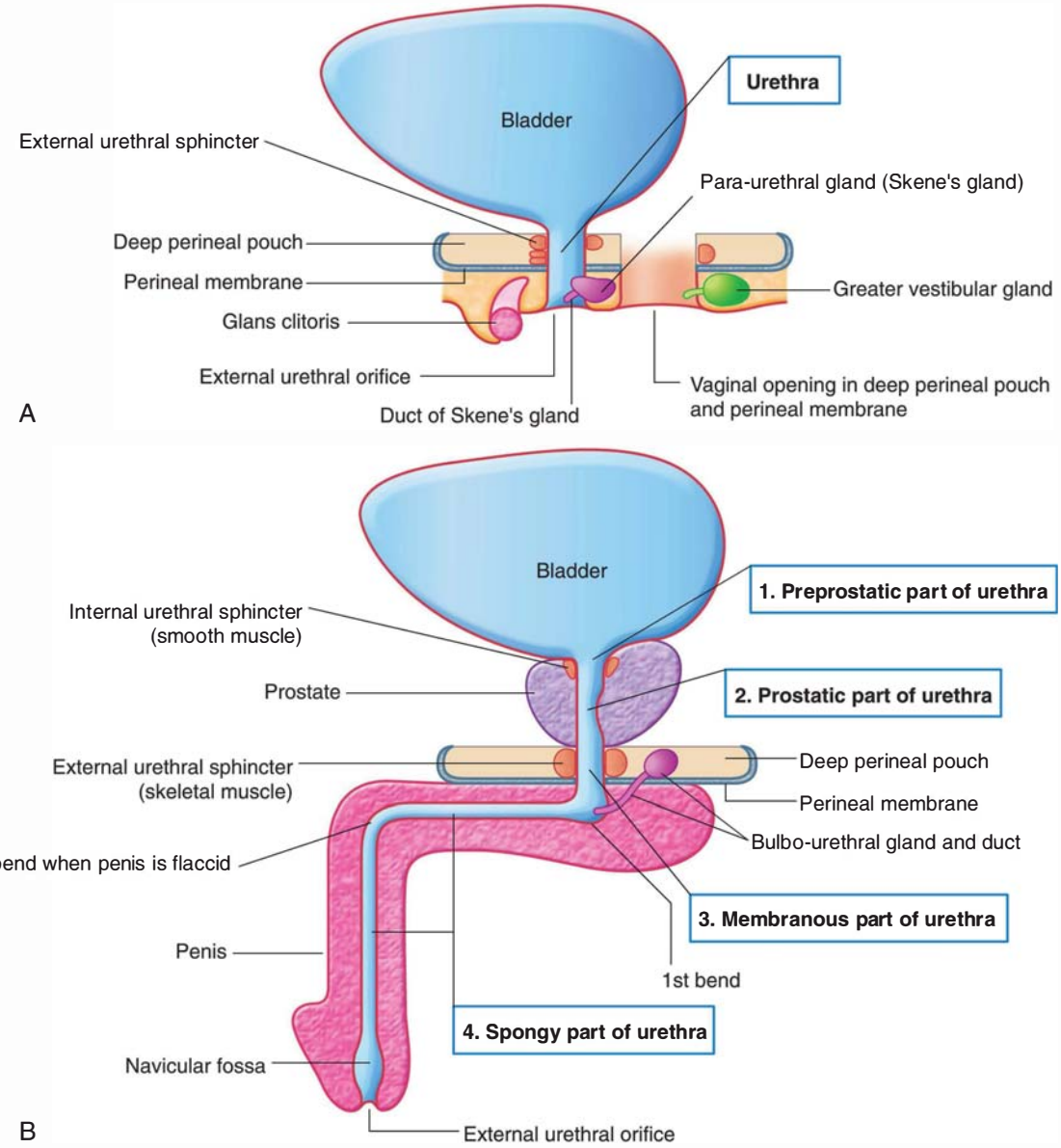


Fig. 5.44 Urethra. **A.** In women. **B.** In men.

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In men

In men, the urethra is long, about 20 cm, and bends twice along its course (Fig. 5.44B). Beginning at the base of the bladder and passing inferiorly through the prostate, it passes through the deep perineal pouch and perineal membrane and immediately enters the root of the penis. As the urethra exits the deep perineal pouch, it bends forward to course anteriorly in the root of the penis. When the penis is flaccid, the urethra makes another bend, this time inferiorly, when passing from the root to the body of the penis. During erection, the bend between the root and body of the penis disappears.

The urethra in men is divided into preprostatic, prostatic, membranous, and spongy parts.

Preprostatic part. The preprostatic part of the urethra is about 1 cm long, extends from the base of the bladder to the prostate, and is associated with a circular cuff of smooth muscle fibers (the **internal urethral sphincter**). Contraction of this sphincter prevents retrograde movement of semen into the bladder during ejaculation.

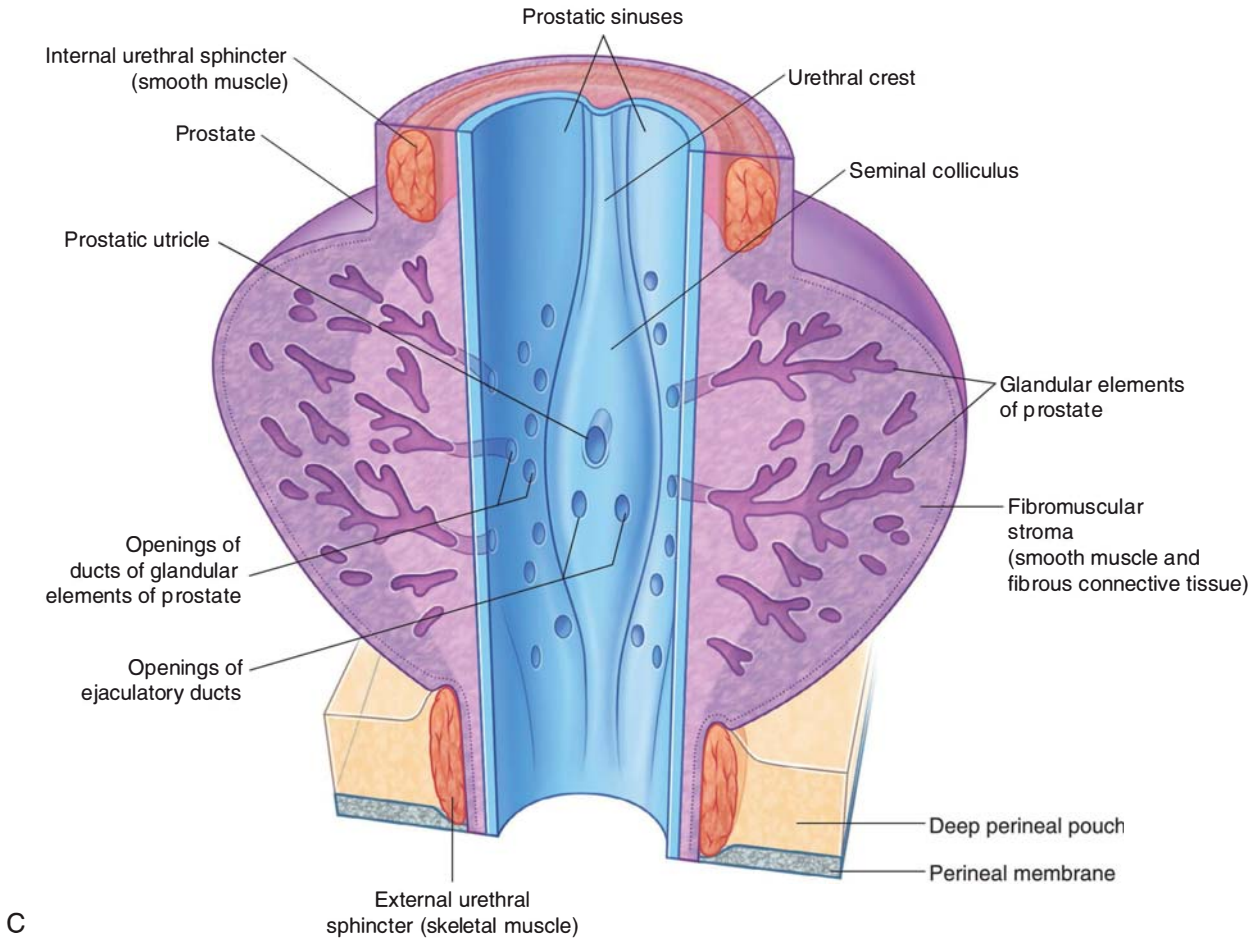


Fig. 5.44, cont'd C. Prostatic part of the urethra in men.

Prostatic part. The prostatic part of the urethra (Fig. 5.44C) is 3 to 4 cm long and is surrounded by the prostate. In this region, the lumen of the urethra is marked by a longitudinal midline fold of mucosa (the **urethral crest**). The depression on each side of the crest is the **prostatic sinus**; the ducts of the prostate empty into these two sinuses.

Midway along its length, the urethral crest is enlarged to form a somewhat circular elevation (the seminal colliculus). In men, the seminal colliculus is used to determine the position of the prostate gland during transurethral transection of the prostate.

A small blind-ended pouch—the **prostatic utricle** (thought to be the homologue of the uterus in women)—opens onto the center of the seminal colliculus. On each side of the prostatic utricle is the opening of the ejaculatory duct of the male reproductive system. Therefore the connection between the urinary and reproductive tracts in men occurs in the prostatic part of the urethra.

Membranous part. The membranous part of the urethra is narrow and passes through the deep perineal pouch (Fig. 5.44B). During its transit through this pouch, the urethra, in both men and women, is surrounded by skeletal muscle of the **external urethral sphincter**.

Spongy urethra. The spongy urethra is surrounded by erectile tissue (the **corpus spongiosum**) of the penis. It is enlarged to form a bulb at the base of the penis and again at the end of the penis to form the **navicular fossa** (Fig. 5.44B). The two bulbo-urethral glands in the deep perineal pouch are part of the male reproductive system and open into the bulb of the spongy urethra. The external urethral orifice is the sagittal slit at the end of the penis.

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In the clinic

Bladder infection

The relatively short length of the urethra in women makes them more susceptible than men to bladder infection. The primary symptom of urinary tract infection in women is usually inflammation of the bladder (cystitis). The infection can be controlled in most instances by oral antibiotics and resolves without complication. In children under 1 year of age, infection from the bladder may spread via the ureters to the kidneys, where it can produce renal damage and ultimately lead to renal failure. Early diagnosis and treatment are necessary.

In the clinic

Urethral catheterization

Urethral catheterization is often performed to drain urine from a patient's bladder when the patient is unable to micturate. When inserting urinary catheters, it is important to appreciate the gender anatomy of the patient.

In men:

- The spongy urethra is surrounded by the erectile tissue of the bulb of the penis immediately inferior to the deep perineal pouch. The wall of this short segment of urethra is relatively thin and angles superiorly to pass through the deep perineal pouch; at this position the urethra is vulnerable to damage, notably during cystoscopy.
- The membranous part of the urethra runs superiorly as it passes through the deep perineal pouch.
- The prostatic part of the urethra takes a slight concave curve anteriorly as it passes through the prostate gland.

In women, it is much simpler to pass catheters and cystoscopes because the urethra is short and straight. Urine may therefore be readily drained from a distended bladder without significant concern for urethral rupture.

Occasionally, it is impossible to pass any form of instrumentation through the urethra to drain the bladder, usually because there is a urethral stricture or prostatic enlargement. In such cases, an ultrasound of the lower abdomen will demonstrate a full bladder (Fig. 5.45) behind the anterior abdominal wall. A suprapubic catheter may be inserted into the bladder with minimal trauma through a small incision under local anesthetic.

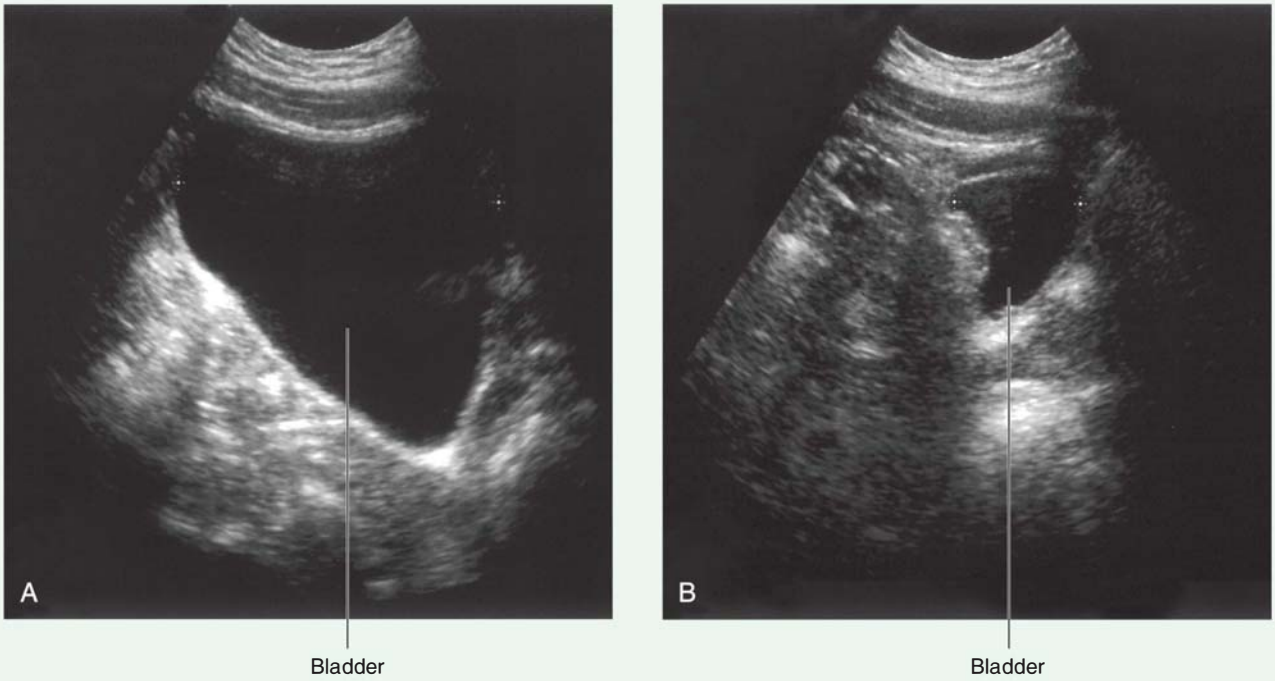


Fig. 5.45 Ultrasound demonstrating the bladder. **A.** Full bladder. **B.** Postmicturition bladder.

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Reproductive system

In men

The reproductive system in men has components in the abdomen, pelvis, and perineum (Fig. 5.46A). The major components are a testis, epididymis, ductus deferens, and ejaculatory duct on each side, and the urethra and penis in the midline. In addition, three types of accessory glands are associated with the system:

- a single prostate,
- a pair of seminal vesicles, and
- a pair of bulbo-urethral glands.

The design of the reproductive system in men is basically a series of ducts and tubules. The arrangement of parts and linkage to the urinary tract reflects its embryological development.

Testes

The **testes** originally develop high on the posterior abdominal wall and then descend, normally before birth, through the inguinal canal in the anterior abdominal wall and into the scrotum of the perineum. During descent, the testes carry their vessels, lymphatics, and nerves, as well as their principal drainage ducts, the **ductus deferens** (**vas deferens**) with them. The lymph drainage of the testes is therefore to the lateral aortic or lumbar nodes and preaortic nodes in the abdomen, and not to the inguinal or pelvic lymph nodes.

Each ellipsoid-shaped testis is enclosed within the end of an elongated musculofascial pouch, which is continuous with the anterior abdominal wall and projects into the scrotum. The **spermatic cord** is the tube-shaped connection between the pouch in the scrotum and the abdominal wall.

The sides and anterior aspect of the testis are covered by a closed sac of peritoneum (the **tunica vaginalis**), which originally connected to the abdominal cavity. Normally after testicular descent, the connection closes, leaving a fibrous remnant.

Each testis (Fig. 5.46B) is composed of seminiferous tubules and interstitial tissue surrounded by a thick connective tissue capsule (the **tunica albuginea**). Spermatozoa are produced by the seminiferous tubules. The 400 to 600 highly coiled seminiferous tubules are modified at each end to become straight tubules, which connect to a collecting chamber (the **rete testis**) in a thick, vertically oriented linear wedge of connective tissue (the **mediastinum testis**), projecting from the capsule into the posterior aspect of the gonad. Approximately 12 to 20 **efferent ductules** originate from the upper end of the rete testis, penetrate the capsule, and connect with the epididymis.

In the clinic

Testicular tumors

Tumors of the testis account for a small percentage of malignancies in men. However, they generally occur in younger patients (between 20 and 40 years of age). When diagnosed at an early stage, most of these tumors are curable by surgery and chemotherapy.

Early diagnosis of testicular tumor is extremely important. Abnormal lumps can be detected by palpation, and diagnosis can be made using ultrasound. Simple ultrasound scanning can reveal the extent of the local tumor, usually at an early stage.

Surgical removal of the malignant testis is often carried out using an inguinal approach. The testis is not usually removed through a scrotal incision, because it is possible to spread tumor cells into the subcutaneous tissues of the scrotum, which has a different lymphatic drainage than the testis.

Epididymis

The **epididymis** courses along the posterolateral side of the testis (Fig. 5.46B). It has two distinct components:

- the **efferent ductules**, which form an enlarged coiled mass that sits on the posterior superior pole of the testis and forms the **head of the epididymis**; and
- the **true epididymis**, which is a single, long coiled duct into which the efferent ductules all drain, and which continues inferiorly along the posterolateral margin of the testis as the **body of the epididymis** and enlarges to form the **tail of the epididymis** at the inferior pole of the testis.

During passage through the epididymis, spermatozoa acquire the ability to move and fertilize an egg. The epididymis also stores spermatozoa until ejaculation. The end of the epididymis is continuous with the ductus deferens.

Ductus deferens

The ductus deferens is a long muscular duct that transports spermatozoa from the tail of the epididymis in the scrotum to the ejaculatory duct in the pelvic cavity (Fig. 5.46A). It ascends in the scrotum as a component of the spermatic cord and passes through the inguinal canal in the anterior abdominal wall.

After passing through the deep inguinal ring, the ductus deferens bends medially around the lateral side of the

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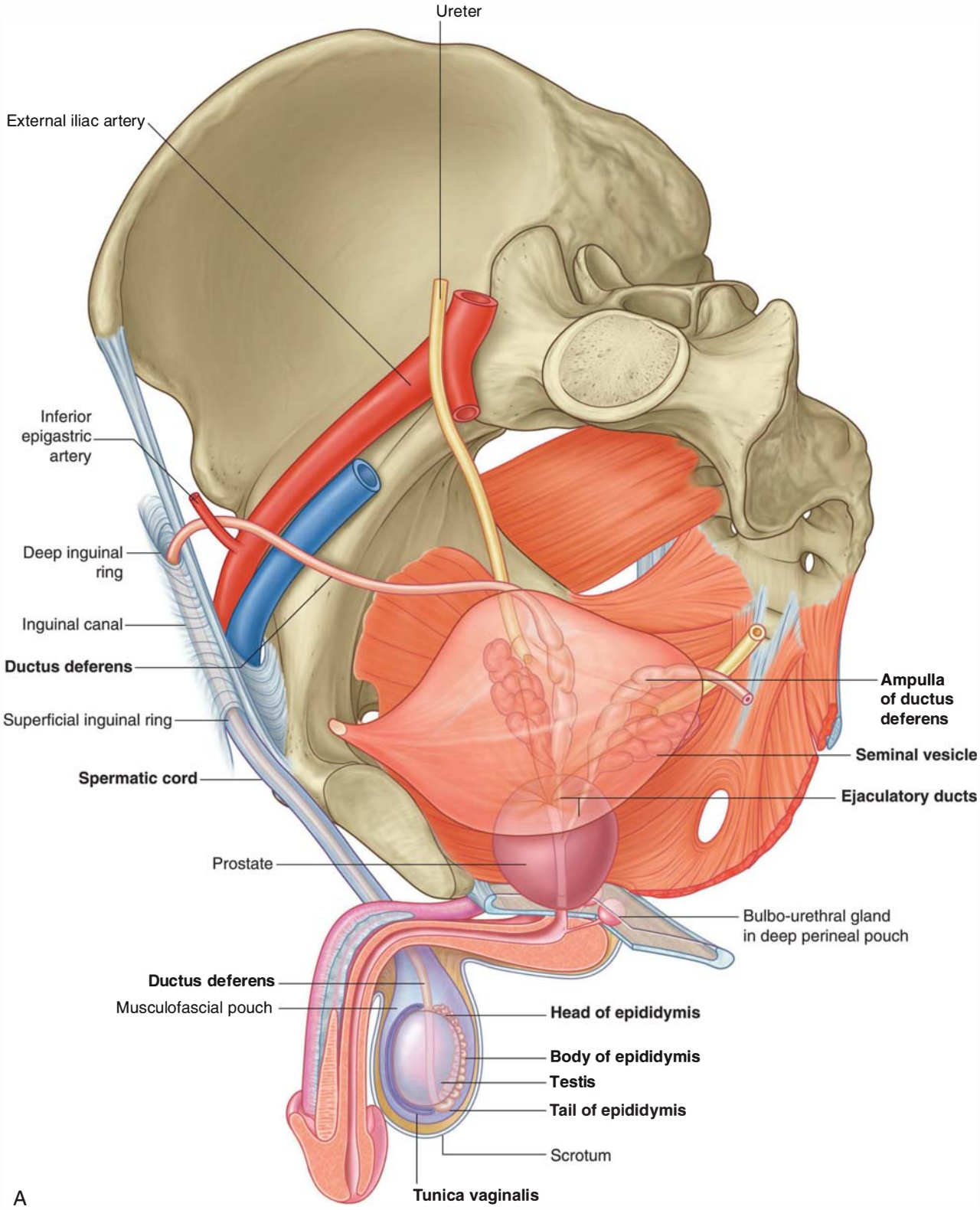


Fig. 5.46 Reproductive system in men. **A.** Overview.

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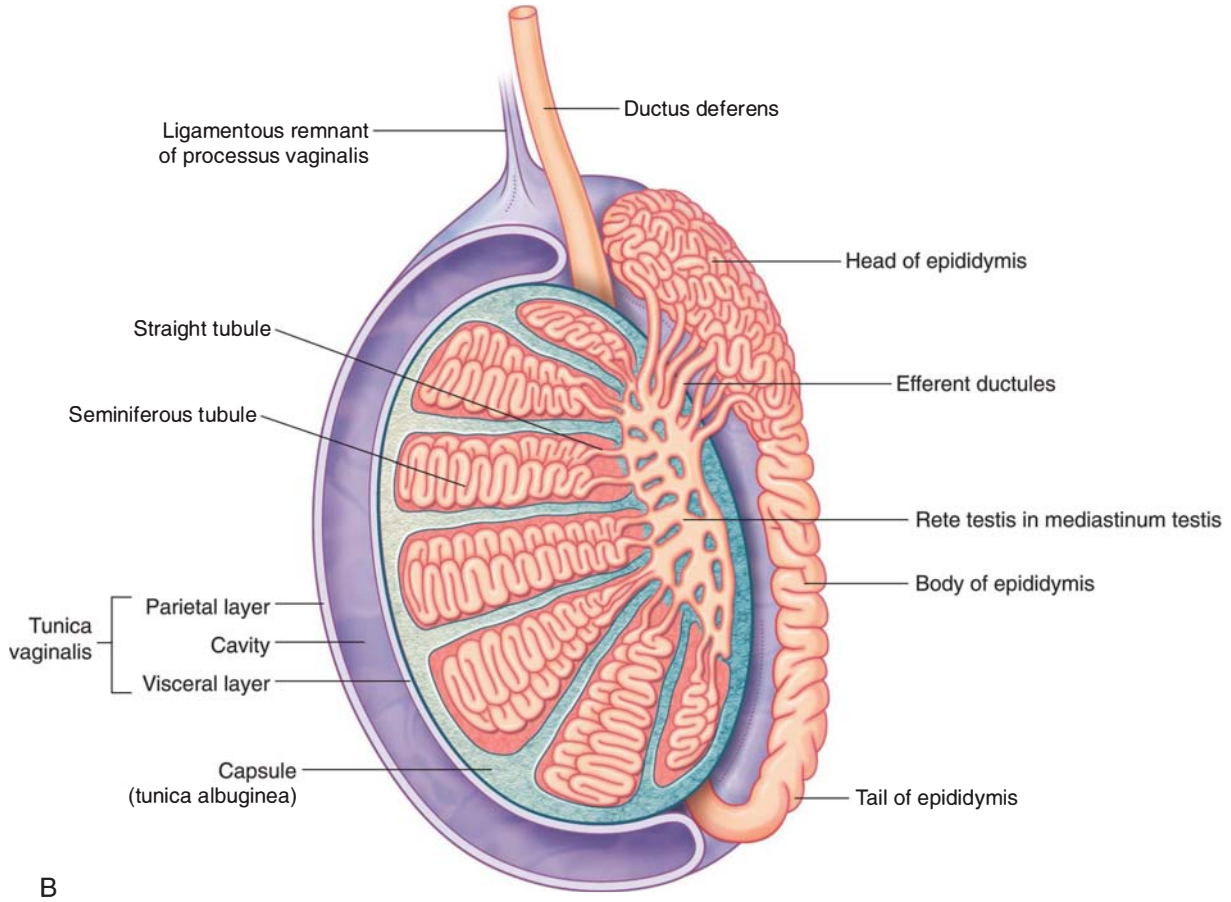


Fig. 5.46, cont'd B. Testis and surrounding structures.

inferior epigastric artery and crosses the external iliac artery and the external iliac vein at the pelvic inlet to enter the pelvic cavity.

The duct descends medially on the pelvic wall, deep to the peritoneum, and crosses the ureter posterior to the bladder. It continues inferomedially along the base of the bladder, anterior to the rectum, almost to the midline, where it is joined by the duct of the seminal vesicle to form the ejaculatory duct.

Between the ureter and ejaculatory duct, the ductus deferens expands to form the ampulla of the ductus deferens. The ejaculatory duct penetrates through the prostate gland to connect with the prostatic urethra.

In the clinic

Vasectomy

The ductus deferens transports spermatozoa from the tail of the epididymis in the scrotum to the ejaculatory duct in the pelvic cavity. Because it has a thick smooth muscle wall, it can be easily palpated in the spermatic cord between the testes and the superficial inguinal ring. Also, because it can be accessed through skin and superficial fascia, it is amenable to surgical dissection and surgical division. When this is carried out bilaterally (vasectomy), the patient is rendered sterile—this is a useful method for male contraception.

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Seminal vesicle

Each **seminal vesicle** is an accessory gland of the male reproductive system that develops as a blind-ended tubular outgrowth from the ductus deferens (Fig. 5.46A). The tube is coiled with numerous pocket-like outgrowths and is encapsulated by connective tissue to form an elongate structure situated between the bladder and rectum. The gland is immediately lateral to and follows the course of the ductus deferens at the base of the bladder.

The duct of the seminal vesicle joins the ductus deferens to form the **ejaculatory duct** (Fig. 5.47). Secretions from the seminal vesicle contribute significantly to the volume of the ejaculate (semen).

Prostate

The **prostate** is an unpaired accessory structure of the male reproductive system that surrounds the urethra in the pelvic cavity (Figs. 5.46A and 5.47). It lies immediately inferior to the bladder, posterior to the pubic symphysis, and anterior to the rectum.

The prostate is shaped like an inverted rounded cone with a larger base, which is continuous above with the neck of the bladder, and a narrower apex, which rests below on the pelvic floor. The inferolateral surfaces of the prostate are in contact with the levator ani muscles that together cradle the prostate between them.

The prostate develops as 30 to 40 individual complex glands, which grow from the urethral epithelium into the surrounding wall of the urethra. Collectively, these glands enlarge the wall of the urethra into what is known as the prostate; however, the individual glands retain their own ducts, which empty independently into the prostatic sinuses on the posterior aspect of the urethral lumen (see Fig. 5.44C).

Secretions from the prostate, together with secretions from the seminal vesicles, contribute to the formation of semen during ejaculation.

The ejaculatory ducts pass almost vertically in an anteroinferior direction through the posterior aspect of the prostate to open into the prostatic urethra.

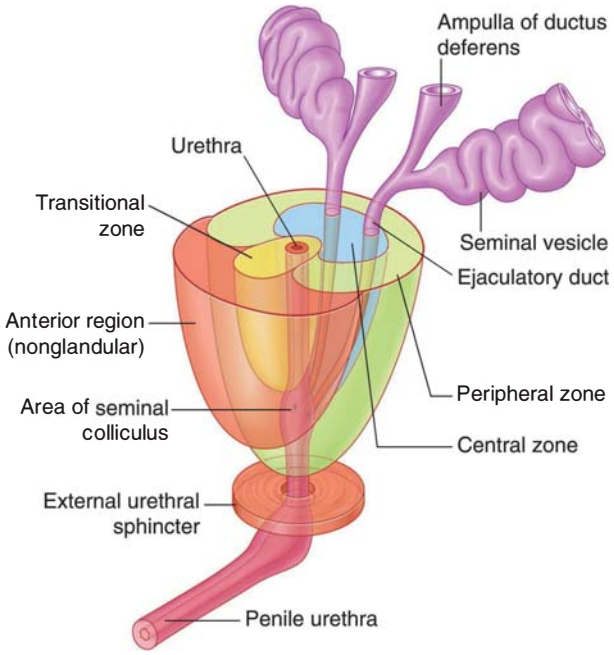


Fig. 5.47 The prostate gland. Zonal anatomy.

Bulbo-urethral glands

The **bulbo-urethral glands** (see Fig. 5.46A), one on each side, are small, pea-shaped mucous glands situated within the deep perineal pouch. They are lateral to the membranous part of the urethra. The duct from each gland passes inferomedially through the **perineal membrane**, to open into the bulb of the spongy urethra at the root of the penis.

Together with small glands positioned along the length of the spongy urethra, the bulbo-urethral glands contribute to lubrication of the urethra and the pre-ejaculatory emission from the penis.

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In the clinic

Prostate problems

Prostate cancer is one of the most commonly diagnosed malignancies in men, and often the disease is advanced at diagnosis. Prostate cancer typically occurs in the peripheral zone of the prostate (see Fig. 5.47) and is relatively asymptomatic. In many cases, it is diagnosed by a digital rectal examination (DRE) (Fig. 5.48A) and by blood tests, which include serum acid phosphatase and serum prostate-specific antigen (PSA). In rectal exams, the tumorous prostate feels "rock" hard. The diagnosis is usually made by obtaining a number of biopsies of the prostate. Ultrasound is used during the biopsy procedure to image the prostate for the purpose of taking measurements and for needle placement.

Benign prostatic hypertrophy is a disease of the prostate that occurs with increasing age in most men (Fig. 5.48B). It generally involves the more central regions of the prostate (see Fig. 5.47), which gradually enlarge. The prostate feels "bulky" on DRE. Owing to the more central hypertrophic change of the prostate, the urethra is compressed, and a urinary outflow obstruction develops in a number of patients. With time, the bladder may become hypertrophied in response to the urinary outflow obstruction. In some male patients, the obstruction becomes so severe that urine cannot be passed and transurethral or suprapubic catheterization is necessary. Despite being a benign disease, benign prostatic hypertrophy can therefore have a marked effect on the daily lives of many patients.

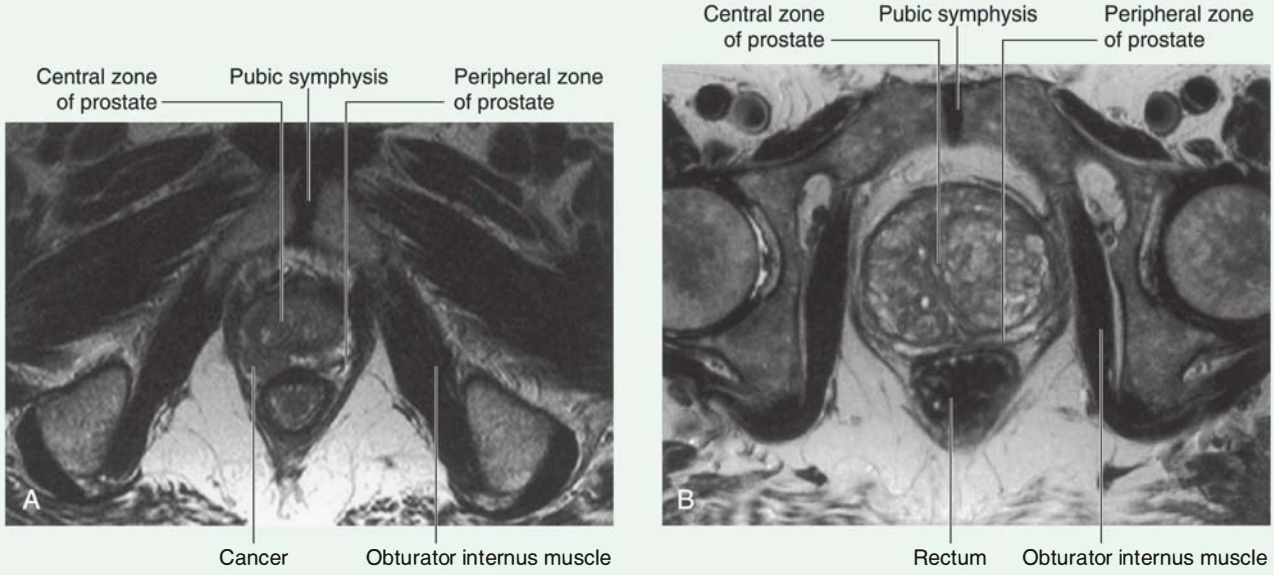


Fig. 5.48 Axial T2-weighted magnetic resonance images of prostate problems. **A.** A small prostatic cancer in the peripheral zone of a normal-sized prostate. **B.** Benign prostatic hypertrophy.

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In women

The reproductive tract in women is contained mainly in the pelvic cavity and perineum, although during pregnancy, the uterus expands into the abdominal cavity. Major components of the system consist of:

- an ovary on each side, and
- a uterus, vagina, and clitoris in the midline (Fig. 5.49).

In addition, a pair of accessory glands (the **greater vestibular glands**) are associated with the tract.

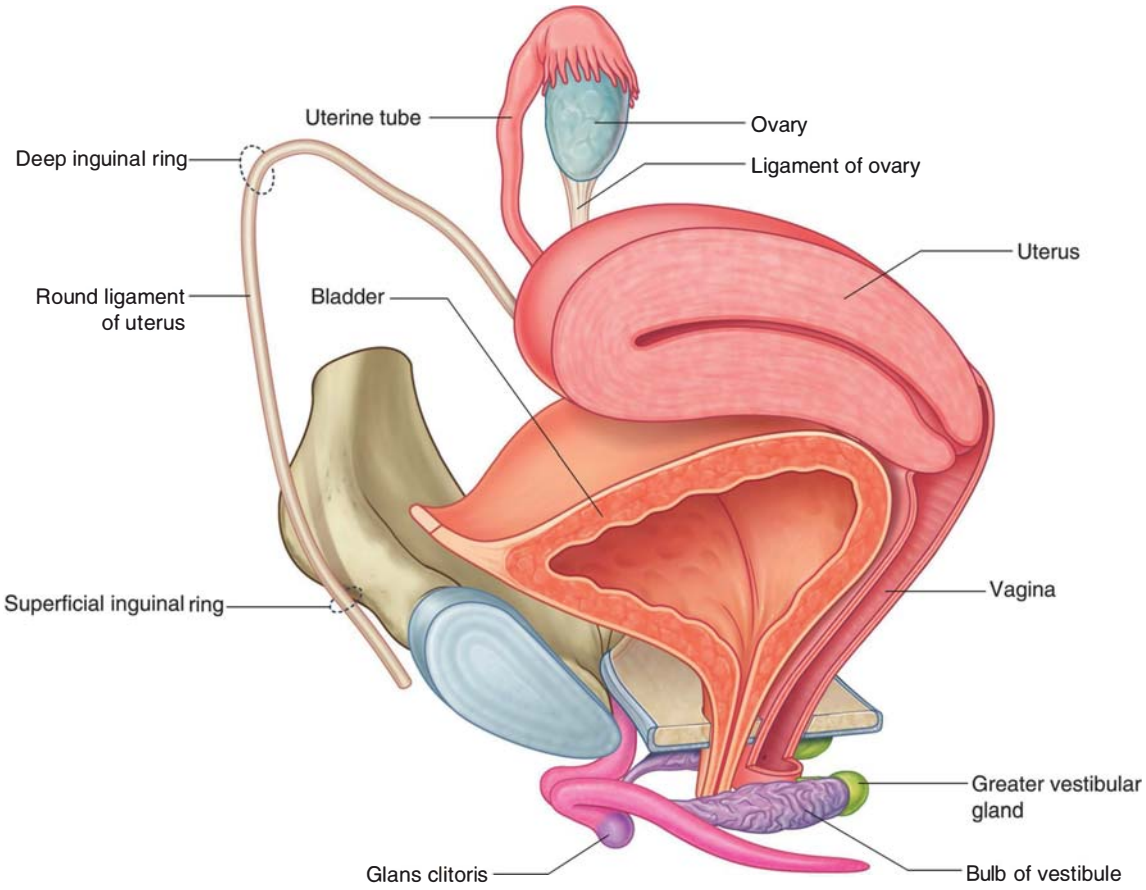


Fig. 5.49 Reproductive system in women.

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Ovaries

Like the testes in men, the **ovaries** develop high on the posterior abdominal wall and then descend before birth, bringing with them their vessels, lymphatics, and nerves. Unlike the testes, the ovaries do not migrate through the inguinal canal into the perineum, but stop short and assume a position on the lateral wall of the pelvic cavity (Fig. 5.50).

The ovaries are the sites of egg production (oogenesis). Mature eggs are ovulated into the peritoneal cavity and normally directed into the adjacent openings of the uterine tubes by cilia on the ends of the uterine tubes.

The ovaries lie adjacent to the lateral pelvic wall just inferior to the pelvic inlet. Each of the two almond-shaped ovaries is about 3 cm long and is suspended by a mesentery (the **mesovarium**) that is a posterior extension of the broad ligament.

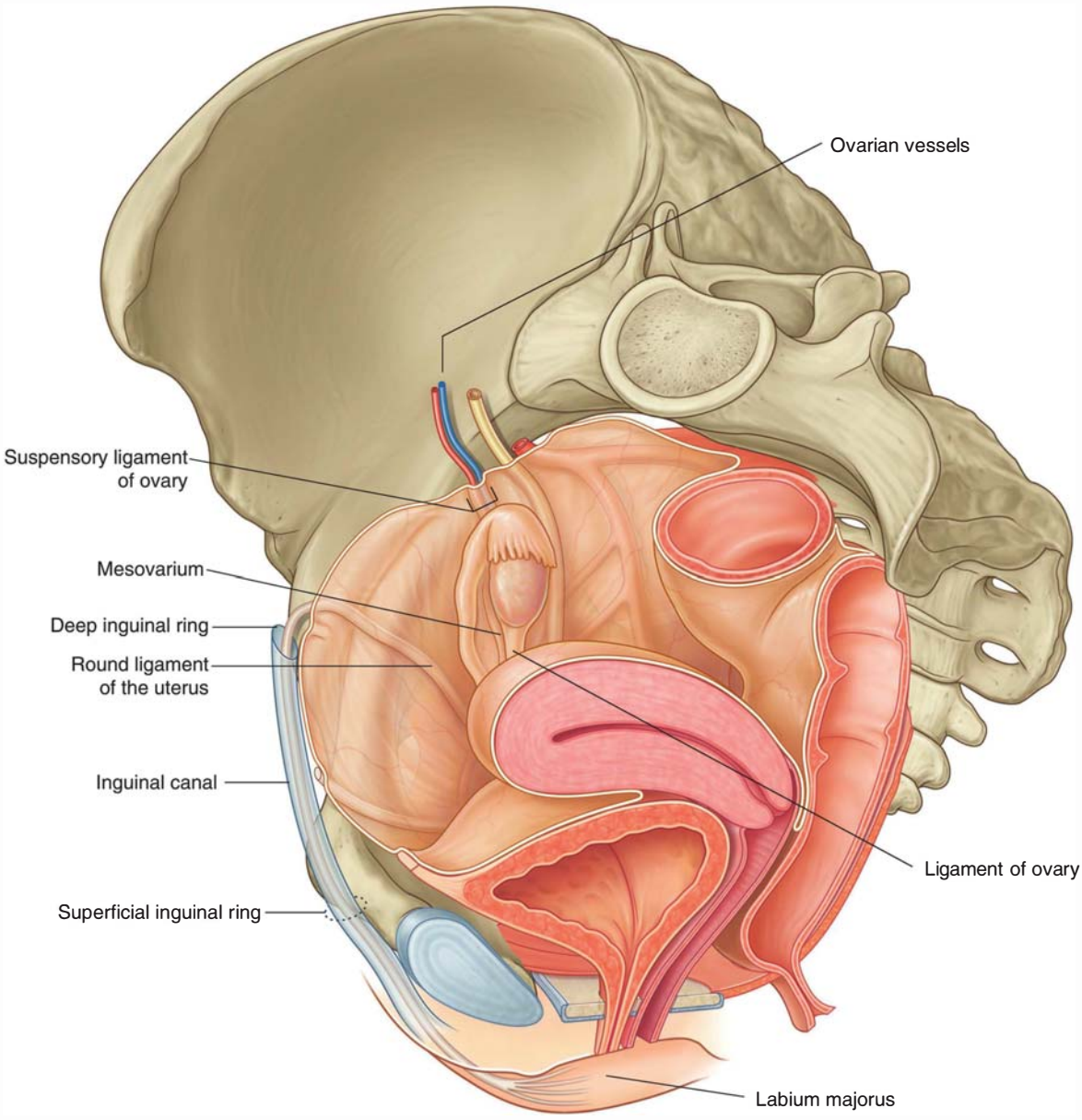


Fig. 5.50 Ovaries and broad ligament.

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In the clinic

Ovarian cancer

Ovarian cancer remains one of the major challenges in oncology. The ovaries contain numerous cell types, all of which can undergo malignant change and require different imaging and treatment protocols and ultimately have different prognoses.

Ovarian tumors most commonly originate from the surface epithelium that covers the ovary and is continuous at a sharp transition zone with the peritoneum of the mesovarium.

Many factors have been linked with the development of ovarian tumors, including a strong family history.

Ovarian cancer may occur at any age, but more typically it occurs in older women.

Cancer of the ovaries may spread via the blood and lymphatics, and frequently metastasizes directly into the peritoneal cavity. Such direct peritoneal cavity spread allows the passage of tumor cells along the paracolic gutters and over the liver from where this disease may disseminate easily. Unfortunately, many patients already have metastatic and diffuse disease (Fig. 5.51) at the time of diagnosis.



Fig. 5.51 Sagittal magnetic resonance image demonstrating ovarian cancer.

In the clinic

Imaging the ovary

The ovaries can be visualized using ultrasound. If the patient drinks enough water, the bladder becomes enlarged and full. This fluid-filled cavity provides an excellent acoustic window, behind which the uterus and ovaries may be identified by transabdominal scanning with ultrasound. This technique also allows obstetricians and technicians to view a fetus and record its growth throughout pregnancy.

Some patients are not suitable for transabdominal scanning, in which case a probe may be passed into the vagina, permitting close visualization of the uterus, the contents of the recto-uterine pouch (pouch of Douglas), and the ovaries. The ovaries can also be visualized laparoscopically.

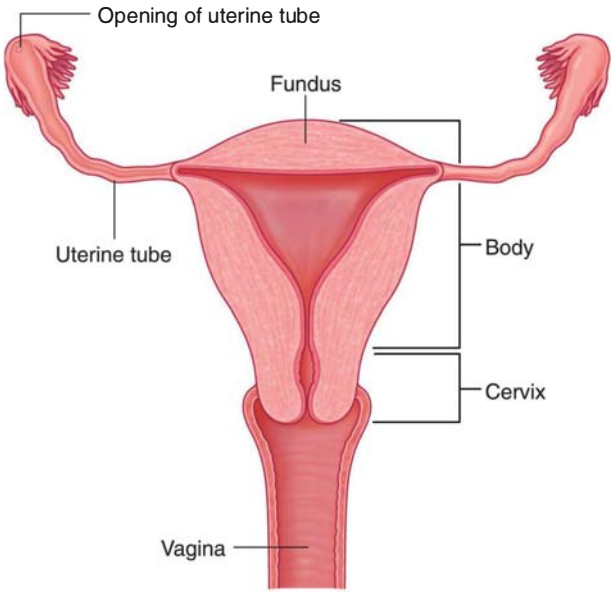


Fig. 5.52 Uterus. Anterior view. The anterior halves of the uterus and vagina have been cut away.

Uterus

The **uterus** is a thick-walled muscular organ in the midline between the bladder and rectum (see Fig. 5.50). It consists of a body and a cervix, and inferiorly it joins the vagina (Fig. 5.52). Superiorly, uterine tubes project laterally from the uterus and open into the peritoneal cavity immediately adjacent to the ovaries.

The body of the uterus is flattened anteroposteriorly and, above the level of origin of the uterine tubes (Fig. 5.52), has a rounded superior end (**fundus of the uterus**). The cavity of the body of the uterus is a narrow slit, when viewed laterally, and is shaped like an inverted triangle, when viewed anteriorly. Each of the superior corners of the cavity is continuous with the lumen of a

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uterine tube; the inferior corner is continuous with the central canal of the cervix.

Implantation of the blastocyst normally occurs in the body of the uterus. During pregnancy, the uterus dramatically expands superiorly into the abdominal cavity.

In the clinic

Hysterectomy

A hysterectomy is the surgical removal of the uterus. This is usually complete excision of the body, fundus, and cervix of the uterus, though occasionally the cervix may be left in situ. In some instances the uterine (fallopian) tubes and ovaries are removed as well. This procedure is called a total abdominal hysterectomy and bilateral salpingo-oophorectomy.

Hysterectomy, oophorectomy, and salpingo-oophorectomy may be performed in patients who have reproductive malignancy, such as uterine, cervical, and ovarian cancers. Other indications include a strong family history of reproductive disorders, endometriosis, and excessive bleeding. Occasionally the uterus may need to be removed postpartum because of excessive postpartum bleeding.

A hysterectomy is performed through a transverse suprapubic incision (Pfannenstiel's incision). During the procedure tremendous care is taken to identify the distal ureters and to ligate the nearby uterine arteries without damage to the ureters.

Uterine tubes

The **uterine tubes** extend from each side of the superior end of the body of the uterus to the lateral pelvic wall and are enclosed within the upper margins of the mesosalpinx portions of the broad ligaments (see p. 483). Because the ovaries are suspended from the posterior aspect of the broad ligaments, the uterine tubes pass superiorly over, and terminate laterally to, the ovaries.

Each uterine tube has an expanded trumpet-shaped end (the **infundibulum**), which curves around the superolateral pole of the related ovary (Fig. 5.53). The margin of the infundibulum is rimmed with small finger-like projections termed **fimbriae**. The lumen of the uterine tube opens into the peritoneal cavity at the narrowed end of the infundibulum. Medial to the infundibulum, the tube expands to form the **ampulla** and then narrows to form the **isthmus**, before joining with the body of the uterus.

The fimbriated infundibulum facilitates the collection of ovulated eggs from the ovary. Fertilization normally occurs in the ampulla.

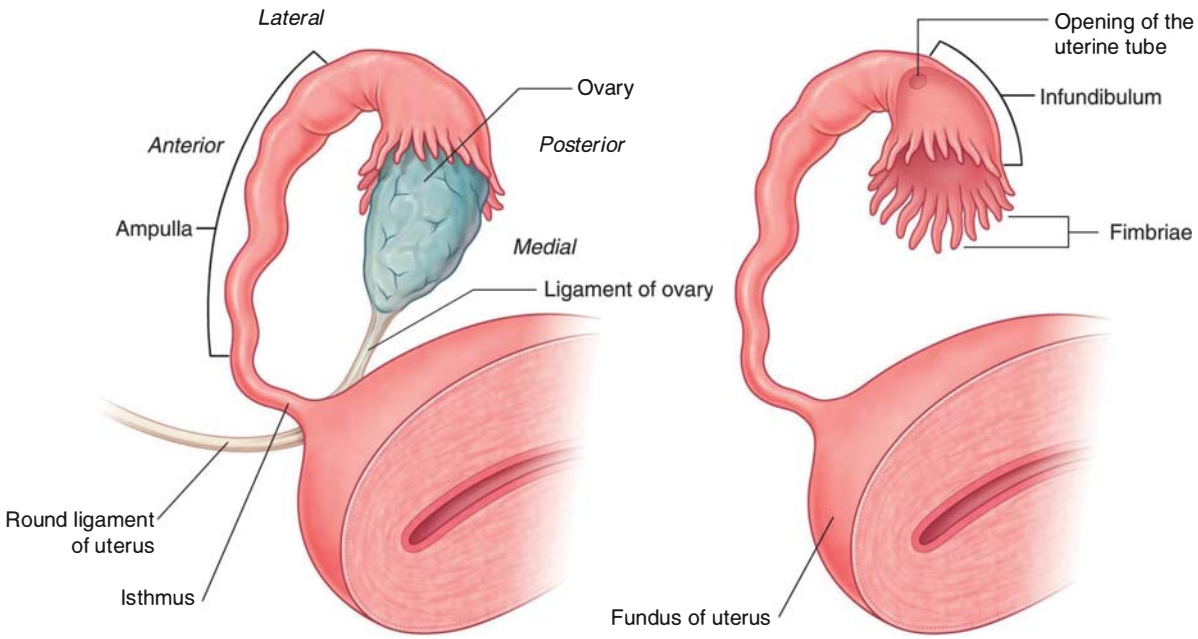


Fig. 5.53 Uterine tubes.

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In the clinic

Tubal ligation

After ovulation, the unfertilized egg is gathered by the fimbriae of the uterine tube. The egg passes into the uterine tube where it is normally fertilized in the ampulla. The zygote then begins development and passes into the uterine cavity where it implants in the uterine wall.

A simple and effective method of birth control is to surgically ligate (clip) the uterine tubes, preventing spermatozoa from reaching ova. This simple short procedure is performed under general anesthetic. A small laparoscope is passed into the peritoneal cavity and special equipment is used to identify the tubes.

Cervix

The **cervix** forms the inferior part of the uterus and is shaped like a short, broad cylinder with a narrow central channel. The body of the uterus normally arches forward (anteflexed on the cervix) over the superior surface of the emptied bladder (Fig. 5.54A). In addition, the cervix is angled forward (anteverted) on the vagina so that the inferior end of the cervix projects into the upper anterior aspect of the vagina. Because the end of the cervix is dome shaped, it bulges into the vagina, and a gutter, or fornix, is formed around the margin of the cervix where it joins the vaginal wall (Fig. 5.54B). The tubular central canal of the cervix opens, below, as the **external os**, into the vaginal cavity and, above, as the **internal os**, into the uterine cavity.

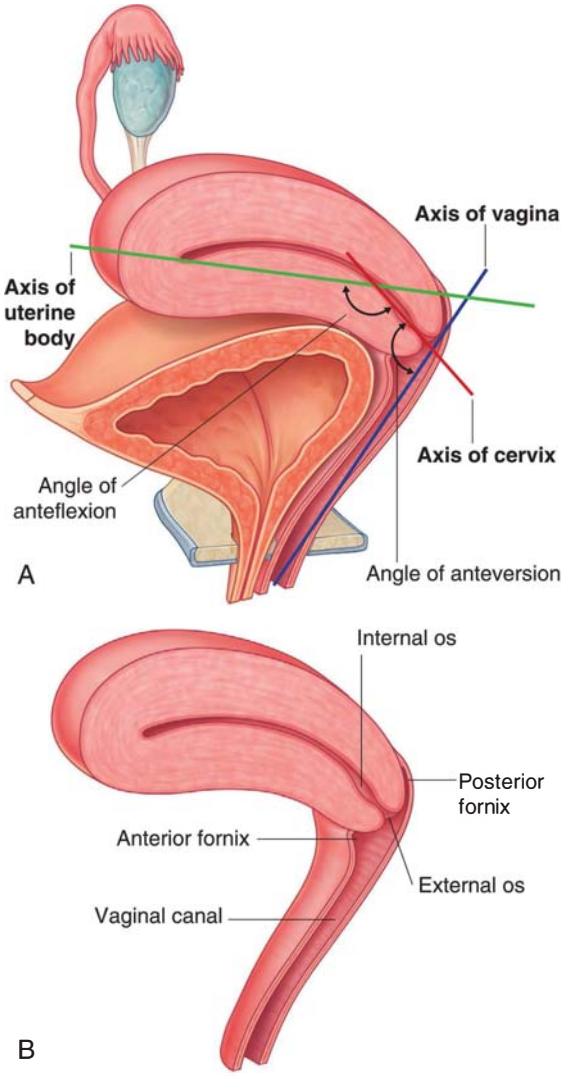


Fig. 5.54 Uterus and vagina. **A.** Angles of anteflexion and anteversion. **B.** The cervix protrudes into the vagina.

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In the clinic

Carcinoma of the cervix and uterus

Carcinoma of the cervix (Fig. 5.55) and uterus is a common disease in women. Diagnosis is by inspection, cytology (examination of the cervical cells), imaging, biopsy, and dilatation and curettage (D&C) of the uterus.

Carcinoma of the cervix and uterus may be treated by local resection, removal of the uterus (hysterectomy), and adjuvant chemotherapy. The tumor spreads via lymphatics to the internal and common iliac lymph nodes.

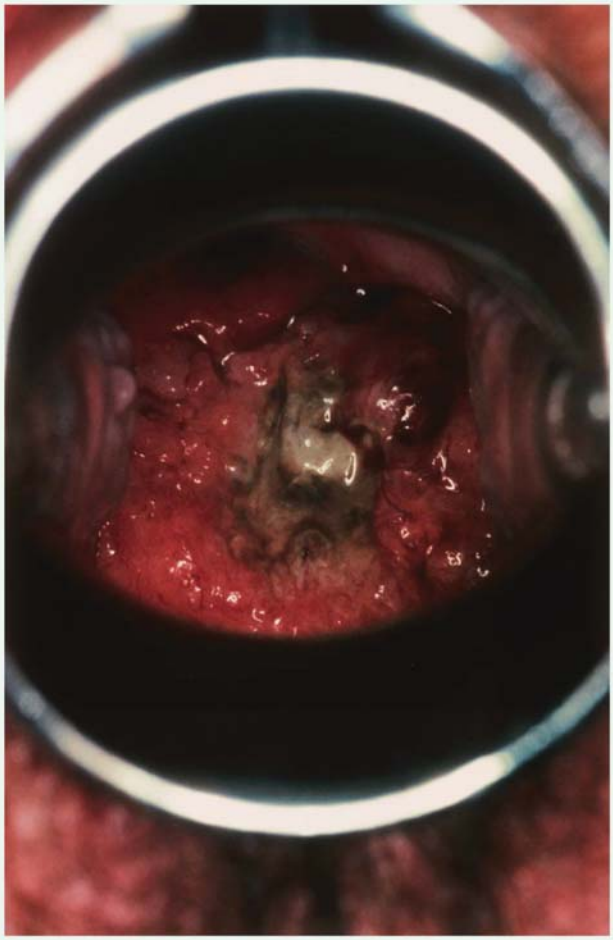


Fig. 5.55 Picture taken through a speculum inserted into the vagina demonstrating cervical cancer. See Fig. 5.83E on p. 524 for a view of the normal cervix.

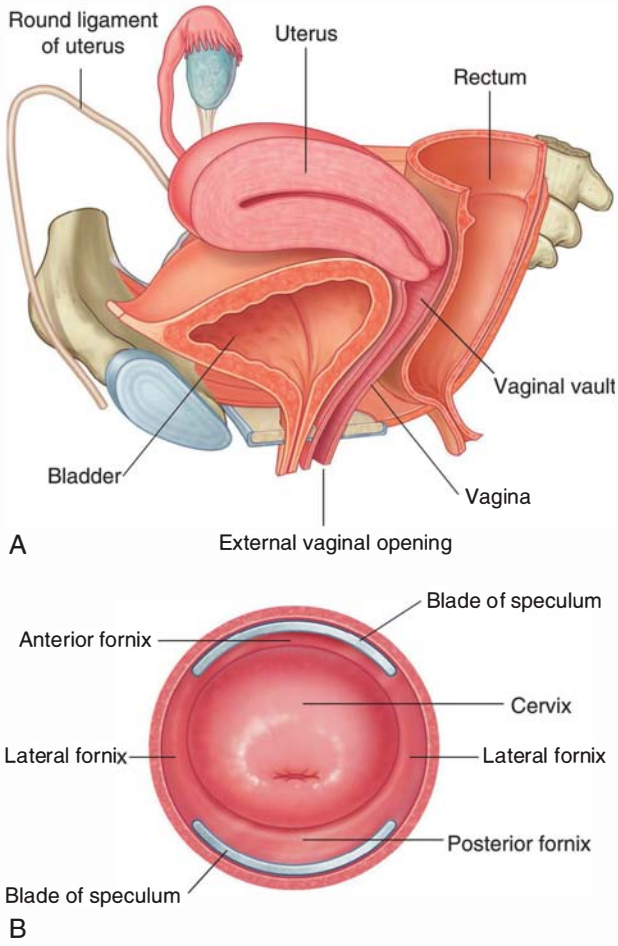


Fig. 5.56 Vagina. **A.** Left half of pelvis cut away. **B.** Vaginal fornices and cervix as viewed through a speculum.

Vagina

The **vagina** is the copulatory organ in women. It is a distensible fibromuscular tube that extends from the perineum through the pelvic floor and into the pelvic cavity (Fig. 5.56A). The internal end of the canal is enlarged to form a region called the **vaginal vault**.

The anterior wall of the vagina is related to the base of the bladder and to the urethra; in fact, the urethra is embedded in, or fused to, the anterior vaginal wall.

Posteriorly, the vagina is related principally to the rectum.

Inferiorly, the vagina opens into the vestibule of the perineum immediately posterior to the external opening of the urethra. From its external opening (the **introitus**), the vagina courses posterosuperiorly through the perineal membrane and into the pelvic cavity, where it is attached by its anterior wall to the circular margin of the cervix.

The **vaginal fornix** is the recess formed between the margin of the cervix and the vaginal wall. Based on position, the fornix is subdivided into a posterior fornix, an anterior fornix, and two lateral fornices (Fig. 5.56A and see Fig. 5.54).

The vaginal canal is normally collapsed so that the anterior wall is in contact with the posterior wall. By using a

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speculum to open the vaginal canal, a physician can see the domed inferior end of the cervix, the vaginal fornices, and the external os of the cervical canal in a patient (Fig. 5.56B).

During intercourse, semen is deposited in the vaginal vault. Spermatozoa make their way into the external os of the cervical canal, pass through the cervical canal into the uterine cavity, and then continue through the uterine cavity into the uterine tubes where fertilization normally occurs in the ampulla.

Fascia

Fascia in the pelvic cavity lines the pelvic walls, surrounds the bases of the pelvic viscera, and forms sheaths around blood vessels and nerves that course medially from the pelvic walls to reach the viscera in the midline. This pelvic fascia is a continuation of the extraperitoneal connective tissue layer found in the abdomen.

In women

In women, a **rectovaginal septum** separates the posterior surface of the vagina from the rectum (Fig. 5.57A). Condensations of fascia form ligaments that extend from the cervix to the anterior (**pubocervical ligament**), lateral (**transverse cervical** or **cardinal ligament**), and posterior (**uterosacral ligament**) pelvic walls (Fig. 5.57A). These ligaments, together with the perineal membrane, the levator ani muscles, and the perineal body, are thought to stabilize the uterus in the pelvic cavity. The most important of these ligaments are the transverse cervical or cardinal ligaments, which extend laterally from each side of the cervix and vaginal vault to the related pelvic wall.

In the clinic

The recto-uterine pouch

The recto-uterine pouch (**pouch of Douglas**) is an extremely important clinical region situated between the rectum and uterus. When the patient is in the supine position, the recto-uterine pouch is the lowest portion of the abdominopelvic cavity and is a site where infection and fluids typically collect. It is impossible to palpate this region transabdominally, but it can be examined by transvaginal and transrectal digital palpation. If an abscess is suspected, it may be drained through the vagina or the rectum without necessitating transabdominal surgery.

In men

In men, a condensation of fascia around the anterior and lateral region of the prostate (**prostatic fascia**) contains and surrounds the prostatic plexus of veins and is continuous posteriorly with the **rectovesical septum**, which separates the posterior surface of the prostate and base of the bladder from the rectum (Fig. 5.57B).

Peritoneum

The peritoneum of the pelvis is continuous at the pelvic inlet with the peritoneum of the abdomen. In the pelvis, the peritoneum drapes over the pelvic viscera in the midline, forming:

- pouches between adjacent viscera, and
- folds and ligaments between viscera and pelvic walls.

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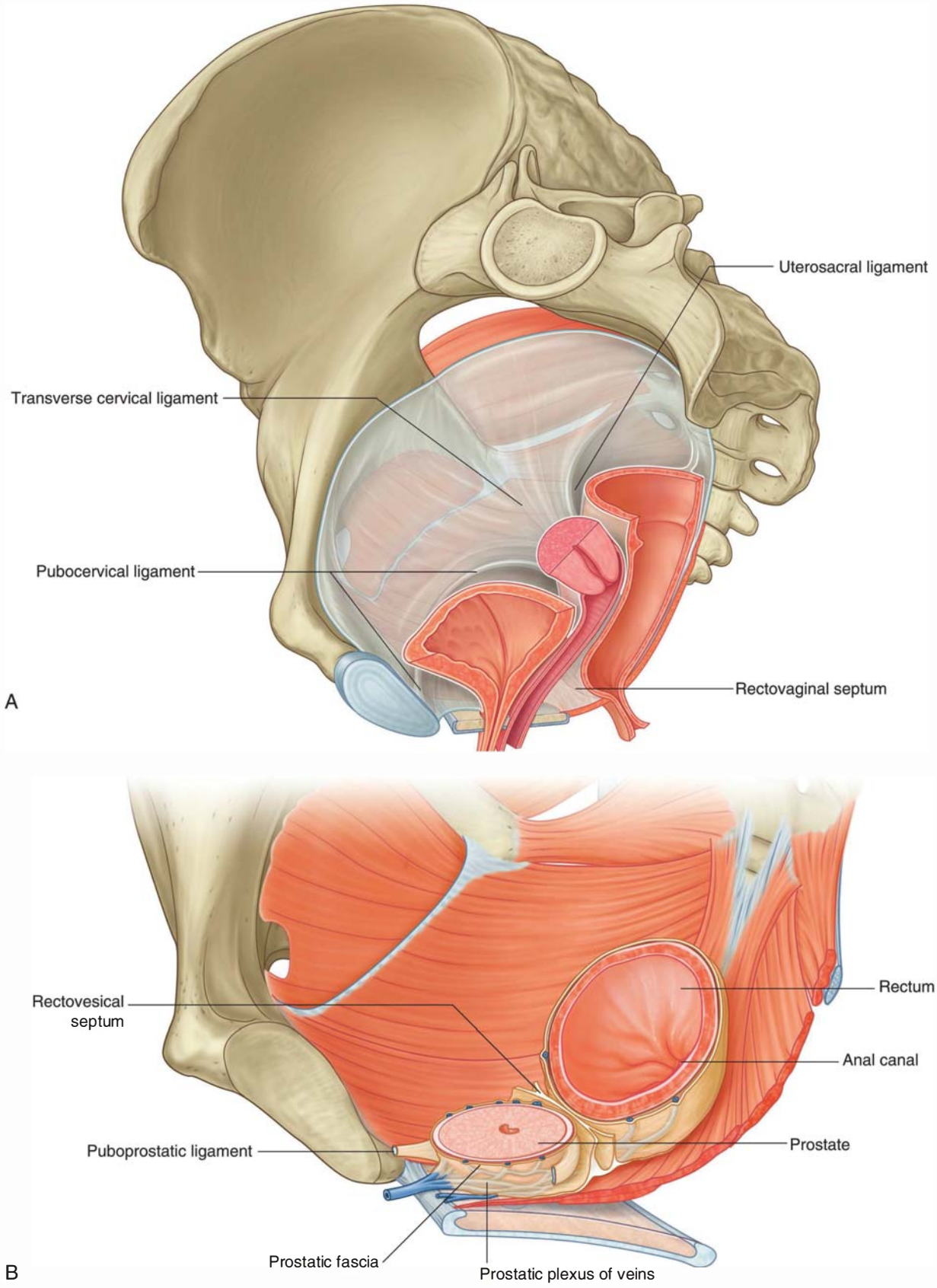


Fig. 5.57 Pelvic fascia. **A.** In women. **B.** In men.

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Anteriorly, median and medial umbilical folds of peritoneum cover the embryological remnants of the urachus and umbilical arteries, respectively (Fig. 5.58). These folds ascend out of the pelvis and onto the anterior abdominal wall. Posteriorly, peritoneum drapes over the anterior and lateral aspects of the upper third of the rectum, but only the anterior surface of the middle third of the rectum is covered by peritoneum; the lower third of the rectum is not covered at all.

In women

In women, the uterus lies between the bladder and rectum, and the uterine tubes extend from the superior aspect of the uterus to the lateral pelvic walls (Fig. 5.58A). As a consequence, a shallow **vesico-uterine pouch** occurs anteriorly, between the bladder and uterus, and a deep **recto-uterine pouch** (pouch of Douglas) occurs posteriorly, between the uterus and rectum. In addition, a large fold of peritoneum (the broad ligament), with a uterine tube enclosed in its superior margin and an ovary attached posteriorly, is located on each side of the uterus and extends to the lateral pelvic walls.

In the midline, the peritoneum descends over the posterior surface of the uterus and cervix and onto the vaginal wall adjacent to the posterior vaginal fornix. It then reflects onto the anterior and lateral walls of the rectum. The deep pouch of peritoneum formed between the anterior surface of the rectum and posterior surfaces of the uterus, cervix, and vagina is the recto-uterine pouch. A sharp sickle-shaped ridge of peritoneum (**recto-uterine fold**) occurs on each side near the base of the recto-uterine pouch. The **recto-uterine folds** overlie the **uterosacral ligaments**, which are condensations of pelvic fascia that extend from the cervix to the posterolateral pelvic walls.

Broad ligament

The **broad ligament** is a sheet-like fold of peritoneum, oriented in the coronal plane that runs from the lateral pelvic wall to the uterus, and encloses the uterine tube in its superior margin and suspends the ovary from its posterior aspect (Fig. 5.58A). The uterine arteries cross the ureters at the base of the broad ligaments, and the ligament of the ovary and round ligament of the uterus are enclosed within the parts of the broad ligament related to the ovary and uterus, respectively. The broad ligament has three parts:

- the mesometrium, the largest part of the broad ligament, which extends from the lateral pelvic walls to the body of the uterus;
- the mesosalpinx, the most superior part of the broad ligament, which suspends the uterine tube in the pelvic cavity; and
- the mesovarium, a posterior extension of the broad ligament, which attaches to the ovary.

The peritoneum of the mesovarium becomes firmly attached to the ovary as the surface epithelium of the ovary. The ovaries are positioned with their long axis in the vertical plane. The ovarian vessels, nerves, and lymphatics enter the superior pole of the ovary from a lateral position and are covered by another raised fold of peritoneum, which with the structures it contains forms the **suspensory ligament of the ovary** (**infundibulopelvic ligament**).

The inferior pole of the ovary is attached to a fibromuscular band of tissue (the **ligament of the ovary**), which courses medially in the margin of the mesovarium to the uterus and then continues anterolaterally as the **round ligament of the uterus** (Fig. 5.58A). The round ligament of the uterus passes over the pelvic inlet to reach the deep inguinal ring and then courses through the inguinal canal to end in connective tissue related to the labium majus in the perineum. Both the ligament of the ovary and the round ligament of the uterus are remnants of the gubernaculum, which attaches the gonad to the labioscrotal swellings in the embryo.

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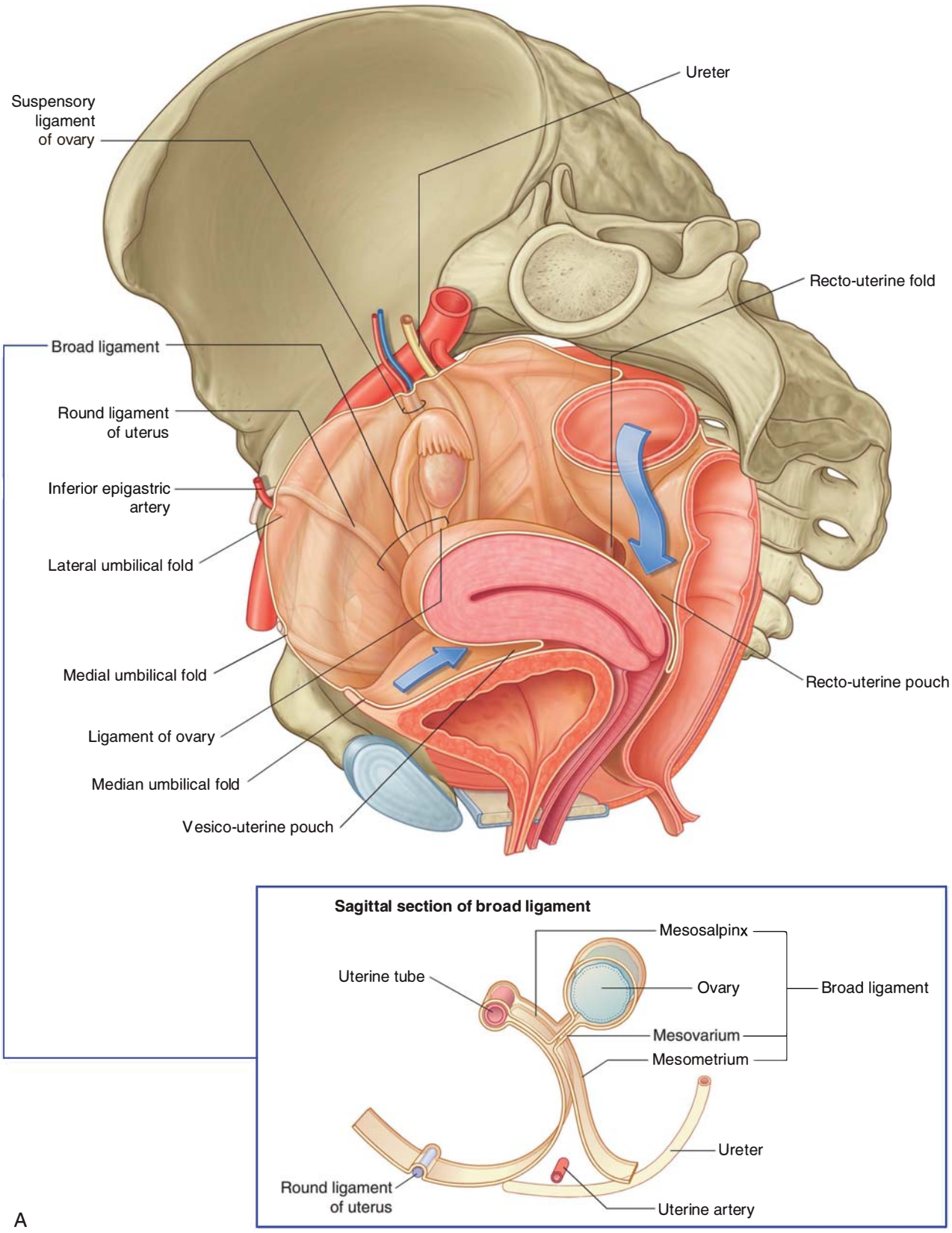


Fig. 5.58 Peritoneum in the pelvis. **A.** In women.

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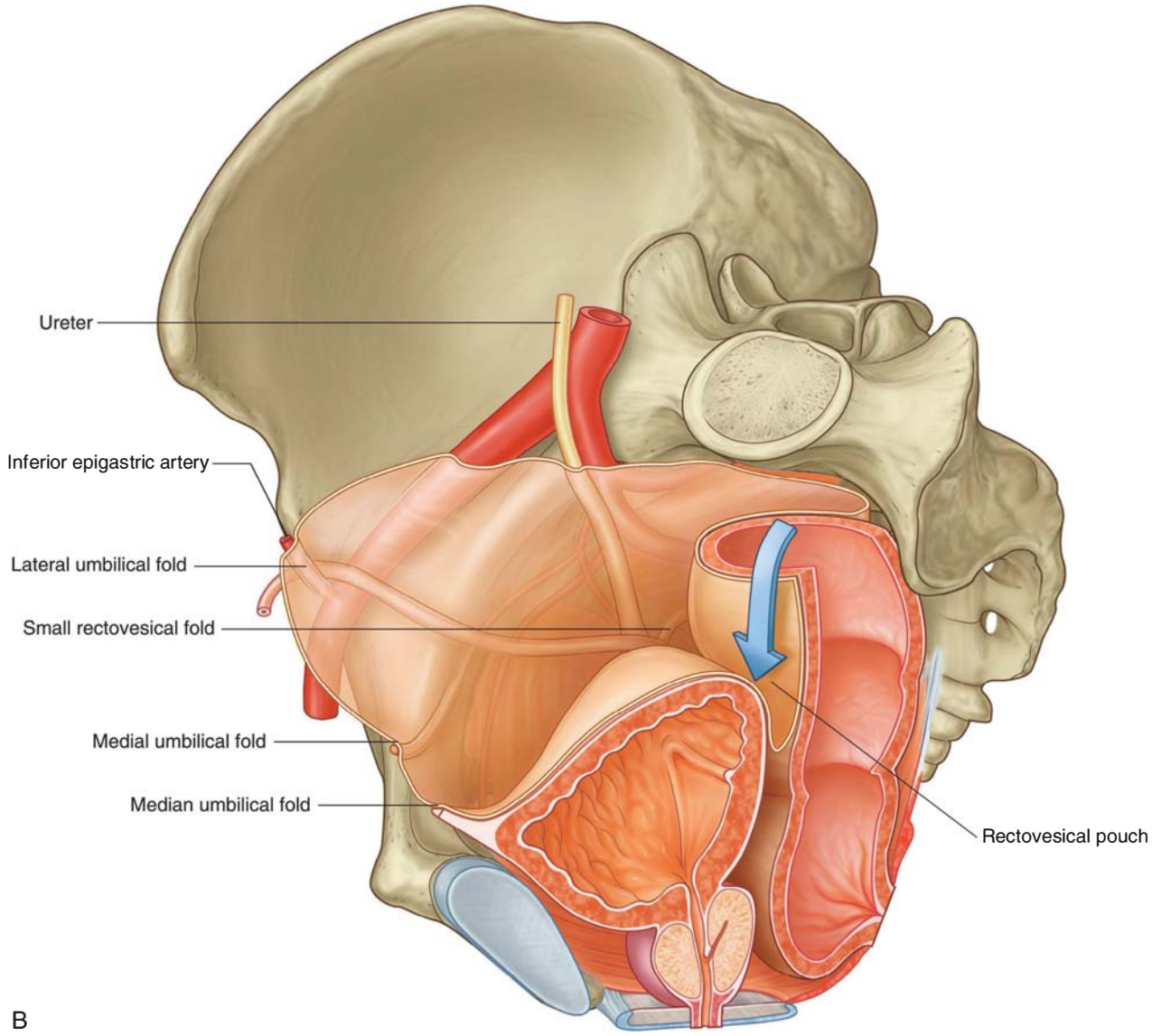


Fig. 5.58, cont'd B. In men.

In men

In men, the visceral peritoneum drapes over the top of the bladder onto the superior poles of the seminal vesicles and then reflects onto the anterior and lateral surfaces of the rectum (Fig. 5.58B). A **rectovesical pouch** occurs between the bladder and rectum.

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Nerves

Somatic plexuses

Sacral and coccygeal plexuses

The sacral and coccygeal plexuses are situated on the posterolateral wall of the pelvic cavity and generally occur in the plane between the muscles and blood vessels. They are formed by the ventral rami of S1 to Co, with a significant contribution from L4 and L5, which enter the pelvis from the lumbar plexus (Fig. 5.59). Nerves from these mainly somatic plexuses contribute to the innervation of the lower limb and muscles of the pelvis and perineum. Cutaneous branches supply skin over the medial side of the foot, the posterior aspect of the lower limb, and most of the perineum.

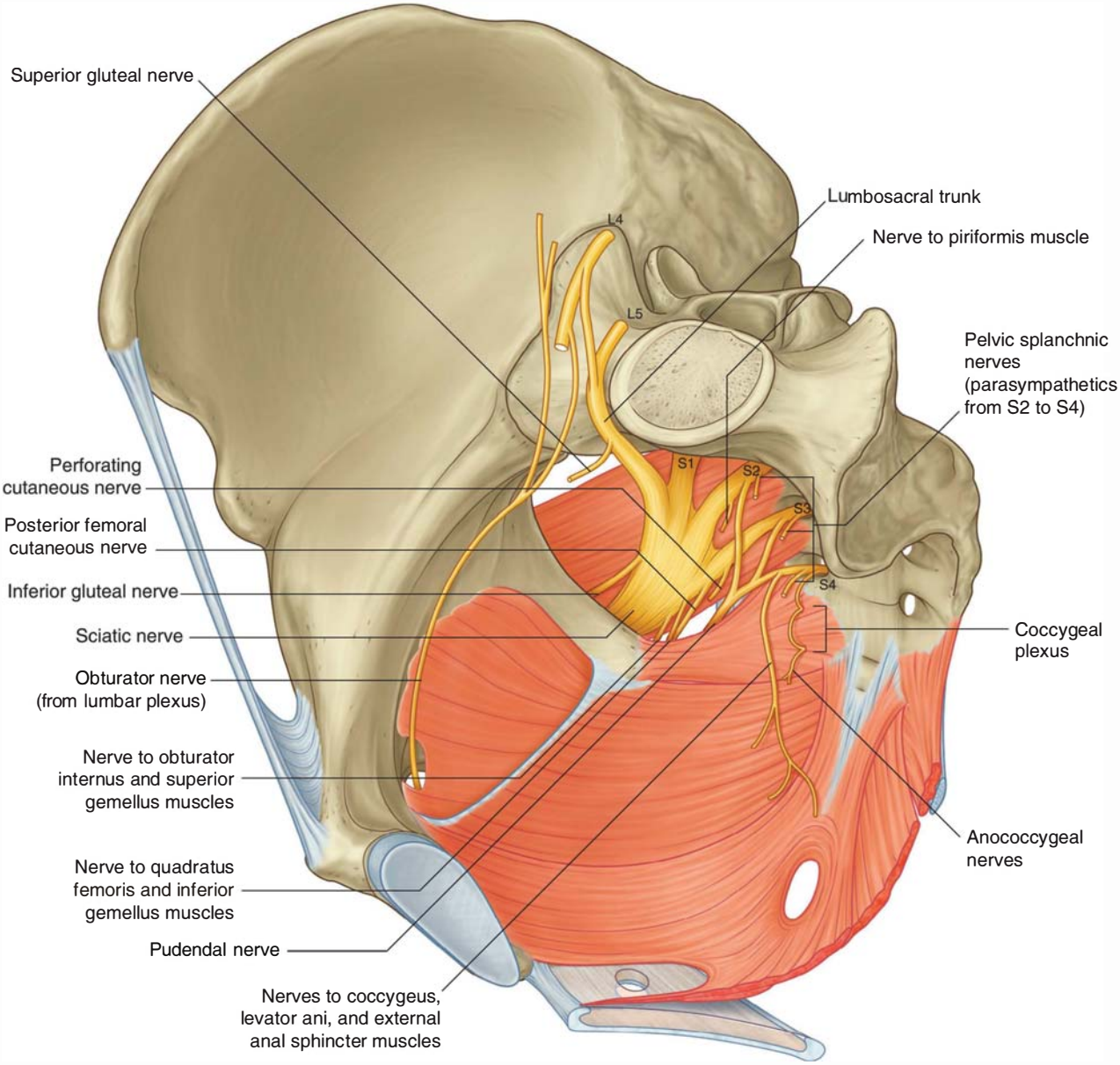


Fig. 5.59 Sacral and coccygeal plexuses.

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Sacral plexus

The sacral plexus on each side is formed by the anterior rami of S1 to S4, and the lumbosacral trunk (L4 and L5) (Fig. 5.60). The plexus is formed in relation to the anterior surface of the piriformis muscle, which is part of the posterolateral pelvic wall. Sacral contributions to the plexus pass out of the anterior sacral foramina and course laterally and inferiorly on the pelvic wall. The lumbosacral trunk, consisting of part of the anterior ramus of L4 and all of the anterior ramus of L5, courses vertically into the pelvic cavity from the abdomen by passing immediately anterior to the sacro-iliac joint.

Gray rami communicantes from ganglia of the sympathetic trunk connect with each of the anterior rami and carry postganglionic sympathetic fibers destined for the

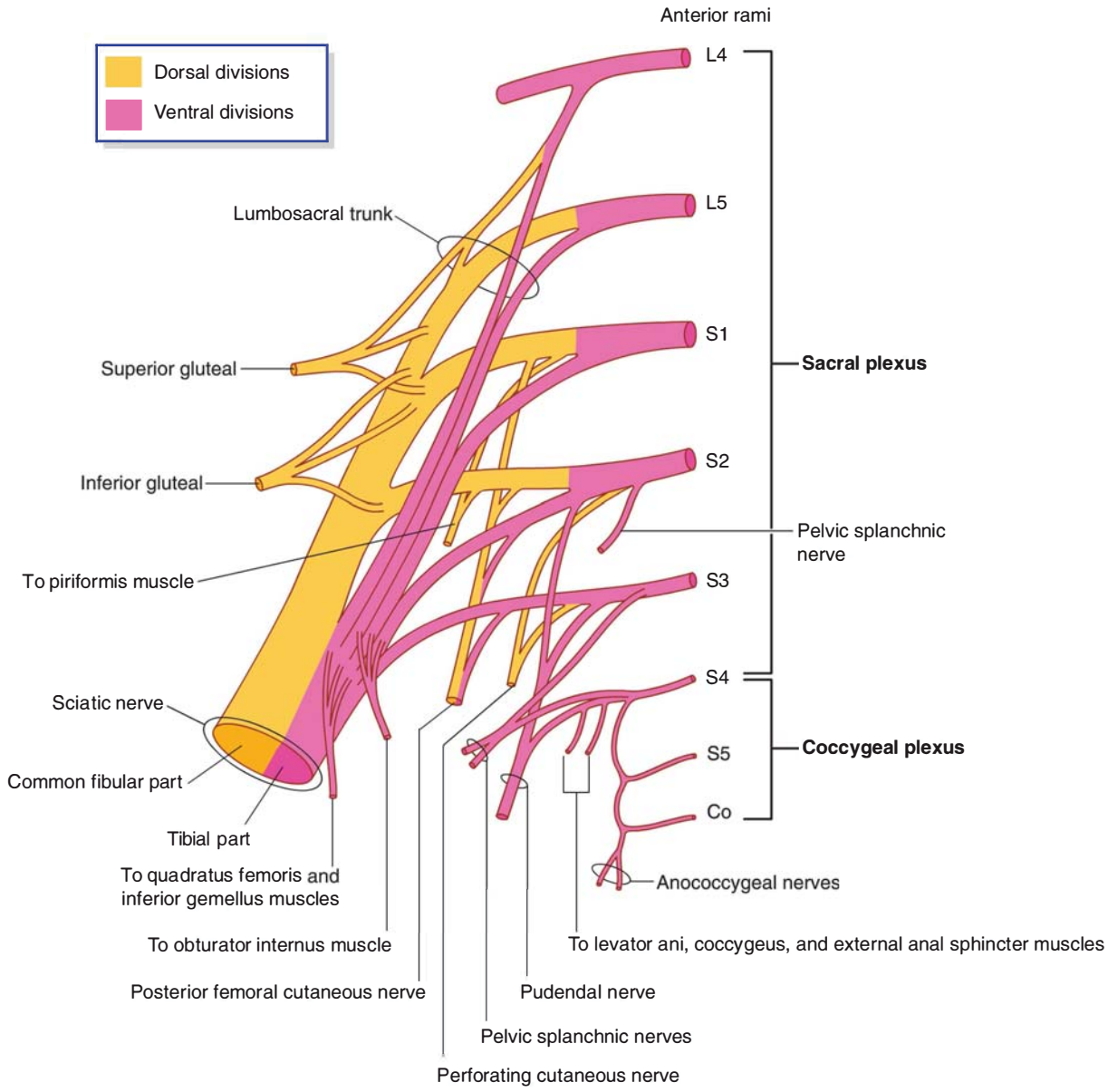


Fig. 5.60 Components and branches of the sacral and coccygeal plexuses.

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periphery to the somatic nerves (Fig. 5.61). In addition, special visceral nerves (**pelvic splanchnic nerves**) originating from S2 to S4 deliver preganglionic parasympathetic fibers to the pelvic part of the prevertebral plexus (Figs. 5.59 and 5.60).

Each anterior ramus has ventral and dorsal divisions that combine with similar divisions from other levels to form terminal nerves (Fig. 5.60). The anterior ramus of S4 has only a ventral division.

Branches of the sacral plexus include the sciatic nerve and gluteal nerves, which are major nerves of the lower limb, and the pudendal nerve, which is the nerve of the perineum (Table 5.4). Numerous smaller branches supply the pelvic wall, floor, and lower limb.

Most nerves originating from the sacral plexus leave the pelvic cavity by passing through the greater sciatic foramen inferior to the piriformis muscle, and enter the gluteal region of the lower limb. Other nerves leave the pelvic cavity using different routes; a few nerves do not leave the pelvic cavity and course directly into the muscles in the pelvic cavity. Finally, two nerves that leave the pelvic cavity through the greater sciatic foramen loop around the ischial spine and sacrospinous ligament and pass medially through the lesser sciatic foramen to supply structures in the perineum and lateral pelvic wall.

Sciatic nerve. The **sciatic nerve** is the largest nerve of the body and carries contributions from L4 to S3 (Figs. 5.59 and 5.60). It:

- forms on the anterior surface of the piriformis muscle and leaves the pelvic cavity through the greater sciatic foramen inferior to the piriformis;

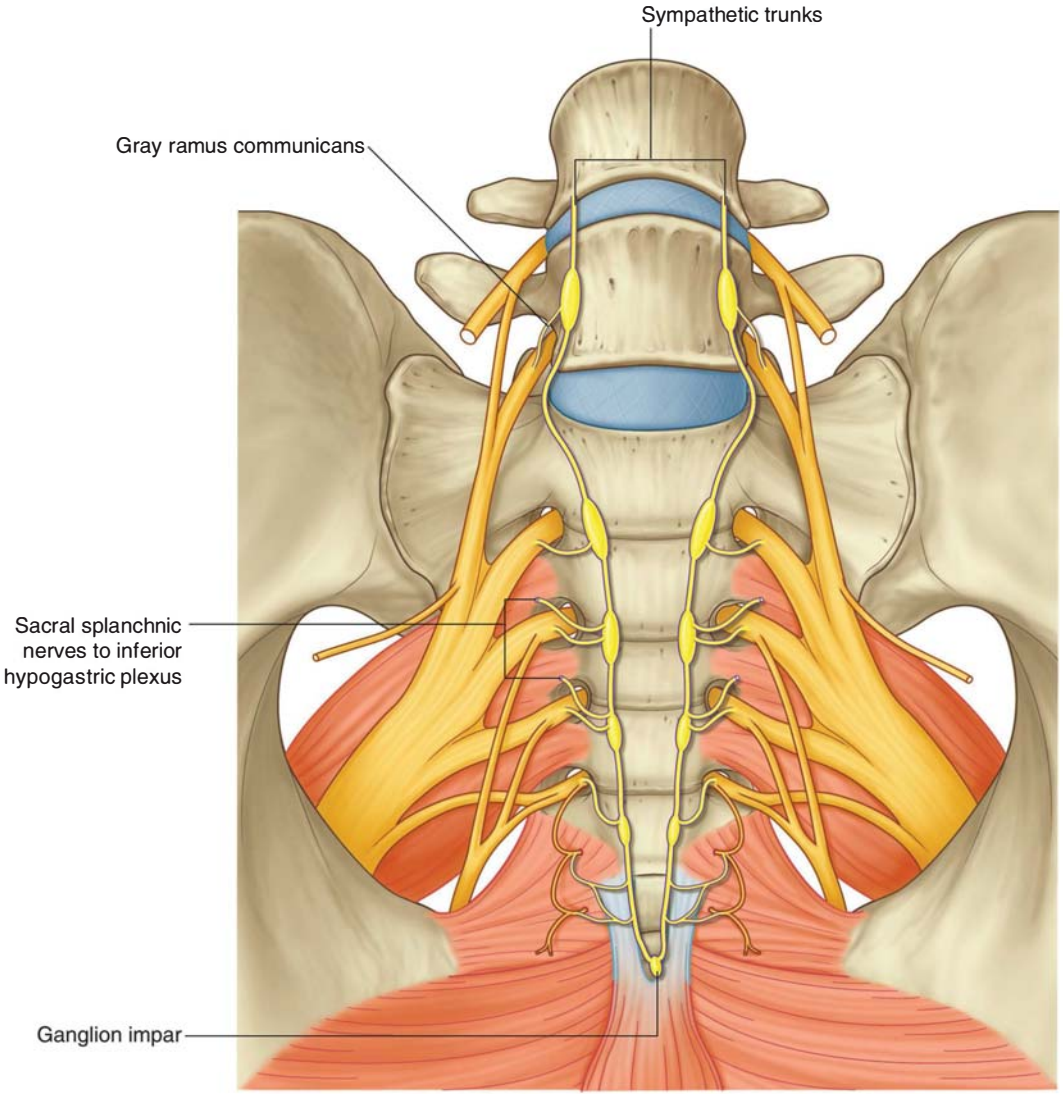
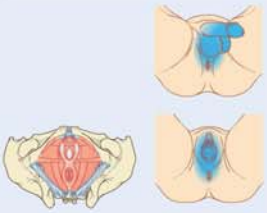


Fig. 5.61 Sympathetic trunks in the pelvis.

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Table 5.4 Branches of the sacral and coccygeal plexuses (spinal segments in parentheses do not consistently participate)

Branch	Spinal segments	Motor function
SACRAL PLEXUS		
Sciatic		
 Tibial part	L4 to S3	All muscles in the posterior or hamstring compartment of the thigh (including the hamstring part of the adductor magnus) except for the short head of the biceps All muscles in the posterior compartment of the leg All muscles in the sole of the foot **Sensory (cutaneous) function** Skin on posterolateral and lateral surfaces of foot and sole of foot
 Common fibular part	L4 to S2	**Motor function** Short head of biceps in the posterior compartment of the thigh All muscles in the anterior and lateral compartments of the leg Extensor digitorum brevis in the foot (also contributes to the supply of the first dorsal interosseous muscle) **Sensory (cutaneous) function** Skin on the anterolateral surface of the leg and dorsal surface of the foot
 **Pudendal**	S2 to S4	**Motor function** Skeletal muscles in the perineum including the external urethral and anal sphincters and levator ani (overlaps in supply of the levator ani and external sphincter with branches directly from ventral division of S4) **Sensory (cutaneous) function** Most skin of the perineum. Penis and clitoris
 **Superior gluteal**	L4 to S1	**Motor function** Gluteus medius, gluteus minimus, and tensor fasciae latae
 **Inferior gluteal**	L5 to S2	**Motor function** Gluteus maximus
 **Nerve to obturator internus and superior gemellus**	L5 to S2	**Motor function** Obturator internus and superior gemellus

(continues)

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Table 5.4 Branches of the sacral and coccygeal plexuses (spinal segments in parentheses do not consistently participate)—cont'd

Branch		
 **Nerve to quadratus femoris and inferior gemellus**	L4 to S1	**Motor function** Quadratus femoris and inferior gemellus
 **Posterior femoral cutaneous (posterior cutaneous nerve of thigh)**	S1, S3	**Sensory (cutaneous) function** Skin on the posterior aspect of the thigh
 Perforating cutaneous	S2, S3	**Sensory (cutaneous) function** Skin over gluteal fold (overlaps with posterior femoral cutaneous)
 **Nerve to piriformis**	S1, S2	**Motor function** Piriformis muscle
 **Nerves to levator ani, coccygeus, and external anal sphincter**	S4	**Motor function** Levator ani, coccygeus, and external anal sphincter. (Overlaps with pudendal nerve) **Sensory (cutaneous) function** Small patch of skin between anus and coccyx
Pelvic splanchnic nerves	S2, S3 (4)	**Motor (visceral) function** Visceral motor (preganglionic parasympathetic) to pelvic part of prevertebral plexus Stimulate erection, modulate mobility in gastrointestinal system distal to the left colic flexure, inhibitory to internal urethral sphincter **Sensory (visceral) function** Visceral afferents (that follow the parasympathetics) from pelvic viscera and distal parts of colon. Pain from cervix and possibly from bladder and proximal urethra
COCCYGEAL PLEXUS		
 **Anococcygeal nerves**	S4 to Co	**Sensory (cutaneous) function** Perianal skin

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- passes through the gluteal region into the thigh, where it divides into its two major branches, the common fibular nerve (common peroneal nerve) and the tibial nerve—dorsal divisions of L4, L5, S1, and S2 are carried in the common fibular part of the nerve and the ventral divisions of L4, L5, S1, S2, and S3 are carried in the tibial part;
- innervates muscles in the posterior compartment of the thigh and muscles in the leg and foot; and
- carries sensory fibers from the skin of the foot and lateral leg.

Pudendal nerve. The **pudendal nerve** forms anteriorly to the lower part of the piriformis muscle from ventral divisions of S2 to S4 (Figs. 5.59 and 5.60). It:

- leaves the pelvic cavity through the greater sciatic foramen, inferior to the piriformis muscle, and enters the gluteal region;
- courses into the perineum by immediately passing around the sacrospinous ligament, where the ligament joins the ischial spine, and through the lesser sciatic foramen (this course takes the nerve out of the pelvic cavity, around the peripheral attachment of the pelvic floor, and into the perineum);
- is accompanied throughout its course by the internal pudendal vessels; and
- innervates skin and skeletal muscles of the perineum, including the external anal and external urethral sphincters.

In the clinic

Pudendal block

Pudendal block anesthesia is performed to relieve the pain associated with childbirth. Although the use of this procedure is less common since the widespread adoption of epidural anesthesia, it provides an excellent option for women who have a contraindication to neuraxial anesthesia (e.g., spinal anatomy, low platelets, too close to delivery). Pudendal blocks are also used for certain types of chronic pelvic pain. The injection is usually given where the pudendal nerve crosses the lateral aspect of the sacrospinous ligament near its attachment to the ischial spine. During childbirth, a finger inserted into the vagina can palpate the ischial spine. The needle is passed transcutaneously to the medial aspect of the ischial spine and around the sacrospinous ligament. Infiltration is performed and the perineum is anesthetized.

Other branches of the sacral plexus. Other branches of the sacral plexus include:

- motor branches to muscles of the gluteal region, pelvic wall, and pelvic floor (superior and inferior gluteal nerves, nerve to obturator internus and superior gemellus, nerve to quadratus femoris and inferior gemellus, nerve to piriformis, nerves to levator ani); and
- sensory nerves to skin over the inferior gluteal region and posterior aspects of the thigh and upper leg (perforating cutaneous nerve and posterior cutaneous nerve of the thigh) (Figs. 5.59 and 5.60).

The **superior gluteal nerve**, formed by branches from the dorsal divisions of L4 to S1, leaves the pelvic cavity through the greater sciatic foramen superior to the piriformis muscle and supplies muscles in the gluteal region—**gluteus medius**, **gluteus minimus**, and **tensor fasciae latae** (**tensor of fascia lata**) **muscles**.

The **inferior gluteal nerve**, formed by branches from the dorsal divisions of L5 to S2, leaves the pelvic cavity through the greater sciatic foramen inferior to the piriformis muscle and supplies the **gluteus maximus**, the largest muscle in the gluteal region.

Both superior and inferior gluteal nerves are accompanied by corresponding arteries.

The **nerve to the obturator internus** and the associated **superior gemellus** muscle originates from the ventral divisions of L5 to S2 and leaves the pelvic cavity through the greater sciatic foramen inferior to the piriformis muscle. Like the pudendal nerve, it passes around the ischial spine and through the lesser sciatic foramen to enter the perineum and supply the obturator internus muscle from the medial side of the muscle, inferior to the attachment of the levator ani muscle.

The **nerve to the quadratus femoris** muscle and the **inferior gemellus** muscle, and the **posterior cutaneous nerve of the thigh** (**posterior femoral cutaneous nerve**) also leave the pelvic cavity through the greater sciatic foramen inferior to the piriformis muscle and course to muscles and skin, respectively, in the lower limb.

Unlike most of the other nerves originating from the sacral plexus, which leave the pelvic cavity through the greater sciatic foramen either above or below the piriformis muscle, the **perforating cutaneous nerve** leaves the pelvic cavity by penetrating directly through the sacrotuberous ligament and then courses to skin over the inferior aspect of the buttocks.

The **nerve to the piriformis** and a number of small nerves to the levator ani and coccygeus muscles originate from the sacral plexus and pass directly into their target muscles without leaving the pelvic cavity.

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The **obturator nerve** (L2 to L4) is a branch of the lumbar plexus. It passes inferiorly along the posterior abdominal wall within the psoas muscle, emerges from the medial surface of the psoas, passes posteriorly to the common iliac artery and medially to the internal iliac artery at the pelvic inlet, and then courses along the lateral pelvic wall. It leaves the pelvic cavity by traveling through the obturator canal and supplies the adductor region of the thigh.

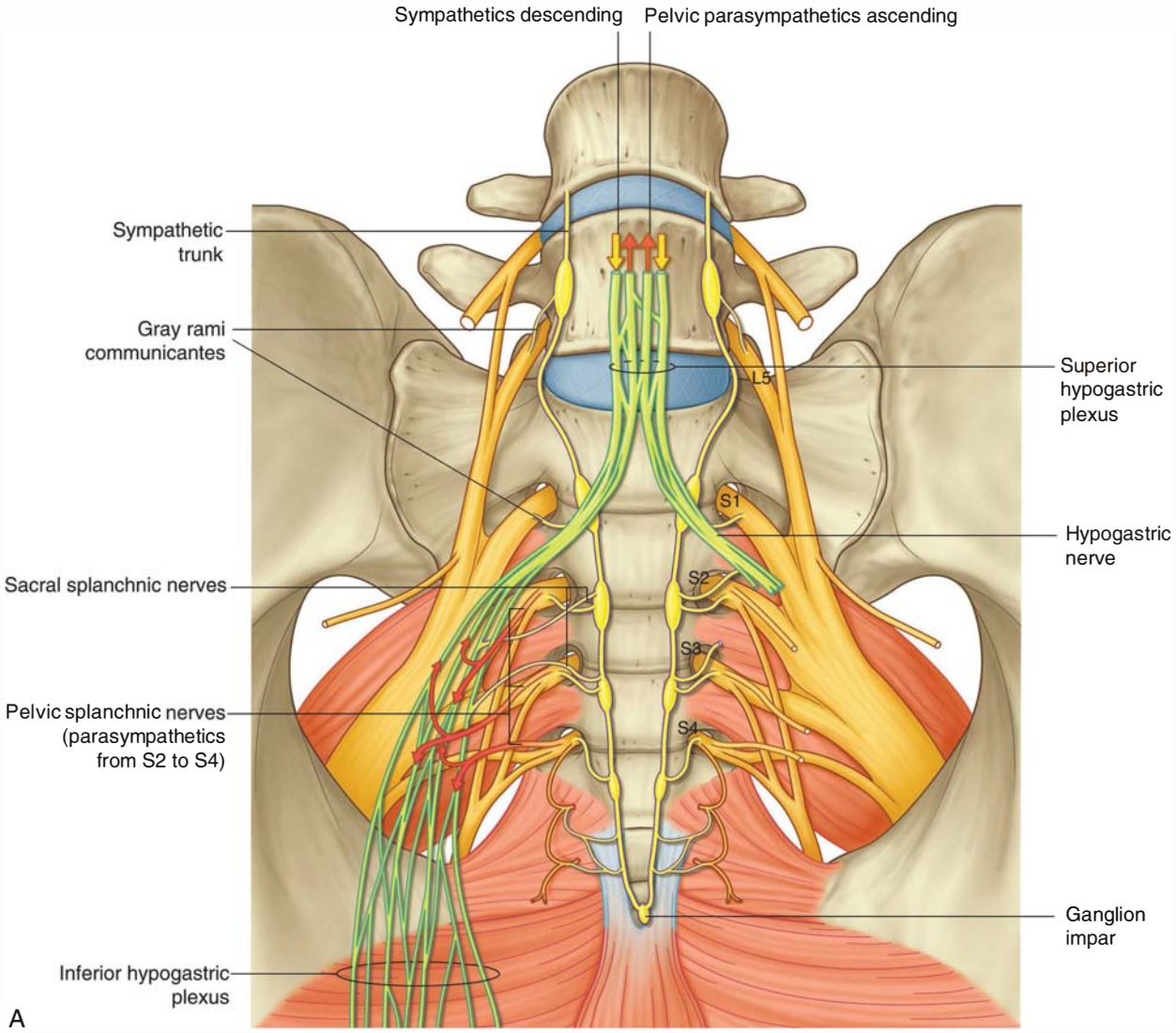
Coccygeal plexus

The small coccygeal plexus has a minor contribution from S4 and is formed mainly by the anterior rami of S5 and Co, which originate inferiorly to the pelvic floor. They penetrate the coccygeus muscle to enter the pelvic cavity and join with the anterior ramus of S4 to form a single trunk, from which small **anococcygeal nerves** originate (Table 5.4). These nerves penetrate the muscle and the overlying sacrospinous and sacrotuberous ligaments and pass superficially to innervate skin in the anal triangle of the perineum.

Visceral plexuses

Paravertebral sympathetic chain

The paravertebral part of the visceral nervous system is represented in the pelvis by the inferior ends of the sympathetic trunks (Fig. 5.62A). Each trunk enters the pelvic cavity from the abdomen by passing over the ala of the sacrum medially to the lumbosacral trunks and posteriorly to the iliac vessels. The trunks course inferiorly along the anterior surface of the sacrum, where they are positioned



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Fig. 5.62 Pelvic extensions of the prevertebral plexus. **A.** Anterior view.

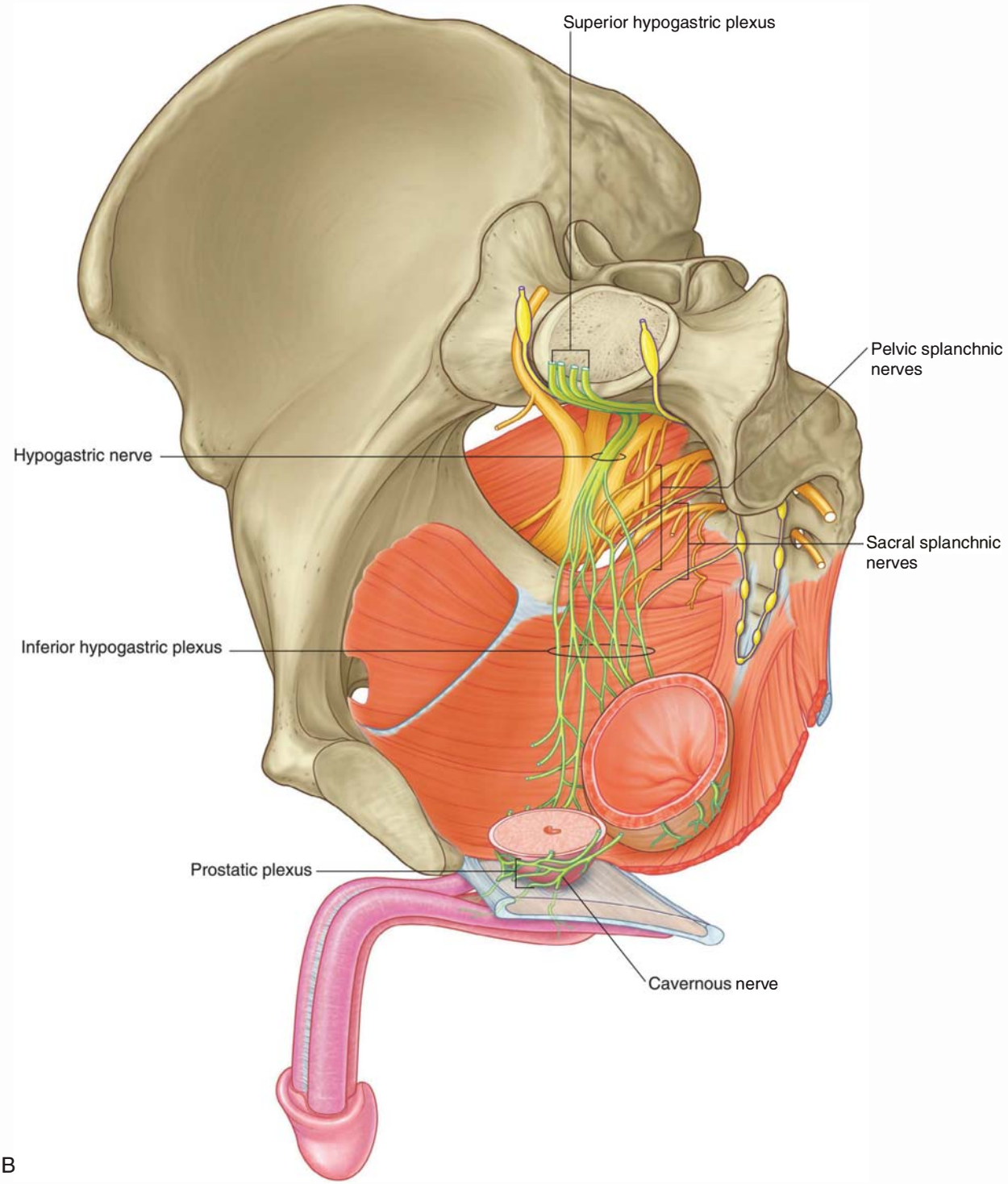


Fig. 5.62, cont'd B. Anteromedial view of right side of plexus.

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medially to the anterior sacral foramina. Four ganglia occur along each trunk. Anteriorly to the coccyx, the two trunks join to form a single small terminal ganglion (the **ganglion impar**).

The principal function of the sympathetic trunks in the pelvis is to deliver postganglionic sympathetic fibers to the anterior rami of sacral nerves for distribution to the periphery, mainly to parts of the lower limb and perineum. This is accomplished by gray rami communicantes, which connect the trunks to the sacral anterior rami.

In addition to gray rami communicantes, other branches (the **sacral splanchnic nerves**) join and contribute to the pelvic part of the prevertebral plexus associated with innervating pelvic viscera (Fig. 5.62A).

Pelvic extensions of the prevertebral plexus

The pelvic parts of the prevertebral plexus carry sympathetic, parasympathetic, and visceral afferent fibers (Fig. 5.62A). Pelvic parts of the plexus are associated with innervating pelvic viscera and erectile tissues of the perineum.

The prevertebral plexus enters the pelvis as two **hypogastric nerves**, one on each side, that cross the pelvic inlet medially to the internal iliac vessels (Fig. 5.62A). The hypogastric nerves are formed by the separation of the fibers in the **superior hypogastric plexus**, into right and left bundles. The superior hypogastric plexus is situated anterior to vertebra LV between the promontory of the sacrum and the bifurcation of the aorta.

When the hypogastric nerves are joined by pelvic splanchnic nerves carrying preganglionic parasympathetic fibers from S2 to S4, the **pelvic plexuses** (**inferior hypogastric plexuses**) are formed (Fig. 5.62). The inferior hypogastric plexuses, one on each side, course in an inferior direction around the pelvic walls, medially to major vessels and somatic nerves. They give origin to the following subsidiary plexuses, which innervate the pelvic viscera:

- the **rectal plexus**,
- the **uterovaginal plexus**,
- the **prostatic plexus**, and
- the **vesical plexus**.

Terminal branches of the inferior hypogastric plexuses penetrate and pass through the deep perineal pouch and innervate erectile tissues of the penis and the clitoris in the perineum (Fig. 5.62B). In men, these nerves, called **cavernous nerves**, are extensions of the prostatic plexus. The pattern of distribution of similar nerves in women is not entirely clear, but they are likely extensions of the uterovaginal plexus.

Sympathetic fibers

Sympathetic fibers enter the inferior hypogastric plexuses from the hypogastric nerves and from branches (sacral splanchnic nerves) of the upper sacral parts of the sympathetic trunks (Fig. 5.62A). Ultimately, these nerves are derived from preganglionic fibers that leave the spinal cord in the anterior roots, mainly of T10 to L2. These fibers:

- innervate blood vessels,
- cause contraction of smooth muscle in the internal urethral sphincter in men and the internal anal sphincters in both men and women,
- cause smooth muscle contraction associated with the reproductive tract and with the accessory glands of the reproductive system, and
- are important in moving secretions from the epididymis and associated glands into the urethra to form semen during ejaculation.

Parasympathetic fibers

Parasympathetic fibers enter the pelvic plexus in pelvic splanchnic nerves that originate from spinal cord levels S2 to S4 (Fig. 5.62A). They:

- are generally vasodilatory,
- stimulate bladder contraction,
- stimulate erection, and
- modulate activity of the enteric nervous system of the colon distal to the left colic flexure (in addition to pelvic viscera, some of the fibers from the pelvic plexus course superiorly in the prevertebral plexus, or as separate nerves, and pass into the inferior mesenteric plexus of the abdomen).

Visceral afferent fibers

Visceral afferent fibers follow the course of the sympathetic and parasympathetic fibers to the spinal cord. Afferent fibers that enter the cord in lower thoracic levels and lumbar levels with sympathetic fibers generally carry pain; however, pain fibers from the cervix and some pain fibers from the bladder and urethra may accompany parasympathetic nerves to sacral levels of the spinal cord.

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In the clinic

Prostatectomy and impotence

It may be necessary to perform radical surgery to cure cancer of the prostate. To do this, the prostate and its attachments around the base of the bladder, including the seminal vesicles, must be removed en masse. Parts of the inferior hypogastric plexus in this region give rise to nerves that innervate the erectile tissues of the penis. Impotence may occur if these nerves cannot be or are not preserved during removal of the prostate.

For the same reasons, women may experience sexual dysfunction if similar nerves are damaged during pelvic surgery, for example, during a total hysterectomy.

Blood vessels

Arteries

The major artery of the pelvis and perineum is the internal iliac artery on each side (Fig. 5.63). In addition to providing a blood supply to most of the pelvic viscera, pelvic walls and floor, and structures in the perineum, including erectile tissues of the clitoris and the penis, this artery gives rise to branches that follow nerves into the gluteal region of the lower limb. Other vessels that originate in the abdomen and contribute to the supply of pelvic structures include the median sacral artery and, in women, the ovarian arteries.

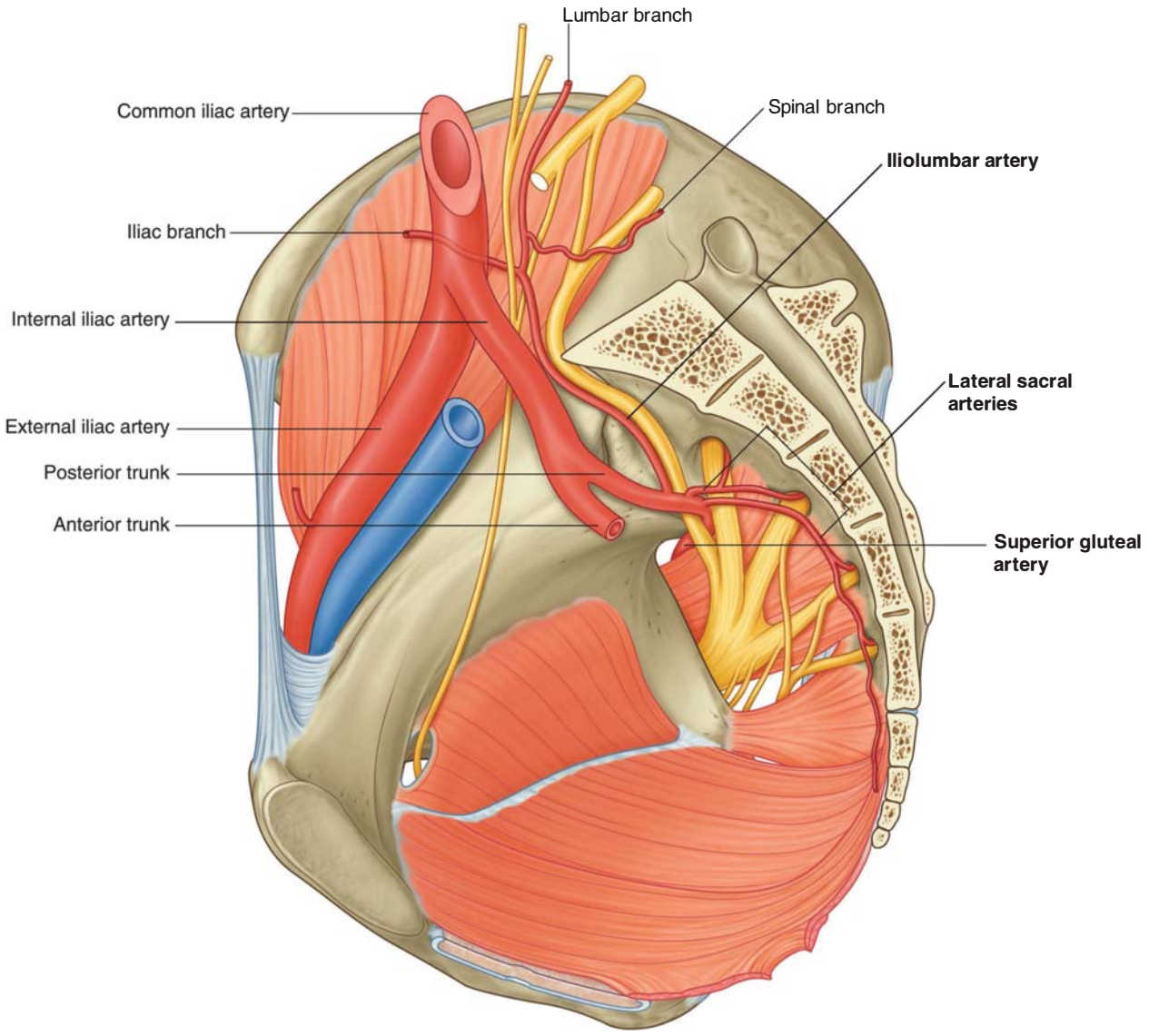


Fig. 5.63 Branches of the posterior trunk of the internal iliac artery.

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Internal iliac artery

The internal iliac artery originates from the common iliac artery on each side, approximately at the level of the intervertebral disc between LV and SI, and lies anteromedial to the sacro-iliac joint (Fig. 5.63). The vessel courses inferiorly over the pelvic inlet and then divides into anterior and posterior trunks at the level of the superior border of the greater sciatic foramen. Branches from the posterior trunk contribute to the supply of the lower posterior abdominal wall, the posterior pelvic wall, and the gluteal region. Branches from the anterior trunk supply the pelvic viscera, the perineum, the gluteal region, the adductor region of the thigh, and, in the fetus, the placenta.

Posterior trunk

Branches of the posterior trunk of the internal iliac artery are the iliolumbar artery, the lateral sacral artery, and the superior gluteal artery (Fig. 5.63).

- The **iliolumbar artery** ascends laterally back out of the pelvic inlet and divides into a lumbar branch and an iliac branch. The lumbar branch contributes to the supply of the posterior abdominal wall, psoas and quadratus lumborum muscles, and cauda equina, via a small spinal branch that passes through the intervertebral foramen between LV and SI. The iliac branch passes laterally into the iliac fossa to supply muscle and bone.
- The **lateral sacral arteries**, usually two, originate from the posterior division of the internal iliac artery and course medially and inferiorly along the posterior pelvic wall. They give rise to branches that pass into the anterior sacral foramina to supply related bone and soft tissues, structures in the vertebral (sacral) canal, and skin and muscle posterior to the sacrum.
- The **superior gluteal artery** is the largest branch of the internal iliac artery and is the terminal continuation of the posterior trunk. It courses posteriorly, usually passing between the lumbosacral trunk and anterior ramus of S1, to leave the pelvic cavity through the greater sciatic foramen above the piriformis muscle and enter the gluteal region of the lower limb. This vessel makes a substantial contribution to the blood supply of muscles and skin in the gluteal region and also supplies branches to adjacent muscles and bones of the pelvic walls.

Anterior trunk

Branches of the anterior trunk of the internal iliac artery include the superior vesical artery, the umbilical artery, the inferior vesical artery, the middle rectal artery, the uterine artery, the vaginal artery, the obturator artery, the internal pudendal artery, and the inferior gluteal artery (Fig. 5.64).

- The first branch of the anterior trunk is the **umbilical artery**, which gives origin to the superior vesical artery and then travels forward just inferior to the margin of the pelvic inlet. Anteriorly, the vessel leaves the pelvic cavity and ascends on the internal aspect of the anterior abdominal wall to reach the umbilicus. In the fetus, the umbilical artery is large and carries blood from the fetus to the placenta. After birth, the vessel closes distally to the origin of the superior vesical artery and eventually becomes a solid fibrous cord. On the anterior abdominal wall, the cord raises a fold of peritoneum termed the **medial umbilical fold**. The fibrous remnant of the umbilical artery itself is the **medial umbilical ligament**.
- The **superior vesical artery** normally originates from the root of the umbilical artery and courses medially and inferiorly to supply the superior aspect of the bladder and distal parts of the ureter. In men, it also may give rise to an artery that supplies the ductus deferens.
- The **inferior vesical artery** occurs in men and supplies branches to the bladder, ureter, seminal vesicle, and prostate. The **vaginal artery** in women is the equivalent of the inferior vesical artery in men and, descending to the vagina, supplies branches to the vagina and to adjacent parts of the bladder and rectum.
- The **middle rectal artery** courses medially to supply the rectum. The vessel anastomoses with the superior rectal artery, which originates from the inferior mesenteric artery in the abdomen, and the inferior rectal artery, which originates from the internal pudendal artery in the perineum.
- The **obturator artery** courses anteriorly along the pelvic wall and leaves the pelvic cavity via the obturator canal. Together with the obturator nerve, above, and obturator vein, below, it enters and supplies the adductor region of the thigh.
- The **internal pudendal artery** courses inferiorly from its origin in the anterior trunk and leaves the pelvic

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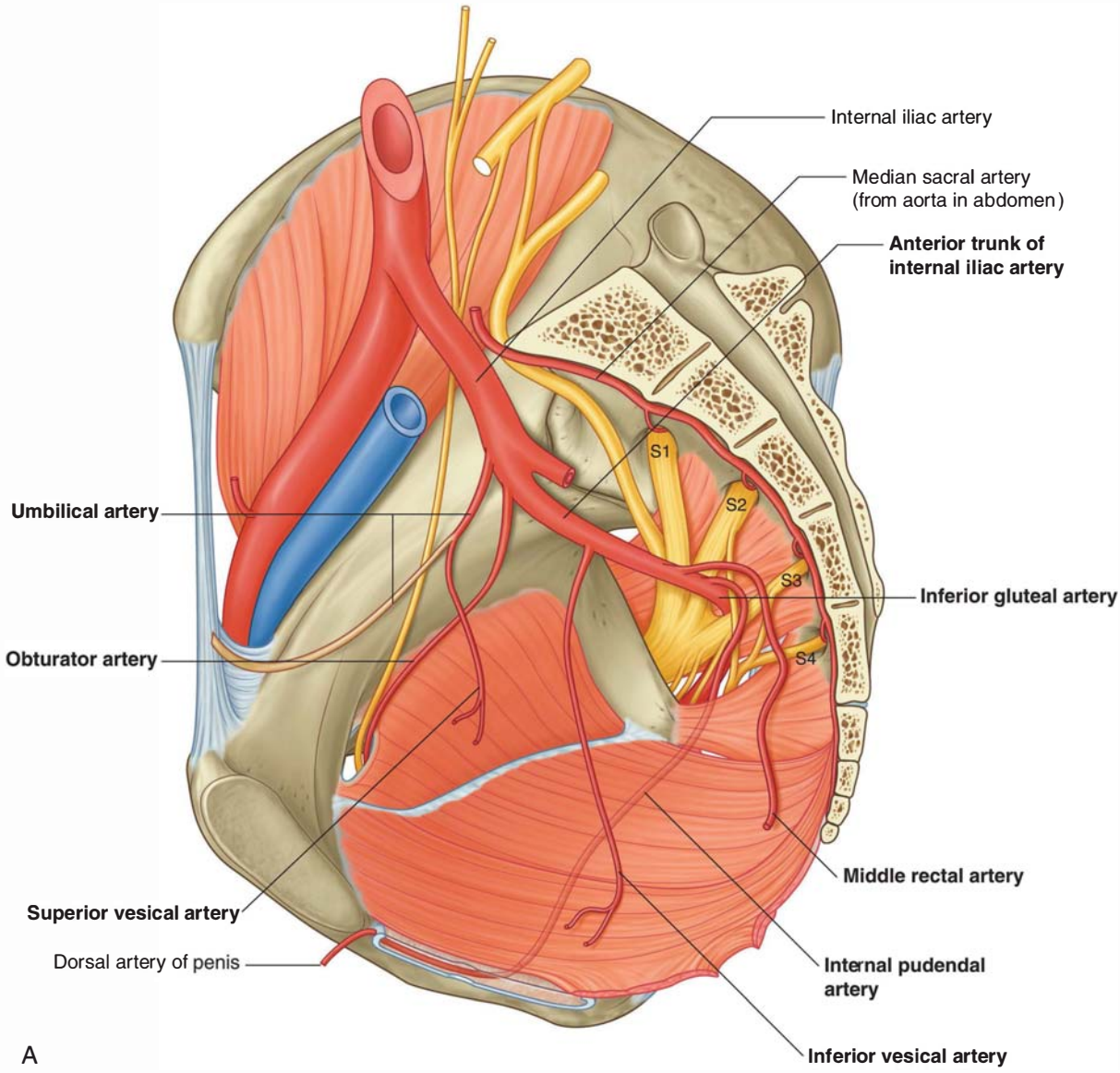


Fig. 5.64 Branches of the anterior trunk of the internal iliac artery. **A.** Male.

cavity through the greater sciatic foramen inferior to the piriformis muscle. In association with the pudendal nerve on its medial side, the vessel passes laterally to the ischial spine and then through the lesser sciatic foramen to enter the perineum. The internal pudendal artery is the main artery of the perineum. Among the structures it supplies are the erectile tissues of the clitoris and the penis.

- The **inferior gluteal artery** is a large terminal branch of the anterior trunk of the internal iliac artery. It passes between the anterior rami S1 and S2 or S2 and S3 of the sacral plexus and leaves the pelvic cavity through the greater sciatic foramen inferior to the piriformis muscle. It enters and contributes to the blood supply of the gluteal region and anastomoses with a network of vessels around the hip joint.

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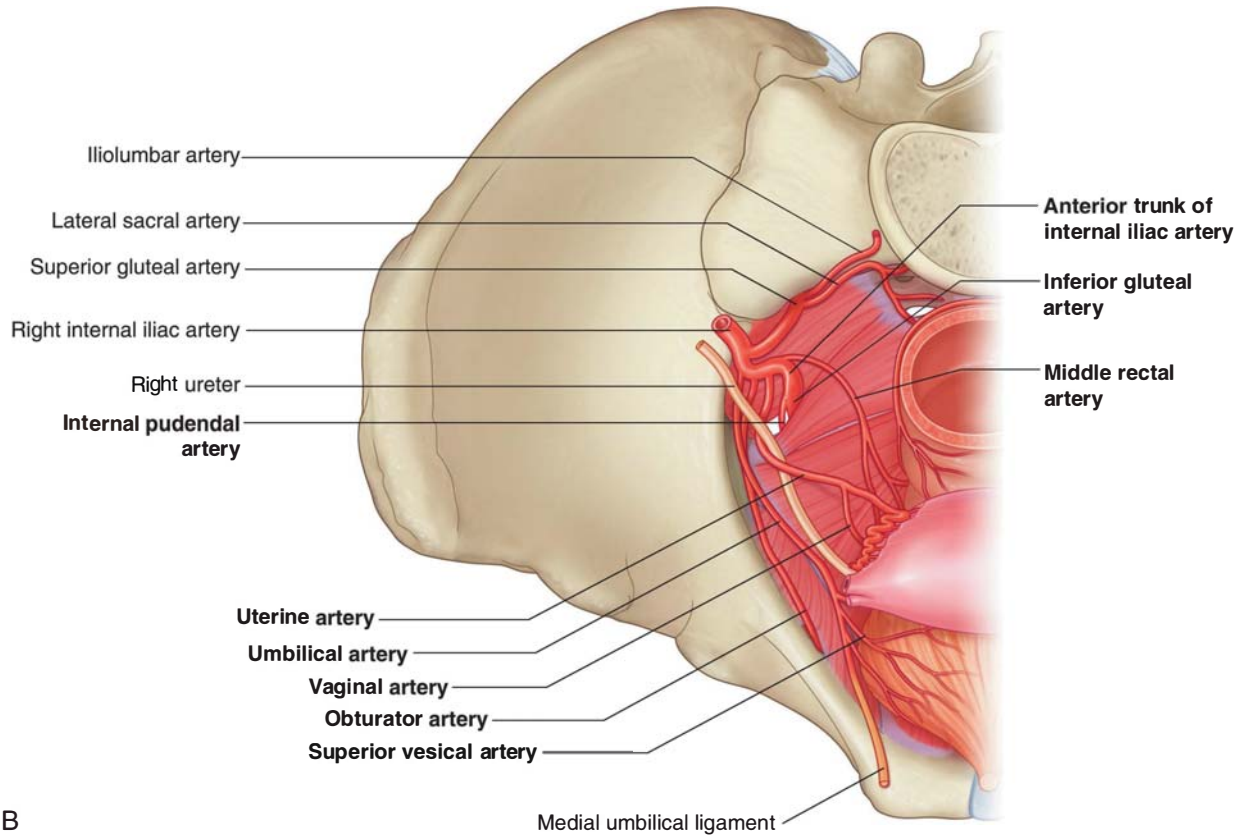


Fig. 5.64, cont'd B. Female.

- The **uterine artery** in women courses medially and anteriorly in the base of the broad ligament to reach the cervix (Figs. 5.64B and 5.65). Along its course, the vessel crosses the ureter and passes superiorly to the lateral vaginal fornix. Once the vessel reaches the cervix, it ascends along the lateral margin of the uterus to reach the uterine tube, where it curves laterally and anastomoses with the ovarian artery. The uterine artery is the major blood supply to the uterus and enlarges significantly during pregnancy. Through anastomoses with other arteries, the vessel contributes to the blood supply of the ovary and vagina as well.

Ovarian arteries

In women, the gonadal (ovarian) vessels originate from the abdominal aorta and then descend to cross the pelvic inlet and supply the ovaries. They anastomose with terminal parts of the uterine arteries (Fig. 5.65). On each side, the vessels travel in the **suspensory ligament of the ovary** (the **infundibulopelvic ligament**) as they cross the pelvic inlet to the ovary. Branches pass through the mesovarium to reach the ovary and through the mesometrium of the broad ligament to anastomose with the uterine artery. The ovarian arteries enlarge significantly during pregnancy to augment the uterine blood supply.

Median sacral artery

The median sacral artery (Figs. 5.64A and 5.65) originates from the posterior surface of the aorta just superior to the aortic bifurcation at vertebral level LIV in the abdomen. It descends in the midline, crosses the pelvic inlet, and then courses along the anterior surface of the sacrum and coccyx. It gives rise to the last pair of lumbar arteries and to branches that anastomose with the iliolumbar and lateral sacral arteries.

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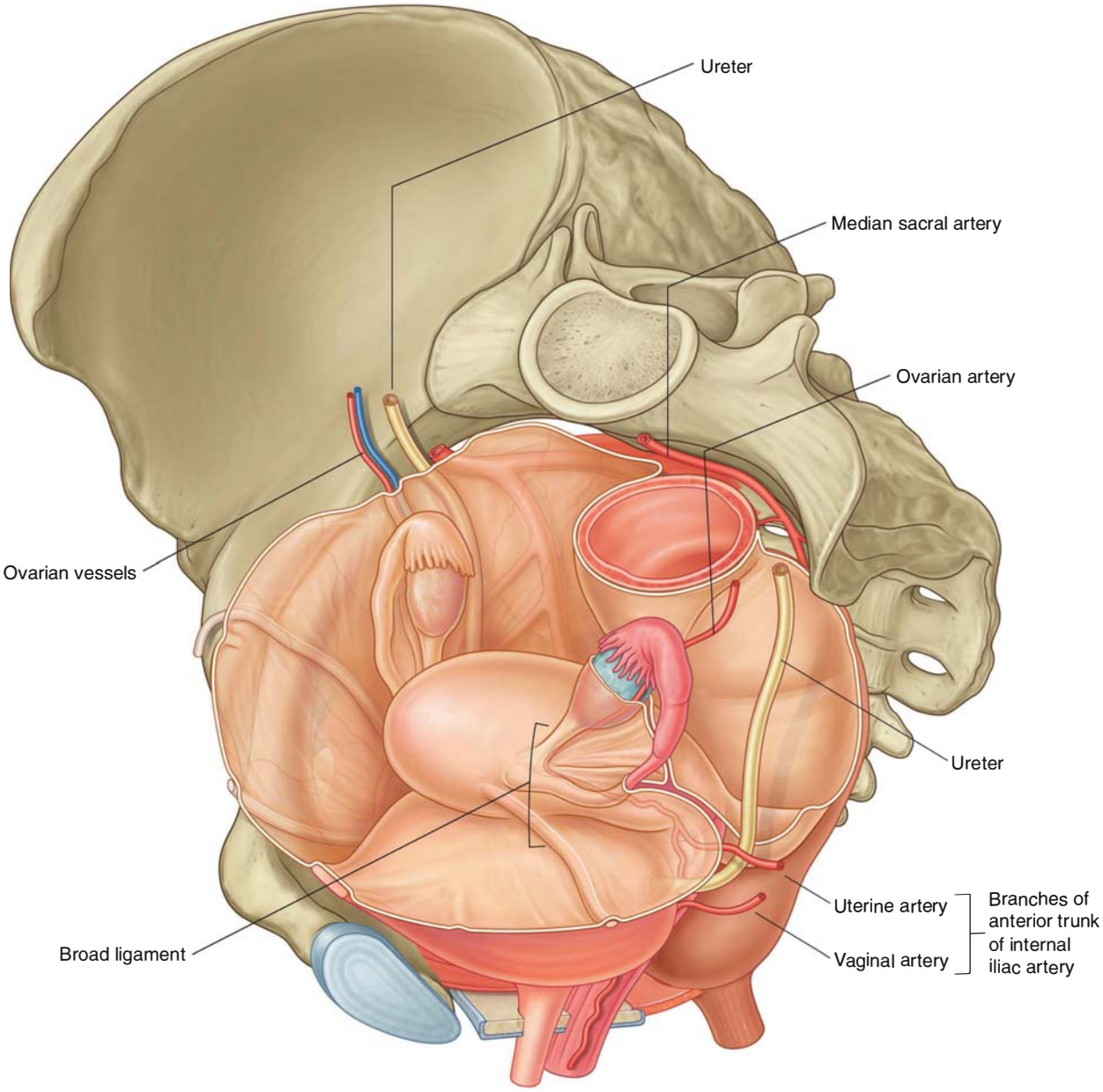


Fig. 5.65 Uterine and vaginal arteries.

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Veins

Pelvic veins follow the course of all branches of the internal iliac artery except for the umbilical artery and the iliolumbar artery (Fig. 5.66A). On each side, the veins drain into internal iliac veins, which leave the pelvic cavity to join common iliac veins situated just superior and lateral to the pelvic inlet.

Within the pelvic cavity, extensive interconnected venous plexuses are associated with the surfaces of the viscera (bladder, rectum, prostate, uterus, and vagina). Together, these plexuses form the pelvic plexus of veins. The part of the venous plexus surrounding the rectum and anal canal drains via superior rectal veins (tributaries of inferior mesenteric veins) into the hepatic portal system, and via middle and inferior rectal veins into the caval system. This pelvic plexus is an important portacaval shunt when the hepatic portal system is blocked (Fig. 5.66B).

The inferior part of the rectal plexus around the anal canal has two parts, an internal and an external. The internal rectal plexus is in connective tissue between the internal anal sphincter and the epithelium lining the canal. This plexus connects superiorly with longitudinally arranged branches of the superior rectal vein that lie one in each

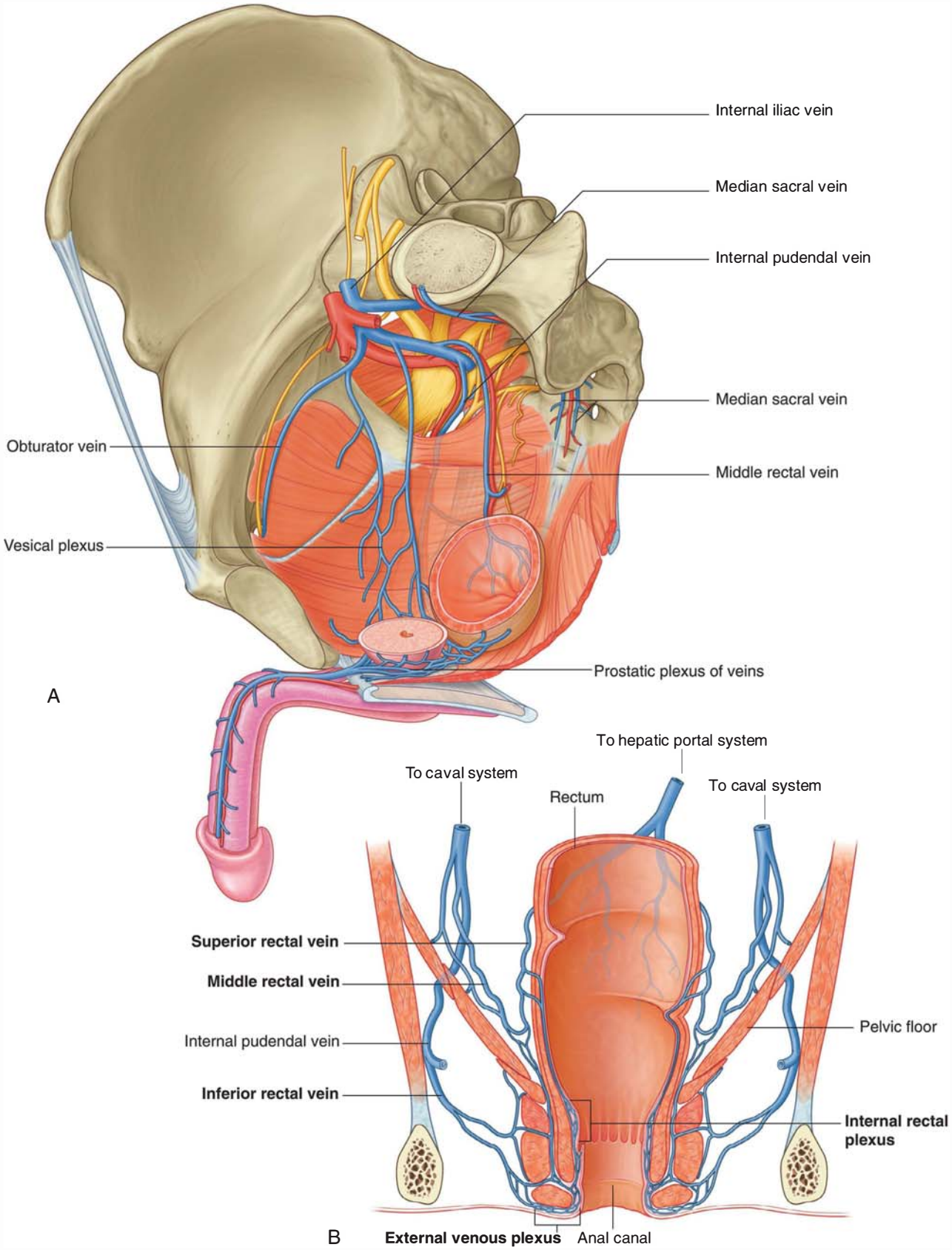


Fig. 5.66 Pelvic veins. **A.** In a man with the left side of the pelvis and most of the viscera removed. **B.** Veins associated with the rectum and anal canal.

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anal column. When enlarged, these branches form internal hemorrhoids, which originate above the pectinate line and are covered by colonic mucosa. The external rectal plexus circles the external anal sphincter and is subcutaneous. Enlargement of vessels in the external rectal plexus results in external hemorrhoids.

The single **deep dorsal vein** that drains erectile tissues of the clitoris and the penis does not follow branches of the internal pudendal artery into the pelvic cavity. Instead, this vein passes directly into the pelvic cavity through a gap formed between the arcuate pubic ligament and the anterior margin of the perineal membrane. The vein joins the prostatic plexus of veins in men and the vesical (bladder) plexus of veins in women. (Superficial veins that drain the skin of the penis and corresponding regions of the clitoris drain into the external pudendal veins, which are tributaries of the great saphenous vein in the thigh.)

In addition to tributaries of the internal iliac vein, median sacral veins and ovarian veins parallel the courses of the median sacral artery and ovarian artery, respectively, and leave the pelvic cavity to join veins in the abdomen:

- The **median sacral veins** coalesce to form a single vein that joins either the left common iliac vein or the junction of the two common iliac veins to form the inferior vena cava.
- The **ovarian veins** follow the course of the corresponding arteries; on the left, they join the left renal vein and, on the right, they join the inferior vena cava in the abdomen.

Lymphatics

Lymphatics from most pelvic viscera drain mainly into lymph nodes distributed along the internal iliac and external iliac arteries and their associated branches (Fig. 5.67), which drain into nodes associated with the common iliac arteries and then into the lateral aortic or lumbar nodes associated with the lateral surfaces of the abdominal aorta. In turn, these lateral aortic or lumbar nodes drain into the lumbar trunks, which continue to the origin of the thoracic duct at approximately vertebral level TXII.

Lymphatics from the ovaries and related parts of the uterus and uterine tubes leave the pelvic cavity superiorly

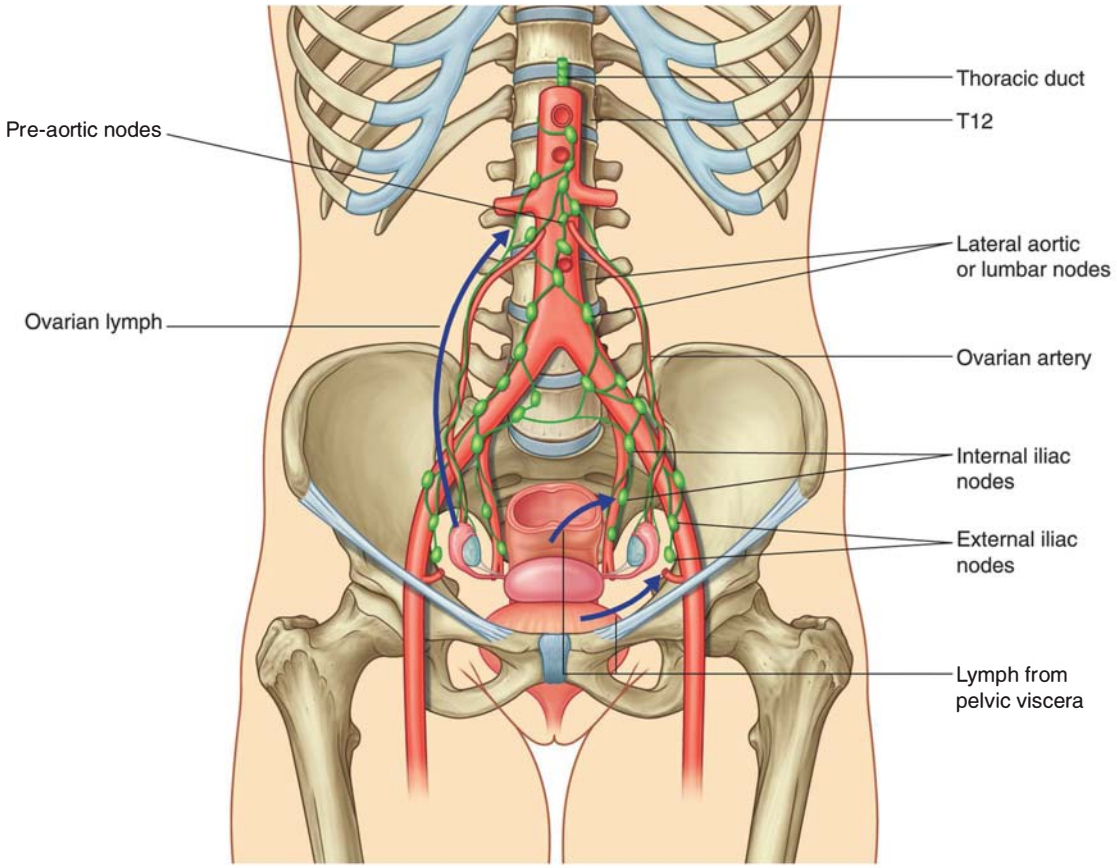


Fig. 5.67 Pelvic lymphatics.

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and drain, via vessels that accompany the ovarian arteries, directly into lateral aortic or lumbar nodes and, in some cases, into the pre-aortic nodes on the anterior surface of the aorta.

In addition to draining pelvic viscera, nodes along the internal iliac artery also receive drainage from the gluteal region of the lower limb and from deep areas of the perineum.

PERINEUM

The perineum is a diamond-shaped region positioned inferiorly to the pelvic floor between the thighs. Its peripheral boundary is the pelvic outlet; its ceiling is the pelvic diaphragm (the levator ani and coccygeus muscles); and its narrow lateral walls are formed by the walls of the pelvic cavity below the attachment of the levator ani muscle (Fig. 5.68A).

The perineum is divided into an anterior urogenital triangle and a posterior anal triangle.

- The urogenital triangle is associated with the openings of the urinary systems and the reproductive systems and functions to anchor the external genitalia.
- The anal triangle contains the anus and the external anal sphincter.

The pudendal nerve (S2 to S4) and the internal pudendal artery are the major nerve and artery of the region.

Borders and ceiling

The margin of the perineum is marked by the inferior border of the pubic symphysis at its anterior point, the tip of the coccyx at its posterior point, and the ischial tuberosities at each of the lateral points (Fig. 5.68A). The lateral margins are formed by the ischiopubic rami anteriorly and by the sacrotuberous ligaments posteriorly. The pubic symphysis, the ischial tuberosities, and the coccyx can be palpated on the patient.

The perineum is divided into two triangles by an imaginary line between the two ischial tuberosities (Fig. 5.68A). Anterior to the line is the urogenital triangle and posterior to the line is the anal triangle. Significantly, the two triangles are not in the same plane. In the anatomical position, the urogenital triangle is oriented in the horizontal plane, whereas the anal triangle is tilted upward at the transtubercular line so that it faces more posteriorly.

The roof of the perineum is formed mainly by the levator ani muscles that separate the pelvic cavity, above, from the perineum, below. These muscles, one on each side, form a cone- or funnel-shaped pelvic diaphragm, with the anal aperture at its inferior apex in the anal triangle.

Anteriorly, in the **urogenital triangle**, a U-shaped defect in the muscles, the **urogenital hiatus**, allows the passage of the urethra and vagina.

Perineal membrane and deep perineal pouch

The perineal membrane (see pp. 457–459) is a thick fibrous sheet that fills the urogenital triangle (Fig. 5.68B). It has a free posterior border, which is anchored in the midline to the perineal body and is attached laterally to the pubic arch. Immediately superior to the perineal membrane is a thin region termed the deep perineal pouch, containing a layer of skeletal muscle and neurovascular tissues. Among the skeletal muscles in the pouch (see p. 459, Fig. 5.36) is the external urethral sphincter.

The perineal membrane and deep perineal pouch provide support for the external genitalia, which are attached to its inferior surface. Also, the parts of the perineal membrane and deep perineal pouch inferior to the urogenital hiatus in the levator ani provide support for the pelvic viscera, above.

The urethra leaves the pelvic cavity and enters the perineum by passing through the deep perineal pouch and perineal membrane. In women, the vagina also passes through these structures posterior to the urethra.

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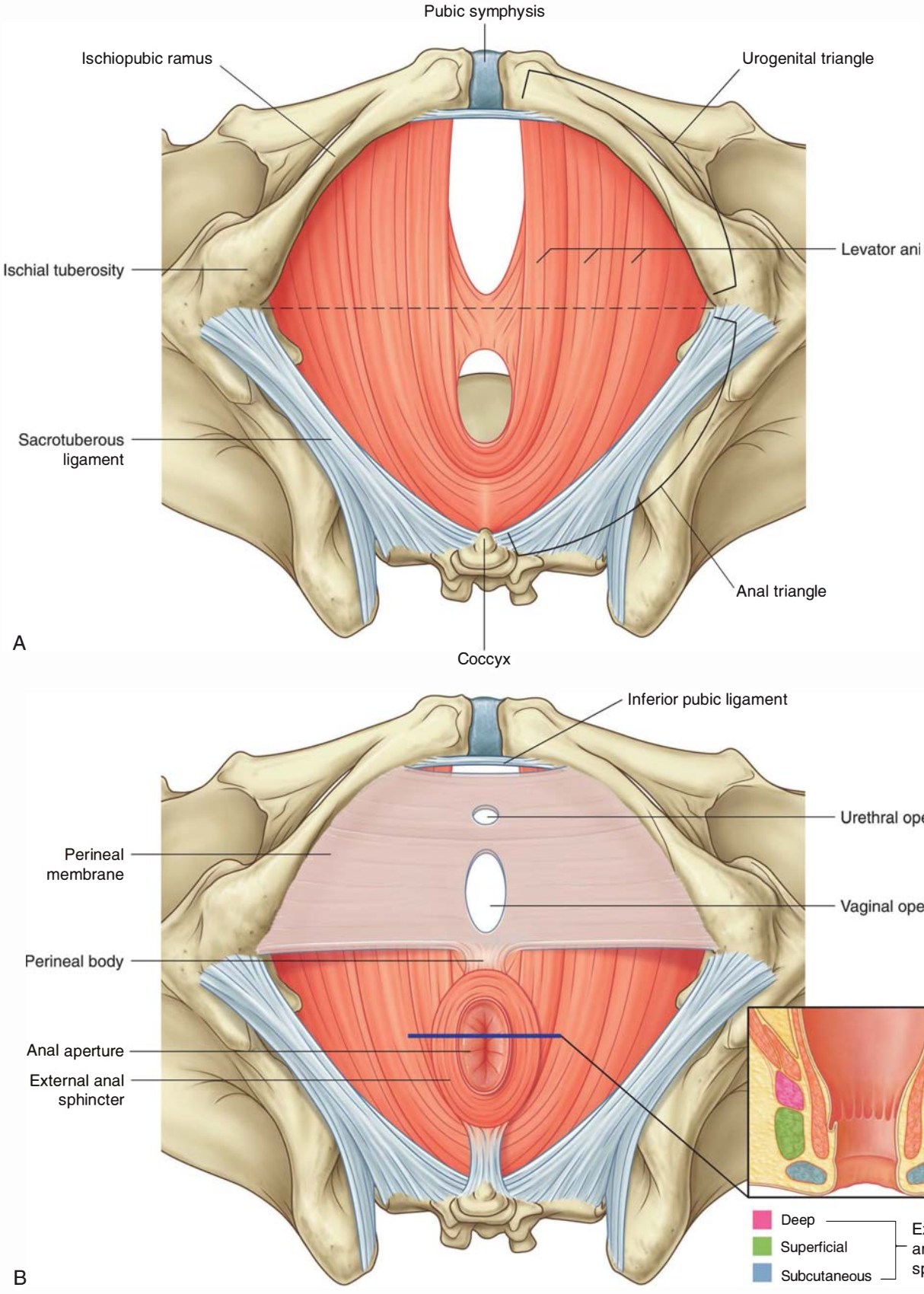


Fig. 5.68 Borders and ceiling of the perineum. **A.** Boundaries of the perineum. **B.** Perineal membrane.

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Ischio-anal fossae and their anterior recesses

Because the levator ani muscles course medially from their origin on the lateral pelvic walls, above, to the anal aperture and urogenital hiatus, below, inverted wedge-shaped gutters occur between the levator ani muscles and adjacent pelvic walls as the two structures diverge inferiorly (Fig. 5.69). In the anal triangle, these gutters, one on each side of the anal aperture, are termed **ischio-anal fossae**. The lateral wall of each fossa is formed mainly by the ischium, obturator internus muscle, and sacrotuberous ligament. The medial wall is the levator ani muscle. The medial and lateral walls converge superiorly where the levator ani muscle attaches to the fascia overlying the obturator internus muscle. The ischio-anal fossae allow movement of the pelvic diaphragm and expansion of the anal canal during defecation.

The ischio-anal fossae of the anal triangle are continuous anteriorly with recesses that project into the urogenital triangle superior to the deep perineal pouch. These anterior recesses of the ischio-anal fossae are shaped like three-sided pyramids that have been tipped onto one of their sides (Fig. 5.69C). The apex of each pyramid is closed and points anteriorly toward the pubis. The base is open and continuous posteriorly with its related ischio-anal fossa. The inferior wall of each pyramid is the deep perineal pouch. The superomedial wall is the levator ani muscle, and the superolateral wall is formed mainly by the obturator internus muscle. The ischio-anal fossae and their anterior recesses are normally filled with fat.

In the clinic

Abscesses in the ischio-anal fossae

The anal mucosa is particularly vulnerable to injury and may be easily torn by hard feces. Occasionally, patients develop inflammation and infection of the anal canal (sinuses or crypts). This infection can spread between the sphincters, producing intersphincteric fistulas. The infection can tract superiorly into the pelvic cavity or laterally into the ischio-anal fossae.

Anal triangle

The anal triangle of the perineum faces posteroinferiorly and is defined laterally by the medial margins of the sacrotuberous ligaments, anteriorly by a horizontal line between the two ischial tuberosities, and posteriorly by the coccyx. The ceiling of the anal triangle is the pelvic diaphragm, which is formed by the levator ani and coccygeus muscles. The anal aperture occurs centrally in the anal triangle and is related on either side to an ischio-anal fossa. The major muscle in the anal triangle is the external anal sphincter.

The **external anal sphincter**, which surrounds the anal canal, is formed by skeletal muscle and consists of three parts—deep, superficial, and subcutaneous—arranged sequentially along the canal from superior to inferior (Fig. 5.68B, Table 5.5). The deep part is a thick ring-shaped muscle that circles the upper part of the anal canal and blends with the fibers of the levator ani muscle. The superficial part also surrounds the anal canal, but is anchored anteriorly to the perineal body and posteriorly to the coccyx and anococcygeal ligament. The subcutaneous part is a horizontally flattened disc of muscle that surrounds the anal aperture just beneath the skin. The external anal sphincter is innervated by inferior rectal branches of the pudendal nerve and by branches directly from the anterior ramus of S4.

In the clinic

Hemorrhoids

A hemorrhoid is an engorgement of the venous plexus at or inside the anal sphincter. It is a common complaint and has prevalence of approximately 4% in the United States. Hemorrhoids have a slight genetic predisposition; however, straining during bowel movements, obesity, and sedentary lifestyle can also produce hemorrhoids. The symptoms include irritation, pain, and swelling. Hemorrhoids occurring at the anal verge (distal boundary of the anal canal) are typically called external hemorrhoids. Internal hemorrhoids occur inside the rectum and have a tendency to bleed. Prolapsed hemorrhoids are internal hemorrhoids that pass outside the anal canal and form lumps, which may undergo thrombosis and become painful.

There are many treatments for hemorrhoids, which include ligation above the pectinate (dentate) line using simple rubber bands or surgical excision. Surgery to this region is not without complications and care must be taken to preserve the internal anal sphincter.

In the back of every physician's mind is the concern that the rectal bleeding or symptoms may not be attributable to hemorrhoids. Therefore, excluding a tumor within the bowel is as important as treating the hemorrhoids.

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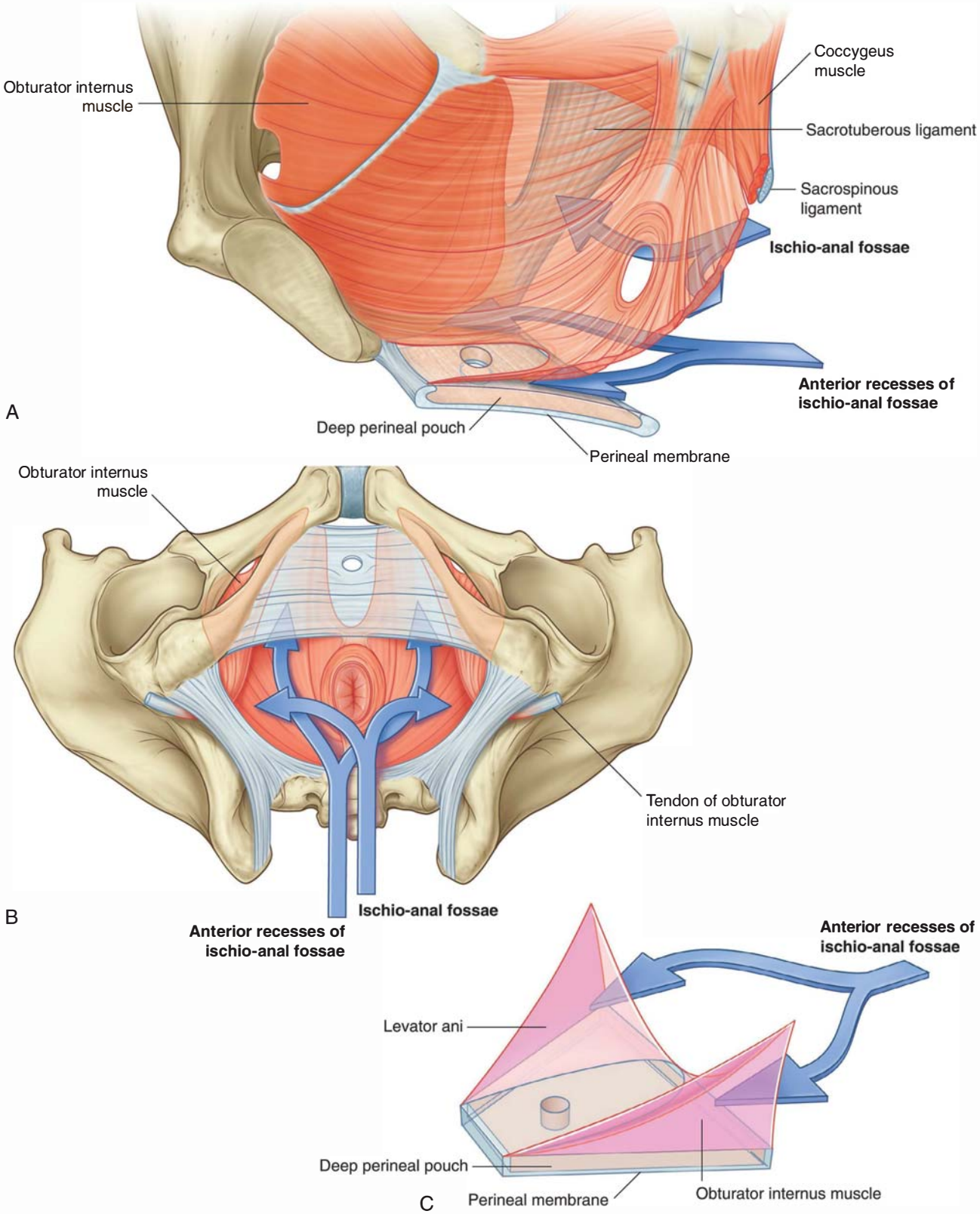


Fig. 5.69 Ischio-anal fossae and their anterior recesses. **A.** Anterolateral view with left pelvic wall removed. **B.** Inferior view. **C.** Anterolateral view with pelvic walls and diaphragm removed.

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Table 5.5 Muscles of the anal triangle

Muscles	Origin	Insertion	Innervation	Function
EXTERNAL ANAL SPHINCTER				
Deep part	Surrounds superior aspect of anal canal		Pudendal nerve (S2 and S3) and branches directly from S4	Closes anal canal
Superficial part	Surrounds lower part of anal canal	Anchored to perineal body and anococcygeal body		
Subcutaneous part	Surrounds anal aperture			

Urogenital triangle

The urogenital triangle of the perineum is the anterior half of the perineum and is oriented in the horizontal plane. It contains the roots of the external genitalia (Fig. 5.70) and the openings of the urogenital system.

The urogenital triangle is defined:

- laterally by the ischiopubic rami,
- posteriorly by an imaginary line between the ischial tuberosities, and
- anteriorly by the inferior margin of the pubic symphysis.

As with the anal triangle, the roof or ceiling of the urogenital triangle is the levator ani muscle.

Unlike the anal triangle, the urogenital triangle contains a strong fibromuscular support platform, the perineal membrane and deep perineal pouch (see pp. 457–459), which is attached to the pubic arch.

Anterior extensions of the ischio-anal fossae occur between the deep perineal pouch and the levator ani muscle on each side.

Between the perineal membrane and the membranous layer of superficial fascia is the **superficial perineal pouch**. The principal structures in this pouch are the erectile tissues of the penis and clitoris and associated skeletal muscles.

Structures in the superficial perineal pouch

The superficial perineal pouch contains:

- erectile structures that join together to form the penis in men and the clitoris in women, and
- skeletal muscles that are associated mainly with parts of the erectile structures attached to the perineal membrane and adjacent bone.

Each erectile structure consists of a central core of expandable vascular tissue and its surrounding connective tissue capsule.

Erectile tissues

Two sets of erectile structures join to form the penis and the clitoris.

A pair of cylindrically shaped **corpora cavernosa**, one on each side of the urogenital triangle, are anchored by their proximal ends to the pubic arch. These attached parts are often termed the **crura** (from the Latin for "legs") of the clitoris or the penis. The distal ends of the corpora, which are not attached to bone, form the body of the clitoris in women and the dorsal parts of the body of the penis in men.

The second set of erectile tissues surrounds the openings of the urogenital system.

- In women, a pair of erectile structures, termed the **bulbs of the vestibule**, are situated, one on each side, at the vaginal opening and are firmly anchored to the perineal membrane (Fig. 5.70A). Small bands of erectile tissues connect the anterior ends of these bulbs to a single, small, pea-shaped erectile mass, the **glans clitoris**, which is positioned in the midline at the end of the body of the clitoris and anterior to the opening of the urethra.
- In men, a single large erectile mass, the **corpus spongiosum**, is the structural equivalent to the bulbs of the vestibule, the glans clitoris, and the interconnecting bands of erectile tissues in women (Fig. 5.70B). The corpus spongiosum is anchored at its base to the perineal membrane. Its proximal end, which is not attached, forms the ventral part of the body of the penis and expands over the end of the body of the penis to form the glans penis. This pattern in men results from the absence of a vaginal opening and from the fusion of structures across the midline during embryological development. As the originally paired erectile structures fuse, they enclose the urethral opening and form an additional channel that ultimately becomes most of the penile part of the urethra. As a consequence of this fusion and growth in men, the urethra is enclosed by the corpus spongiosum and opens at the end of the

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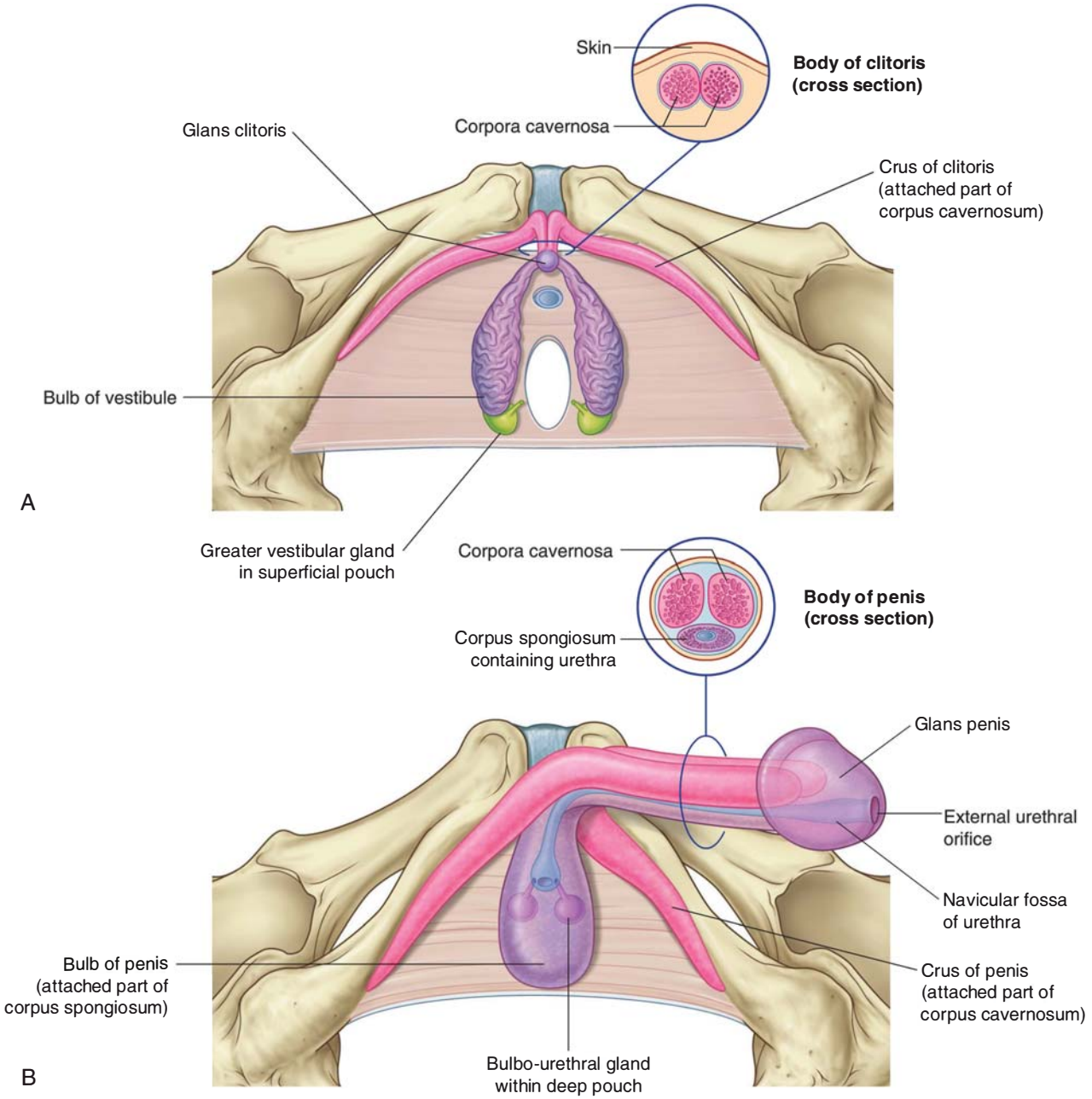


Fig. 5.70 Erectile tissues of clitoris and penis. **A.** Clitoris. **B.** Penis.

penis. This is unlike the situation in women, where the urethra is not enclosed by erectile tissue of the clitoris and opens directly into the vestibule of the perineum.

Clitoris

The clitoris is composed of two corpora cavernosa and the **glans clitoris** (Fig. 5.70A). As in the penis, it has an attached part (root) and a free part (body).

- Unlike the root of the penis, the **root of the clitoris** technically consists only of the two crura. (Although the bulbs of the vestibule are attached to the glans clitoris by thin bands of erectile tissue, they are not included in the attached part of the clitoris.)
- The **body of the clitoris**, which is formed only by the unattached parts of the two corpora cavernosa, angles posteriorly and is embedded in the connective tissues of the perineum.

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The body of the clitoris is supported by a suspensory ligament that attaches superiorly to the pubic symphysis. The glans clitoris is attached to the distal end of the body and is connected to the bulbs of the vestibule by small bands of erectile tissue. The glans clitoris is exposed in the perineum and the body of the clitoris can be palpated through skin.

Penis

The penis is composed mainly of the two corpora cavernosa and the single corpus spongiosum, which contains the urethra (Fig. 5.70B.) As in the clitoris, it has an attached part (root) and a free part (body):

- The **root of the penis** consists of the two crura, which are proximal parts of the corpora cavernosa attached to the pubic arch, and the **bulb of the penis**, which is the proximal part of the corpus spongiosum anchored to the perineal membrane.
- The **body of the penis**, which is covered entirely by skin, is formed by the tethering of the two proximal free parts of the corpora cavernosa and the related free part of the corpus spongiosum.

The base of the body of the penis is supported by two ligaments: the **suspensory ligament of the penis** (attached superiorly to the pubic symphysis), and the more superficially positioned **fundiform ligament of the penis** (attached above to the linea alba of the anterior abdominal wall and split below into two bands that pass on each side of the penis and unite inferiorly).

Because the anatomical position of the penis is erect, the paired corpora are defined as dorsal in the body of the penis and the single corpus spongiosum as ventral, even though the positions are reversed in the nonerect (flaccid) penis.

The corpus spongiosum expands to form the head of the penis (**glans penis**) over the distal ends of the corpora cavernosa (Fig. 5.70B).

Erection

Erection of the penis and clitoris is a vascular event generated by parasympathetic fibers carried in pelvic splanchnic nerves from the anterior rami of S2 to S4, which enter the inferior hypogastric part of the prevertebral plexus and ultimately pass through the deep perineal pouch and perineal membrane to innervate the erectile tissues. Stimulation of these nerves causes specific arteries in the erectile tissues to relax. This allows blood to fill the tissues, causing the penis and clitoris to become erect.

Arteries supplying the penis and clitoris are branches of the internal pudendal artery; branches of the pudendal nerve (S2 to S4) carry general sensory nerves from the penis and clitoris.

Greater vestibular glands

The greater vestibular glands (**Bartholin's glands**) are seen in women. They are small, pea-shaped mucous glands that lie posterior to the bulbs of the vestibule on each side of the vaginal opening and are the female homologues of the bulbo-urethral glands in men (Fig. 5.70). However, the bulbo-urethral glands are located within the deep perineal pouch, whereas the greater vestibular glands are in the superficial perineal pouch.

The duct of each greater vestibular gland opens into the vestibule of the perineum along the posterolateral margin of the vaginal opening.

Like the bulbo-urethral glands in men, the greater vestibular glands produce secretion during sexual arousal.

Muscles

The superficial perineal pouch contains three pairs of muscles: the ischiocavernosus, bulbospongiosus, and superficial transverse perineal muscles (Fig. 5.71 and Table 5.6). Two of these three pairs of muscles are associated with the roots of the penis and clitoris; the other pair is associated with the perineal body.

Ischiocavernosus

The two **ischiocavernosus muscles** cover the crura of the penis and clitoris (Fig. 5.71). Each muscle is anchored to the medial margin of the ischial tuberosity and related ischial ramus and passes forward to attach to the sides and inferior surface of the related crus, and forces blood from the crus into the body of the erect penis and clitoris.

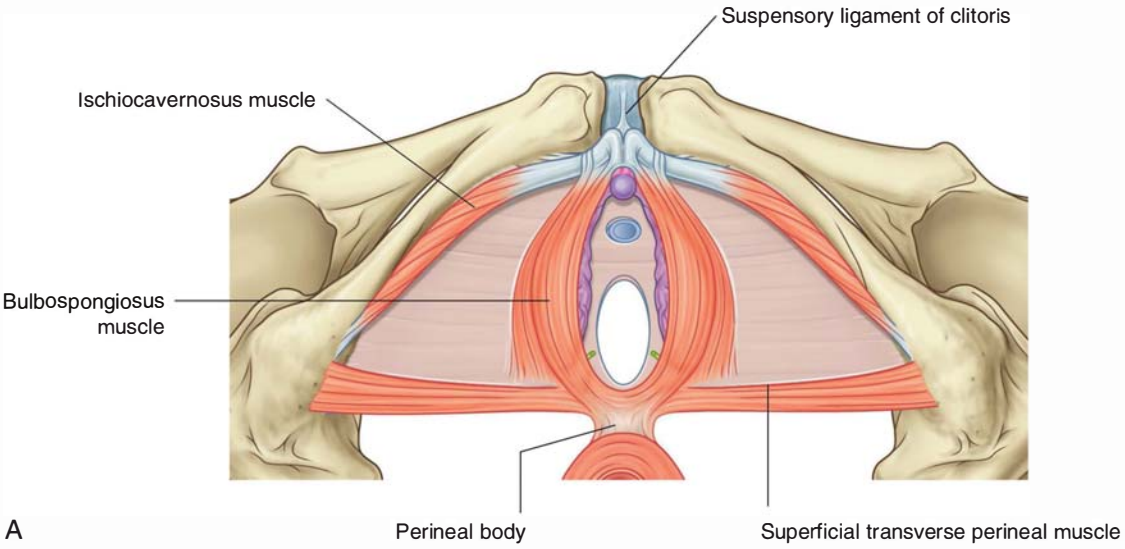
Bulbospongiosus

The two **bulbospongiosus muscles** are associated mainly with the bulbs of the vestibule in women and with the attached part of the corpus spongiosum in men (Fig. 5.71).

In women, each bulbospongiosus muscle is anchored posteriorly to the perineal body and courses anterolaterally over the inferior surface of the related greater vestibular gland and the bulb of the vestibule to attach to the surface of the bulb and to the perineal membrane (Fig. 5.71A). Other fibers course anterolaterally to blend with the fibers of the ischiocavernosus muscle, and still others travel anteriorly and arch over the body of the clitoris.

In men, the bulbospongiosus muscles are joined in the midline to a raphe on the inferior surface of the bulb of the penis. The raphe is anchored posteriorly to the perineal body. Muscle fibers course anterolaterally, on each side, from the raphe and perineal body to cover each side of the

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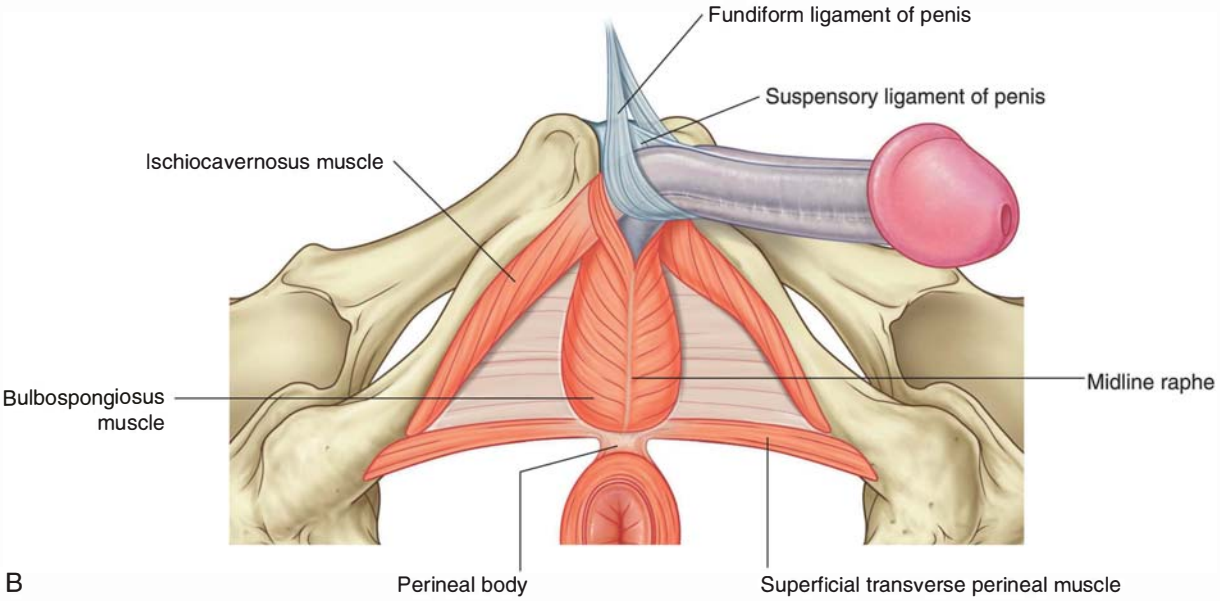


Fig. 5.71 Muscles in the superficial perineal pouch. **A.** In women. **B.** In men.

Table 5.6 Muscles of the superficial perineal pouch

Muscles	Origin	Insertion	Innervation	Function
Ischiocavernosus	Ischial tuberosity and ramus	Crus of penis and clitoris	Pudendal nerve (S2 to S4)	Move blood from crura into the body of the erect penis and clitoris
Bulbospongiosus	In women: perineal body In men: perineal body, midline raphe	In women: bulb of vestibule, perineal membrane, body of clitoris, and corpus cavernosum In men: bulbospongiosus, perineal membrane, corpus cavernosum	Pudendal nerve (S2 to S4)	Move blood from attached parts of the clitoris and penis into the glans In men: removal of residual urine from urethra after urination; pulsatile emission of semen during ejaculation
Superficial transverse perineal	Ischial tuberosity and ramus	Perineal body	Pudendal nerve (S2 to S4)	Stabilize the perineal body

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bulb of the penis and attach to the perineal membrane and connective tissue of the bulb. Others extend anterolaterally to associate with the crura and attach anteriorly to the ischiocavernosus muscles.

In both men and women, the bulbospongiosus muscles compress attached parts of the erect corpus spongiosum and bulbs of the vestibule and force blood into more distal regions, mainly the glans. In men, the bulbospongiosus muscles have two additional functions:

- They facilitate emptying of the bulbous part of the penile urethra following urination (micturition).
- Their reflex contraction during ejaculation is responsible for the pulsatile emission of semen from the penis.

Superficial transverse perineal muscles

The paired **superficial transverse perineal muscles** follow a course parallel to the posterior margin of the inferior surface of the perineal membrane (Fig. 5.71). These flat band-shaped muscles, which are attached to ischial tuberosities and rami, extend medially to the perineal body in the midline and stabilize the perineal body.

Superficial features of the external genitalia

In women

In women, the clitoris and vestibular apparatus, together with a number of skin and tissue folds, form the **vulva** (Fig. 5.72). On either side of the midline are two thin folds

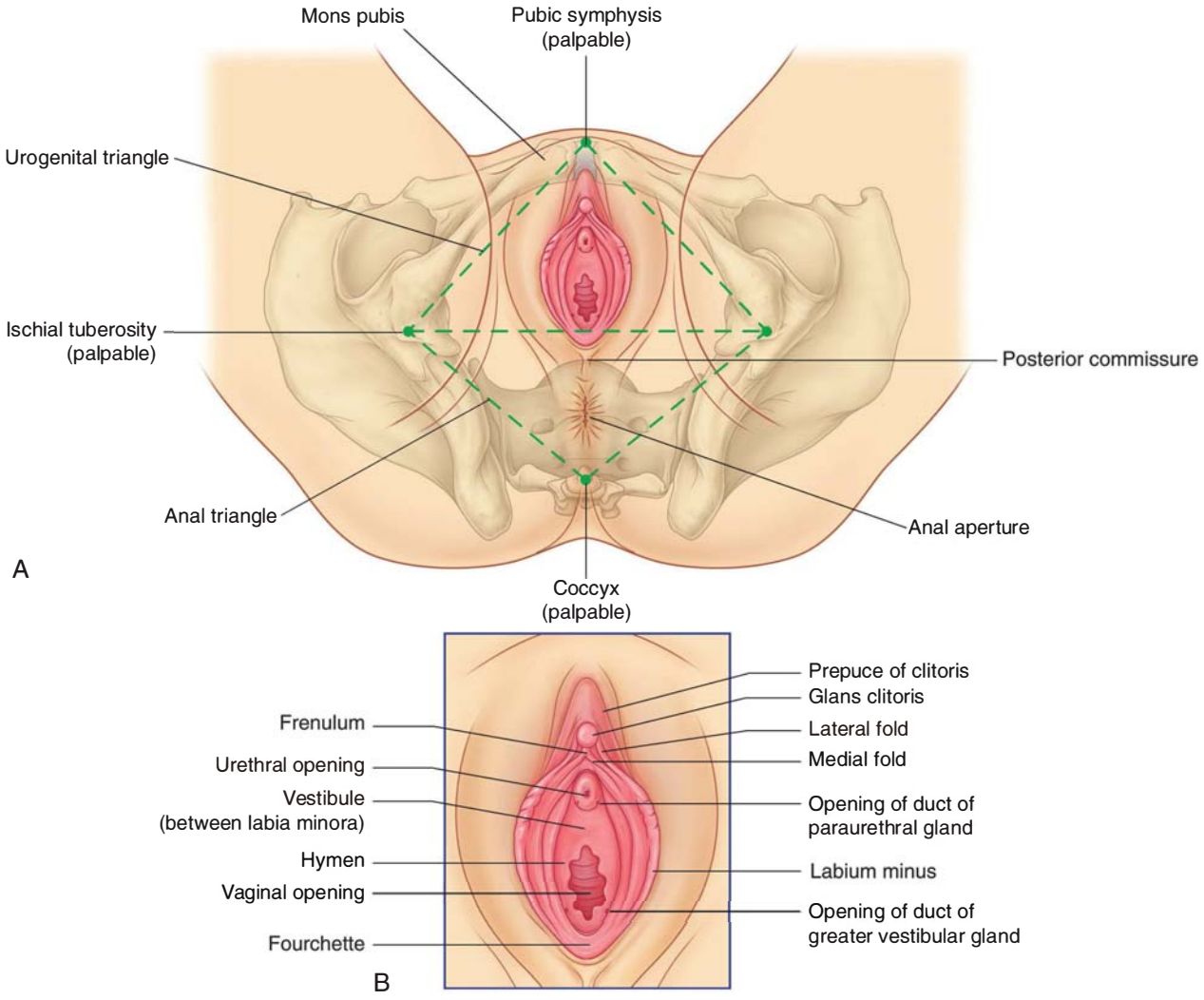


Fig. 5.72 Superficial features of the perineum in women. **A.** Overview. **B.** Close-up of external genitalia.

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of skin termed the **labia minora**. The region enclosed between them, and into which the urethra and vagina open, is the **vestibule**. Anteriorly, the labia minora each bifurcate, forming a medial and a lateral fold. The medial folds unite to form the **frenulum of the clitoris**, that joins the glans clitoris. The lateral folds unite ventrally over the glans clitoris and the body of the clitoris to form the **prepuce of the clitoris** (hood). The body of the clitoris extends anteriorly from the glans clitoris and is palpable deep to the prepuce and related skin. Posterior to the vestibule, the labia minora unite, forming a small transverse fold, the **frenulum of the labia minora** (the **fourchette**).

Within the vestibule, the vaginal orifice is surrounded to varying degrees by a ring-like fold of membrane, the **hymen**, which may have a small central perforation or may completely close the vaginal opening. Following rupture of the hymen (resulting from first sexual intercourse or injury), irregular remnants of the hymen fringe the vaginal opening.

The orifices of the urethra and the vagina are associated with the openings of glands. The ducts of the para-urethral glands (**Skene's glands**) open into the vestibule, one on each side of the lateral margin of the urethra. The ducts of the greater vestibular glands (Bartholin's glands) open adjacent to the posterolateral margin of the vaginal opening in the crease between the vaginal orifice and remnants of the hymen.

Lateral to the labia minora are two broad folds, the **labia majora**, which unite anteriorly to form the mons pubis. The **mons pubis** overlies the inferior aspect of the pubic symphysis and is anterior to the vestibule and the clitoris. Posteriorly, the labia majora do not unite and are separated by a depression termed the **posterior commissure**, which overlies the position of the perineal body.

In men

Superficial components of the genital organs in men consist of the scrotum and the penis (Fig. 5.73). The **scrotum** is the male homologue of the labia majora in

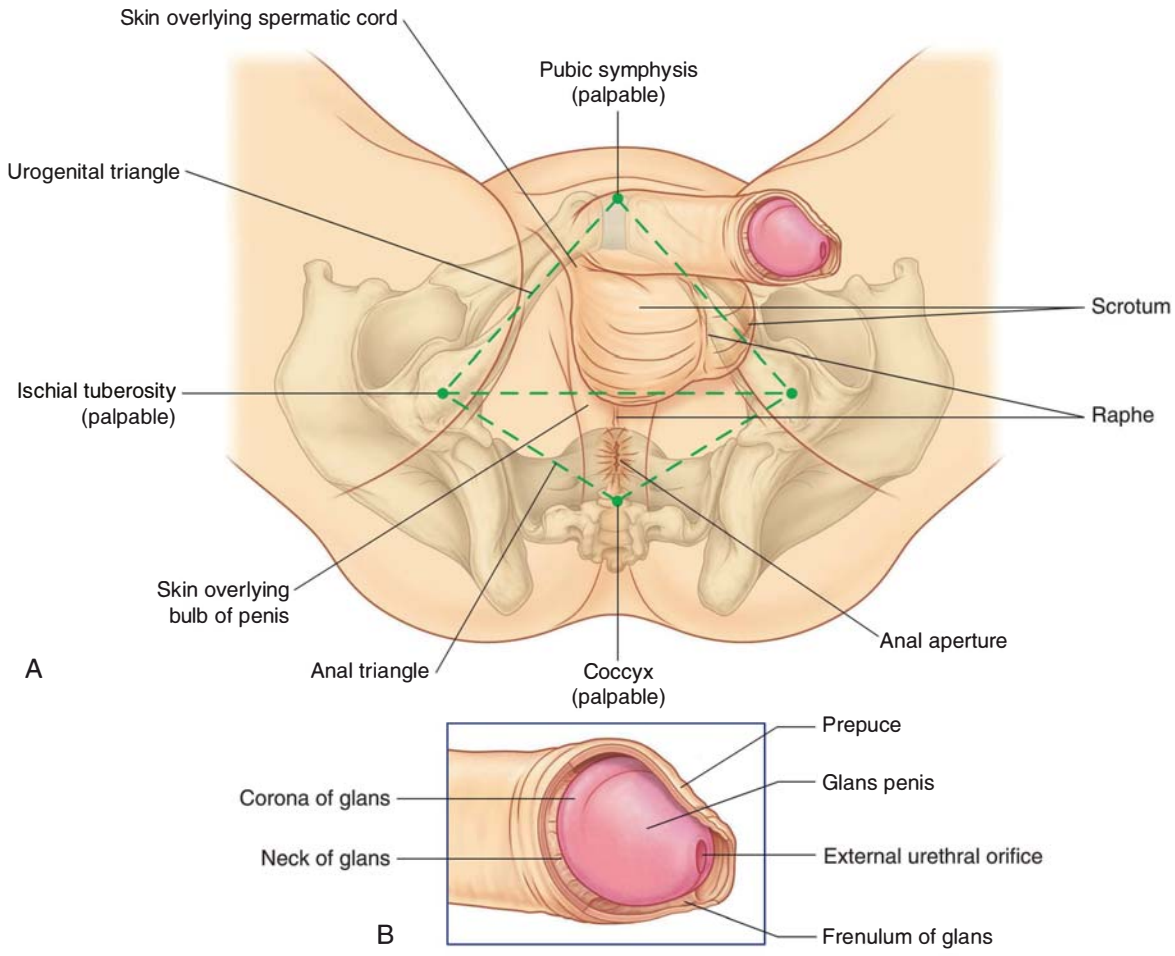


Fig. 5.73 Superficial features of the perineum in men. **A.** Overview. **B.** Close-up of external genitalia.

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women. In the fetus, labioscrotal swellings fuse across the midline, resulting in a single scrotum into which the testes and their associated musculofascial coverings, blood vessels, nerves, lymphatics, and drainage ducts descend from the abdomen. The remnant of the line of fusion between the labioscrotal swellings in the fetus is visible on the skin of the scrotum as a longitudinal midline **raphe** that extends from the anus, over the scrotal sac, and onto the inferior aspect of the body of the penis.

The **penis** consists of a root and body. The attached root of the penis is palpable posterior to the scrotum in the urogenital triangle of the perineum. The pendulous part of the penis (body of penis) is entirely covered by skin; the tip of the body is covered by the glans penis.

The external urethral orifice is a sagittal slit, normally positioned at the tip of the glans. The inferior margin of the urethral orifice is continuous with a midline **raphe of the penis**, which represents a line of fusion formed in the glans as the urethra develops in the fetus. The base of this raphe is continuous with the **frenulum** of the glans, which is a median fold of skin that attaches the glans to more loosely attached skin proximal to the glans. The base of the glans is expanded to form a raised circular margin (the **corona of the glans**); the two lateral ends of the corona join inferiorly at the midline raphe of the glans. The depression posterior to the corona is the neck of the glans. Normally, a fold of skin at the neck of the glans is continuous anteriorly with thin skin that tightly adheres to the glans and posteriorly with thicker skin loosely attached to the body. This fold, known as the prepuce, extends forward to cover the glans. The prepuce is removed during male circumcision, leaving the glans exposed.

Superficial fascia of the urogenital triangle

The superficial fascia of the urogenital triangle is continuous with similar fascia on the anterior abdominal wall.

As with the superficial fascia of the abdominal wall, the perineal fascia has a membranous layer on its deep surface. This membranous layer (**Colles' fascia**), is attached:

- posteriorly to the perineal membrane and therefore does not extend into the anal triangle (Fig. 5.74), and
- to the ischiopubic rami that form the lateral borders of the urogenital triangle and therefore does not extend into the thigh (Fig. 5.74).

It defines the external limits of the superficial perineal pouch, lines the scrotum or labia, and extends around the body of the penis and clitoris.

Anteriorly, the membranous layer of fascia is continuous over the pubic symphysis and pubic bones with the membranous layer of fascia on the anterior abdominal wall. In the lower lateral abdominal wall, the membranous layer of abdominal fascia is attached to the deep fascia of the thigh just inferior to the inguinal ligament.

Because the membranous layer of fascia encloses the superficial perineal pouch and continues up the anterior abdominal wall, fluids or infectious materials that accumulate in the pouch can track out of the perineum and onto the lower abdominal wall. This material will not track into the anal triangle or the thigh because the fascia fuses with deep tissues at the borders of these regions.

In the clinic

Urethral rupture

Urethral rupture may occur at a series of well-defined anatomical points.

The commonest injury is a rupture of the proximal spongy urethra below the perineal membrane. The urethra is usually torn when structures of the perineum are caught between a hard object (e.g., a steel beam or crossbar of a bicycle) and the inferior pubic arch. Urine escapes through the rupture into the superficial perineal pouch and descends into the scrotum and up onto the anterior abdominal wall deep to the superficial fascia.

In association with severe pelvic fractures, urethral rupture may occur at the prostatomembranous junction above the deep perineal pouch. The urine will extravasate into the true pelvis.

The worst and most serious urethral rupture is related to serious pelvic injuries where there is complete disruption of the puboprostatic ligaments. The prostate is dislocated superiorly not only by the ligamentous disruption but also by the extensive hematoma formed within the true pelvis. The diagnosis can be made by palpating the elevated prostate during a digital rectal examination.

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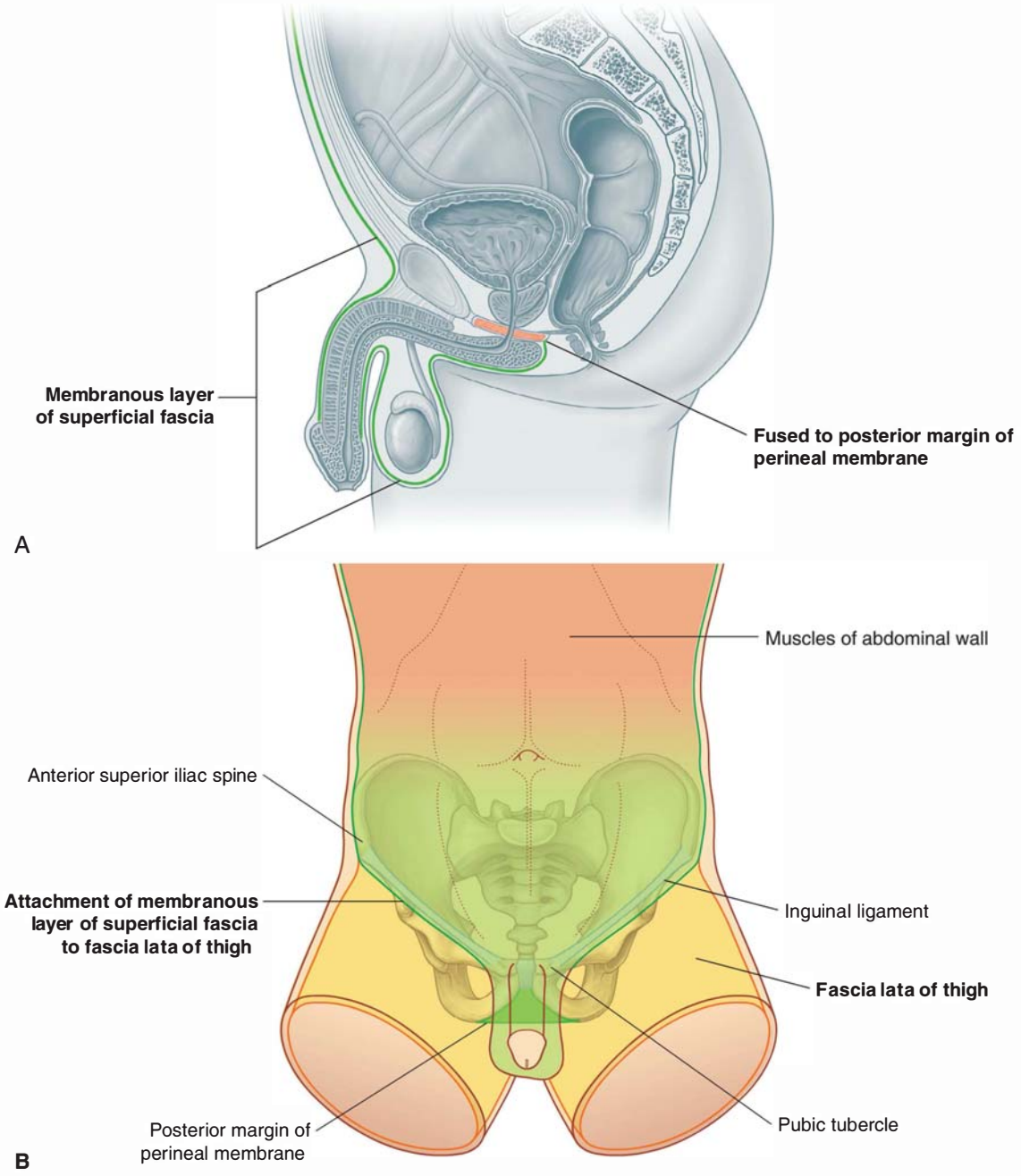


Fig. 5.74 Superficial fascia. **A.** Lateral view. **B.** Anterior view.

Somatic nerves

Pudendal nerve

The major somatic nerve of the perineum is the pudendal nerve. This nerve originates from the sacral plexus and carries fibers from spinal cord levels S2 to S4. It leaves the pelvic cavity through the greater sciatic foramen inferior to the piriformis muscle, passes around the sacrospinous ligament, and then enters the anal triangle of the perineum by passing medially through the lesser sciatic foramen. As it enters and courses through the perineum, it travels along the lateral wall of the ischio-anal fossa in the **pudendal canal**, which is a tubular compartment formed in the fascia that covers the obturator internus muscle. This pudendal canal also contains the internal pudendal artery and accompanying veins.

The pudendal nerve (Fig. 5.75) has three major terminal branches—the inferior rectal and perineal nerves and the dorsal nerve of the penis or clitoris—which are

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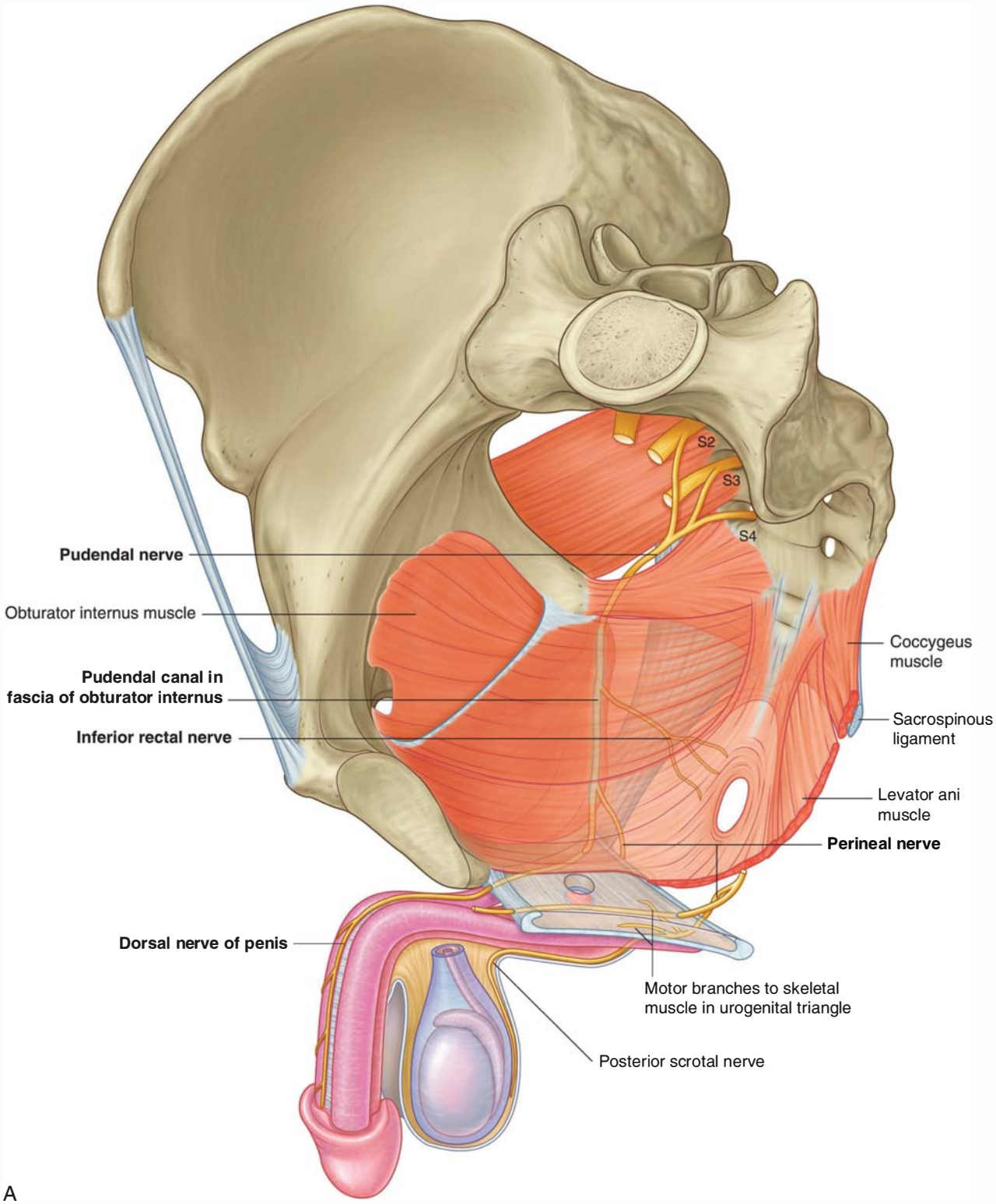


Fig. 5.75 Pudendal nerve. **A.** In men.

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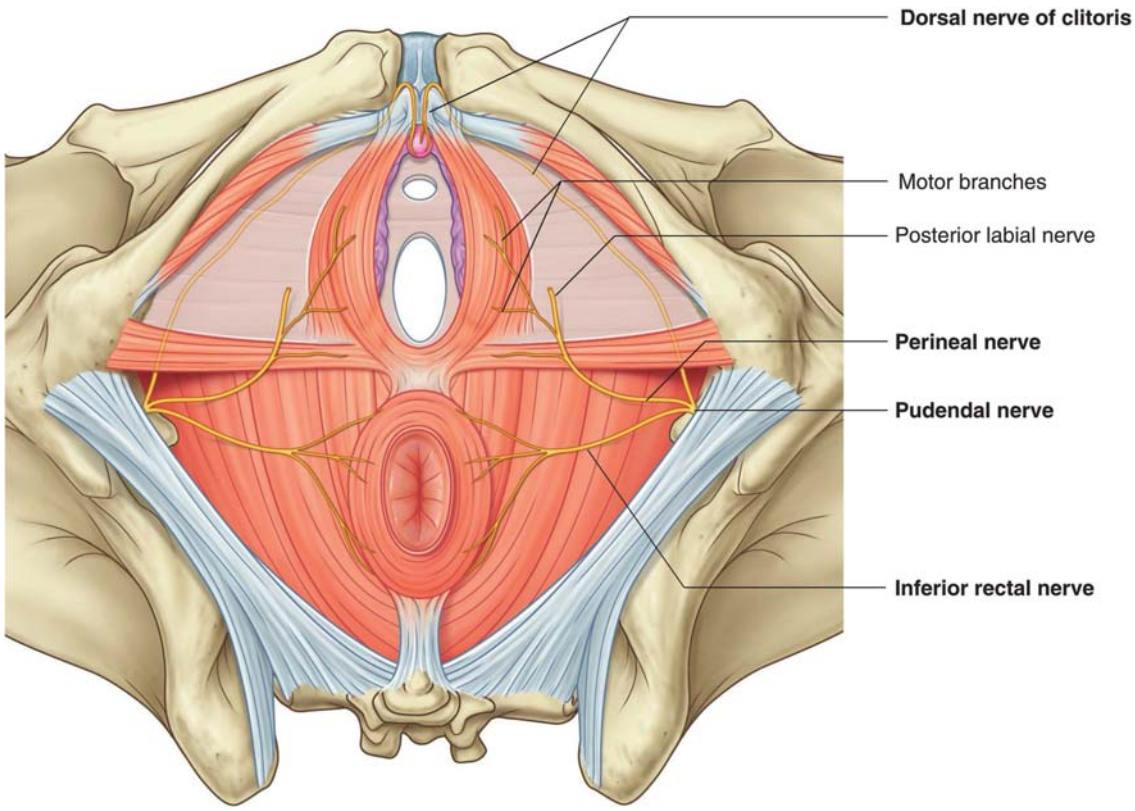


Fig. 5.75, cont'd B. In women.

accompanied by branches of the internal pudendal artery (Fig. 5.76).

- The **inferior rectal nerve** is often multiple, penetrates through the fascia of the pudendal canal, and courses medially across the ischio-anal fossa to innervate the external anal sphincter and related regions of the levator ani muscles. The nerve is also general sensory for the skin of the anal triangle.
- The **perineal nerve** passes into the urogenital triangle and gives rise to motor and cutaneous branches. The motor branches supply skeletal muscles in the superficial and deep perineal pouches. The largest of the sensory branches is the posterior scrotal nerve in men and the posterior labial nerve in women.
- The **dorsal nerve of the penis** and **clitoris** enters the deep perineal pouch (Fig. 5.75). It passes along the lateral margin of the pouch and then exits by passing inferiorly through the perineal membrane in a position just inferior to the pubic symphysis where it meets the body of the clitoris or the penis. It courses along the dorsal surface of the body to reach the glans. The dorsal nerve is sensory to the penis and clitoris, particularly to the glans.

Other somatic nerves

Other somatic nerves that enter the perineum are mainly sensory and include branches of the ilio-inguinal, genitofemoral, posterior femoral cutaneous, and anococcygeal nerves.

Visceral nerves

Visceral nerves enter the perineum by two routes:

- Those to the skin, which consist mainly of postganglionic sympathetics, are delivered into the region along the pudendal nerve. These fibers join the pudendal nerve from gray rami communicantes that connect pelvic parts of the sympathetic trunks to the anterior rami of the sacral spinal nerves (see p. 487 and Fig. 5.61).
- Those to erectile tissues enter the region mainly by passing through the deep perineal pouch from the inferior hypogastric plexus in the pelvic cavity (see p. 494 and Fig. 5.62B). The fibers that stimulate erection are parasympathetic fibers, which enter the inferior hypogastric plexus via pelvic splanchnic nerves from spinal cord levels of S2 to S4 (see Fig. 5.62A,B).

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Blood vessels

Arteries

The most significant artery of the perineum is the internal pudendal artery (Fig. 5.76). Other arteries entering the area include the external pudendal, the testicular, and the cremasteric arteries.

Internal pudendal artery

The **internal pudendal artery** originates as a branch of the anterior trunk of the internal iliac artery in the pelvis (Fig. 5.76). Along with the pudendal nerve, it leaves the pelvis through the greater sciatic foramen inferior to the piriformis muscle. It passes around the ischial spine, where the artery lies lateral to the nerve, enters the perineum by coursing through the lesser sciatic foramen, and accompanies the pudendal nerve in the pudendal canal on the lateral wall of the ischio-anal fossa.

The branches of the internal pudendal artery are similar to those of the pudendal nerve in the perineum and include the inferior rectal and perineal arteries, and branches to the erectile tissues of the penis and clitoris (Fig. 5.76).

Inferior rectal arteries

One or more **inferior rectal arteries** originate from the internal pudendal artery in the anal triangle and cross the ischio-anal fossa medially to branch and supply muscle and related skin (Fig. 5.76). They anastomose with middle and superior rectal arteries from the internal iliac artery and the inferior mesenteric artery, respectively, to form a network of vessels that supply the rectum and anal canal.

Perineal artery

The **perineal artery** originates near the anterior end of the pudendal canal and gives off a transverse perineal branch, and a posterior scrotal or labial artery to surrounding tissues and skin (Fig. 5.76).

Terminal part of the internal pudendal artery

The terminal part of the internal pudendal artery accompanies the dorsal nerve of the penis or clitoris into the deep perineal pouch and supplies branches to the tissues in the deep perineal pouch and erectile tissues.

Branches that supply the erectile tissues in men include the artery to the bulb of the penis, the urethral artery, the deep artery of the penis, and the dorsal artery of the penis (Fig. 5.76).

- The **artery of the bulb of the penis** has a branch that supplies the bulbo-urethral gland and then penetrates the perineal membrane to supply the corpus spongiosum.
- A **urethral artery** also penetrates the perineal membrane and supplies the penile urethra and surrounding erectile tissue to the glans.
- Near the anterior margin of the deep perineal pouch, the internal pudendal artery bifurcates into two terminal branches. A **deep artery of the penis** penetrates the perineal membrane to enter the crus and supply the crus and corpus cavernosum of the body. The **dorsal artery of the penis** penetrates the anterior margin of the perineal membrane to meet the dorsal surface of the body of the penis. The vessel courses along the dorsal surface of the penis, medial to the dorsal nerve, and supplies the glans penis and superficial tissues of the penis; it also anastomoses with branches of the deep artery of the penis and the urethral artery.

Branches that supply the erectile tissues in women are similar to those in men.

- **Arteries of the bulb of the vestibule** supply the bulb of the vestibule and related vagina.
- **Deep arteries of the clitoris** supply the crura and corpus cavernosum of the body.
- **Dorsal arteries of the clitoris** supply surrounding tissues and the glans.

External pudendal arteries

The **external pudendal arteries** consist of a superficial vessel and a deep vessel, which originate in the femoral artery in the thigh. They course medially to enter the perineum anteriorly and supply related skin of the penis and scrotum or the clitoris and labia majora.

Testicular and cremasteric arteries

In men, the **testicular arteries** originate from the abdominal aorta and descend into the scrotum through the inguinal canal to supply the testes. Also, **cremasteric arteries**, which originate from the inferior epigastric branch of the external iliac artery, accompany the spermatic cord into the scrotum.

In women, small cremasteric arteries follow the round ligament of the uterus through the inguinal canal.

Veins

Veins in the perineum generally accompany the arteries and join the **internal pudendal veins** that connect with the **internal iliac vein** in the pelvis (Fig. 5.77). The exception is the **deep dorsal vein of the penis or clitoris** that drains mainly the glans and the corpora cavernosa. The deep dorsal vein courses along the midline between the

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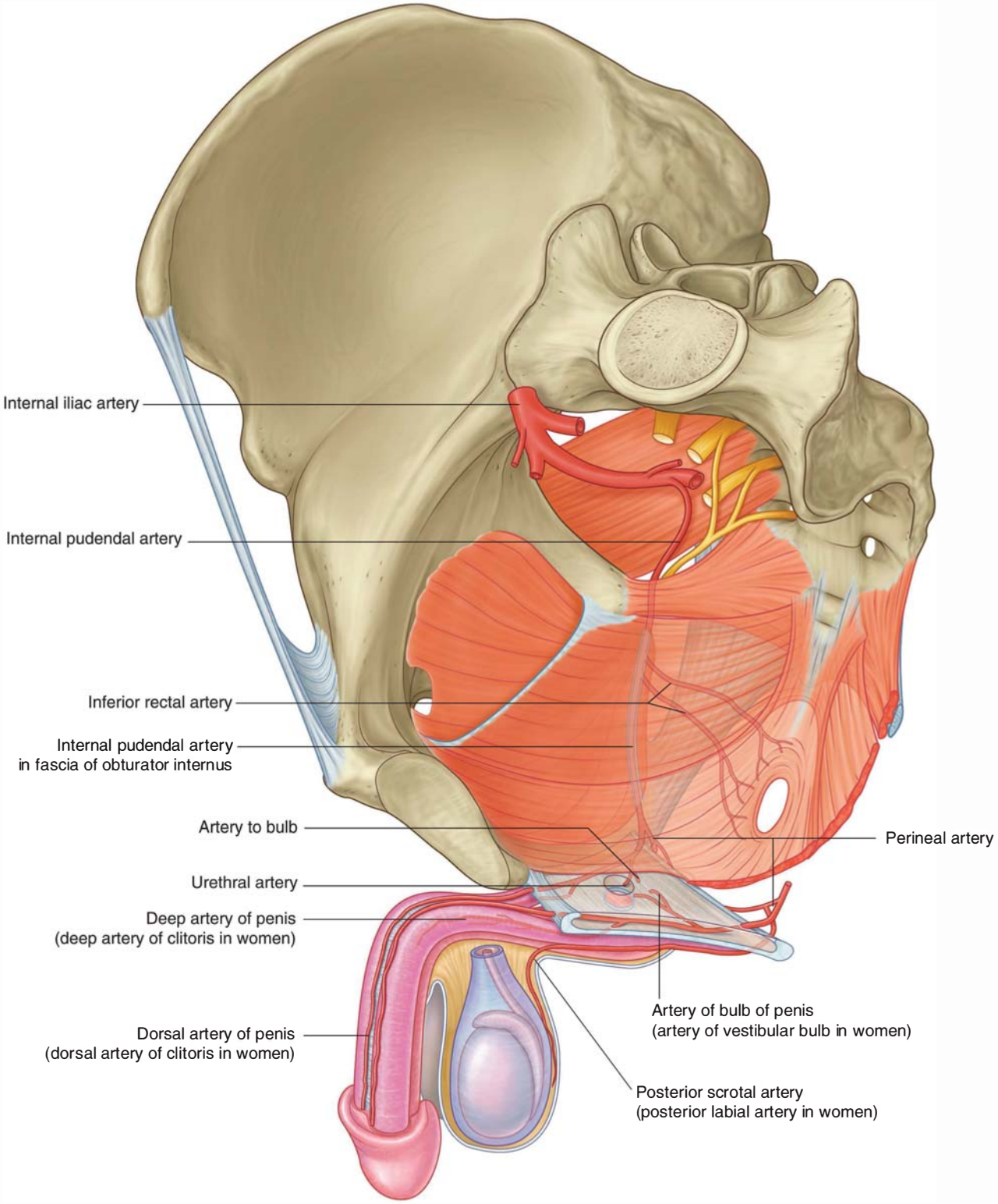


Fig. 5.76 Arteries in the perineum.

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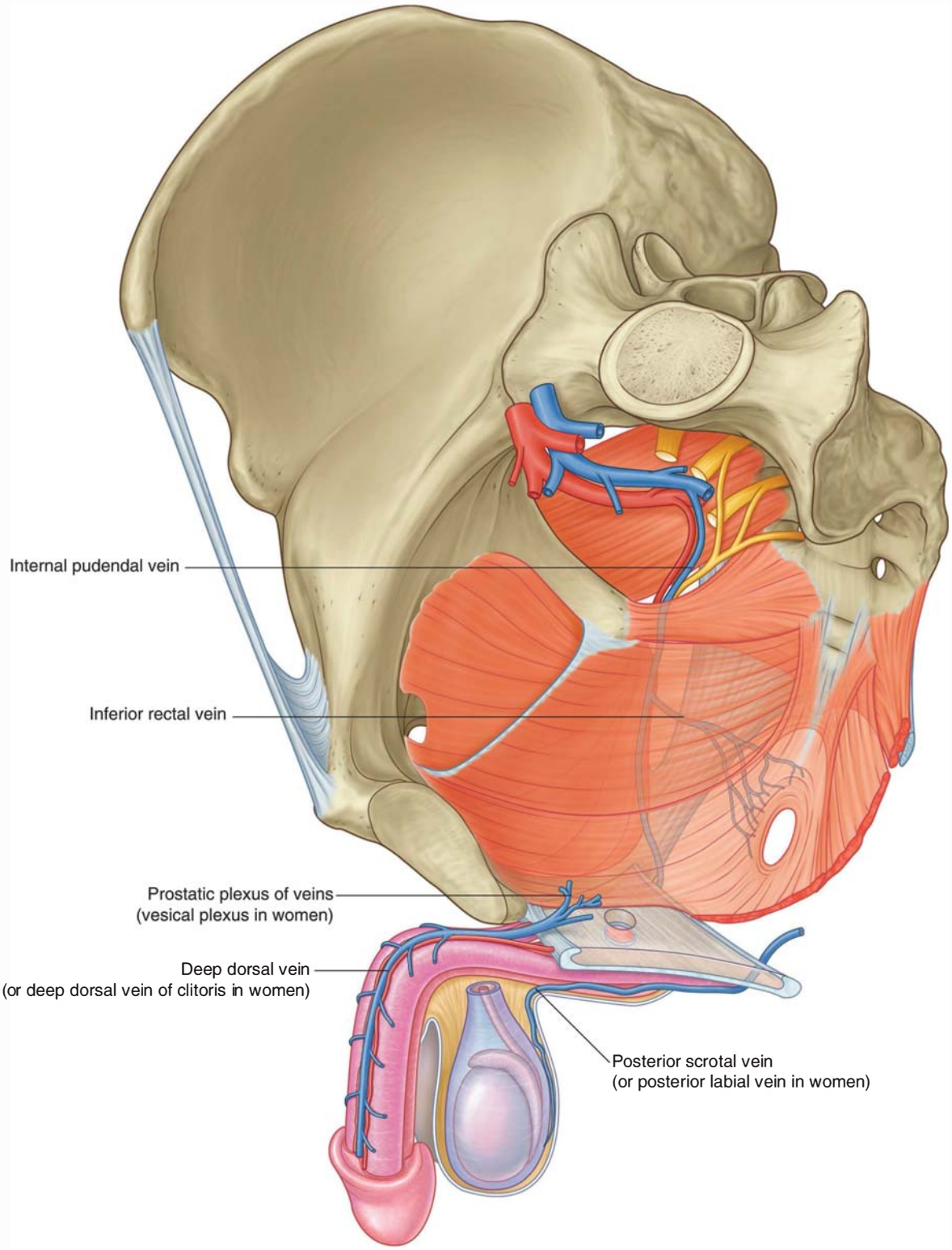


Fig. 5.77 Perineal veins.

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dorsal arteries on each side of the body of the penis or clitoris, passes though the gap between the inferior pubic ligament and the deep perineal pouch, and connects with the plexus of veins surrounding the prostate in men or bladder in women.

External pudendal veins, which drain anterior parts of the labia majora or the scrotum and overlap with the area of drainage of the internal pudendal veins, connect with the femoral vein in the thigh. Superficial dorsal veins of the penis or clitoris that drain skin are tributaries of the external pudendal veins.

Lymphatics

Lymphatic vessels from deep parts of the perineum accompany the internal pudendal blood vessels and drain mainly into **internal iliac nodes** in the pelvis.

Lymphatic channels from superficial tissues of the penis or the clitoris accompany the superficial external pudendal blood vessels and drain mainly into **superficial inguinal nodes**, as do lymphatic channels from the scrotum or labia majora (Fig. 5.78). The glans penis, glans clitoris, labia minora, and terminal inferior end of the vagina drain into **deep inguinal nodes** and **external iliac nodes**.

Lymphatics from the testes drain via channels that ascend in the spermatic cord, pass through the inguinal canal, and course up the posterior abdominal wall to connect directly with **lateral aortic** or **lumbar nodes** and **pre-aortic nodes** around the aorta, at approximately vertebral levels LI and LII. Therefore disease from the testes tracks superiorly to nodes high in the posterior abdominal wall and not to inguinal or iliac nodes.

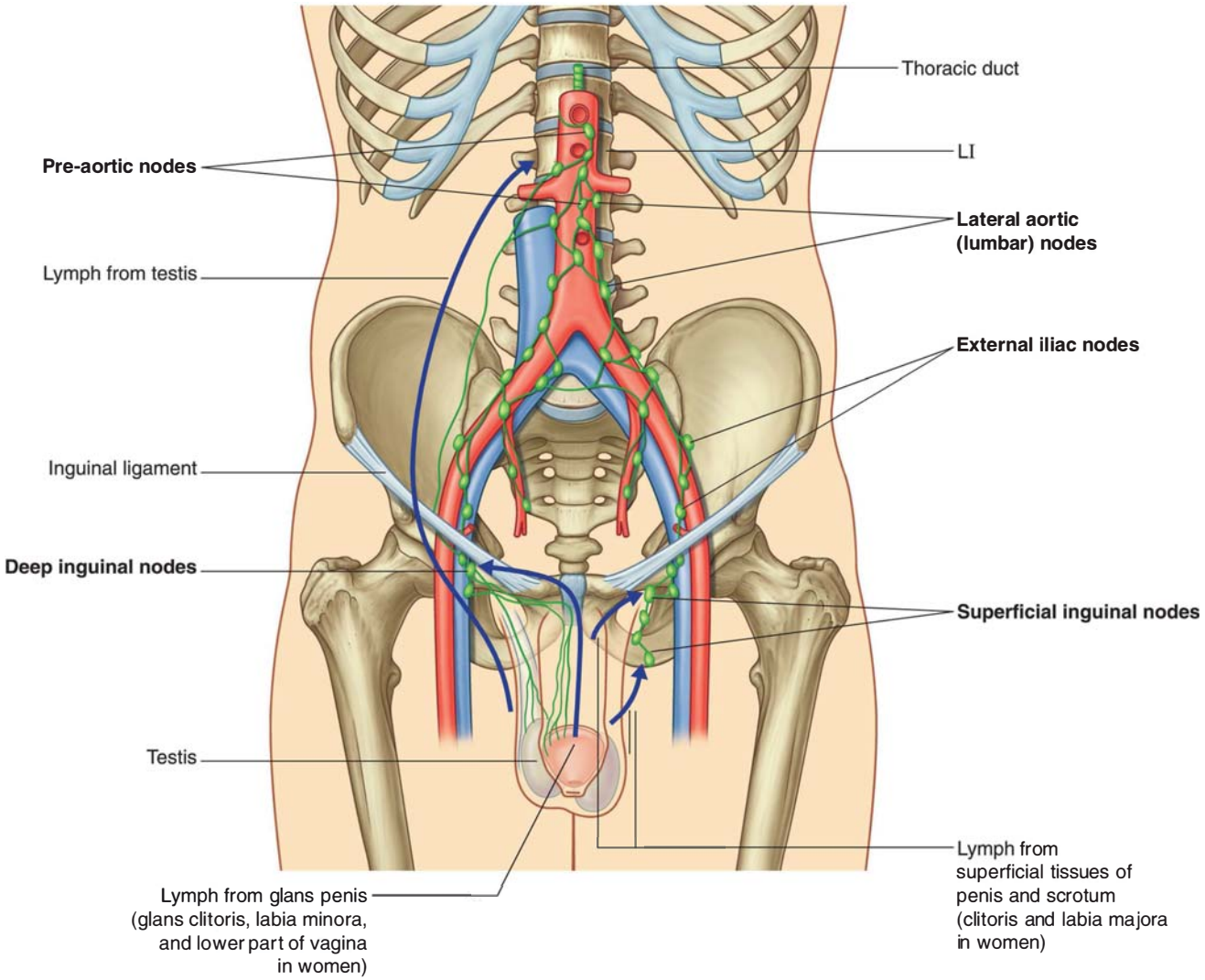


Fig. 5.78 Lymphatic drainage of the perineum.

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Surface anatomy

Surface anatomy of the pelvis and perineum

Palpable bony features of the pelvis are used as landmarks for:

- locating soft tissue structures,
- visualizing the orientation of the pelvic inlet, and
- defining the margins of the perineum.

The ability to recognize the normal appearance of structures in the perineum is an essential part of a physical examination.

In women, the cervix can be visualized directly by opening the vaginal canal using a speculum.

In men, the size and texture of the prostate in the pelvic cavity can be assessed by digital palpation through the anal aperture.

Orientation of the pelvis and perineum in the anatomical position

In the anatomical position, the anterior superior iliac spines and the anterior superior edge of the pubic symphysis lie in the same vertical plane. The pelvic inlet faces anterosuperiorly. The urogenital triangle of the perineum is oriented in an almost horizontal plane and faces inferiorly, whereas the anal triangle is more vertical and faces posteriorly (Figs. 5.79 and 5.80).

How to define the margins of the perineum

The pubic symphysis, ischial tuberosities, and tip of the sacrum are palpable on patients and can be used to define the boundaries of the perineum. This is best done with patients lying on their backs with their thighs flexed and abducted in the lithotomy position (Fig. 5.81).

- The ischial tuberosities are palpable on each side as large bony masses near the crease of skin (gluteal fold) between the thigh and gluteal region. They mark the lateral corners of the diamond-shaped perineum.
- The tip of the coccyx is palpable in the midline posterior to the anal aperture and marks the most posterior limit of the perineum.
- The anterior limit of the perineum is the pubic symphysis. In women, this is palpable in the midline deep to the

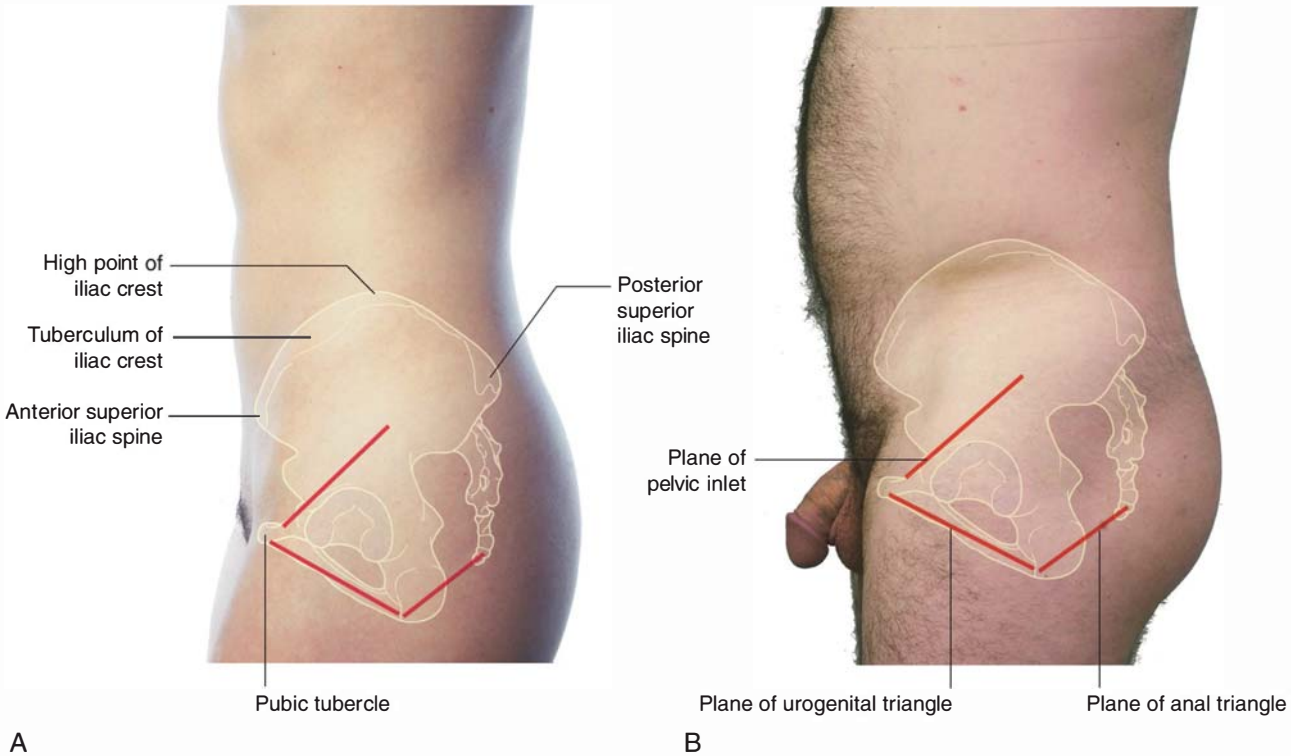


Fig. 5.79 Lateral view of the pelvic area with the position of the skeletal features indicated. The orientation of the pelvic inlet, urogenital triangle, and anal triangle is also shown. **A.** In a woman. **B.** In a man.

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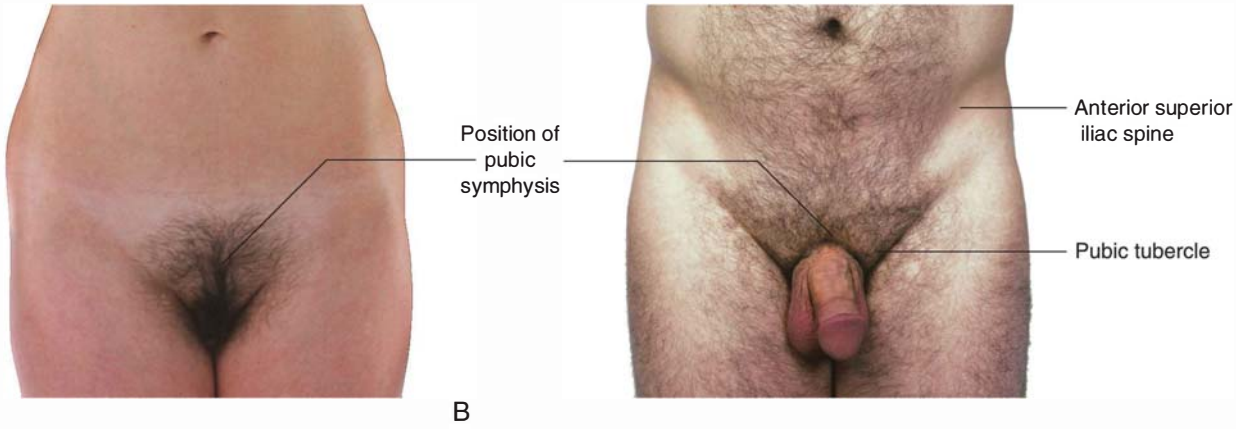
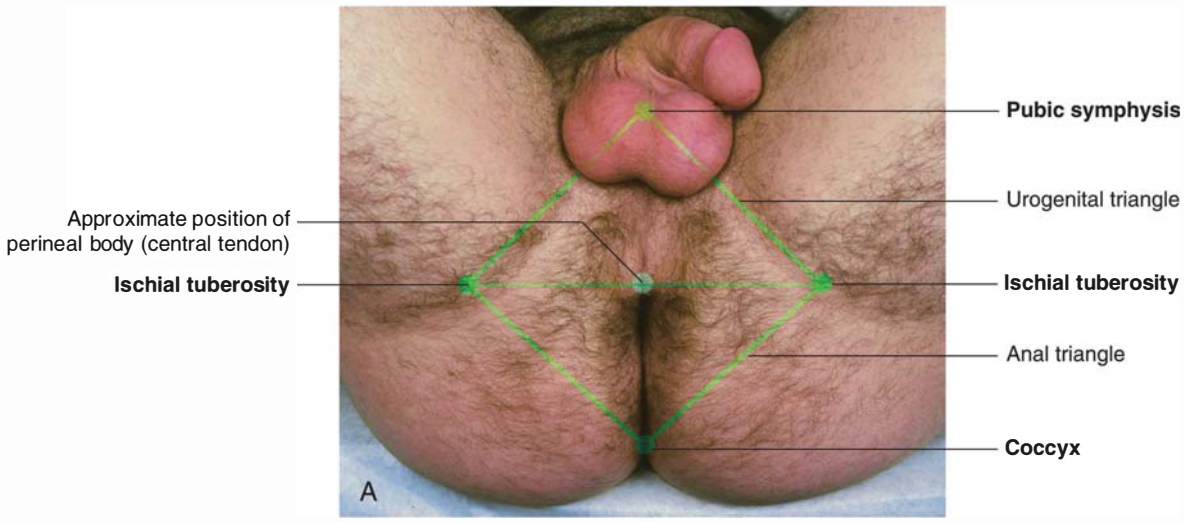


Fig. 5.80 Anterior view of the pelvic area. **A.** In a woman showing the position of the pubic symphysis. **B.** In a man showing the position of the pubic tubercle, pubic symphysis, and anterior superior iliac spine.



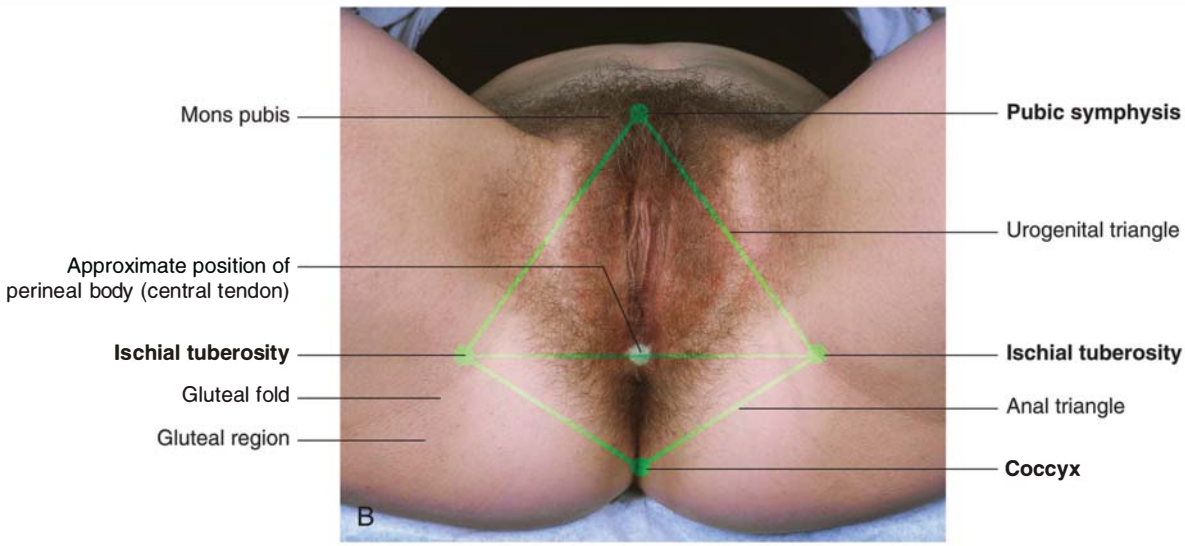


Fig. 5.81 Inferior view of the perineum in the lithotomy position. Boundaries, subdivisions, and palpable landmarks are indicated. **A.** In a man. **B.** In a woman.

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mons pubis. In men, the pubic symphysis is palpable immediately superior to where the body of the penis joins the lower abdominal wall.

Imaginary lines that join the ischial tuberosities with the pubic symphysis in front, and with the tip of the coccyx behind, outline the diamond-shaped perineum. An additional line between the ischial tuberosities divides the perineum into two triangles, the urogenital triangle anteriorly and anal triangle posteriorly. This line also approximates the position of the posterior margin of the perineal membrane. The midpoint of this line marks the location of the perineal body or central tendon of the perineum.

Identification of structures in the anal triangle

The anal triangle is the posterior half of the perineum. The base of the triangle faces anteriorly and is an imaginary line joining the two ischial tuberosities. The apex of the triangle is the tip of the coccyx; the lateral margins can be approximated by lines joining the coccyx to the ischial tuberosities. In both women and men, the major feature of the anal triangle is the anal aperture in the center of the triangle. Fat fills the ischio-anal fossa on each side of the anal aperture (Fig. 5.82).

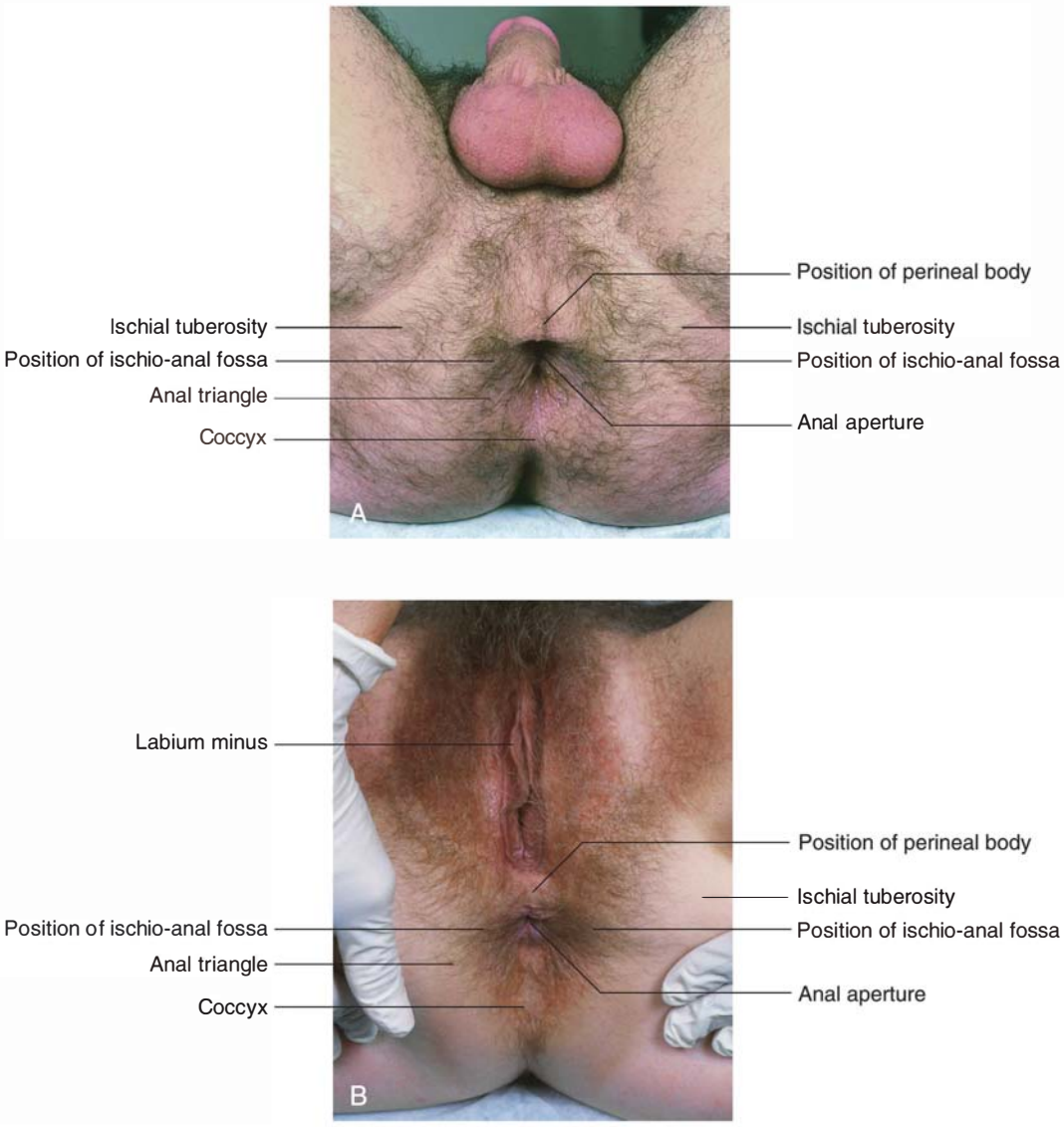


Fig. 5.82 Anal triangle with the anal aperture and position of the ischio-anal fossae indicated. **A.** In a man. **B.** In a woman.

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Identification of structures in the urogenital triangle of women

The urogenital triangle is the anterior half of the perineum. The base of the triangle faces posteriorly and is an imaginary line joining the two ischial tuberosities. The apex of the triangle is the pubic symphysis. The lateral margins can be approximated by lines joining the pubic symphysis to the ischial tuberosities. These lines overlie the ischiopubic rami, which can be felt on deep palpation.

In women, the major contents of the urogenital triangle are the clitoris, the vestibule, and skin folds that together form the vulva (Fig. 5.83A,B).

Two thin skin folds, the labia minora, enclose between them a space termed the vestibule into which the vagina and the urethra open (Fig. 5.83C). Gentle lateral traction on the labia minora opens the vestibule and reveals a soft tissue mound on which the urethra opens. The para-urethral (Skene's) glands, one on each side, open into the skin crease between the urethra and the labia minora (Fig. 5.83D).

Posterior to the urethra is the vaginal opening. The vaginal opening (introitus) is ringed by remnants of the hymen that originally closes the vaginal orifice and is usually ruptured during the first sexual intercourse. The ducts of the greater vestibular (Bartholin's) glands, one on

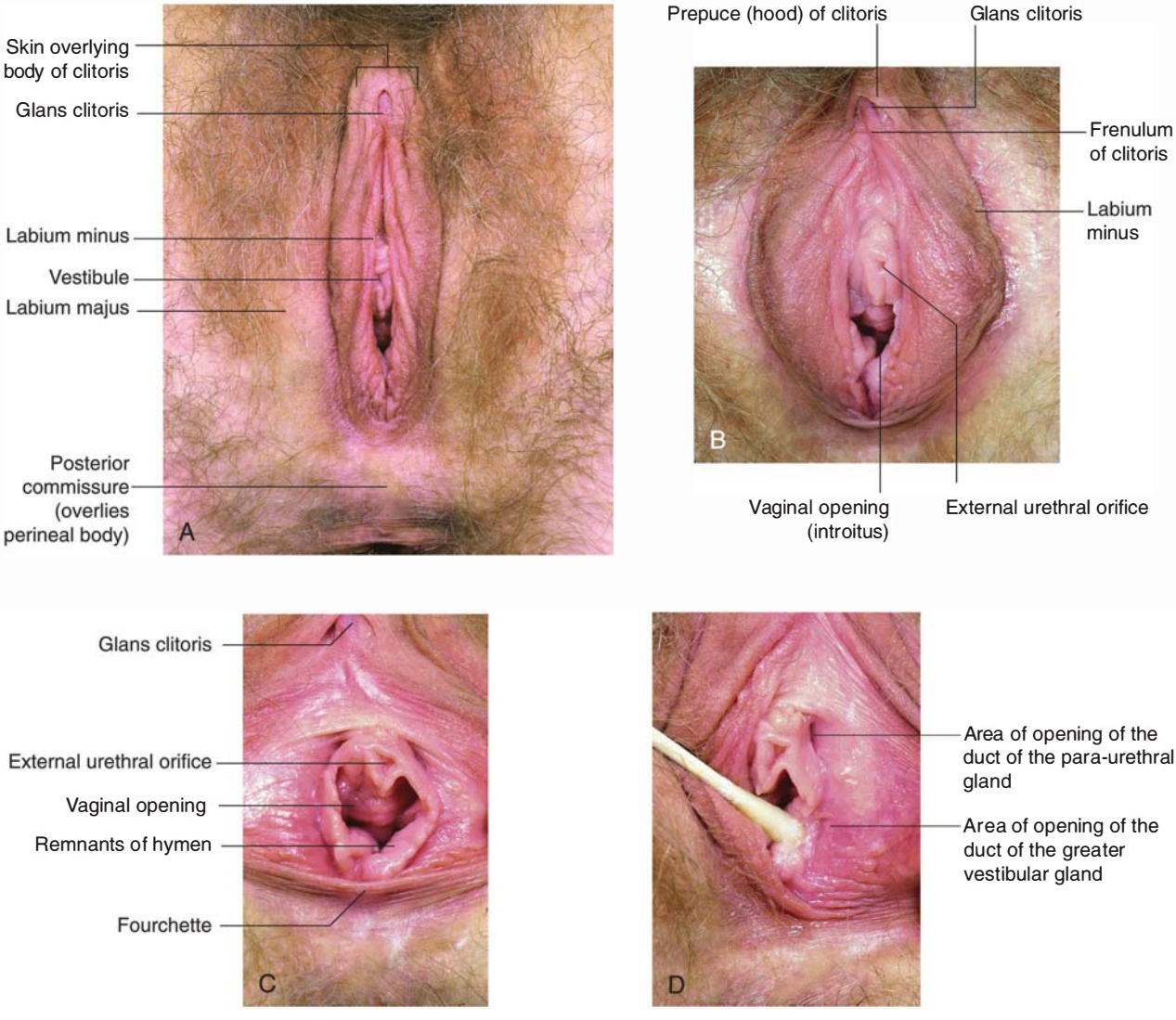


Fig. 5.83 Structures in the urogenital triangle of a woman. **A.** Inferior view of the urogenital triangle of a woman with major features indicated. **B.** Inferior view of the vestibule. The labia minora have been pulled apart to open the vestibule. Also indicated are the glans clitoris, the clitoral hood, and the frenulum of the clitoris. **C.** Inferior view of the vestibule showing the urethral and vaginal orifices and the hymen. The labia minora have been pulled further apart than in Figure 5.83B. **D.** Inferior view of the vestibule with the left labium minus pulled to the side to show the regions of the vestibule into which the greater vestibular and para-urethral glands open.

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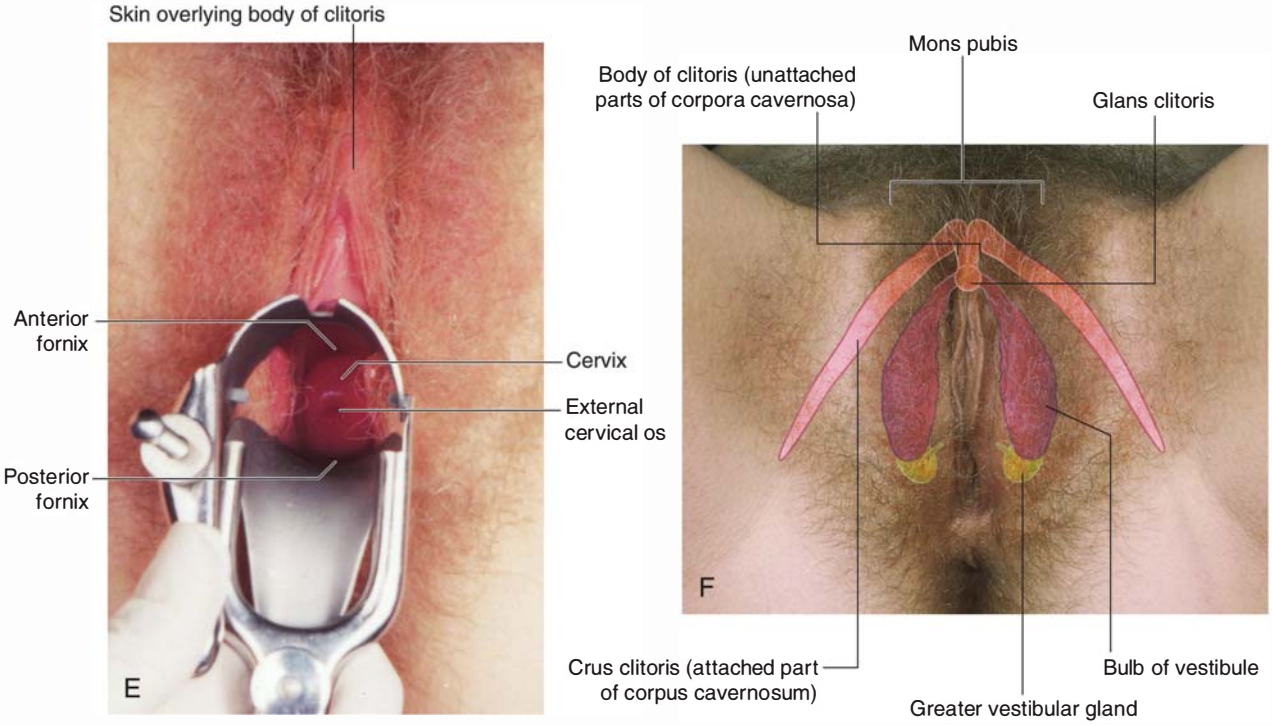


Fig. 5.83, cont'd **E.** View through the vaginal canal of the cervix. **F.** Inferior view of the urogenital triangle of a woman with the erectile tissues of the clitoris and vestibule and the greater vestibular glands indicated with overlays.

each side, open into the skin crease between the hymen and the adjacent labium minus (Fig. 5.83D).

The labia minora each bifurcate anteriorly into medial and lateral folds. The medial folds unite at the midline to form the frenulum of the clitoris. The larger lateral folds also unite across the midline to form the clitoral hood or prepuce that covers the glans clitoris and distal parts of the body of the clitoris. Posterior to the vaginal orifice, the labia minora join, forming a transverse skin fold (the fourchette).

The labia majora are broad folds positioned lateral to the labia minora. They come together in front to form the mons pubis, which overlies the inferior aspect of the pubic symphysis. The posterior ends of the labia majora are separated by a depression termed the posterior commissure, which overlies the position of the perineal body.

The cervix is visible when the vaginal canal is opened with a speculum (Fig. 5.83E). The external cervical os opens onto the surface of the dome-shaped cervix. A recess or gutter, termed the fornix, occurs between the cervix and the vaginal wall and is further subdivided, based on location, into anterior, posterior, and lateral fornices.

The roots of the clitoris occur deep to surface features of the perineum and are attached to the ischiopubic rami and the perineal membrane.

The bulbs of the vestibule (Fig. 5.83F), composed of erectile tissues, lie deep to the labia minora on either side of the vestibule. These erectile masses are continuous, via thin bands of erectile tissues, with the glans clitoris, which is visible under the clitoral hood. The greater vestibular glands occur posterior to the bulbs of the vestibule on either side of the vaginal orifice.

The crura of the clitoris are attached, one on each side, to the ischiopubic rami. Each crus is formed by the attached part of the corpus cavernosum. Anteriorly, these erectile corpora detach from bone, curve posteroinferiorly, and unite to form the body of the clitoris.

The body of the clitoris underlies the ridge of skin immediately anterior to the clitoral hood (prepuce). The glans clitoris is positioned at the end of the body of the clitoris.

Identification of structures in the urogenital triangle of men

In men, the urogenital triangle contains the root of the penis. The testes and associated structures, although they migrate into the scrotum from the abdomen, are generally evaluated with the penis during a physical examination.

The scrotum in men is homologous to the labia majora in women. Each oval testis is readily palpable through the

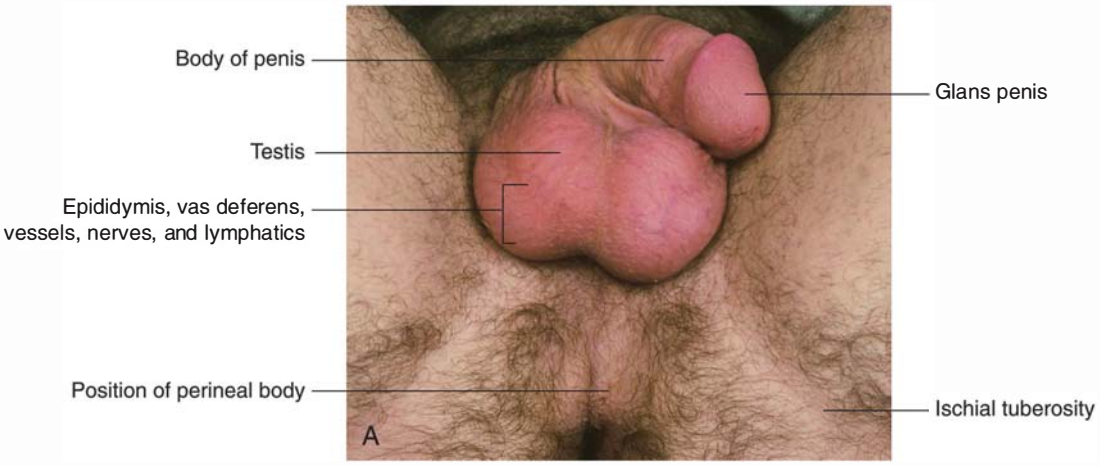
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skin of the scrotum (Fig. 5.84A). Posterolateral to the testis is an elongated mass of tissue, often visible as a raised ridge that contains lymphatics and blood vessels of the testis, and the epididymis and ductus deferens. A midline raphe (Fig. 5.84B) is visible on the skin separating left and right sides of the scrotum. In some individuals, this raphe is prominent and extends from the anal aperture, over the scrotum and along the ventral surface of the body of the penis, to the frenulum of the glans.

The root of the penis is formed by the attached parts of the corpus spongiosum and the corpora cavernosa. The corpus spongiosum is attached to the perineal membrane and can be easily palpated as a large mass anterior to the perineal body. This mass, which is covered by the bulbospongiosus muscles, is the bulb of penis.

The corpus spongiosum detaches from the perineal membrane anteriorly, becomes the ventral part of the body of the penis (shaft of penis), and eventually terminates as the expanded glans penis (Fig. 5.84C,D).

The crura of the penis, one crus on each side, are the attached parts of the corpora cavernosa and are anchored to the ischiopubic rami (Fig. 5.84E). The corpora cavernosa are unattached anteriorly and become the paired erectile masses that form the dorsal part of the body of the penis. The glans penis caps the anterior ends of the corpora cavernosa.



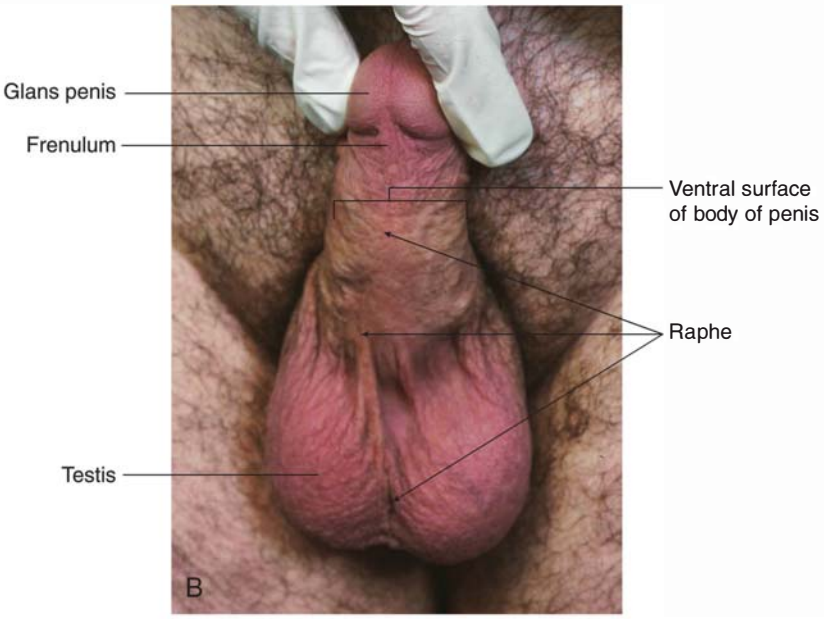


Fig. 5.84 Structures in the urogenital triangle of a man. **A.** Inferior view. **B.** Ventral surface of the body of the penis.

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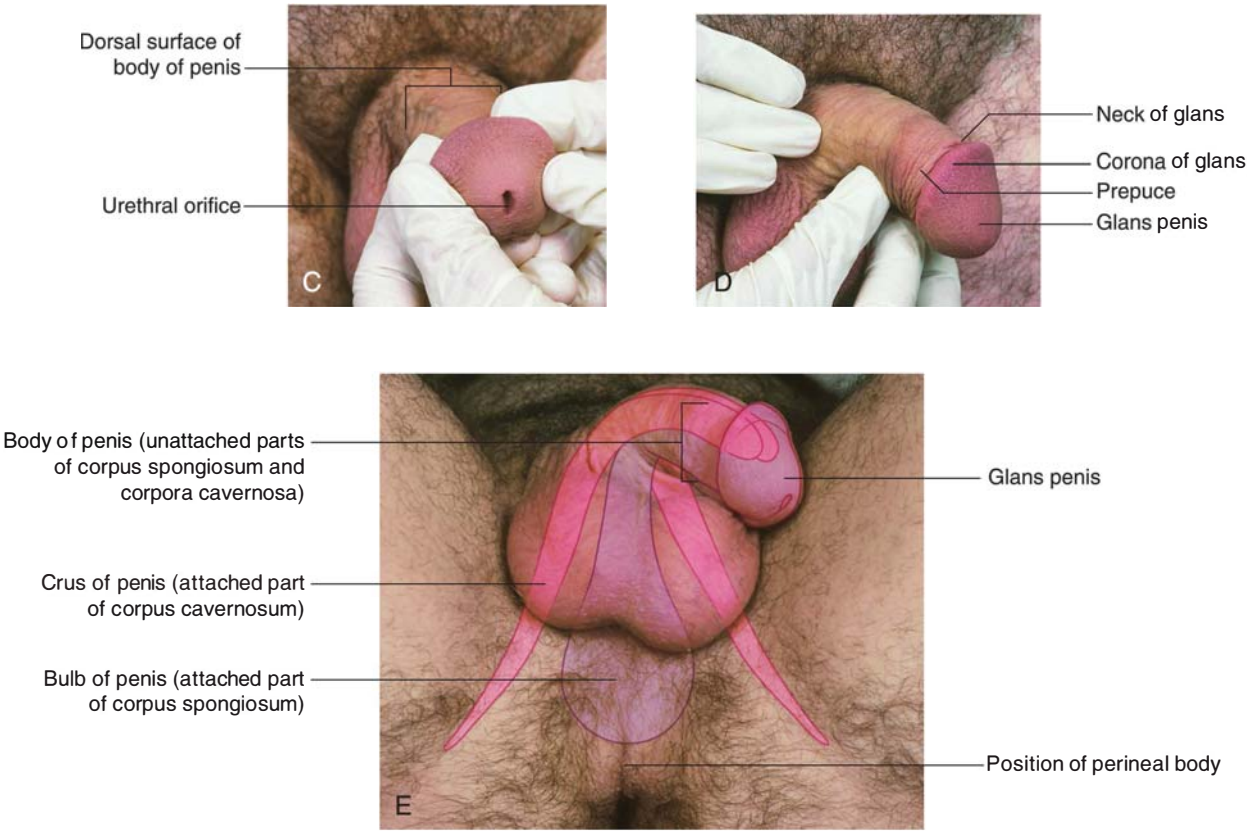


Fig. 5.84, cont'd C. Anterior view of the glans penis showing the urethral opening. **D.** Lateral view of the body of the penis and glans. **E.** Inferior view of the urogenital triangle of a man with the erectile tissues of the penis indicated with overlays.

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Clinical cases

Case 1

VARICOCELE

A 25-year-old man visited his family physician because he had a "dragging feeling" in the left side of his scrotum. He was otherwise healthy and had no other symptoms. During examination, the physician palpated the left testis, which was normal, although he noted soft nodular swelling around the superior aspect of the testes and the epididymis. In his clinical notes, he described these findings as a "bag of worms" (Fig. 5.85). The bag of worms was a varicocele.

The venous drainage of the testis is via the pampiniform plexus of veins that runs within the spermatic cord. A varicocele is a collection of dilated veins that arise from the pampiniform plexus. In many ways, they are similar to varicose veins that develop in the legs. Typically, the patient complains of a dragging feeling in the scrotum and around the testis, which is usually worse toward the end of the day.

The family physician recommended surgical treatment, with a recommendation for surgery through an inguinal incision.

A simple surgical technique divides the skin around the inguinal ligament. The aponeurosis of the external oblique muscle is divided in the anterior abdominal wall to display the spermatic cord. Careful inspection of the spermatic cord reveals the veins, which are surgically ligated.

Another option is to embolize the varicocele.

In this technique, a small catheter is placed via the right femoral vein. The catheter is advanced along the external iliac vein and the common iliac vein and into the inferior vena cava. The catheter is then positioned in the left renal vein, and a venogram is performed to demonstrate the origin of the left testicular vein. The catheter is advanced down the left testicular vein into the veins of the inguinal canal and the pampiniform plexus. Metal coils to occlude the vessels are injected, and the catheter is withdrawn.

The patient asked how blood would drain from the testis after the operation.

Although the major veins of the testis had been occluded, small collateral veins running within the scrotum and around the outer aspect of the spermatic cord permitted drainage without recurrence of the varicocele.



Fig. 5.85 Left testicular venogram demonstrating the pampiniform plexus of veins.

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Case 2

SCIATIC NERVE COMPRESSION

A young man developed pain in his right gluteal region, in the posterior aspect of the thigh and around the posterior and lateral aspects of the leg. On further questioning, he reported that the pain also radiated over the lateral part of the foot, particularly around the lateral malleolus.

The areas of pain correspond to dermatomes L4 to S3 nerves.

Over the following weeks, the patient began to develop muscular weakness, predominantly footdrop.

These findings are consistent with loss of the motor function and sensory change in the common fibular nerve, which is a branch of the sciatic nerve in the lower limb.

A computed tomography (CT) scan of the abdomen and pelvis revealed a mass in the posterior aspect of the right side of the pelvis. The mass was anterior to the piriformis muscle and adjacent to the rectum.

On the anterior belly of the piriformis muscle, the sciatic nerve is formed from the roots of L4 to S3 nerves. The mass in the patient's pelvis compressed this nerve, producing his sensory and motor dysfunction.

During surgery, the mass was found to be a benign nerve tumor and was excised. This patient had no long-standing neurological deficit.

Case 3

PELVIC KIDNEY

A young woman visited her family practitioner because she had mild upper abdominal pain. An ultrasound demonstrated gallstones within the gallbladder, which explained the patient's pain. However, when the technician assessed the pelvis, she noted a mass behind the bladder, which had sonographic findings similar to a kidney (Fig. 5.86).

What did the sonographer do next? Having demonstrated this pelvic mass behind the bladder, the sonographer assessed both kidneys. The patient had a normal right kidney. However, the left kidney could not be found in its usual place. The technician diagnosed a pelvic kidney.

A pelvic kidney can be explained by the embryology. The kidneys develop from a complex series of structures that originate adjacent to the bladder within the fetal pelvis. As development proceeds and the functions of the various parts of the developing kidneys change, they attain a superior position in the upper abdomen adjacent to the abdominal aorta and inferior vena cava, on the posterior abdominal wall. A developmental arrest or complication may prevent the kidney from obtaining its usual position. Fortunately, it is unusual for patients to have any symptoms relating to a pelvic kidney.

This patient had no symptoms attributable to the pelvic kidney and she was discharged.

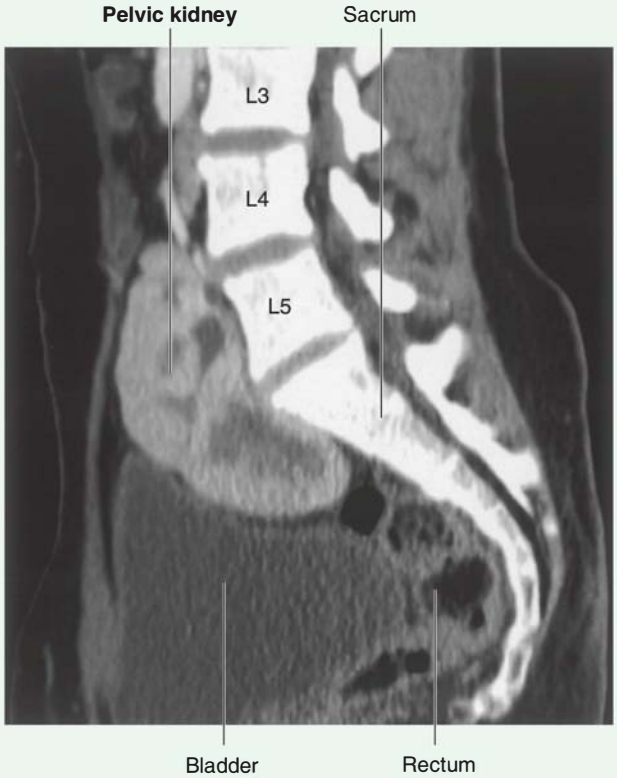


Fig. 5.86 Sagittal computed tomogram demonstrating a pelvic kidney.

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Case 4

LEFT COMMON ILIAC ARTERY OBSTRUCTION

A 65-year-old man was examined by a surgical intern because he had a history of buttock pain and impotence. On examination he had a reduced peripheral pulse on the left foot compared to the right. On direct questioning, the patient revealed that he experienced severe left-sided buttock pain after walking 100 yards. After a short period of rest, he could walk another 100 yards before the same symptoms recurred. He also noticed that over the past year he was unable to obtain an erection. He smoked heavily and was on no other drugs or treatment.

The pain in the left buttock is ischemic in nature. He gives a typical history relating to lack of blood flow to the muscles. A similar finding is present when muscular branches of the femoral artery are occluded or stenosed. Such patients develop similar (ischemic) pain in the calf muscles called intermittent claudication.

How does the blood get to the gluteal muscles?

Blood arrives at the aortic bifurcation and then passes into the common iliac arteries, which divide into the internal and external iliac vessels. The internal iliac artery then divides into anterior and posterior divisions, which in turn give rise to vessels that leave the pelvis by passing through the greater sciatic foramen and supply the gluteal muscles. The internal pudendal artery also arises from the anterior division of the internal iliac artery and supplies the penis.

The patient's symptoms occur on the left side, suggesting that an obstruction exists on that side only.

Because the patient's symptoms occur on the left side only, the lesion is likely in the left common iliac artery (Fig. 5.87) and is preventing blood flow into the external and internal iliac arteries on the left side.

"How will I be treated?" asked the patient.

The patient was asked to stop smoking and begin regular exercise. Other treatment options include unblocking the lesion by ballooning the blockage to reopen the vessels or by a surgical bypass graft.

Stopping smoking and regular exercise improved the patient's walking distance. The patient underwent the less invasive procedure of ballooning the vessel (angioplasty) and as a result he was able to walk unhindered and to have an erection.

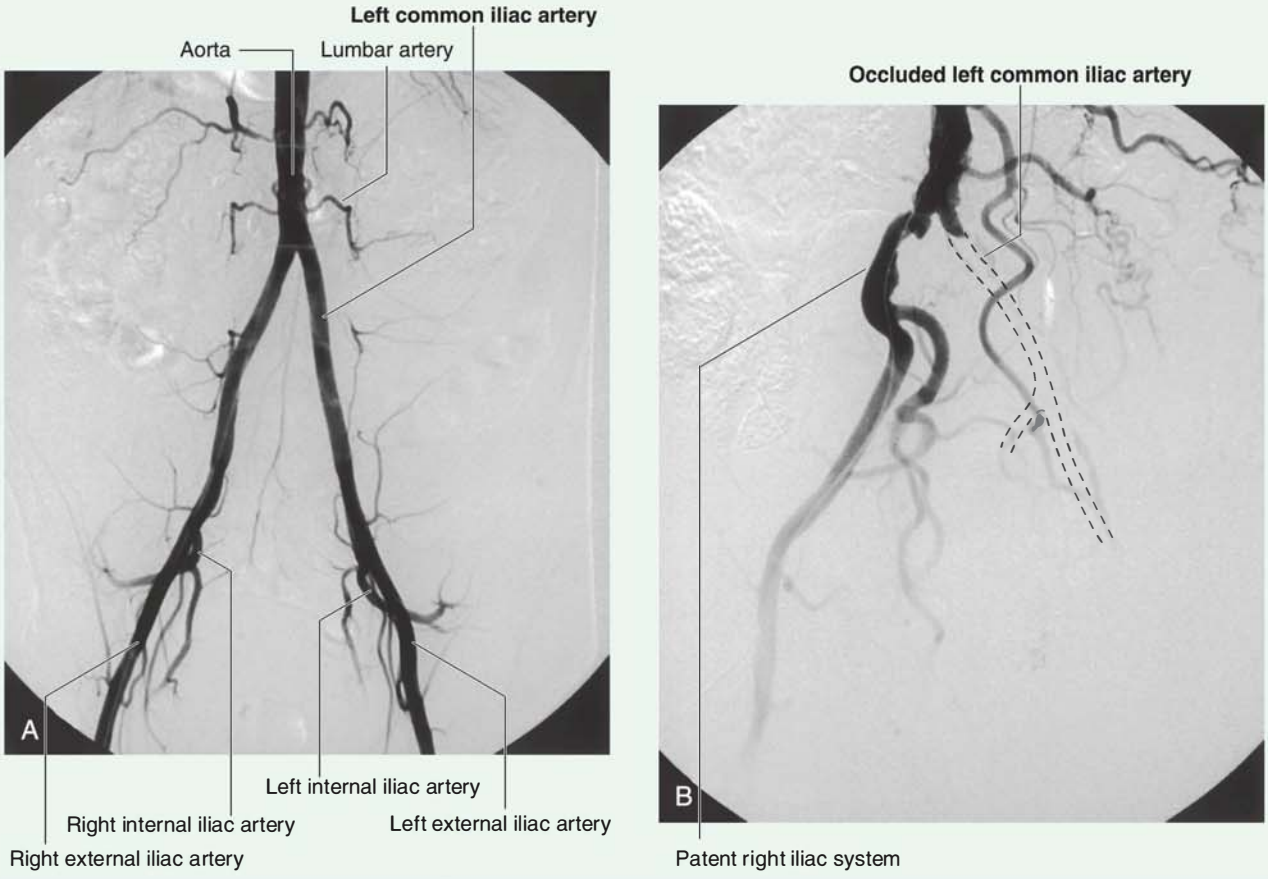


Fig. 5.87 Digital subtraction aorto-iliac angiogram. **A.** Normal circulation pattern. **B.** Occluded left common iliac artery.

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Case 5

IATROGENIC URETERIC INJURY

A 50-year-old woman was admitted to hospital for surgical resection of the uterus (hysterectomy) for cancer. The surgeon was also going to remove all the pelvic lymph nodes and carry out a bilateral salpingo-oophorectomy (removal of uterine tubes and ovaries). The patient was prepared for this procedure and underwent routine surgery. Twenty-five hours after surgery, it was noted that the patient had passed no urine and her abdomen was expanding. An ultrasound scan demonstrated a considerable amount of fluid within the abdomen. Fluid withdrawn from the abdomen was tested and found to be urine.

It was postulated that this patient's ureters had been damaged during the surgery.

The pelvic part of the ureter courses posteroinferiorly and external to the parietal peritoneum on the lateral wall of the pelvis anterior to the internal iliac artery. It continues in its course to a point approximately 2 cm superior to the ischial spine and then passes anteromedially and superior to the levator ani muscles. Importantly, the ureter closely adheres to the peritoneum. The only structure that passes between the ureter and the peritoneum in men is the ductus deferens. In women, however, as the ureter descends on the pelvic wall, it passes under the uterine artery. The ureter continues close to the lateral fornix of the vagina, especially on the left, and enters the posterosuperior angle of the bladder. It was at this point that the ureter was inadvertently damaged.

Knowing the anatomy and recognizing the possibility of ureteric damage enabled the surgeons to reestablish continuity of the ureter surgically. The patient was hospitalized a few days longer than expected and made an uneventful recovery.

Case 6

ECTOPIC PREGNANCY

A 25-year-old woman was admitted to the emergency department with a complaint of pain in her right iliac fossa. The pain had developed rapidly over approximately 40 minutes and was associated with cramps and vomiting. The surgical intern made an initial diagnosis of appendicitis.

The typical history for appendicitis is a central abdominal, colicky (intermittent waxing and waning) pain, which over a period of hours localizes to become a constant pain in the right iliac fossa. The central colicky pain is typical for a poorly localized visceral type of pain. As the parietal peritoneum becomes inflamed, the pain becomes localized. Although this patient does have right iliac fossa pain, the history is not typical for appendicitis (although it must be remembered that patients may not always have a classical history for appendicitis).

The surgical intern asked a more senior colleague for an opinion.

The senior colleague considered other anatomical structures that lie within the right iliac fossa as a potential cause of pain. These include the appendix, the cecum, and the small bowel. Musculoskeletal pain and referred pain could also be potential causes. In women, pain may also arise from the ovary, fallopian tube, and uterus. In a young patient, diseases of these organs are rare. Infection and pelvic inflammatory disease may occur in the younger patient and need to be considered.

The patient gave no history of these disorders.

Upon further questioning, however, the patient revealed that her last menstrual period was 6 weeks before this examination. The senior physician realized that a potential cause of the abdominal pain was a pregnancy outside the uterus (ectopic pregnancy). The patient was rushed for an abdominal ultrasound, which revealed no fetus or sac in the uterus. She was also noted to have a positive pregnancy test. The patient underwent surgery and was found to have a ruptured fallopian tube caused by an ectopic pregnancy.

Whenever a patient has apparent pelvic pain, it is important to consider the gender-related anatomical differences. Ectopic pregnancy should always be considered in women of childbearing age.

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Case 7

UTERINE TUMOR

A 35-year-old woman visited her family practitioner because she had a "bloating" feeling and an increase in abdominal girth. The family practitioner examined the lower abdomen, which revealed a mass that extended from the superior pubic rami to the level of the umbilicus. The superior margin of the mass was easily palpated, but the inferior margin appeared to be less well defined.

This patient has a pelvic mass.

When examining a patient in the supine position, the observer should uncover the whole of the abdomen.

Inspection revealed a bulge in the lower abdomen to the level of the umbilicus. Palpation revealed a hard and slightly irregular mass with well-defined superior and lateral borders and a less well-defined inferior border, giving the impression that the mass continued into the pelvis. The lesion was dull to percussion. Auscultation did not reveal any abnormal sounds.

The doctor pondered which structures this mass may be arising from. When examining the pelvis, it is important to remember the sex differences. Common to both men and women are the rectum, bowel, bladder, and musculature. Certain pathological states are also common to both sexes, including the development of pelvic abscesses and fluid collections.

In men, the prostate cannot be palpated transabdominally, and it is extremely rare for it to enlarge to such an extent in benign diseases. Aggressive prostate cancer can spread throughout the whole of the pelvis, although this is often associated with bowel obstruction and severe bladder symptoms.

In women, a number of organs can develop large masses, including the ovaries (solid and cystic tumors), the embryological remnants within the broad ligaments, and the uterus (pregnancy and fibroids).

The physician asked further questions.

It is always important to establish whether the patient is pregnant (occasionally, pregnancy may come as a surprise to the patient).

This patient's pregnancy test was negative. After the patient emptied her bladder, there was no change in the mass. The physician thought the mass might be a common benign tumor of the uterus (fibroid). To establish the diagnosis, he obtained an ultrasound scan of the pelvis, which confirmed that the mass stemmed from the uterus.

The patient was referred to a gynecologist, and after a long discussion regarding her symptomatology, fertility, and risks, the surgeon and the patient agreed that a hysterectomy (surgical removal of the uterus) would be an appropriate course of therapy.

The patient sought a series of opinions from other gynecologists, all of whom agreed that surgery was the appropriate option.

The fibroid was removed with no complications.

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Case 8

UTERINE FIBROIDS

A 52-year-old woman was referred to a gynecologist. Magnetic resonance imaging (MRI) indicated the presence of uterine fibroids. After a long discussion regarding her symptomatology, fertility, and risks, she was offered the choice between a hysterectomy (surgical removal of the uterus) or uterine artery embolization.

A uterine artery embolization is a procedure where an interventional radiologist uses a catheter to inject small particles into the uterine arteries. This reduces the blood supply to the fibroids and causes them to shrink.

The patient opted for the uterine artery embolization.

An MRI performed 6 months after the embolization procedure showed a favorable reduction in the size of the uterine fibroids (Fig. 5.88).

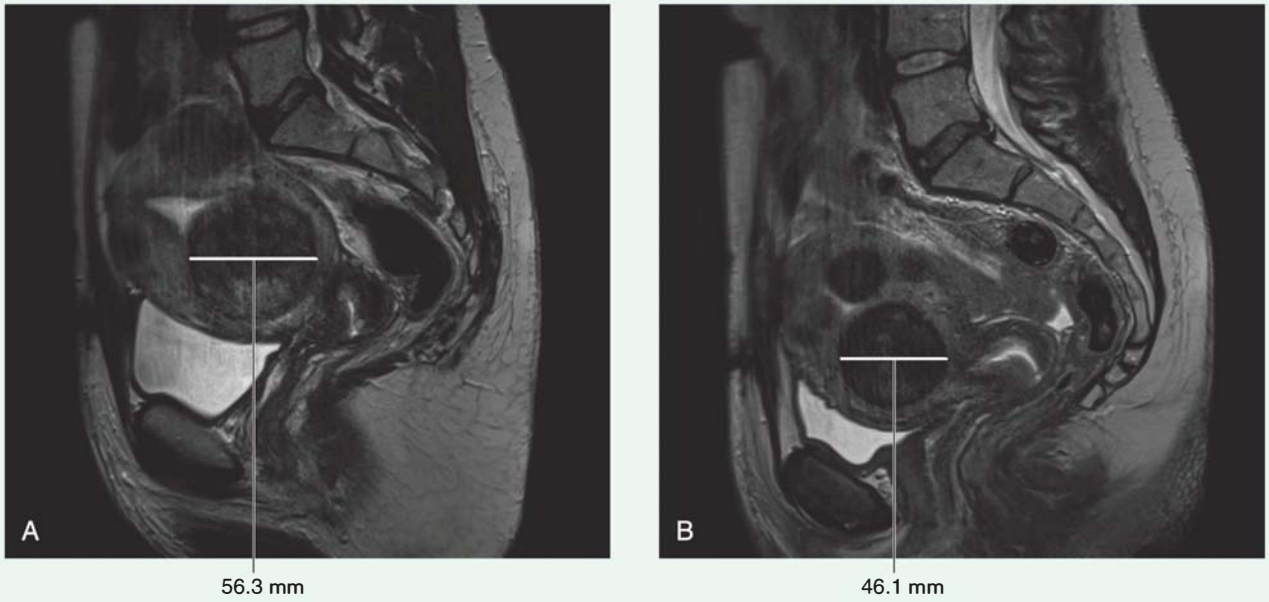


Fig. 5.88 Sagittal MRI of the pelvic cavity. **A.** Measurement of a fibroid before the uterine artery embolization. **B.** Measurement of a fibroid 6 months after the embolization. The size of the fibroid has decreased.

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6 Lower Limb

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ADDITIONAL LEARNING RESOURCES for Chapter 6, Lower Limb, on STUDENT CONSULT (www.studentconsult.com):

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- Interactive Surface Anatomy—interactive surface animations, Chapter 6
- PT Case Studies, Chapter 6
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  - Achilles tendinitis and tendinosis
  - Eversion ankle sprain
  - High ankle sprain
  - Patellofemoral pain syndrome
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  - Groin injury
  - Iliopsoas tendinitis
  - Iliotibial band syndrome
  - Nerve entrapment syndrome
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  - Knee joint injury
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Conceptual overview

GENERAL INTRODUCTION

The lower limb is directly anchored to the axial skeleton by a sacroiliac joint and by strong ligaments, which link the pelvic bone to the sacrum. It is separated from the abdomen, back, and perineum by a continuous line (Fig. 6.1), which:

- joins the pubic tubercle with the anterior superior iliac spine (position of the inguinal ligament) and then continues along the iliac crest to the posterior superior iliac spine to separate the lower limb from the anterior and lateral abdominal walls;
- passes between the posterior superior iliac spine and along the dorsolateral surface of the sacrum to the coccyx to separate the lower limb from the muscles of the back; and
- joins the medial margin of the sacrotuberous ligament, the ischial tuberosity, the ischiopubic ramus, and the pubic symphysis to separate the lower limb from the perineum.

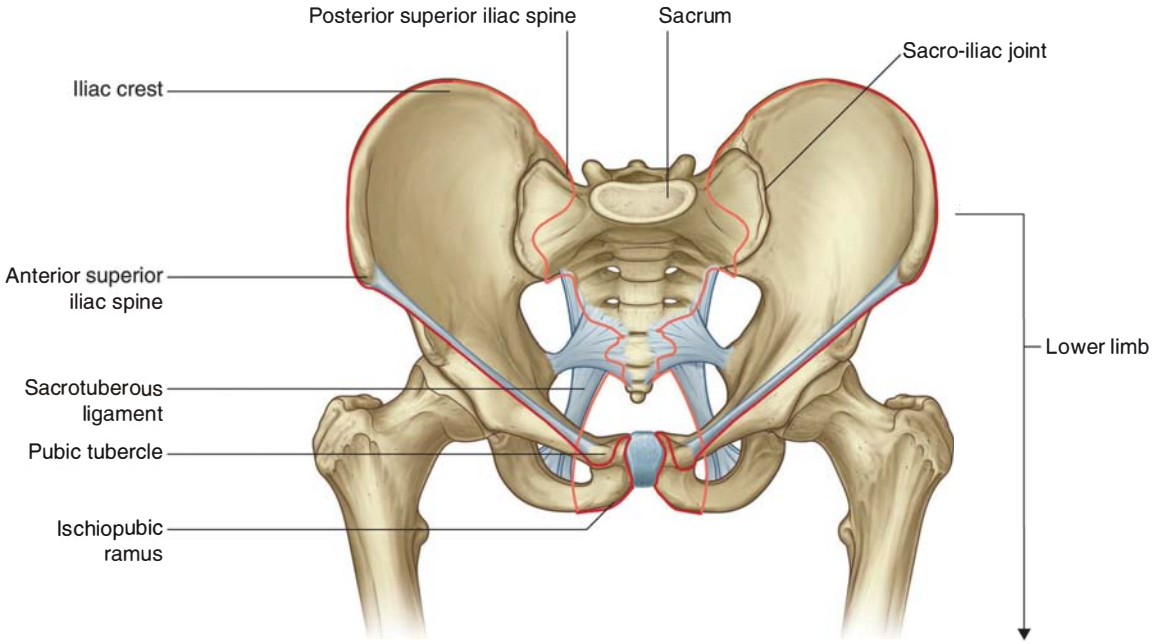


Fig. 6.1 Upper margin of the lower limb.

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The lower limb is divided into the gluteal region, thigh, leg, and foot on the basis of major joints, component bones, and superficial landmarks (Fig. 6.2):

- The **gluteal region** is posterolateral and between the iliac crest and the fold of skin (gluteal fold) that defines the lower limit of the buttocks.
- Anteriorly, the **thigh** is between the inguinal ligament and the knee joint—the hip joint is just inferior to the middle third of the inguinal ligament, and the posterior thigh is between the gluteal fold and the knee.
- The **leg** is between the knee and ankle joint.
- The **foot** is distal to the ankle joint.

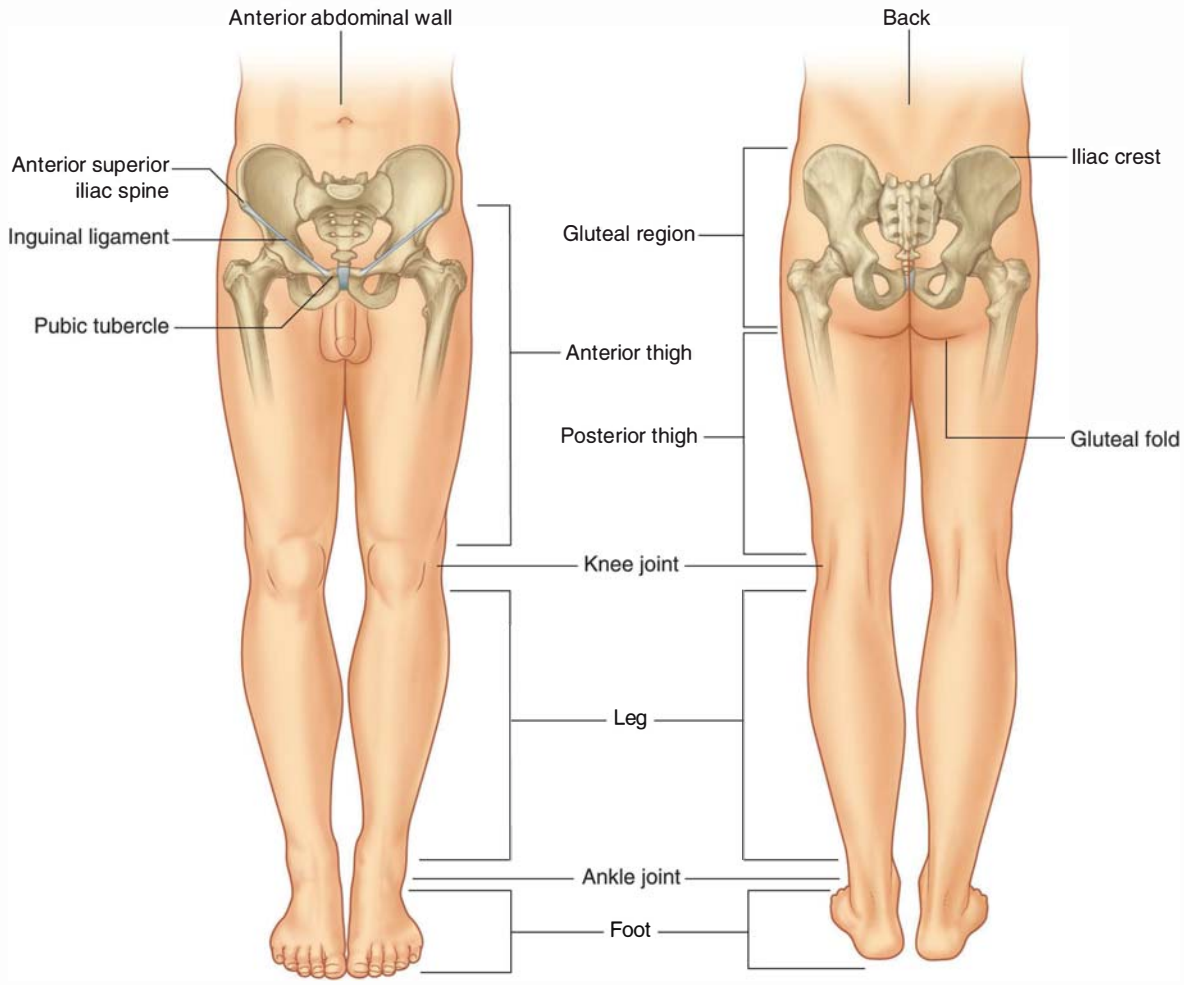


Fig. 6.2 Regions of the lower limb.

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The femoral triangle and popliteal fossa, as well as the posteromedial side of the ankle, are important areas of transition through which structures pass between regions (Fig. 6.3).

The **femoral triangle** is a pyramid-shaped depression formed by muscles in the proximal regions of the thigh and by the inguinal ligament, which forms the base of the triangle. The major blood supply and one of the nerves of the limb (femoral nerve) enter into the thigh from the abdomen by passing under the inguinal ligament and into the femoral triangle.

The **popliteal fossa** is posterior to the knee joint and is a diamond-shaped region formed by muscles of the thigh and leg. Major vessels and nerves pass between the thigh and leg through the popliteal fossa.

Most nerves, vessels, and flexor tendons that pass between the leg and foot pass through a series of canals (collectively termed the tarsal tunnel) on the posteromedial side of the ankle. The canals are formed by adjacent bones and a flexor retinaculum, which holds the tendons in position.

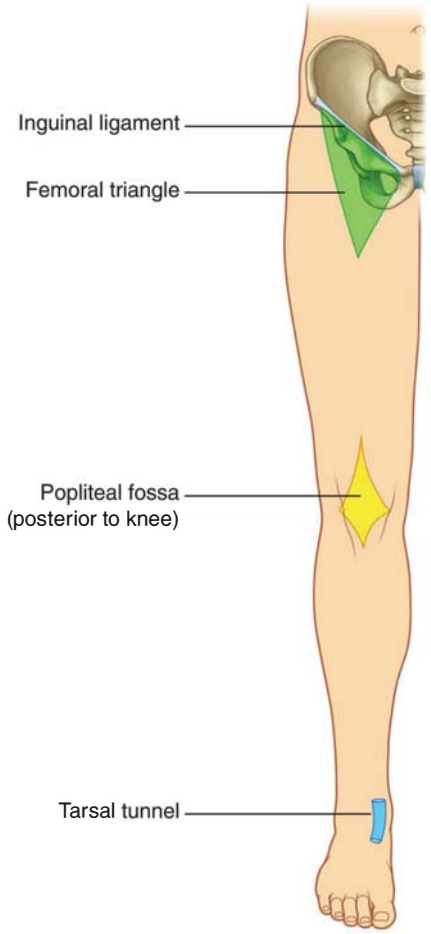


Fig. 6.3 Areas of transition.

FUNCTION

Support the body weight

A major function of the lower limb is to support the weight of the body with minimal expenditure of energy. When standing erect, the center of gravity is anterior to the edge of the SII vertebra in the pelvis (Fig. 6.4). The vertical line through the center of gravity is slightly posterior to the hip joints, anterior to the knee and ankle joints, and directly over the almost circular support base formed by the feet on the ground and holds the knee and hip joints in extension.

The organization of ligaments at the hip and knee joints, together with the shape of the articular surfaces, particularly at the knee, facilitates "locking" of these joints into position when standing, thereby reducing the muscular energy required to maintain a standing position.

Locomotion

A second major function of the lower limbs is to move the body through space. This involves the integration of movements at all joints in the lower limb to position the foot on the ground and to move the body over the foot.

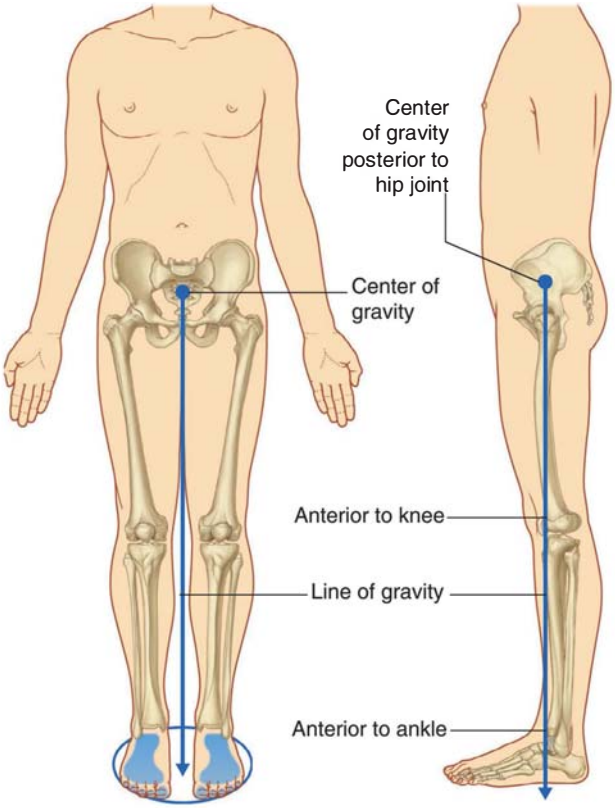


Fig. 6.4 Center and line of gravity.

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Movements at the hip joint are flexion, extension, abduction, adduction, medial and lateral rotation, and circumduction (Fig. 6.5).

The knee and ankle joints are primarily hinge joints. Movements at the knee are mainly flexion and extension (Fig. 6.6A). Movements at the ankle are dorsiflexion (movement of the dorsal side of the foot toward the leg) and plantarflexion (Fig. 6.6B).

During walking, many anatomical features of the lower limbs contribute to minimizing fluctuations in the body's center of gravity and thereby reduce the amount of energy needed to maintain locomotion and produce a smooth,

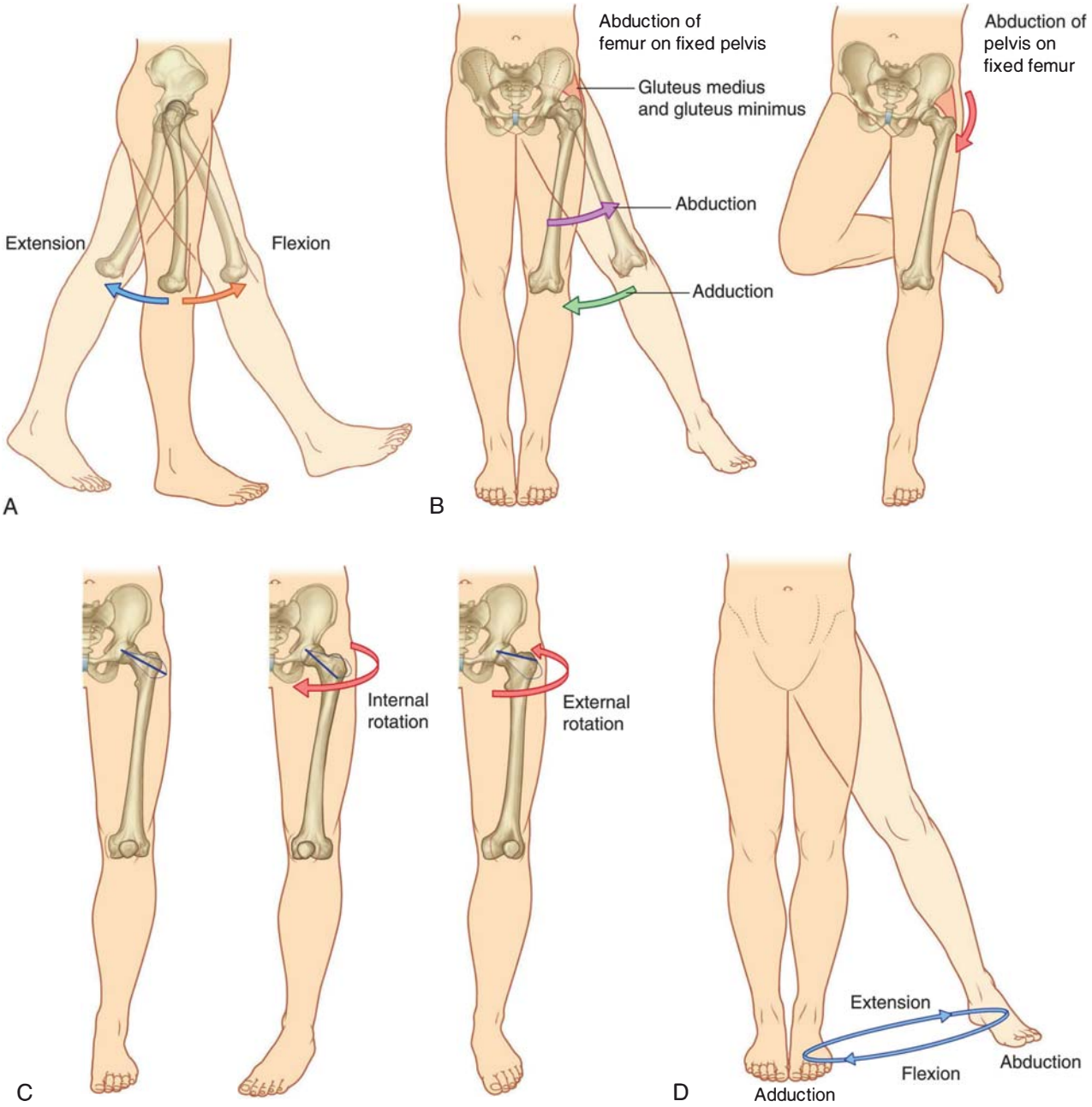


Fig. 6.5 Movements of the hip joint. **A.** Flexion and extension. **B.** Abduction and adduction. **C.** External and internal rotation. **D.** Circumduction.

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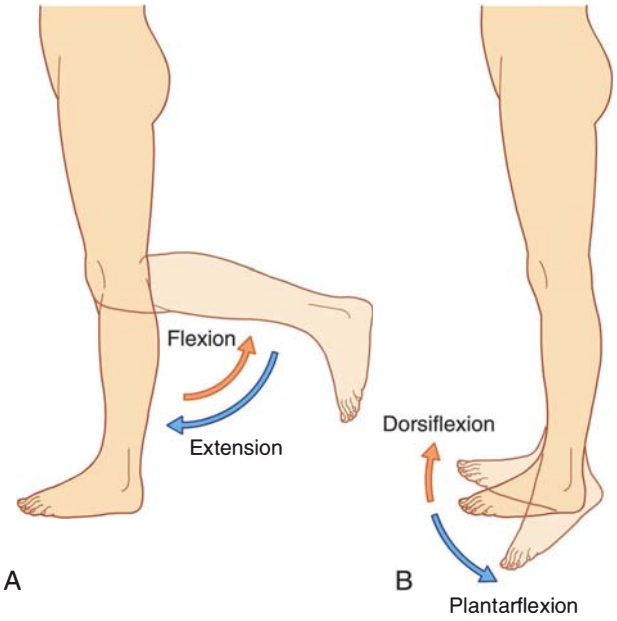


Fig. 6.6 Movements of the knee and ankle. **A.** Knee flexion and extension. **B.** Ankle dorsiflexion and plantarflexion.

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efficient gait (Fig. 6.7). They include pelvic tilt in the coronal plane, pelvic rotation in the transverse plane, movement of the knees toward the midline, flexion of the knees, and complex interactions between the hip, knee, and ankle. As a result, during walking, the body's center of gravity normally fluctuates only 5 cm in both vertical and lateral directions.

COMPONENT PARTS

Bones and joints

The bones of the gluteal region and the thigh are the pelvic bone and the femur (Fig. 6.8). The large ball and socket joint between these two bones is the hip joint.

The femur is the bone of the thigh. At its distal end, its major weight-bearing articulation is with the tibia, but it also articulates anteriorly with the patella (knee cap). The patella is the largest sesamoid bone in the body and is embedded in the quadriceps femoris tendon.

The joint between the femur and tibia is the principal articulation of the knee joint, but the joint between the patella and femur shares the same articular cavity. Although the main movements at the knee are flexion and extension, the knee joint also allows the femur to rotate on the tibia. This rotation contributes to "locking" of the knee when fully extended, particularly when standing.

The leg contains two bones:

- The tibia is medial in position, is larger than the laterally positioned fibula, and is the weight-bearing bone.
- The fibula does not take part in the knee joint and forms only the most lateral part of the ankle joint—proximally, it forms a small synovial joint (superior tibiofibular joint) with the inferolateral surface of the head of the tibia.

The tibia and fibula are linked along their lengths by an interosseous membrane, and at their distal ends by a fibrous inferior tibiofibular joint, and little movement occurs between them. The distal surfaces of the tibia and fibula together form a deep recess. The ankle joint is formed by this recess and part of one of the tarsal bones of the foot (talus), which projects into the recess. The ankle is most stable when dorsiflexed.

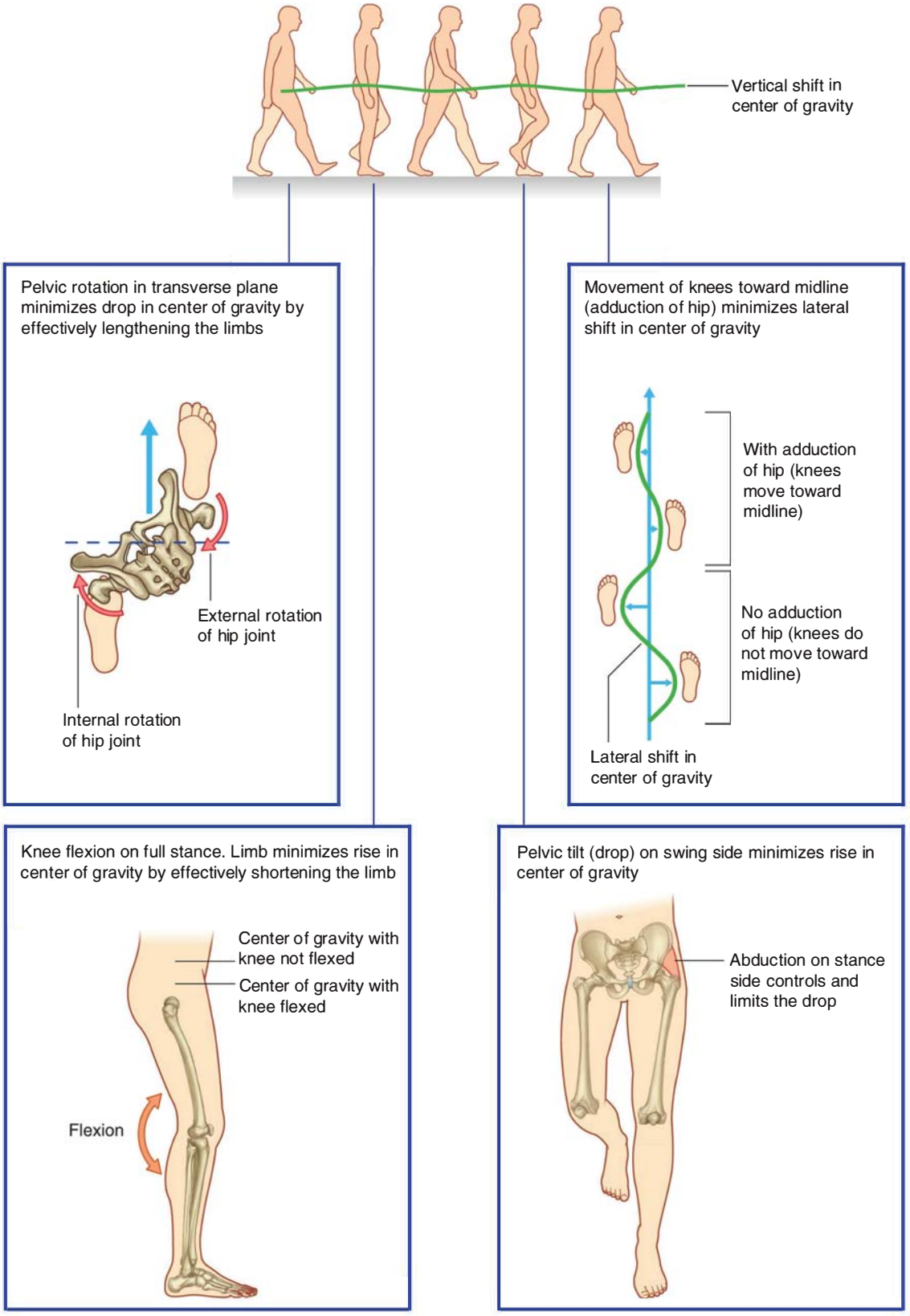


Fig. 6.7 Some of the determinants of gait.

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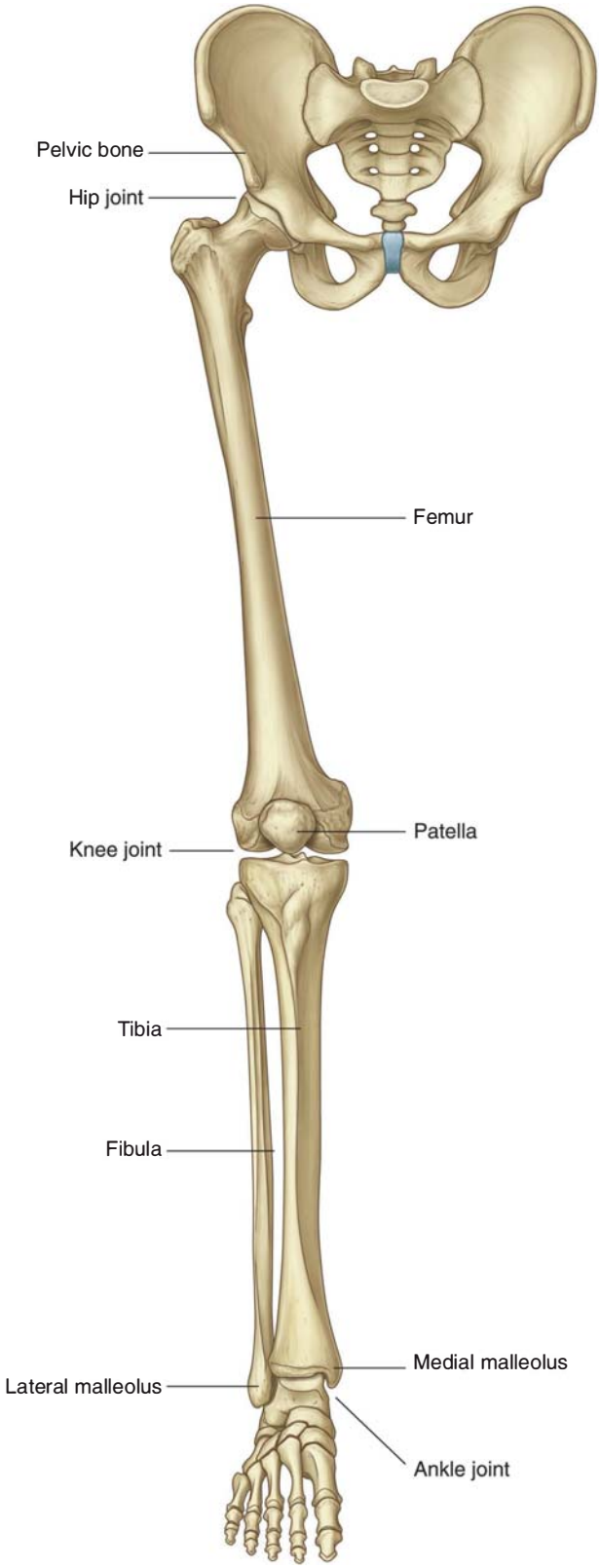


Fig. 6.8 Bones and joints of the lower limb.

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The bones of the foot consist of the tarsal bones, the metatarsals, and the phalanges (Fig. 6.9). There are seven tarsal bones, which are organized in two rows with an intermediate bone between the two rows on the medial side. Inversion and eversion of the foot, or turning the sole of the foot inward and outward, respectively, occur at joints between the tarsal bones.

The tarsal bones articulate with the metatarsals at tarsometatarsal joints, which allow only limited sliding movements.

Independent movements of the metatarsals are restricted by deep transverse metatarsal ligaments, which effectively link together the distal heads of the bones at the metatarsophalangeal joints. There is a metatarsal for each of the five digits, and each digit has three phalanges except for the great toe (digit I), which has only two.

The metatarsophalangeal joints allow flexion, extension, abduction, and adduction of the digits, but the range of movement is more restricted than in the hand.

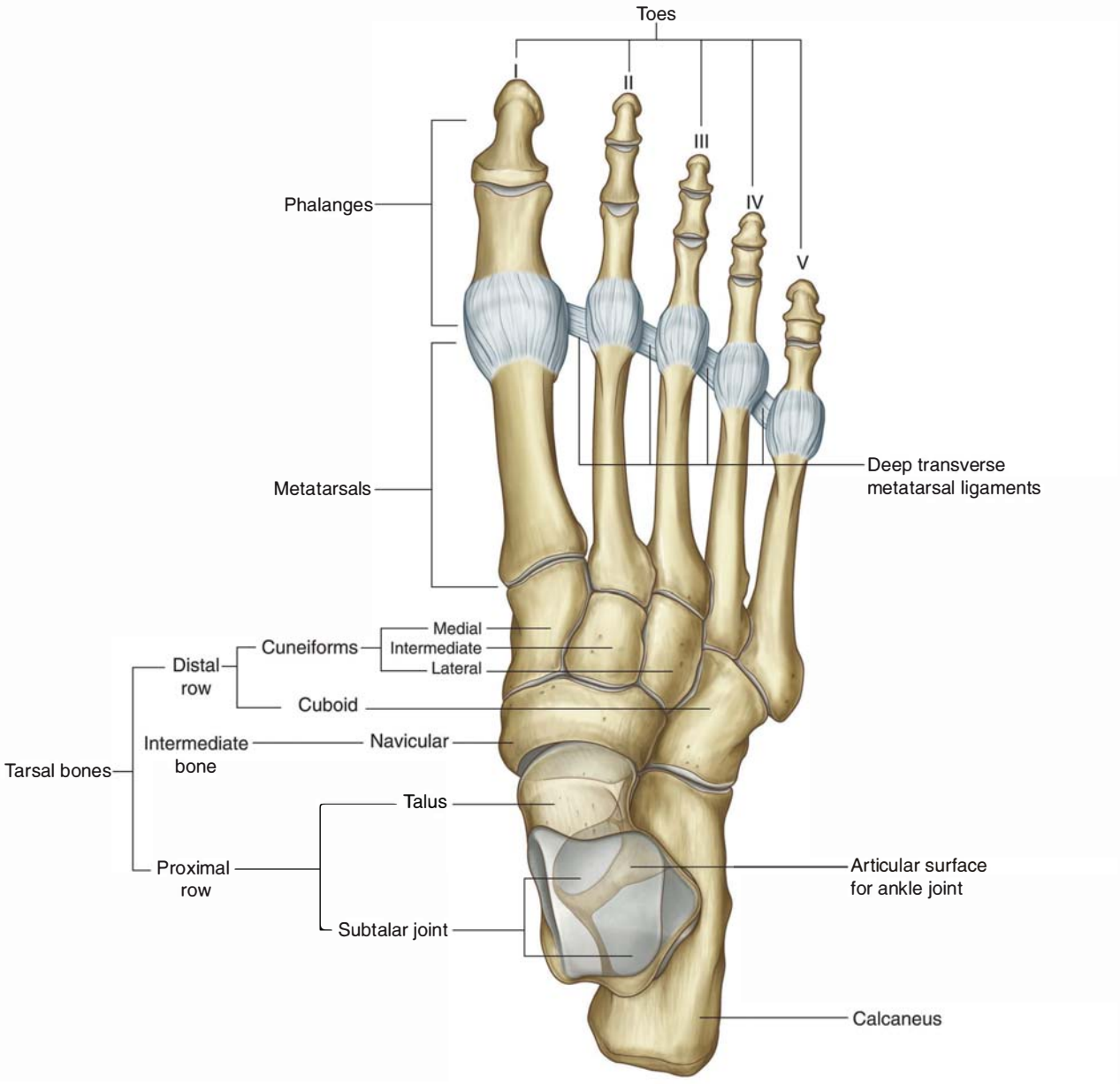


Fig. 6.9 Bones of the foot.

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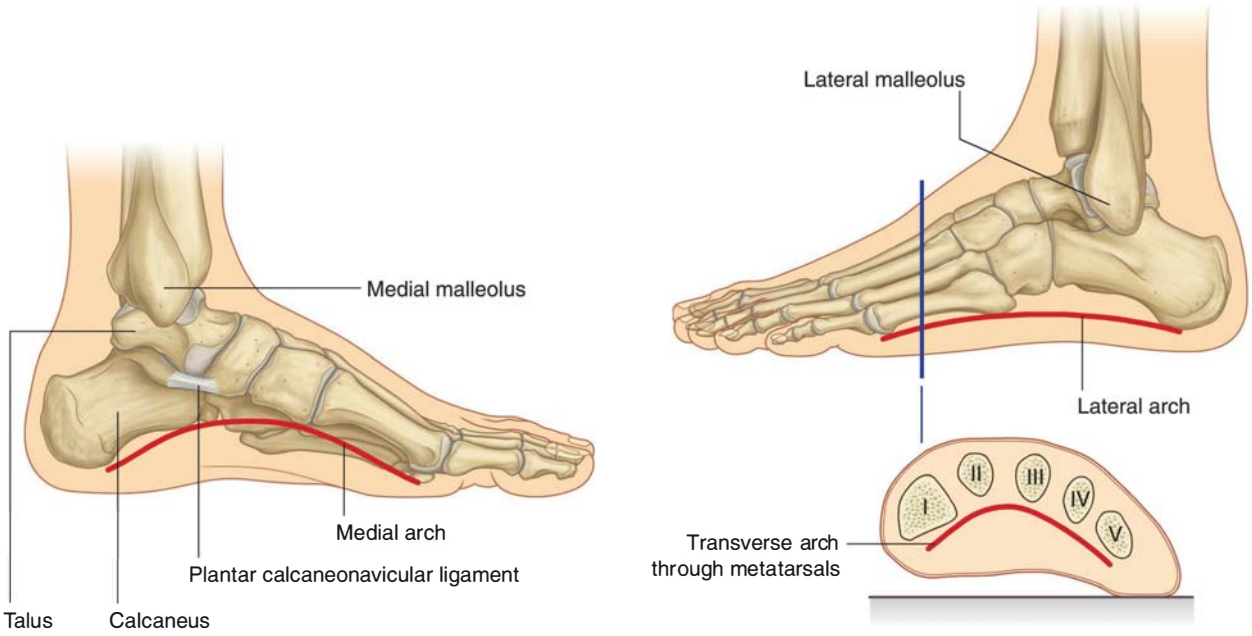


Fig. 6.10 Longitudinal and transverse arches of the foot.

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The interphalangeal joints are hinge joints and allow flexion and extension.

The bones of the foot are not organized in a single plane so that they lie flat on the ground. Rather, the metatarsals and tarsals form longitudinal and transverse arches (Fig. 6.10). The longitudinal arch is highest on the medial side of the foot. The arches are flexible in nature and are supported by muscles and ligaments. They absorb and transmit forces during walking and standing.

Muscles

Muscles of the gluteal region consist predominantly of extensors, rotators, and abductors of the hip joint (Fig. 6.11). In addition to moving the thigh on a fixed pelvis, these muscles also control the movement of the pelvis relative to the limb bearing the body's weight (weight-bearing or stance limb) while the other limb swings forward (swing limb) during walking.

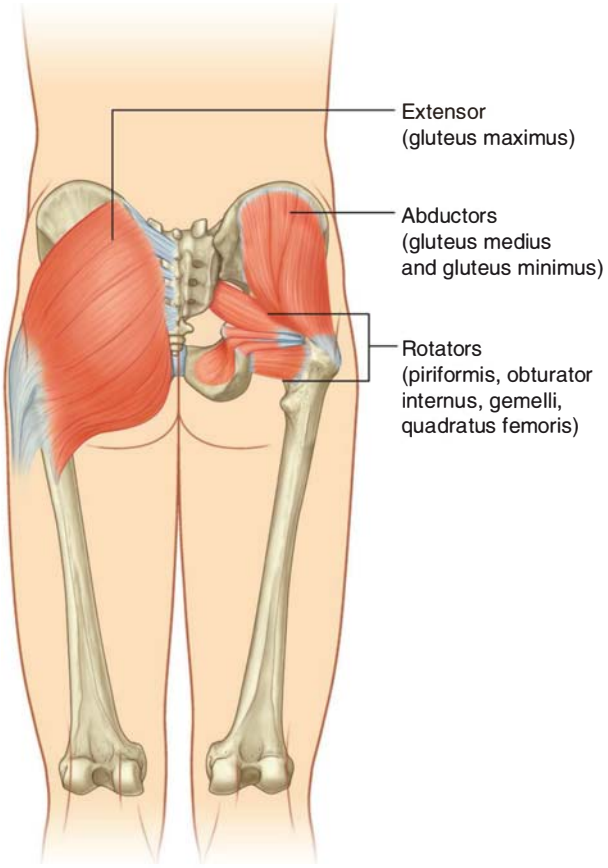


Fig. 6.11 Muscles of the gluteal region.

Major flexor muscles of the hip (iliopsoas—psoas major and iliacus) do not originate in the gluteal region or the thigh. Instead, they are attached to the posterior abdominal wall and descend through the gap between the inguinal ligament and pelvic bone to attach to the proximal end of the femur (Fig. 6.12).

Muscles in the thigh and leg are separated into three compartments by layers of fascia, bones, and ligaments (Fig. 6.13).

In the thigh, there are medial (adductor), anterior (extensor), and posterior (flexor) compartments:

- Most muscles in the medial compartment act mainly on the hip joint.
- The large muscles (hamstrings) in the posterior compartment act on the hip (extension) and knee (flexion) because they attach to both the pelvis and bones of the leg.
- Muscles in the anterior compartment (quadriceps femoris) predominantly extend the knee.

Muscles in the leg are divided into lateral (fibular), anterior, and posterior compartments:

- Muscles in the lateral compartment predominantly evert the foot.
- Muscles in the anterior compartment dorsiflex the foot and extend the digits.
- Muscles in the posterior compartment plantarflex the foot and flex the digits; one of the muscles can also flex the knee because it attaches superiorly to the femur.

Specific muscles in each of the three compartments in the leg also provide dynamic support for the arches of the foot.

Muscles found entirely in the foot (intrinsic muscles) modify the forces produced by tendons entering the toes from the leg and provide dynamic support for the longitudinal arches of the foot when walking, particularly when levering the body forward on the stance limb just before toe-off.

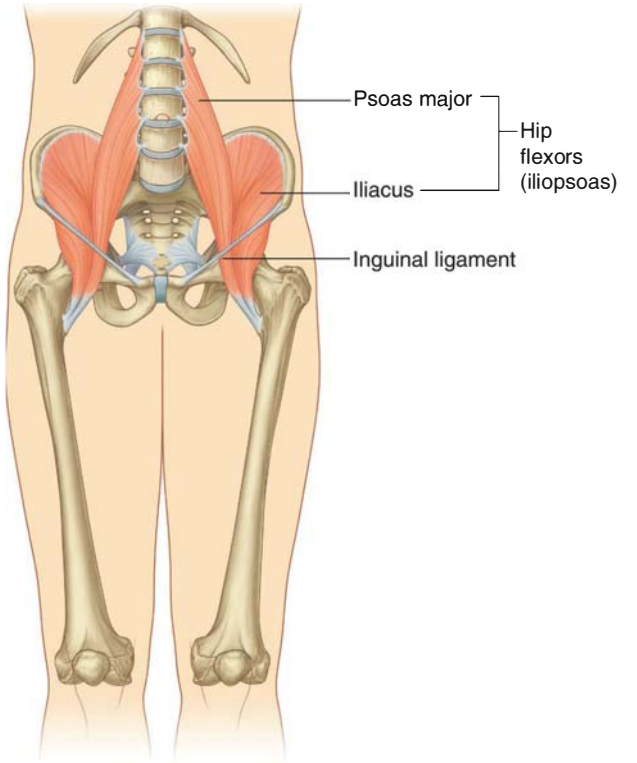


Fig. 6.12 Major flexors of the hip.

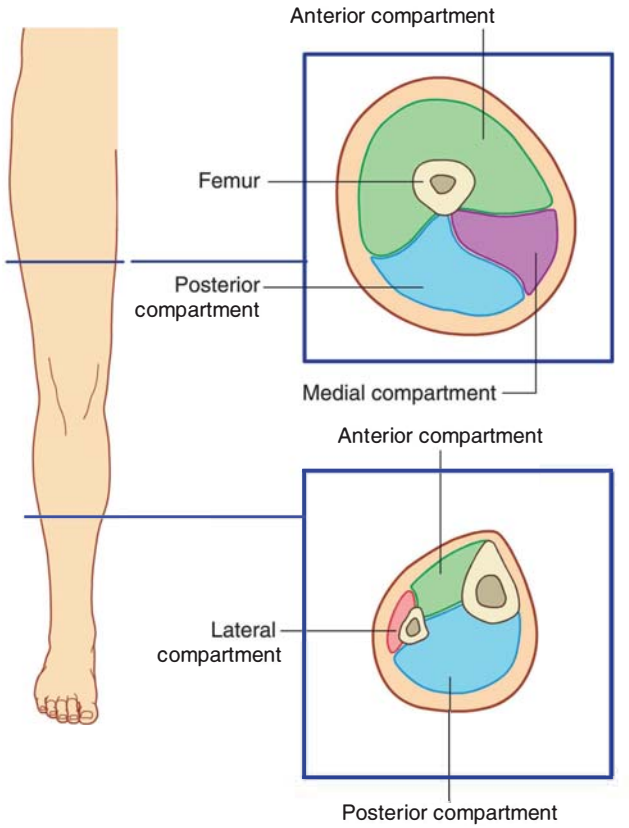


Fig. 6.13 Muscle compartments in the thigh and leg.

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RELATIONSHIP TO OTHER REGIONS

Unlike in the upper limb where most structures pass between the neck and limb through a single axillary inlet, in the lower limb, there are four major entry and exit points between the lower limb and the abdomen, pelvis, and perineum (Fig. 6.14). These are:

- the gap between the inguinal ligament and pelvic bone,
- the greater sciatic foramen,
- the obturator canal (at the top of the obturator foramen), and
- the lesser sciatic foramen.

Abdomen

The lower limb communicates directly with the abdomen through a gap between the pelvic bone and the inguinal ligament (Fig. 6.14). Structures passing though this gap include:

- muscles—psoas major, iliacus, and pectineus;
- nerves—femoral and femoral branch of the genitofemoral nerves, and the lateral cutaneous nerve of the thigh;
- vessels—femoral artery and vein; and
- lymphatics.

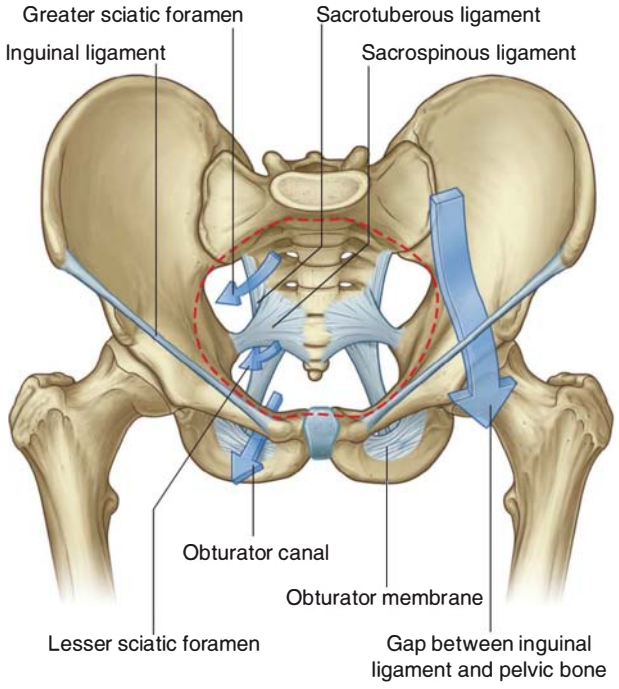


Fig. 6.14 Apertures of communication between the lower limb and other regions.

This gap between the pelvic bone and the inguinal ligament is a weak area in the abdominal wall and often associated with abnormal protrusion of the abdominal cavity and contents into the thigh (femoral hernia). This type of hernia usually occurs where the lymphatic vessels pass through the gap (the femoral canal).

Pelvis

Structures within the pelvis communicate with the lower limb through two major apertures (Fig. 6.14).

Posteriorly, structures communicate with the gluteal region through the greater sciatic foramen and include:

- a muscle—piriformis;
- nerves—sciatic, superior and inferior gluteal, and pudendal nerves; and
- vessels—superior and inferior gluteal arteries and veins, and the internal pudendal artery.

The sciatic nerve is the largest peripheral nerve of the body and is the major nerve of the lower limb.

Anteriorly, the obturator nerve and vessels pass between the pelvis and thigh through the obturator canal. This canal is formed between bone at the top of the obturator foramen and the obturator membrane, which closes most of the foramen during life.

Perineum

Structures pass between the perineum and gluteal region through the lesser sciatic foramen (Fig. 6.14). The most important with respect to the lower limb is the tendon of the obturator internus muscle.

The nerve and artery of the perineum (the internal pudendal artery and pudendal nerve) pass out of the pelvis through the greater sciatic foramen into the gluteal region and then immediately pass around the ischial spine and sacrospinous ligament and through the lesser sciatic foramen to enter the perineum.

KEY POINTS

Innervation is by lumbar and sacral spinal nerves

Somatic motor and general sensory innervation of the lower limb is by peripheral nerves emanating from the lumbar and sacral plexuses on the posterior abdominal and pelvic walls. These plexuses are formed by the anterior rami of L1 to L3 and most of L4 (lumbar plexus) and L4 to S5 (sacral plexus).

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Nerves originating from the lumbar and sacral plexuses and entering the lower limb carry fibers from spinal cord levels L1 to S3 (Fig. 6.15). Nerves from lower sacral segments innervate the perineum. Terminal nerves exit the abdomen and pelvis through a number of apertures and foramina and enter the limb. As a consequence of this innervation, lumbar and upper sacral nerves are tested clinically by examining the lower limb. In addition, clinical signs (such as pain, pins-and-needles sensations, paresthesia, and fascicular muscle twitching) resulting from any disorder affecting these spinal nerves (e.g., herniated intervertebral disc in the lumbar region) appear in the lower limb.

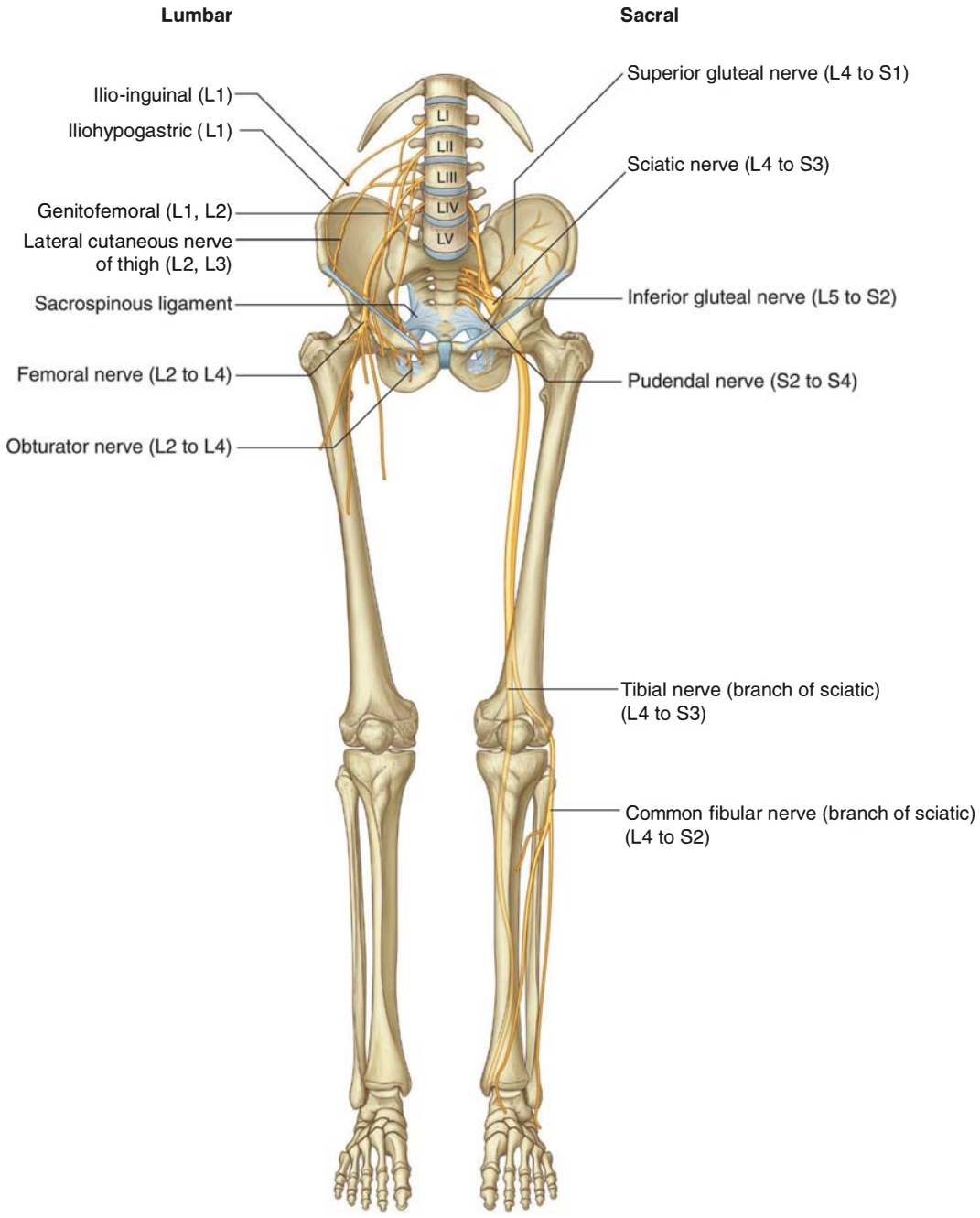


Fig. 6.15 Innervation of the lower limb.

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Dermatomes in the lower limb are shown in Fig. 6.16. Regions that can be tested for sensation and are reasonably autonomous (have minimal overlap) are:

- over the inguinal ligament—L1,
- lateral side of the thigh—L2,
- lower medial side of the thigh—L3,
- medial side of the great toe (digit I)—L4,
- medial side of digit II—L5,
- little toe (digit V)—S1,
- back of the thigh—S2, and
- skin over the gluteal fold—S3.

The dermatomes of S4 and S5 are tested in the perineum.

Selected joint movements are used to test myotomes (Fig. 6.17). For example:

- Flexion of the hip is controlled primarily by L1 and L2.
- Extension of the knee is controlled mainly by L3 and L4.
- Knee flexion is controlled mainly by L5 to S2.
- Plantarflexion of the foot is controlled predominantly by S1 and S2.
- Adduction of the digits is controlled by S2 and S3.

In an unconscious patient, both somatic sensory and somatic motor functions of spinal cord levels can be tested using tendon reflexes:

- A tap on the patellar ligament at the knee tests predominantly L3 and L4.
- A tendon tap on the calcaneal tendon posterior to the ankle (tendon of gastrocnemius and soleus) tests S1 and S2.

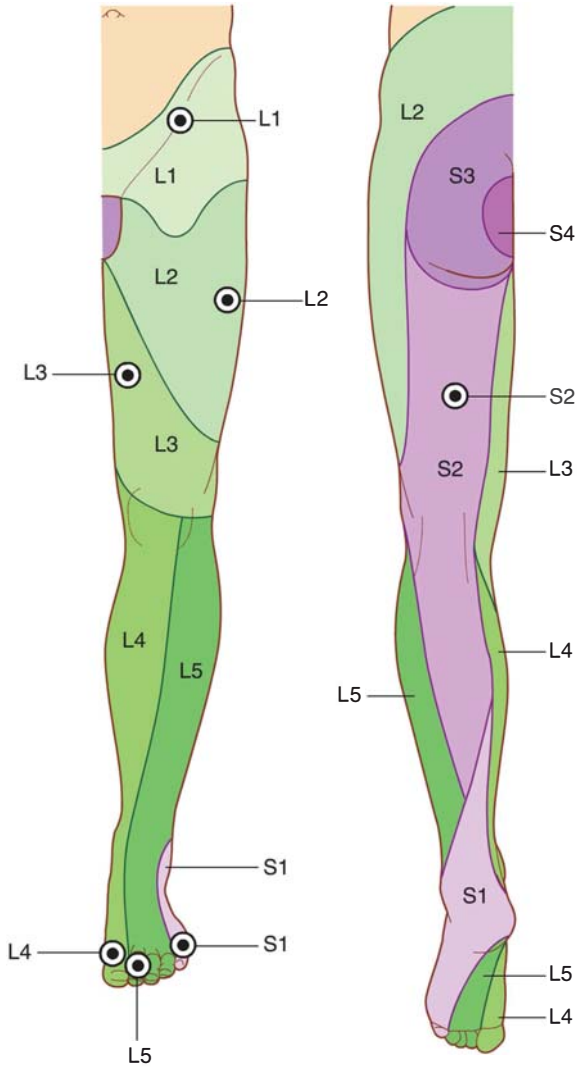


Fig. 6.16 Dermatomes of the lower limb. Dots indicate autonomous zones (i.e., with minimal overlap).

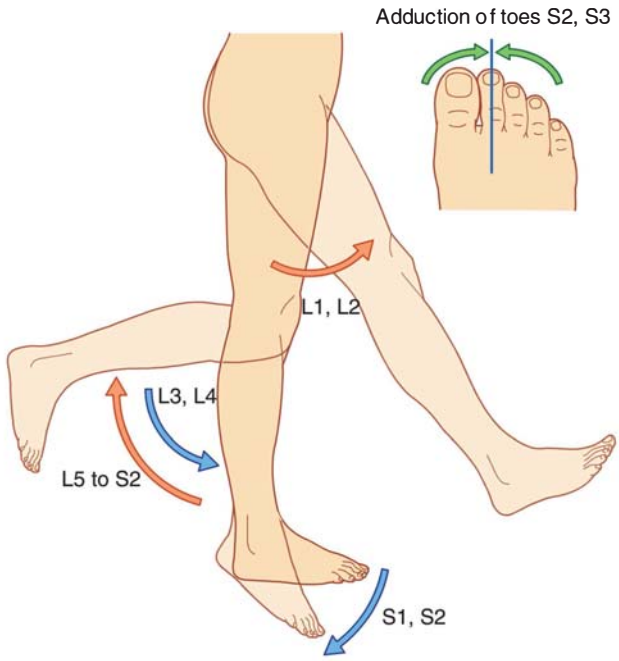


Fig. 6.17 Movements generated by myotomes.

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Each of the major muscle groups or compartments in the lower limb is innervated primarily by one or more of the major nerves that originate from the lumbar and sacral plexuses (Fig. 6.18):

- Large muscles in the gluteal region are innervated by the superior and inferior gluteal nerves.

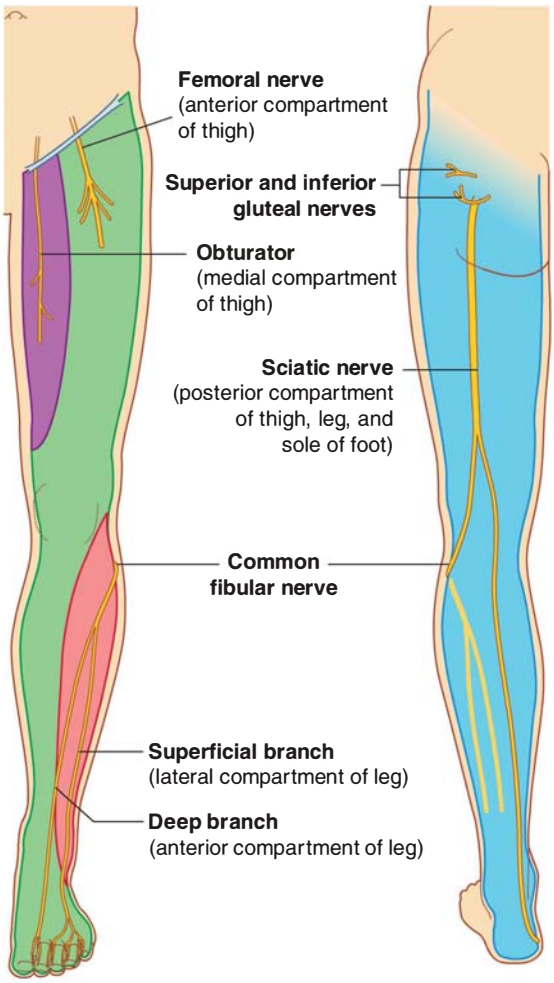


Fig. 6.18 Major nerves of the lower limb (colors indicate regions of motor innervation).

- Most muscles in the anterior compartment of the thigh are innervated by the femoral nerve (except the tensor fasciae latae, which are innervated by the superior gluteal nerve).
- Most muscles in the medial compartment are innervated mainly by the obturator nerve (except the pectineus, which is innervated by the femoral nerve, and part of the adductor magnus, which is innervated by the tibial division of the sciatic nerve).
- Most muscles in the posterior compartment of the thigh and the leg and in the sole of the foot are innervated by the tibial part of the sciatic nerve (except the short head of the biceps femoris in the posterior thigh, which is innervated by the common fibular division of the sciatic nerve).
- The anterior and lateral compartments of the leg and muscles associated with the dorsal surface of the foot are innervated by the common fibular part of the sciatic nerve.

In addition to innervating major muscle groups, each of the major peripheral nerves originating from the lumbar and sacral plexuses carries general sensory information from patches of skin (Fig. 6.19). Sensation from these areas can be used to test for peripheral nerve lesions:

- The femoral nerve innervates skin on the anterior thigh, medial side of the leg, and medial side of the ankle.
- The obturator nerve innervates the medial side of the thigh.
- The tibial part of the sciatic nerve innervates the lateral side of the ankle and foot.
- The common fibular nerve innervates the lateral side of the leg and the dorsum of the foot.

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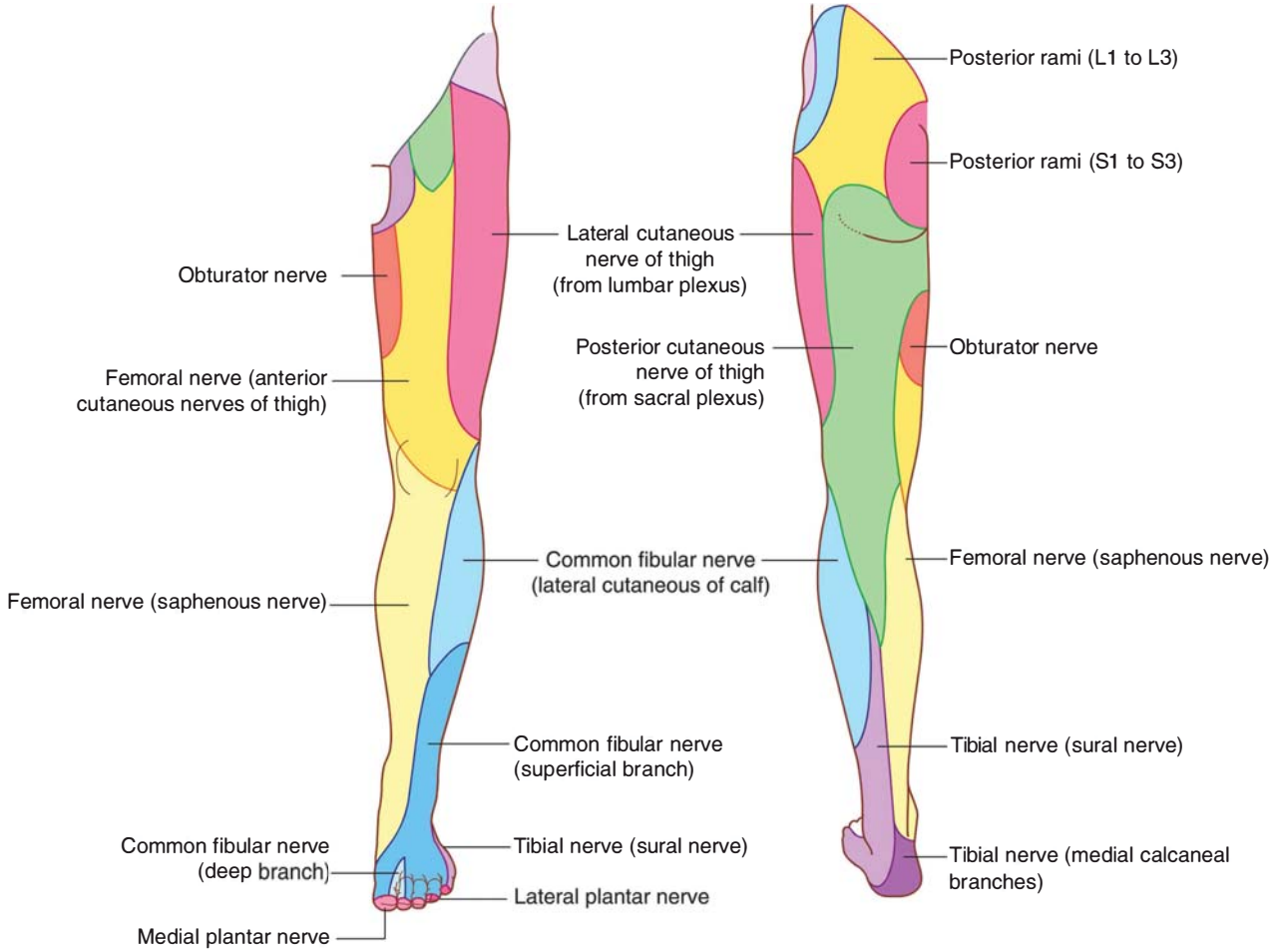


Fig. 6.19 Regions of skin innervated by peripheral nerves.

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Nerves related to bone

The common fibular branch of the sciatic nerve curves laterally around the neck of the fibula when passing from the popliteal fossa into the leg (Fig. 6.20). The nerve can be rolled against bone just distal to the attachment of biceps femoris to the head of the fibula. In this location, the nerve can be damaged by impact injuries, fractures to the bone, or leg casts that are placed too high.

Superficial veins

Large veins embedded in the subcutaneous (superficial) fascia of the lower limb (Fig. 6.21) often become distended (varicose). These vessels can also be used for vascular transplantation.

The most important superficial veins are the great and small saphenous veins, which originate from the medial and lateral sides, respectively, of a dorsal venous arch in the foot.

- The great saphenous vein passes up the medial side of the leg, knee, and thigh to pass through an opening in deep fascia covering the femoral triangle and join with the femoral vein.
- The small saphenous vein passes behind the distal end of the fibula (lateral malleolus) and up the back of the leg to penetrate deep fascia and join the popliteal vein posterior to the knee.

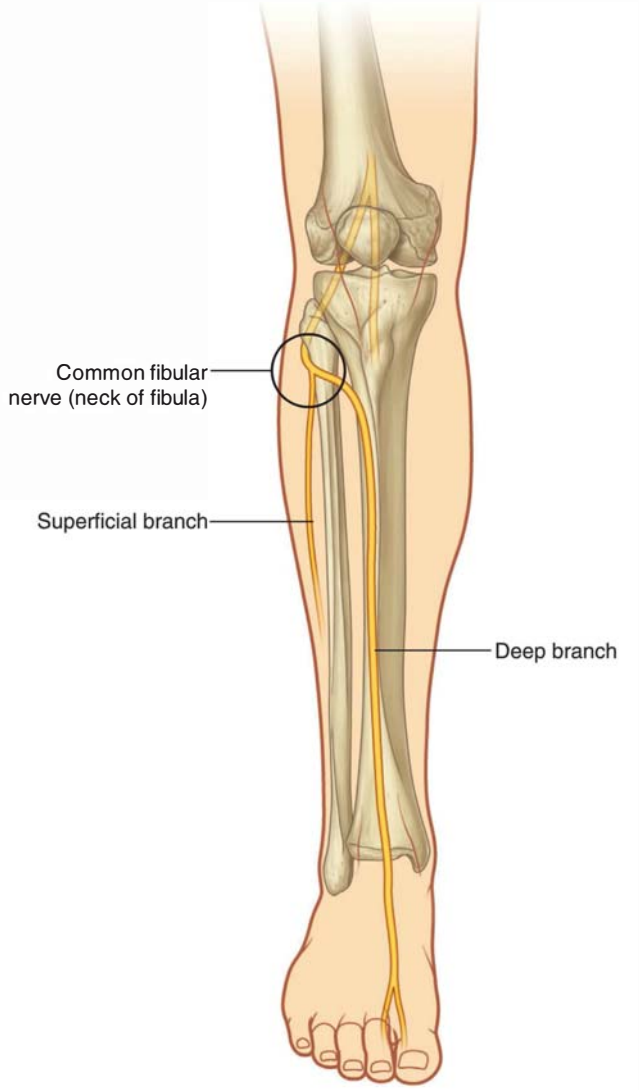


Fig. 6.20 Nerves related to bone.

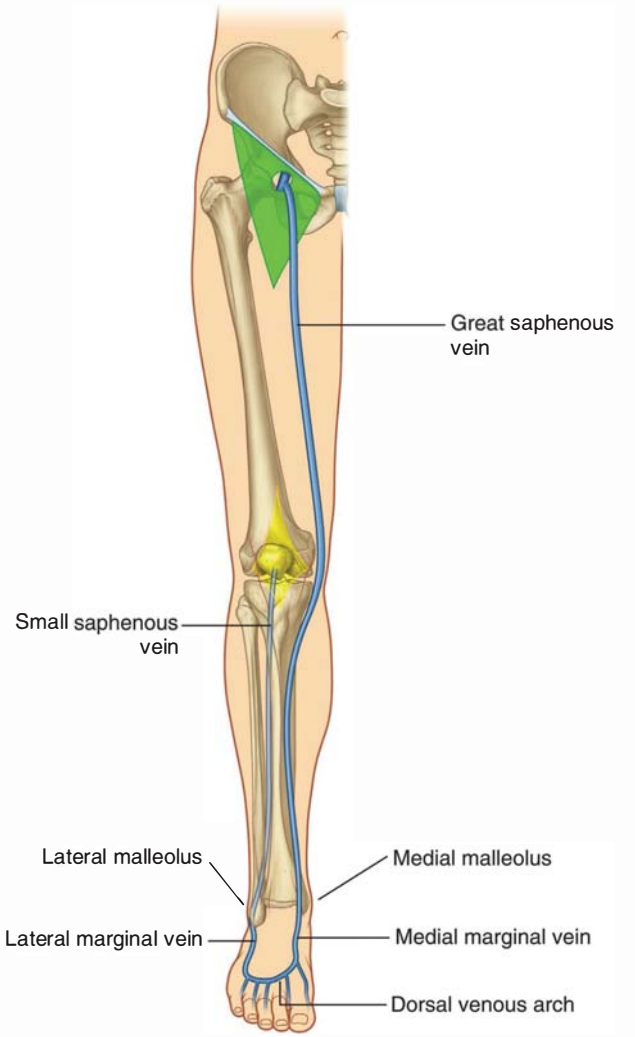


Fig. 6.21 Superficial veins.

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Regional anatomy

Bony pelvis

The external surfaces of the pelvic bones, sacrum, and coccyx are predominantly the regions of the pelvis associated with the lower limb, although some muscles do originate from the deep or internal surfaces of these bones and from the deep surfaces of the lumbar vertebrae, above (Fig. 6.22).

Each pelvic bone is formed by three bones (ilium, ischium, and pubis), which fuse during childhood. The **ilium** is superior and the **pubis** and **ischium** are anteroinferior and posteroinferior, respectively.

The ilium articulates with the sacrum. The pelvic bone is further anchored to the end of the vertebral column (sacrum and coccyx) by the sacrotuberous and sacrospinous ligaments, which attach to a tuberosity and spine on the ischium.

The outer surface of the ilium, and the adjacent surfaces of the sacrum, coccyx, and sacrotuberous ligament, are associated with the gluteal region of the lower limb and provide extensive muscle attachment. The ischial tuberosity provides attachment for many of the muscles in the posterior compartment of the thigh, and the ischiopubic ramus and body of the pubis are associated mainly with muscles in the medial compartment of the thigh. The head of the femur articulates with the acetabulum on the lateral surface of the pelvic bone.

Ilium

The upper fan-shaped part of the ilium is associated on its inner side with the abdomen and on its outer side with the lower limb. The top of this region is the **iliac crest**, which

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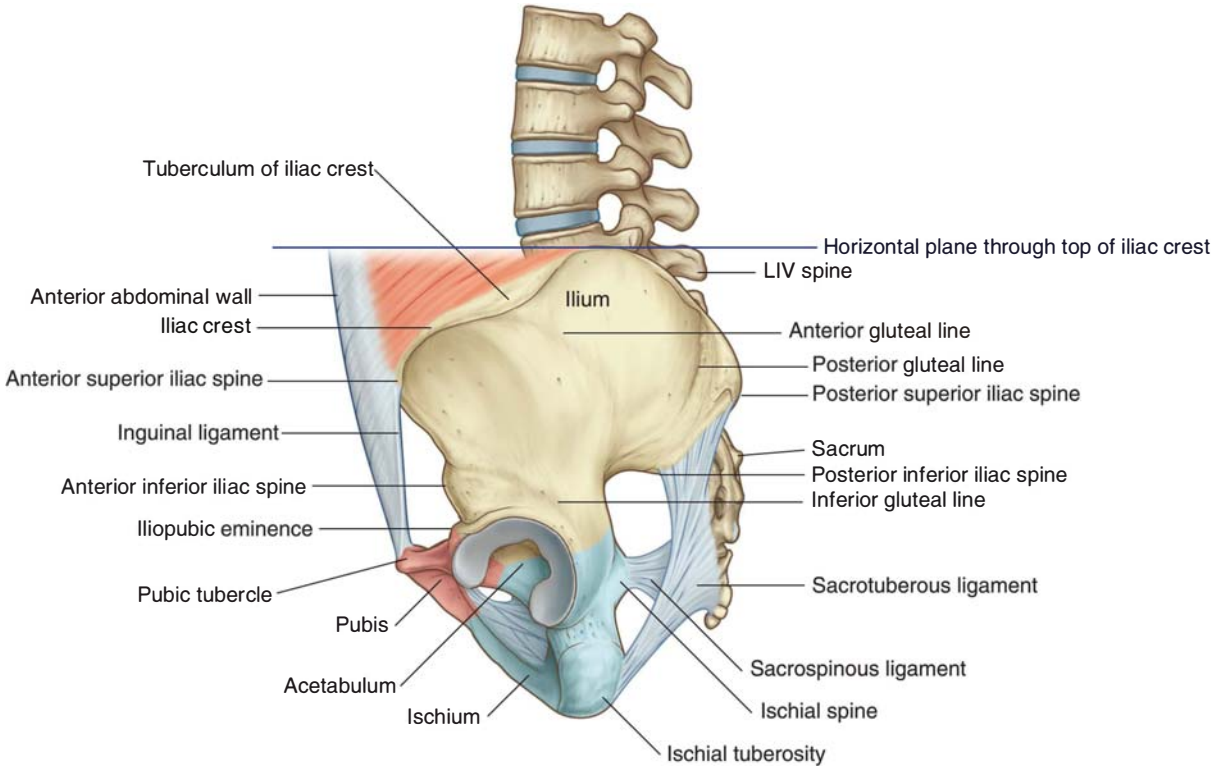


Fig. 6.22 External surface of the bony pelvis. Lateral view.

ends anteriorly as the **anterior superior iliac spine** and posteriorly as the **posterior superior iliac spine**. A prominent lateral expansion of the crest just posterior to the anterior superior iliac spine is the **tuberculum of the iliac crest**.

The anterior inferior iliac spine is on the anterior margin of the ilium, and below this, where the ilium fuses with the pubis, is a raised area of bone (the **iliopubic eminence**).

The gluteal surface of the ilium faces posterolaterally and lies below the iliac crest. It is marked by three curved lines (inferior, anterior, and posterior gluteal lines), which divide the surface into four regions:

- The **inferior gluteal line** originates just superior to the anterior inferior iliac spine and curves inferiorly across the bone to end near the posterior margin of the acetabulum—the rectus femoris muscle attaches to the anterior inferior iliac spine and to a roughened patch of bone between the superior margin of the acetabulum and the inferior gluteal line.
- The **anterior gluteal line** originates from the lateral margin of the iliac crest between the anterior superior iliac spine and the tuberculum of the iliac crest, and arches inferiorly across the ilium to disappear just superior to the upper margin of the greater sciatic foramen—the gluteus minimus muscle originates from between the inferior and anterior gluteal lines.
- The **posterior gluteal line** descends almost vertically from the iliac crest to a position near the posterior inferior iliac spine—the gluteus medius muscle attaches to bone between the anterior and posterior gluteal lines, and the gluteus maximus muscle attaches posterior to the posterior gluteal line.

Ischial tuberosity

The **ischial tuberosity** is posteroinferior to the acetabulum and is associated mainly with the hamstring muscles of the posterior thigh (Fig. 6.23). It is divided into upper and lower areas by a transverse line.

The upper area of the ischial tuberosity is oriented vertically and is further subdivided into two parts by an oblique line, which descends, from medial to lateral, across the surface:

- The more medial part of the upper area is for the attachment of the combined origin of the semitendinosus muscle and the long head of the biceps femoris muscle.
- The lateral part is for the attachment of the semimembranosus muscle.

The lower area of the ischial tuberosity is oriented horizontally and is divided into medial and lateral regions by a ridge of bone:

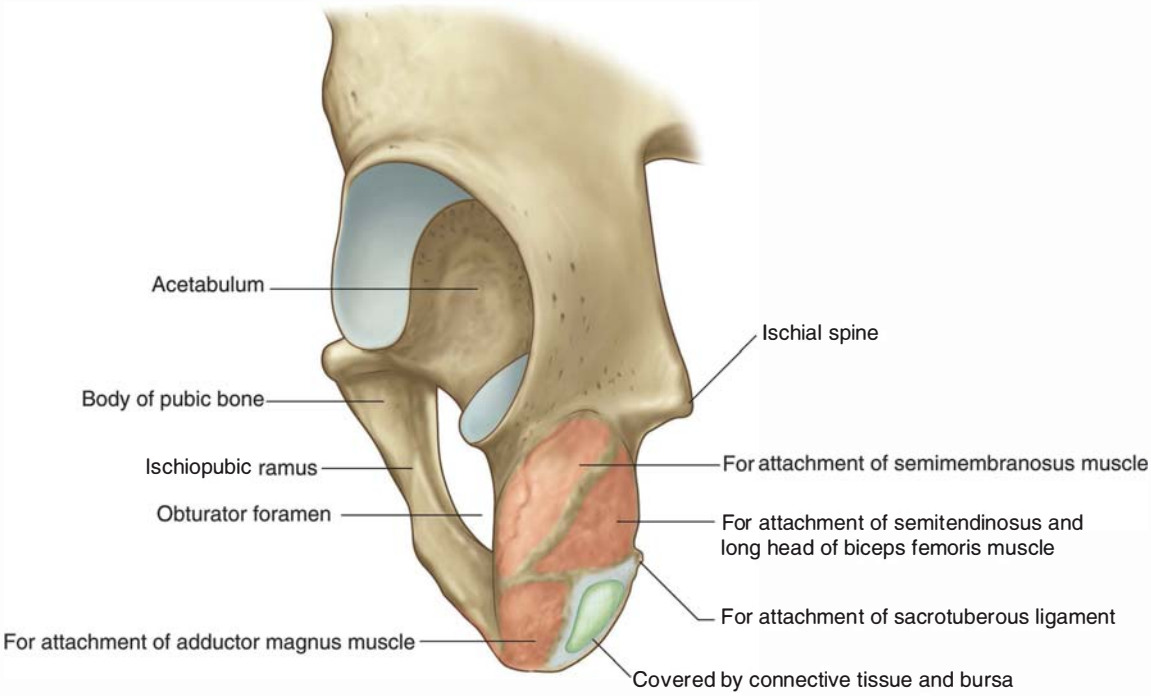


Fig. 6.23 Ischial tuberosity. Posterolateral view.

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- The lateral region provides attachment for part of the adductor magnus muscle.
- The medial part faces inferiorly and is covered by connective tissue and by a bursa.

When sitting, this medial part supports the body weight.

The sacrotuberous ligament is attached to a sharp ridge on the medial margin of the ischial tuberosity.

Ischiopubic ramus and pubic bone

The external surfaces of the ischiopubic ramus anterior to the ischial tuberosity and the body of the pubis provide attachment for muscles of the medial compartment of the thigh (Fig. 6.23). These muscles include the adductor longus, adductor brevis, adductor magnus, pectineus, and gracilis.

Acetabulum

The large cup-shaped **acetabulum** for articulation with the head of the femur is on the lateral surface of the pelvic bone in the region where the ilium, pubis, and ischium fuse (Fig. 6.24).

The margin of the acetabulum is marked inferiorly by a prominent notch (**acetabular notch**).

The wall of the acetabulum consists of nonarticular and articular parts:

- The nonarticular part is rough and forms a shallow circular depression (the **acetabular fossa**) in central and inferior parts of the acetabular floor—the acetabular notch is continuous with the acetabular fossa.
- The articular surface is broad and surrounds the anterior, superior, and posterior margins of the acetabular fossa.

The smooth crescent-shaped articular surface (the **lunate surface**) is broadest superiorly where most of the body's weight is transmitted through the pelvis to the femur. The lunate surface is deficient inferiorly at the acetabular notch.

The acetabular fossa provides attachment for the ligament of the head of the femur, whereas blood vessels and nerves pass through the acetabular notch.

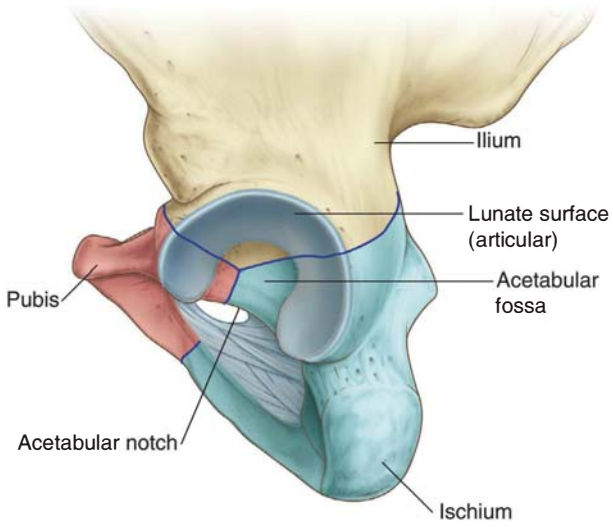


Fig. 6.24 Acetabulum.

In the clinic

Pelvic fractures

The pelvic bones, sacrum, and associated joints form a bony ring surrounding the pelvic cavity. Soft tissue and visceral organ damage must be suspected when the pelvis is fractured. Patients with multiple injuries and evidence of chest, abdominal, and lower limb trauma should also be investigated for pelvic trauma.

Pelvic fractures can be associated with appreciable blood loss (concealed exsanguination) and blood transfusion is often required. In addition, this bleeding tends to form a significant pelvic hematoma, which can compress nerves, press on organs, and inhibit pelvic visceral function (Fig. 6.25).

There are many ways of classifying pelvic fractures, which enable the surgeon to determine the appropriate treatment and the patient's prognosis. Pelvic fractures are generally of four types.

- Type 1 injuries occur without disruption of the bony pelvic ring (e.g., a fracture of the iliac crest). These types of injuries are unlikely to represent significant trauma, though in the case of a fracture of the iliac crest, blood loss needs to be assessed.
- Type 2 injuries occur with a single break in the bony pelvic ring. An example of this would be a single fracture with diastasis (separation) of the symphysis pubis. Again, these injuries are relatively benign in nature, but it may be appropriate to assess for blood loss.

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In the clinic—cont'd

- Type 3 injuries occur with double breaks in the bony pelvic ring. These include bilateral fractures of the pubic rami, which may produce urethral damage.
- Type 4 injuries occur at and around the acetabulum.

Other types of pelvic ring injuries include fractures of the pubic rami and disruption of the sacro-iliac joint with or without dislocation. This may involve significant visceral pelvic trauma and hemorrhage.

Other general pelvic injuries include stress fractures and insufficiency fractures, as seen in athletes and elderly patients with osteoporosis, respectively.

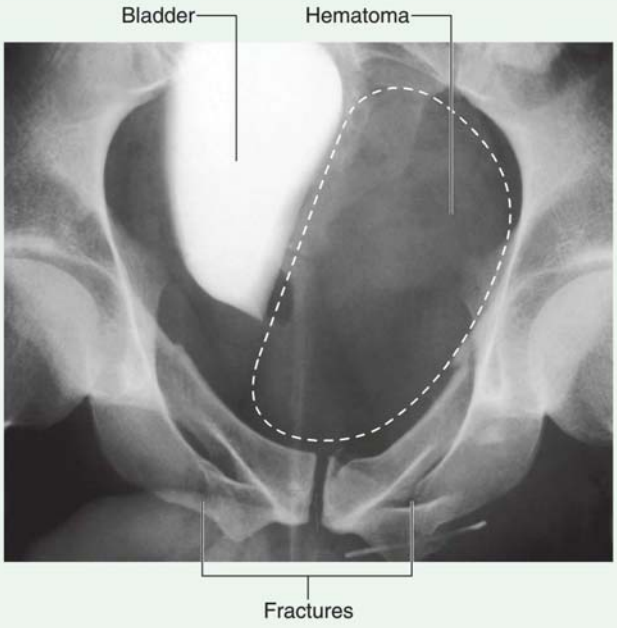


Fig. 6.25 Multiple fractures of the pelvis. Radiograph with contrast in the bladder. A large accumulation of blood is deforming the bladder.

Proximal femur

The femur is the bone of the thigh and the longest bone in the body. Its proximal end is characterized by a head and neck, and two large projections (the greater and lesser trochanters) on the upper part of the shaft (Fig. 6.26).

The **head** of the femur is spherical and articulates with the acetabulum of the pelvic bone. It is characterized by a nonarticular pit (**fovea**) on its medial surface for the attachment of the ligament of the head.

The **neck** of the femur is a cylindrical strut of bone that connects the head to the shaft of the femur. It projects superomedially from the shaft at an angle of approximately 125°, and projects slightly forward. The orientation of the neck relative to the shaft increases the range of movement of the hip joint.

The upper part of the **shaft** of the femur bears a greater and lesser trochanter, which are attachment sites for muscles that move the hip joint.

Greater and lesser trochanters

The **greater trochanter** extends superiorly from the shaft of the femur just lateral to the region where the shaft joins the neck of the femur (Fig. 6.26). It continues posteriorly where its medial surface is deeply grooved to form the **trochanteric fossa**. The lateral wall of this fossa bears a distinct oval depression for attachment of the obturator externus muscle.

The greater trochanter has an elongate ridge on its anterolateral surface for attachment of the gluteus minimus and a similar ridge more posteriorly on its lateral surface for attachment of the gluteus medius. Between these two points, the greater trochanter is palpable.

On the medial side of the superior aspect of the greater trochanter and just above the trochanteric fossa is a small impression for attachment of the obturator internus and its associated gemelli muscles, and immediately above and behind this feature is an impression on the margin of the trochanter for attachment of the piriformis muscle.

The **lesser trochanter** is smaller than the greater trochanter and has a blunt conical shape. It projects posteromedially from the shaft of the femur just inferior to the junction with the neck (Fig. 6.26). It is the attachment site for the combined tendons of psoas major and iliacus muscles.

Extending between the two trochanters and separating the shaft from the neck of the femur are the intertrochanteric line and intertrochanteric crest.

Intertrochanteric line

The **intertrochanteric line** is a ridge of bone on the anterior surface of the upper margin of the shaft that descends

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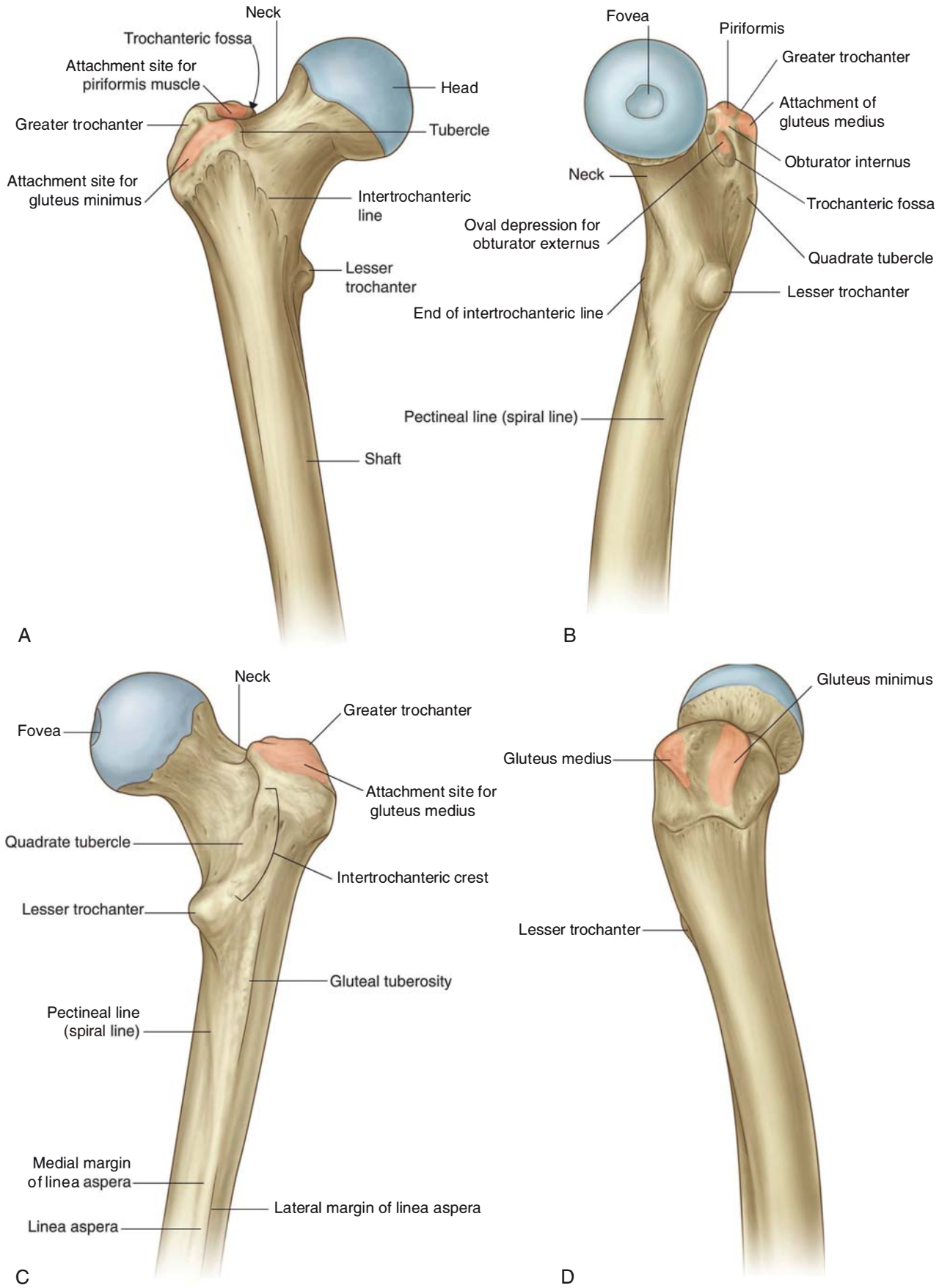


Fig. 6.26 Proximal end of the femur (*right*). **A.** Anterior view. **B.** Medial view. **C.** Posterior view. **D.** Lateral view.

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medially from a tubercle on the anterior surface of the base of the greater trochanter to a position just anterior to the base of the lesser trochanter (Fig. 6.26). It is continuous with the **pectineal line** (spiral line), which curves medially under the lesser trochanter and around the shaft of the femur to merge with the medial margin of the **linea aspera** on the posterior aspect of the femur.

Intertrochanteric crest

The **intertrochanteric crest** is on the posterior surface of the femur and descends medially across the bone from the posterior margin of the greater trochanter to the base of the lesser trochanter (Fig. 6.26). It is a broad smooth ridge of bone with a prominent tubercle (the **quadrate tubercle**) on its upper half, which provides attachment for the quadratus femoris muscle.

Shaft of the femur

The shaft of the femur descends from lateral to medial in the coronal plane at an angle of 7° from the vertical axis (Fig. 6.27). The distal end of the femur is therefore closer to the midline than the upper end of the shaft.

The middle third of the shaft of the femur is triangular in shape with smooth lateral and medial margins between anterior, lateral (posterolateral), and medial (posteromedial) surfaces. The posterior margin is broad and forms a prominent raised crest (the linea aspera).

The linea aspera is a major site of muscle attachment in the thigh. In the proximal third of the femur, the medial and lateral margins of the linea aspera diverge and continue superiorly as the pectineal line and gluteal tuberosity, respectively (Fig. 6.27):

- The pectineal line curves anteriorly under the lesser trochanter and joins the intertrochanteric line.
- The gluteal tuberosity is a broad linear roughening that curves laterally to the base of the greater trochanter.

The gluteus maximus muscle is attached to the gluteal tuberosity.

The triangular area enclosed by the pectineal line, the gluteal tuberosity, and the intertrochanteric crest is the posterior surface of the proximal end of the femur.

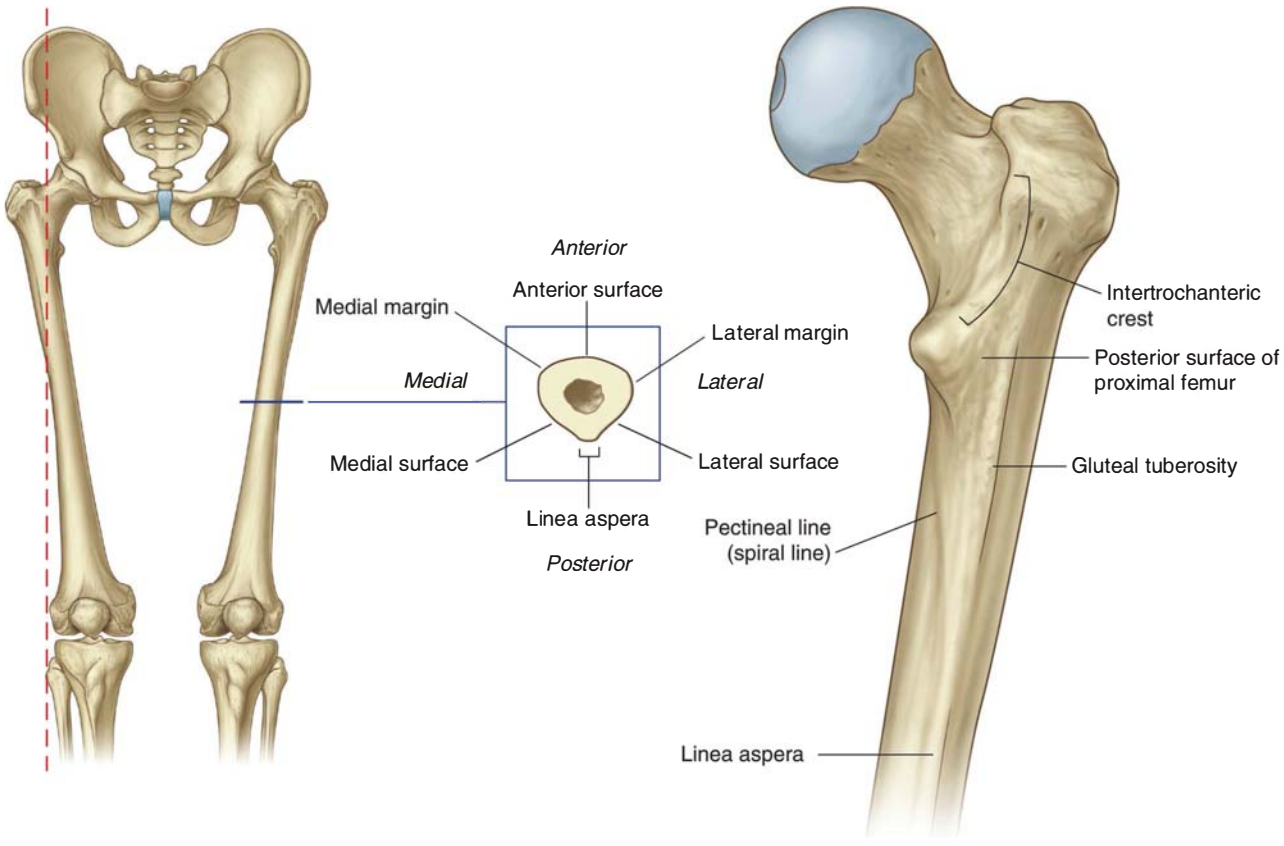


Fig. 6.27 Shaft of the femur. On the right is a posterior view of proximal shaft of right femur.

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In the clinic

Femoral neck fractures

Femoral neck fractures (Fig. 6.28) can interrupt the blood supply to the femoral head. The blood supply to the head and neck is primarily from an arterial ring formed around the base of the femoral neck. From here, vessels course along the neck, penetrate the capsule, and supply the femoral head. The blood supply to the femoral head and femoral neck is further enhanced by the artery of the ligamentum teres, which is generally small and variable. Femoral neck fractures may disrupt associated vessels and lead to necrosis of the femoral head.



Fig. 6.28 This radiograph of the pelvis, anteroposterior view, demonstrates a fracture of the neck of the femur.

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In the clinic

Intertrochanteric fractures

In these fractures, the break usually runs from the greater trochanter through to the lesser trochanter and does not involve the femoral neck. Intertrochanteric fractures preserve the femoral neck blood supply and do not render the femoral head ischemic.

In the clinic

Femoral shaft fractures

An appreciable amount of energy is needed to fracture the femoral shaft. This type of injury is therefore accompanied by damage to the surrounding soft tissues, which include the muscle compartments and the structures they contain.

Hip joint

The hip joint is a synovial articulation between the head of the femur and the acetabulum of the pelvic bone (Fig. 6.29A). The joint is a multiaxial ball and socket joint designed for stability and weight-bearing at the expense of mobility. Movements at the joint include flexion, extension, abduction, adduction, medial and lateral rotation, and circumduction.

When considering the effects of muscle action on the hip joint, the long neck of the femur and the angulation of the neck on the shaft of the femur must be borne in mind. For example, medial and lateral rotation of the femur involves muscles that move the greater trochanter forward and backward, respectively, relative to the acetabulum (Fig. 6.29B).

The articular surfaces of the hip joint are:

- the spherical head of the femur, and
- the lunate surface of the acetabulum of the pelvic bone.

The acetabulum almost entirely encompasses the hemispherical head of the femur and contributes substantially to joint stability. The nonarticular acetabular fossa contains loose connective tissue. The lunate surface is covered by hyaline cartilage and is broadest superiorly.

Except for the fovea, the head of the femur is also covered by hyaline cartilage.

The rim of the acetabulum is raised slightly by a fibrocartilaginous collar (the acetabular labrum). Inferiorly, the labrum bridges across the acetabular notch as the

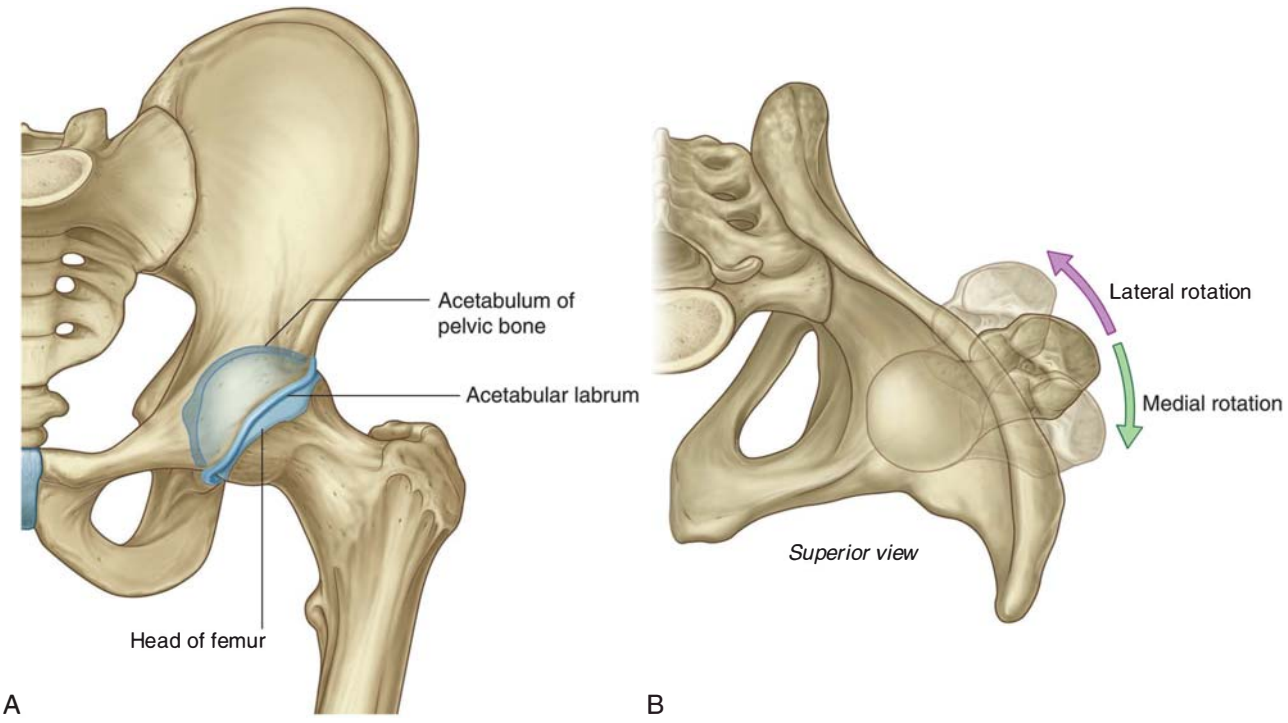


Fig. 6.29 Hip joint. **A.** Articular surfaces. Anterior view. **B.** Movement of the neck of the femur during medial and lateral rotation. Superior view.

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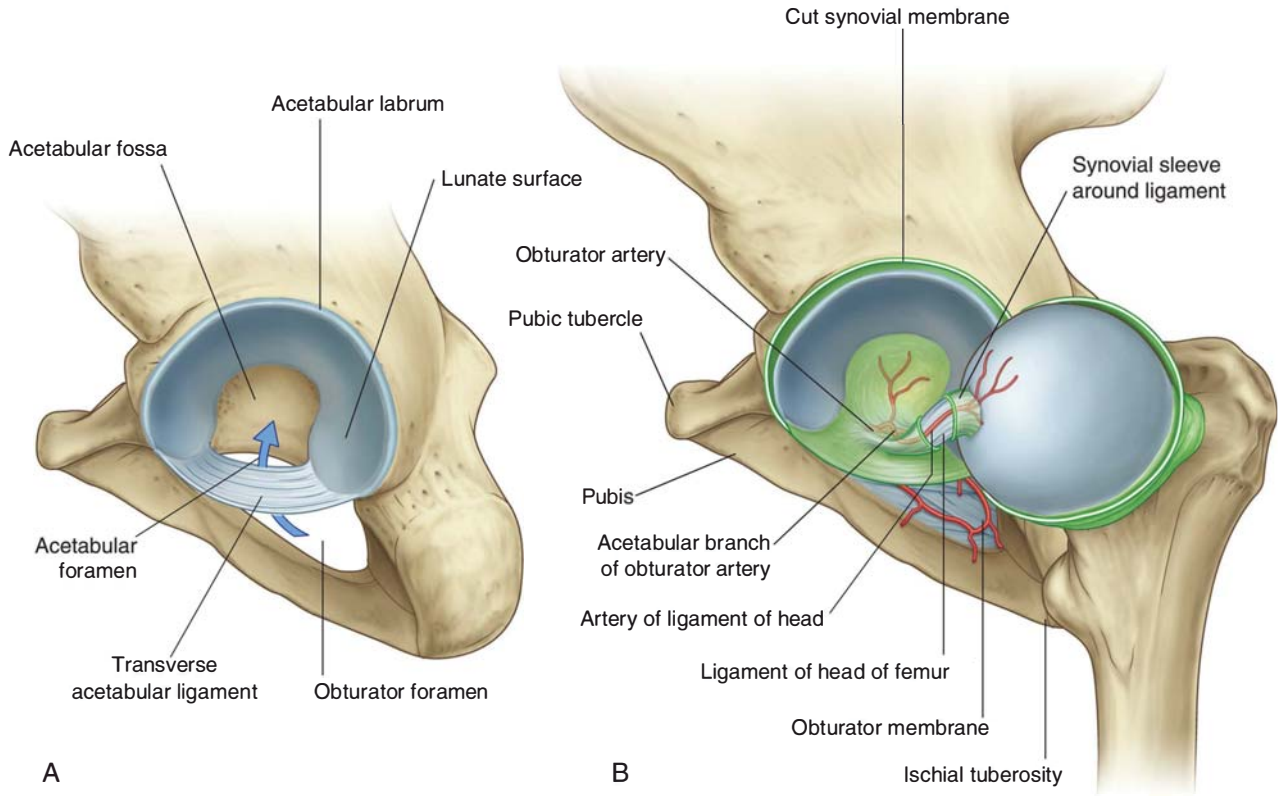


Fig. 6.30 Hip joint. **A.** Transverse acetabular ligament. **B.** Ligament of the head of the femur. The head of the femur has been laterally rotated out of the acetabulum to show the ligament.

transverse acetabular ligament and converts the notch into a foramen (Fig. 6.30A).

The **ligament of the head of the femur** is a flat band of delicate connective tissue that attaches at one end to the fovea on the head of the femur and at the other end to the acetabular fossa, transverse acetabular ligament, and margins of the acetabular notch (Fig. 6.30B). It carries a small branch of the obturator artery, which contributes to the blood supply of the head of the femur.

The synovial membrane attaches to the margins of the articular surfaces of the femur and acetabulum, forms a tubular covering around the ligament of the head of the femur, and lines the fibrous membrane of the joint (Figs. 6.30B and 6.31). From its attachment to the margin of the head of the femur, the synovial membrane covers the neck of the femur before reflecting onto the fibrous membrane (Fig. 6.31).

The fibrous membrane that encloses the hip joint is strong and generally thick. Medially, it is attached to the margin of the acetabulum, the transverse acetabular

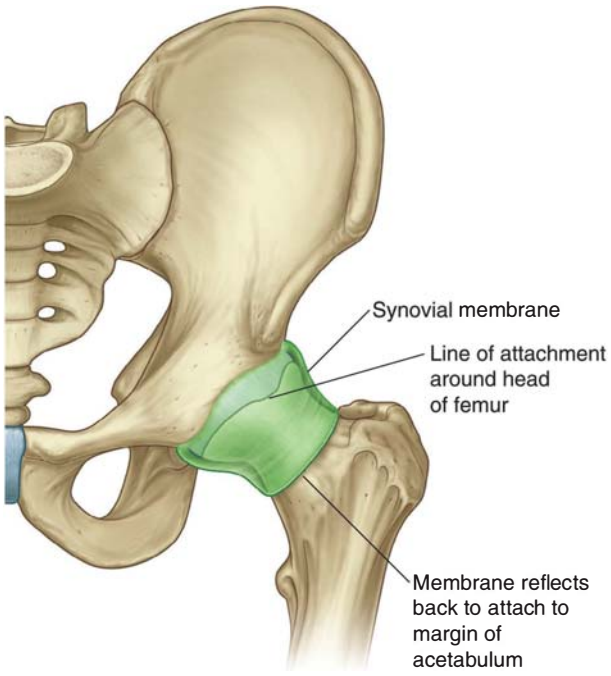


Fig. 6.31 Synovial membrane of the hip joint.

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ligament, and the adjacent margin of the obturator foramen (Fig. 6.32A). Laterally, it is attached to the intertrochanteric line on the anterior aspect of the femur and to the neck of the femur just proximal to the intertrochanteric crest on the posterior surface.

Ligaments

Three ligaments reinforce the external surface of the fibrous membrane and stabilize the joint: the iliofemoral, pubofemoral, and ischiofemoral ligaments.

- The **iliofemoral ligament** is anterior to the hip joint and is triangular shaped (Fig. 6.32B). Its apex is attached to the ilium between the anterior inferior iliac spine and the margin of the acetabulum and its base is attached along the intertrochanteric line of the femur. Parts of the ligament attached above and below the intertrochanteric line are thicker than the part attached to the central part of the line. This results in the ligament having a Y appearance.
- The **pubofemoral ligament** is anteroinferior to the hip joint (Fig. 6.32B). It is also triangular in shape, with its base attached medially to the iliopubic eminence, adjacent bone, and obturator membrane. Laterally, it blends with the fibrous membrane and with the deep surface of the iliofemoral ligament.
- The **ischiofemoral ligament** reinforces the posterior aspect of the fibrous membrane (Fig. 6.32C). It is attached medially to the ischium, just posteroinferior to the acetabulum, and laterally to the greater trochanter deep to the iliofemoral ligament.

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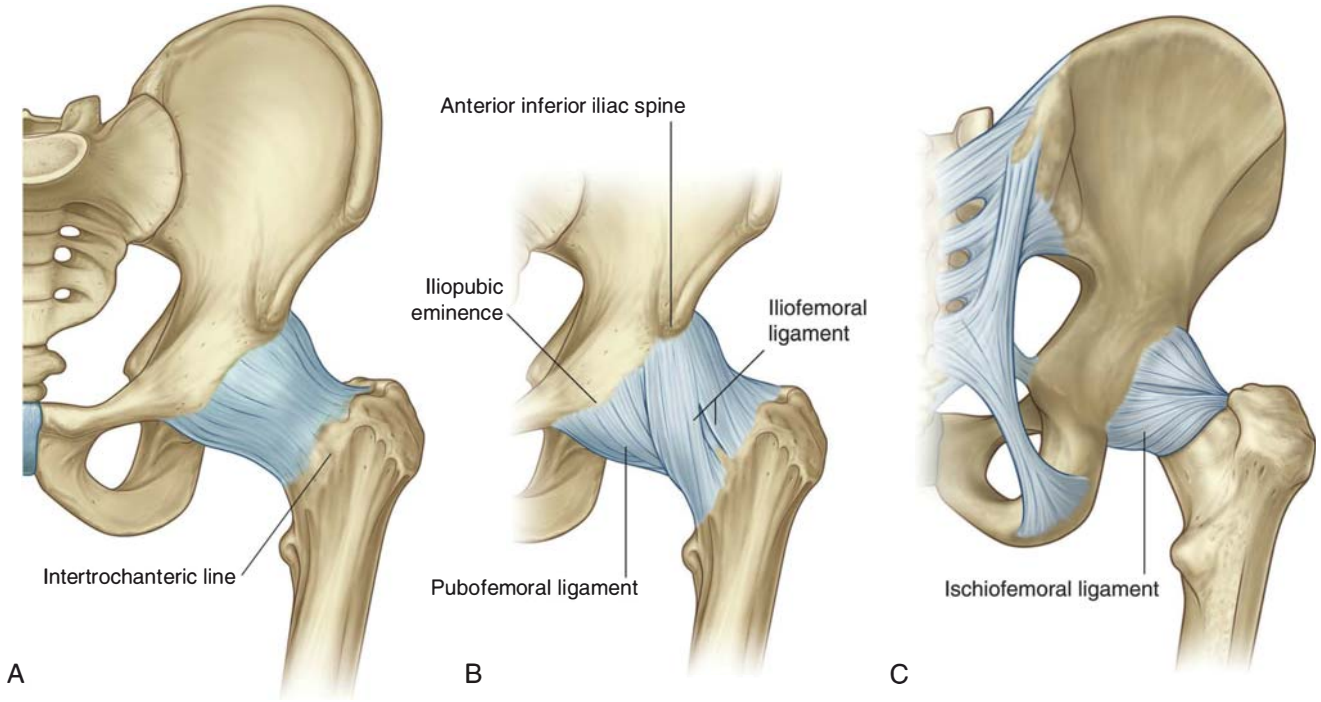


Fig. 6.32 Fibrous membrane and ligaments of the hip joint. **A.** Fibrous membrane of the joint capsule. Anterior view. **B.** Iliofemoral and pubofemoral ligaments. Anterior view. **C.** Ischiofemoral ligament. Posterior view.

The fibers of all three ligaments are oriented in a spiral fashion around the hip joint so that they become taut when the joint is extended. This stabilizes the joint and reduces the amount of muscle energy required to maintain a standing position.

Vascular supply to the hip joint is predominantly through branches of the obturator artery, medial and lateral circumflex femoral arteries, superior and inferior gluteal arteries, and the first perforating branch of the deep artery of the thigh. The articular branches of these vessels form a network around the joint (Fig. 6.33).

The hip joint is innervated by articular branches from the femoral, obturator, and superior gluteal nerves, and the nerve to the quadratus femoris.

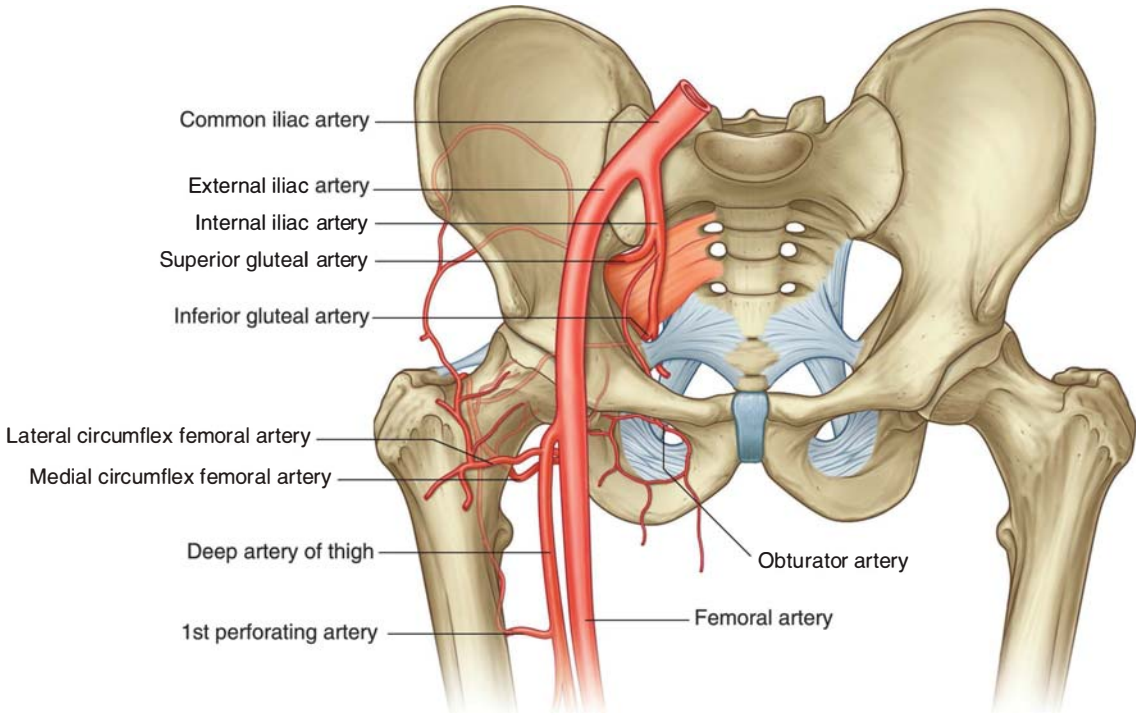


Fig. 6.33 Blood supply of the hip joint.

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Gateways to the lower limb

There are four major routes by which structures pass from the abdomen and pelvis into and out of the lower limb. These are the obturator canal, the greater sciatic foramen, the lesser sciatic foramen, and the gap between the inguinal ligament and the anterosuperior margin of the pelvis (Fig. 6.34).

Obturator canal

The **obturator canal** is an almost vertically oriented passageway at the anterosuperior edge of the obturator foramen (Fig. 6.34). It is bordered:

- above by a groove (**obturator groove**) on the inferior surface of the superior ramus of the pubic bone, and
- below by the upper margin of the obturator membrane, which fills most of the obturator foramen, and by muscles (obturator internus and externus) attached to the inner and outer surfaces of the obturator membrane and surrounding bone.

The obturator canal connects the abdominopelvic region with the medial compartment of the thigh. The obturator nerve and vessels pass through the canal.

Greater sciatic foramen

The **greater sciatic foramen** is formed on the posterolateral pelvic wall and is the major route for structures to pass between the pelvis and the gluteal region of the lower limb (Fig. 6.34). The margins of the foramen are formed by:

- the greater sciatic notch,
- parts of the upper borders of the sacrospinous and sacrotuberous ligaments, and
- the lateral border of the sacrum.

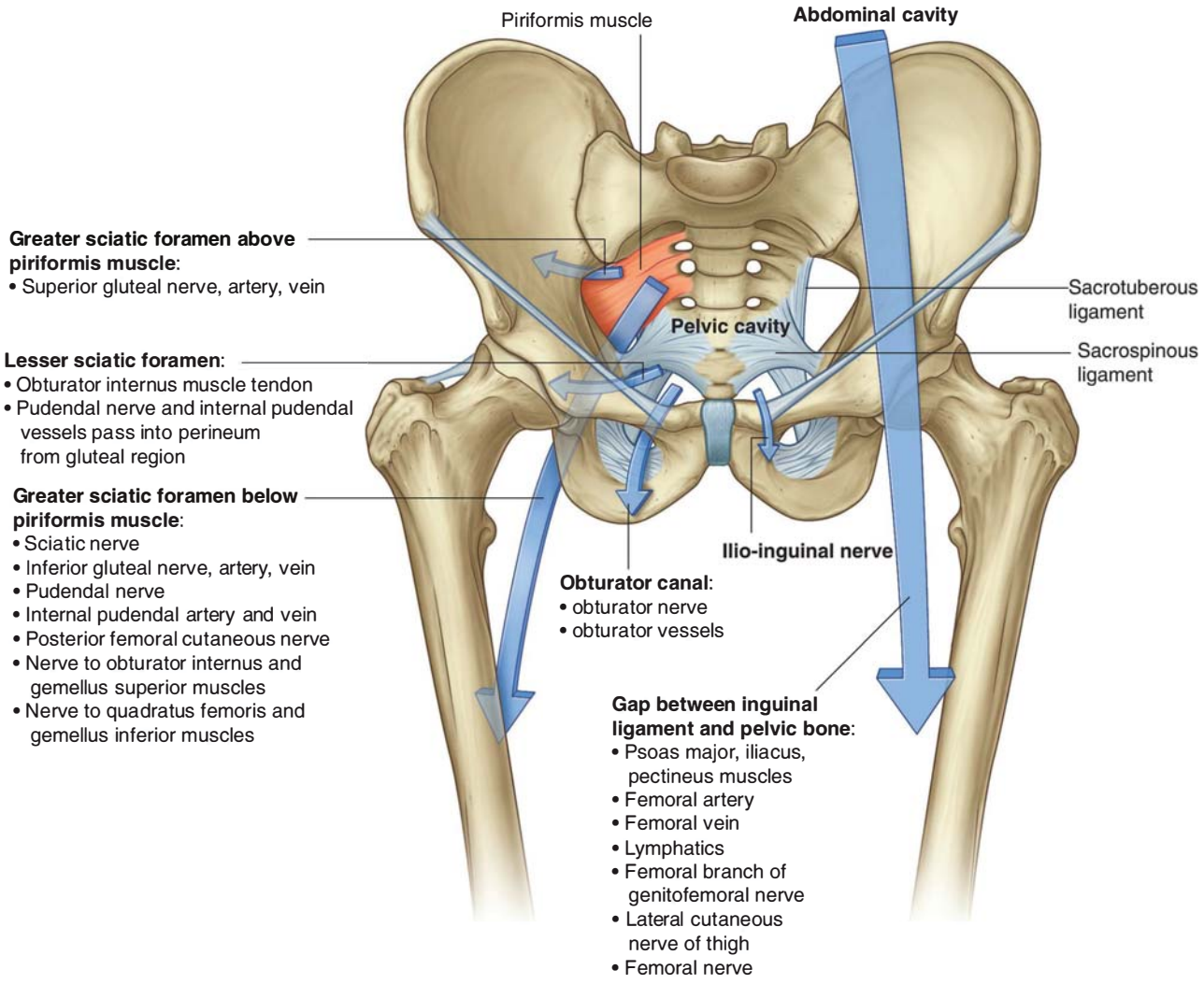


Fig. 6.34 Gateways to the lower limb.

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The piriformis muscle passes out of the pelvis into the gluteal region through the greater sciatic foramen and separates the foramen into two parts, a part above the muscle and a part below:

- The superior gluteal nerve and vessels pass through the greater sciatic foramen above the piriformis.
- The sciatic nerve, inferior gluteal nerve and vessels, pudendal nerve and internal pudendal vessels, posterior cutaneous nerve of the thigh, nerve to the obturator internus and gemellus superior, and nerve to the quadratus femoris and gemellus inferior pass through the greater sciatic foramen below the muscle.

Lesser sciatic foramen

The **lesser sciatic foramen** is inferior to the greater sciatic foramen on the posterolateral pelvic wall (Fig. 6.34). It is also inferior to the lateral attachment of the pelvic floor (levator ani and coccygeus muscles) to the pelvic wall and therefore connects the gluteal region with the perineum:

- The tendon of the obturator internus passes from the lateral pelvic wall through the lesser sciatic foramen into the gluteal region to insert on the femur.
- The pudendal nerve and internal pudendal vessels, which first exit the pelvis by passing through the greater sciatic foramen below the piriformis muscle, enter the perineum below the pelvic floor by passing around the ischial spine and sacrospinous ligament and medially through the lesser sciatic foramen.

Gap between the inguinal ligament and pelvic bone

The large crescent-shaped gap between the inguinal ligament above and the anterosuperior margin of the pelvic bone below is the major route of communication between the abdomen and the anteromedial aspect of the thigh (Fig. 6.34). The psoas major, iliacus, and pectineus muscles pass through this gap to insert onto the femur. The major blood vessels (femoral artery and vein) and lymphatics of the lower limb also pass through it, as does the femoral nerve, to enter the femoral triangle of the thigh.

Nerves

Nerves that enter the lower limb from the abdomen and pelvis are terminal branches of the lumbosacral plexus on the posterior wall of the abdomen and the posterolateral walls of the pelvis (Fig. 6.35 and Table 6.1).

The **lumbar plexus** is formed by the anterior rami of spinal nerves L1 to L3 and part of L4 (see Chapter 4, pp. 398–401). The rest of the anterior ramus of L4 and the anterior ramus of L5 combine to form the **lumbosacral trunk**, which enters the pelvic cavity and joins with the anterior rami of S1 to S3 and part of S4 to form the **sacral plexus** (see Chapter 5, pp. 486–492).

Major nerves that originate from the lumbosacral plexus and leave the abdomen and pelvis to enter the lower limb include the femoral nerve, obturator nerve, sciatic nerve, superior gluteal nerve, and inferior gluteal nerve. Other nerves that also originate from the plexus and enter the lower limb to supply skin or muscle include the lateral cutaneous nerve of the thigh, nerve to the obturator internus, nerve to the quadratus femoris, posterior cutaneous nerve of the thigh, perforating cutaneous nerve, and branches of the ilio-inguinal and genitofemoral nerves.

Femoral nerve

The **femoral nerve** carries contributions from the anterior rami of L2 to L4 and leaves the abdomen by passing through the gap between the inguinal ligament and superior margin of the pelvis to enter the femoral triangle on the anteromedial aspect of the thigh (Fig. 6.34 and Table 6.1). In the femoral triangle it is lateral to the femoral artery. The femoral nerve:

- innervates all muscles in the anterior compartment of the thigh,
- in the abdomen, gives rise to branches that innervate the iliacus and pectineus muscles, and
- innervates skin over the anterior aspect of the thigh, the anteromedial side of the knee, the medial side of the leg, and the medial side of the foot.

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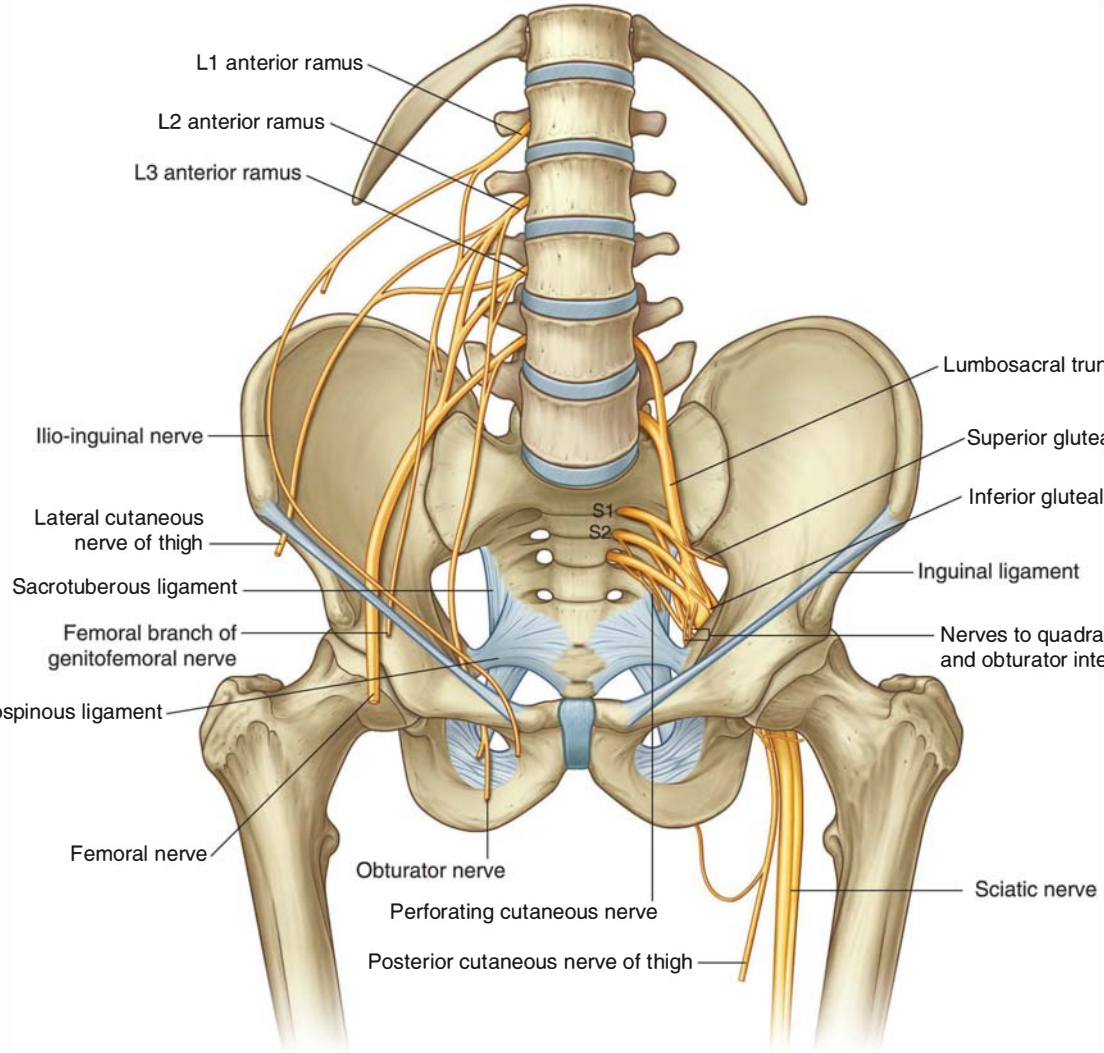


Fig. 6.35 Branches of the lumbosacral plexus.

Table 6.1 Branches of the lumbosacral plexus associated with the lower limb

Branch	Spinal segments	Function: motor	Function: sensory (cutaneous)
Ilio-inguinal	L1	No motor function in lower limb, but innervates muscles of the abdominal wall	Skin over anteromedial part of upper thigh and adjacent skin of perineum
Genitofemoral	L1, L2	No motor function in lower limb, but genital branch innervates cremaster muscle in the wall of the spermatic cord in men	Femoral branch innervates skin on anterior central part of upper thigh; the genital branch innervates skin in anterior part of perineum (anterior scrotum in men, and mons pubis and anterior labia majora in women)
Femoral	L2 to L4	All muscles in the anterior compartment of thigh; in the abdomen, also gives rise to branches that supply iliacus and pectineus	Skin over the anterior thigh, anteromedial knee, medial side of the leg, and the medial side of the foot
Obturator	L2 to L4	All muscles in the medial compartment of thigh (except pectineus and the part of adductor magnus attached to the ischium); also innervates obturator externus	Skin over upper medial aspect of thigh

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Table 6.1 Branches of the lumbosacral plexus associated with the lower limb—cont'd

Branch	Spinal segments	Function: motor	Function: sensory (cutaneous)
Sciatic	L4 to S3	All muscles in the posterior compartment of thigh and the part of adductor magnus attached to the ischium; all muscles in the leg and foot	Skin over lateral side of leg and foot, and over the sole and dorsal surface of foot
Superior gluteal	L4 to S1	Muscles of the gluteal region (gluteus medius, gluteus minimus, tensor fasciae latae)	
Inferior gluteal	L5 to S2	Muscle of the gluteal region (gluteus maximus)	
Lateral cutaneous nerve of thigh	L2, L3		Parietal peritoneum in iliac fossa; skin over anterolateral thigh
Posterior cutaneous nerve of thigh	S1 to S3		Skin over gluteal fold and upper medial aspect of thigh and adjacent perineum, posterior aspect of thigh and upper posterior leg
Nerve to quadratus femoris	L4 to S1	Muscles of gluteal region (quadratus femoris and gemellus inferior)	
Nerve to obturator internus	L5 to S2	Muscles of gluteal region (obturator internus and gemellus superior)	
Perforating cutaneous nerve	S2, S3		Skin over medial aspect of gluteal fold

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Obturator nerve

The **obturator nerve**, like the femoral nerve, originates from L2 to L4. It descends along the posterior abdominal wall, passes through the pelvic cavity and enters the thigh by passing through the obturator canal (Fig. 6.35 and Table 6.1). The obturator nerve innervates:

- all muscles in the medial compartment of the thigh, except the part of the adductor magnus muscle that originates from the ischium and the pectineus muscle, which are innervated by the sciatic and the femoral nerves, respectively;
- the obturator externus muscle; and
- skin on the medial side of the upper thigh.

Sciatic nerve

The **sciatic nerve** is the largest nerve of the body and carries contributions from L4 to S3. It leaves the pelvis through the greater sciatic foramen inferior to the piriformis muscle, enters and passes through the gluteal region (Fig. 6.35 and Table 6.1), and then enters the posterior compartment of the thigh where it divides into its two major branches:

- the common fibular nerve, and
- the tibial nerve.

Posterior divisions of L4 to S2 are carried in the common fibular part of the nerve and the anterior divisions of L4 to S3 are carried in the tibial part.

The sciatic nerve innervates:

- all muscles in the posterior compartment of the thigh,
- the part of the adductor magnus originating from the ischium,
- all muscles in the leg and foot, and
- skin on the lateral side of the leg and the lateral side and sole of the foot.

Gluteal nerves

The gluteal nerves are major motor nerves of the gluteal region.

The **superior gluteal nerve** (Fig. 6.35 and Table 6.1) carries contributions from the anterior rami of L4 to S1, leaves the pelvis through the greater sciatic foramen above the piriformis muscle, and innervates:

- the gluteus medius and minimus muscles, and
- the tensor fasciae latae muscle.

The **inferior gluteal nerve** (Fig. 6.35 and Table 6.1) is formed by contributions from L5 to S2, leaves the pelvis through the greater sciatic foramen inferior to the

piriformis muscle, and enters the gluteal region to supply the gluteus maximus.

Ilio-inguinal and genitofemoral nerves

Terminal sensory branches of the ilio-inguinal nerve (L1) and the genitofemoral nerve (L1, L2) descend into the upper thigh from the lumbar plexus.

The **ilio-inguinal nerve** originates from the superior part of the lumbar plexus, descends around the abdominal wall in the plane between the transversus abdominis and internal oblique muscles, and then passes through the inguinal canal to leave the abdominal wall through the superficial inguinal ring (Fig. 6.35 and Table 6.1). Its terminal branches innervate skin on the medial side of the upper thigh and adjacent parts of the perineum.

The **genitofemoral nerve** passes anteroinferiorly through the psoas major muscle on the posterior abdominal wall and descends on the anterior surface of the psoas major (Fig. 6.35 and Table 6.1). Its femoral branch passes into the thigh by crossing under the inguinal ligament where it is lateral to the femoral artery. It passes superficially to innervate skin over the upper central part of the anterior thigh.

Lateral cutaneous nerve of thigh

The **lateral cutaneous nerve of the thigh** originates from L2 and L3. It leaves the abdomen either by passing through the gap between the inguinal ligament and the pelvic bone just medial to the anterior superior iliac spine or by passing directly through the inguinal ligament (Fig. 6.35 and Table 6.1). It supplies skin on the lateral side of the thigh.

Nerve to quadratus femoris and nerve to obturator internus

The **nerve to the quadratus femoris** (L4 to S1) and the **nerve to the obturator internus** (L5 to S2) are small motor nerves that originate from the sacral plexus. Both nerves pass through the greater sciatic foramen inferior to the piriformis muscle and enter the gluteal region (Fig. 6.35 and Table 6.1):

- The nerve to the obturator internus supplies the gemellus superior muscle in the gluteal region and then loops around the ischial spine and enters the perineum through the lesser sciatic foramen to penetrate the perineal surface of the obturator internus muscle.
- The nerve to the quadratus femoris supplies the gemellus inferior and quadratus femoris muscles.

Posterior cutaneous nerve of thigh

The **posterior cutaneous nerve of the thigh** is formed by contributions from S1 to S3 and leaves the pelvic cavity through the greater sciatic foramen inferior to the piriformis muscle (Fig. 6.35 and Table 6.1). It passes vertically through the gluteal region deep to the gluteus maximus and enters the posterior thigh and innervates:

- a longitudinal band of skin over the posterior aspect of the thigh that continues into the upper leg, and
- skin over the gluteal fold, over the upper medial part of the thigh and in the adjacent regions of the perineum.

Perforating cutaneous nerve

The **perforating cutaneous nerve** is a small sensory nerve formed by contributions from S2 and S3. It leaves the pelvic cavity by penetrating directly through the sacrotuberous ligament (Fig. 6.35 and Table 6.1) and passes inferiorly around the lower border of the gluteus maximus where it overlaps with the posterior cutaneous nerve of the thigh in innervating skin over the medial aspect of the gluteal fold.

Arteries

Femoral artery

The major artery supplying the lower limb is the **femoral artery** (Fig. 6.36), which is the continuation of the external iliac artery in the abdomen. The external iliac artery becomes the femoral artery as the vessel passes under the inguinal ligament to enter the femoral triangle in the anterior aspect of the thigh. Branches supply most of the thigh and all of the leg and foot.

Superior and inferior gluteal arteries and the obturator artery

Other vessels supplying parts of the lower limb include the superior and inferior gluteal arteries and the obturator artery (Fig. 6.36).

The **superior and inferior gluteal arteries** originate in the pelvic cavity as branches of the internal iliac artery (see Chapter 5, pp. 495–498) and supply the gluteal region. The superior gluteal artery leaves the pelvis through the greater sciatic foramen above the piriformis muscle, and the inferior gluteal artery leaves through the same foramen but below the piriformis muscle.

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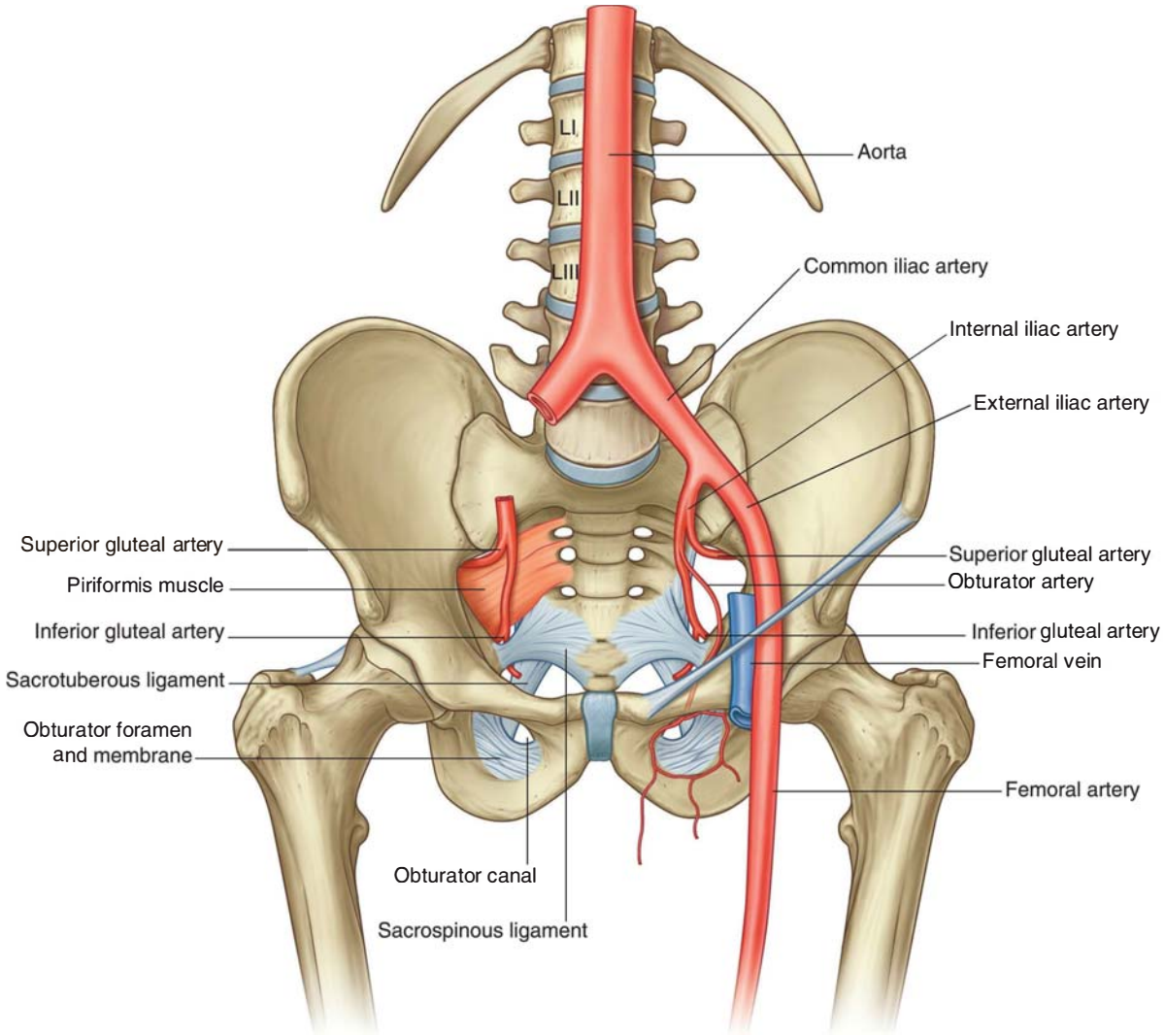


Fig. 6.36 Arteries of the lower limb.

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The **obturator artery** is also a branch of the internal iliac artery in the pelvic cavity (see Chapter 5, pp. 496–497) and passes through the obturator canal to enter and supply the medial compartment of the thigh.

Branches of the femoral, inferior gluteal, superior gluteal, and obturator arteries, together with branches from the internal pudendal artery of the perineum, interconnect to form an anastomotic network in the upper thigh and gluteal region. The presence of these anastomotic channels may provide collateral circulation when one of the vessels is interrupted.

Veins

Veins draining the lower limb form superficial and deep groups.

The deep veins generally follow the arteries (femoral, superior gluteal, inferior gluteal, and obturator). The major deep vein draining the limb is the **femoral vein** (Fig. 6.37). It becomes the external iliac vein when it passes under the inguinal ligament to enter the abdomen.

The superficial veins are in the subcutaneous connective tissue and are interconnected with and ultimately drain into the deep veins. The superficial veins form two major channels—the great saphenous vein and the small saphenous vein. Both veins originate from a dorsal venous arch in the foot:

- The **great saphenous vein** originates from the medial side of the dorsal venous arch and then ascends up the medial side of the leg, knee, and thigh to connect with the femoral vein just inferior to the inguinal ligament.
- The **small saphenous vein** originates from the lateral side of the dorsal venous arch, ascends up the posterior surface of the leg, and then penetrates deep fascia to join the popliteal vein posterior to the knee; proximal to the knee, the popliteal vein becomes the femoral vein.

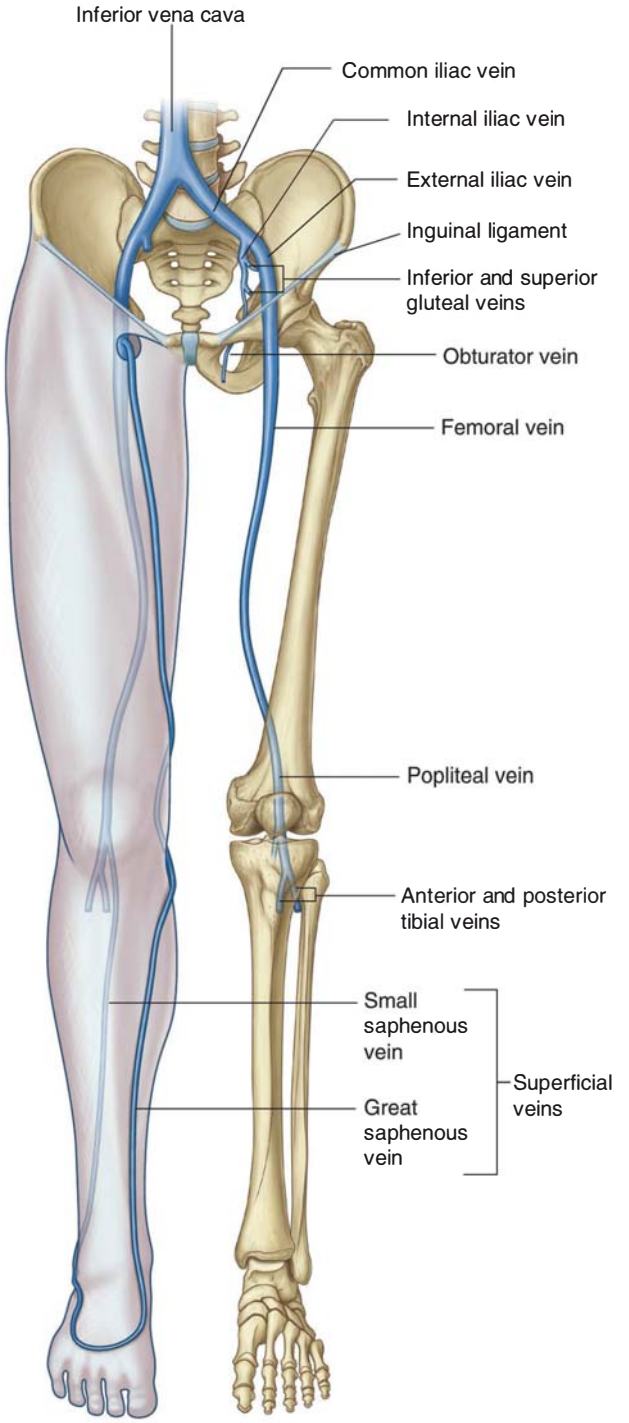


Fig. 6.37 Veins of the lower limb.

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In the clinic

Varicose veins

The normal flow of blood in the lower limbs is from the skin and subcutaneous tissues to the superficial veins, which drain via perforating veins to the deep veins, which in turn drain into the iliac veins and inferior vena cava.

The normal flow of blood in the venous system depends upon the presence of competent valves, which prevent reflux. Venous return is supplemented with contraction of the muscles in the lower limb, which pump the blood toward the heart. When venous valves become incompetent they tend to place extra pressure on more distal valves, which may also become incompetent. This condition produces dilated tortuous superficial veins (varicose veins) in the distribution of the great (long) and small (short) saphenous venous systems.

Varicose veins occur more commonly in women than in men, and symptoms are often aggravated by pregnancy. Some individuals have a genetic predisposition to developing varicose veins. Valves may also be destroyed when a deep vein thrombosis occurs if the clot incorporates the valve into its interstices; during the process of healing and recanalization the valve is destroyed, rendering it incompetent.

Typical sites for valvular incompetence include the junction between the great (long) saphenous vein and the femoral vein, perforating veins in the midthigh, and the junction between the small (short) saphenous vein and the popliteal vein.

Varicose veins may be unsightly, and soft tissue changes may occur with chronic venous incompetence. As the venous pressure rises, increased venular and capillary pressure damages the cells, and blood and blood products extrude into the soft tissue. This may produce a brown pigmentation in the skin, and venous eczema may develop. Furthermore, if the pressure remains high the skin may break down and ulcerate, and many weeks of hospitalization may be needed for this to heal.

Treatments for varicose veins include tying off the valve, "stripping" (removing) the great (long) and small (short) saphenous systems, and in some cases valvular reconstruction.

In the clinic

Deep vein thrombosis

Thrombosis may occur in the deep veins of the lower limb and within the pelvic veins. Its etiology was eloquently described by Virchow, who described the classic triad (venous stasis, injury to the vessel wall, and hypercoagulable states) that precipitates thrombosis.

In some patients a deep vein thrombosis (DVT) in the calf veins may propagate into the femoral veins. This clot may break off and pass through the heart to enter the pulmonary circulation, resulting in occlusion of the pulmonary artery, cardiopulmonary arrest, and death.

A significant number of patients undergoing surgery are likely to develop a DVT, so most surgical patients are given specific prophylactic treatment to prevent thrombosis. A typical DVT prophylactic regimen includes anticoagulant injections and graduated stockings (to prevent deep venous stasis and facilitate emptying of the deep veins).

Although physicians aim to prevent the formation of DVT, it is not always possible to detect it because there may be no clinical signs. Calf muscle tenderness, postoperative pyrexia, and limb swelling can be helpful clues. The diagnosis is made by duplex Doppler sonography or ascending venography.

If DVT is confirmed, intravenous and oral anticoagulation are started to prevent extension of the thrombus.

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Lymphatics

Most lymphatic vessels in the lower limb drain into superficial and deep inguinal nodes located in the fascia just inferior to the inguinal ligament (Fig. 6.38).

Superficial inguinal nodes

The **superficial inguinal nodes**, approximately ten in number, are in the superficial fascia and parallel the course of the inguinal ligament in the upper thigh. Medially, they extend inferiorly along the terminal part of the great saphenous vein.

Superficial inguinal nodes receive lymph from the gluteal region, lower abdominal wall, perineum, and superficial regions of the lower limb. They drain, via vessels that accompany the femoral vessels, into **external iliac nodes** associated with the external iliac artery in the abdomen.

Deep inguinal nodes

The **deep inguinal nodes**, up to three in number, are medial to the femoral vein (Fig. 6.38).

The deep inguinal nodes receive lymph from deep lymphatics associated with the femoral vessels and from the glans penis (or clitoris) in the perineum. They interconnect

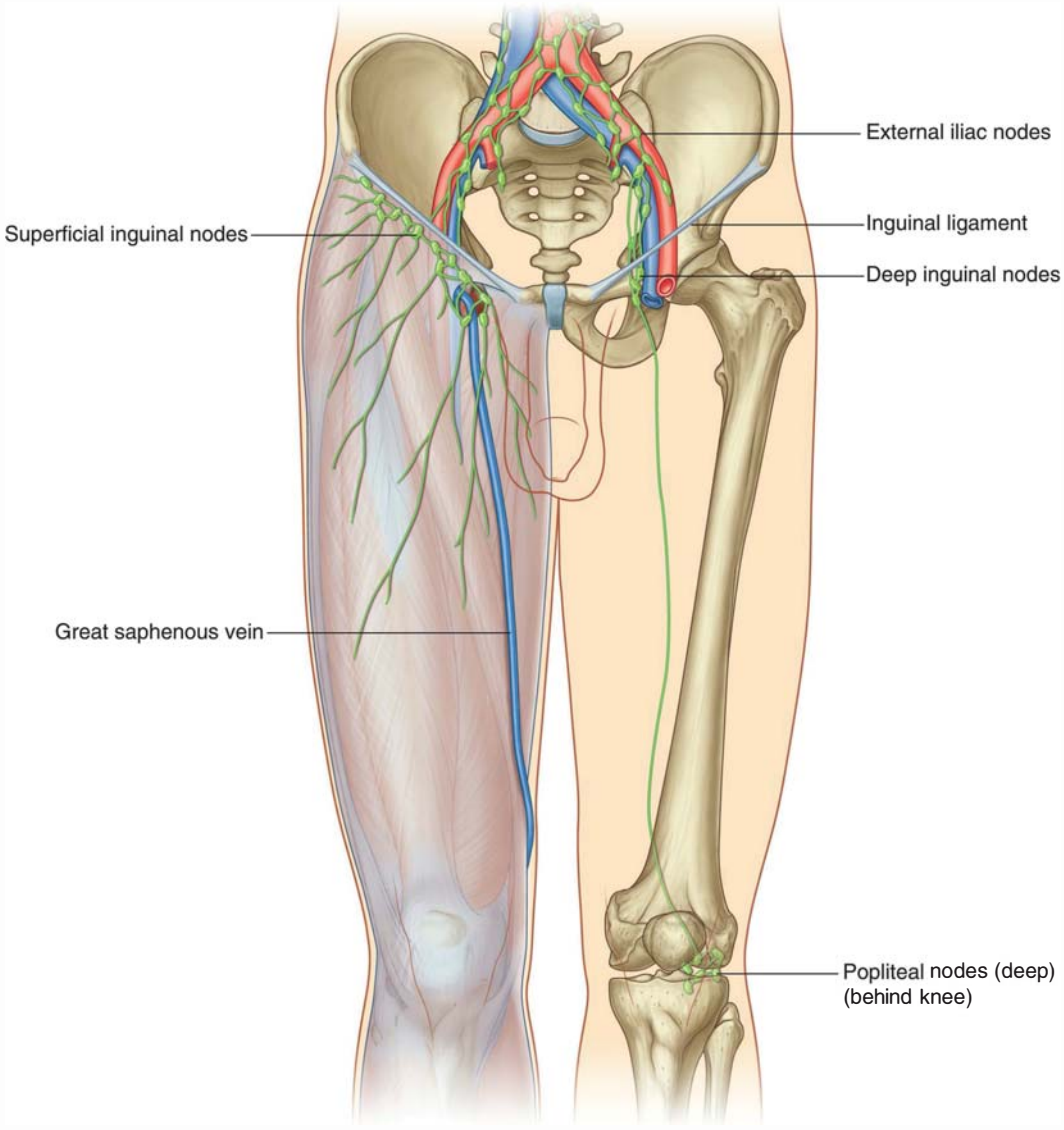


Fig. 6.38 Lymphatic drainage of the lower limb.

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with the superficial inguinal nodes and drain into the external iliac nodes via vessels that pass along the medial side of the femoral vein as it passes under the inguinal ligament. The space through which the lymphatic vessels pass under the inguinal ligament is the femoral canal.

Popliteal nodes

In addition to the inguinal nodes, there is a small collection of deep nodes posterior to the knee close to the popliteal vessels (Fig. 6.38). These **popliteal nodes** receive lymph from superficial vessels, which accompany the small saphenous vein, and from deep areas of the leg and foot. They ultimately drain into the deep and superficial inguinal nodes.

Deep fascia and the saphenous opening

Fascia lata

The outer layer of deep fascia in the lower limb forms a thick "stocking-like" membrane, which covers the limb and lies beneath the superficial fascia (Fig. 6.39A). This deep fascia is particularly thick in the thigh and gluteal region and is termed the **fascia lata**.

The fascia lata is anchored superiorly to bone and soft tissues along a line of attachment that defines the upper margin of the lower limb. Beginning anteriorly and circling laterally around the limb, this line of attachment includes the inguinal ligament, iliac crest, sacrum, coccyx, sacrotuberous ligament, inferior ramus of the pubic bone, body of the pubic bone, and superior ramus of the pubic bone.

Inferiorly, the fascia lata is continuous with the deep fascia of the leg.

Iliotibial tract

The fascia lata is thickened laterally into a longitudinal band (the **iliotibial tract**), which descends along the lateral margin of the limb from the tuberculum of the iliac crest to a bony attachment just below the knee (Fig. 6.39B).

The superior aspect of the fascia lata in the gluteal region splits anteriorly to enclose the tensor fasciae latae muscle and posteriorly to enclose the gluteus maximus muscle:

- The tensor fasciae latae muscle is partially enclosed by and inserts into the superior and anterior aspects of the iliotibial tract.
- Most of the gluteus maximus muscle inserts into the posterior aspect of the iliotibial tract.

The tensor fasciae latae and gluteus maximus muscles, working through their attachments to the iliotibial tract, hold the leg in extension once other muscles have extended the leg at the knee joint. The iliotibial tract and its two associated muscles also stabilize the hip joint by preventing lateral displacement of the proximal end of the femur away from the acetabulum.

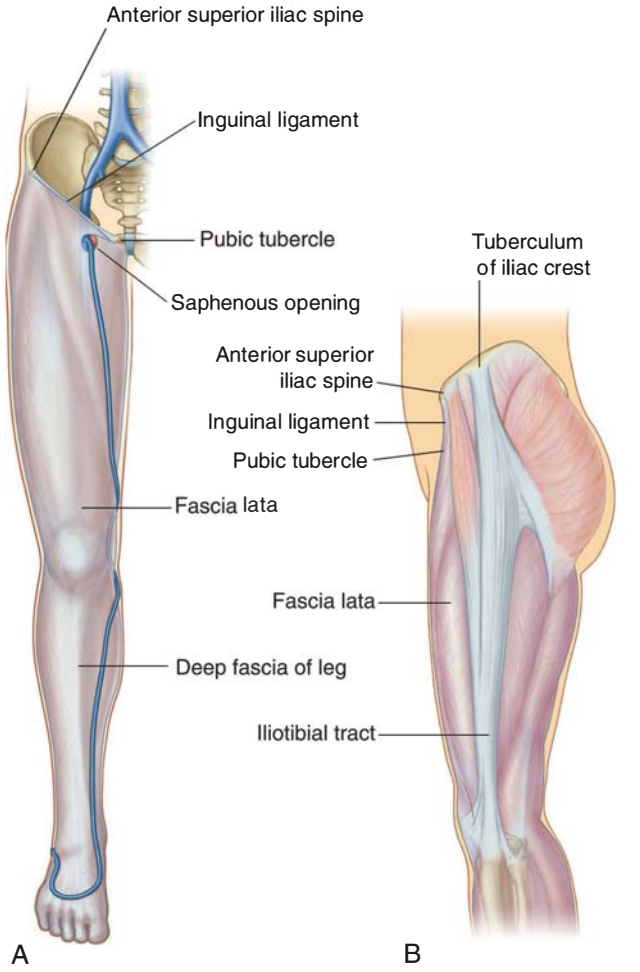


Fig. 6.39 Fascia lata. **A.** Right limb. Anterior view. **B.** Lateral view.

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Saphenous opening

The fascia lata has one prominent aperture on the anterior aspect of the thigh just inferior to the medial end of the inguinal ligament (the **saphenous opening**), which allows the great saphenous vein to pass from superficial fascia through the deep fascia to connect with the femoral vein (Fig. 6.40).

The margin of the saphenous opening is formed by the free medial edge of the fascia lata as it descends from the inguinal ligament and spirals around the lateral side of the great saphenous vein and medially under the femoral vein to attach to the pectineal line (pecten pubis) of the pelvic bone.

Femoral triangle

The femoral triangle is a wedge-shaped depression formed by muscles in the upper thigh at the junction between the anterior abdominal wall and the lower limb (Fig. 6.41):

- The base of the triangle is the inguinal ligament.
- The medial border is the medial margin of the adductor longus muscle in the medial compartment of the thigh.
- The lateral margin is the medial margin of the sartorius muscle in the anterior compartment of the thigh.
- The floor of the triangle is formed medially by the pectineus and adductor longus muscles in the medial compartment of the thigh and laterally by the iliopsoas muscle descending from the abdomen.
- The apex of the femoral triangle points inferiorly and is continuous with a fascial canal (**adductor canal**), which descends medially down the thigh and posteriorly through an aperture in the lower end of one of the largest of the adductor muscles in the thigh (the adductor magnus muscle) to open into the popliteal fossa behind the knee.

The femoral nerve, artery, and vein and lymphatics pass between the abdomen and lower limb under the

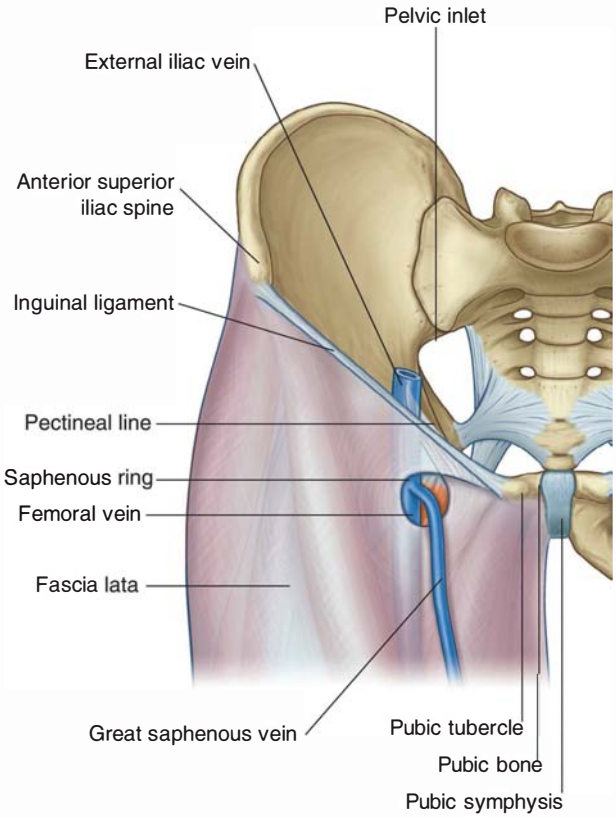


Fig. 6.40 Saphenous ring. Anterior view.

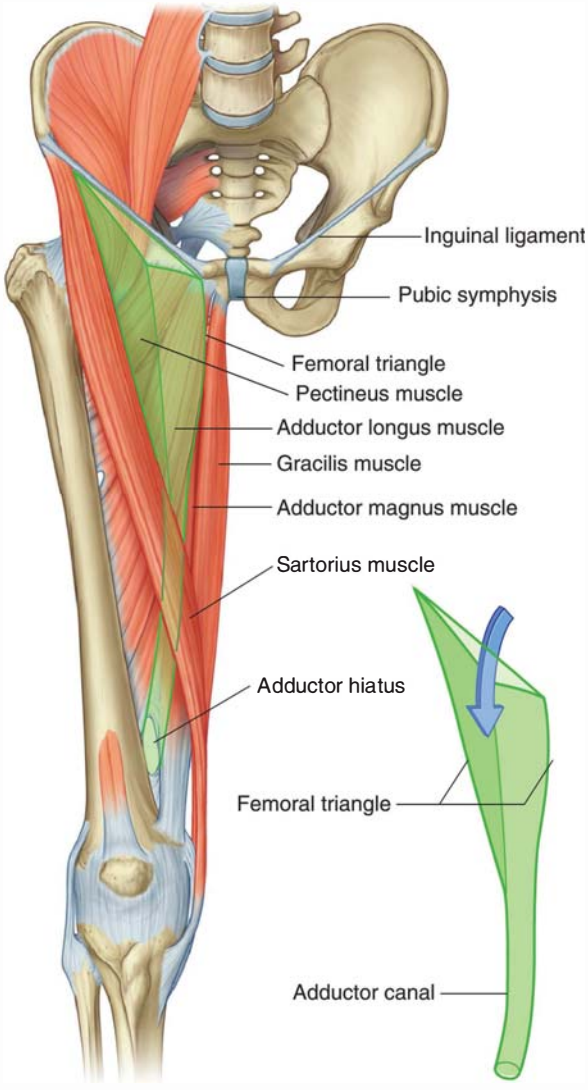


Fig. 6.41 Boundaries of the femoral triangle.

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